A SHARP-GROWTH EXTENSION OF THE FRACTIONAL DE GIORGI METHOD THROUGH DIMENSION EIGHT

Abstract. We develop an extension of the finite-layer method from a three-dimensional sheet base to bases of dimension at most six, assuming the sharp center-uniform variation bound for stable solutions. An augmented excess supplies escaping relative scales before the interaction estimate is known, and complete short-gap counting subsequently removes its artificial part. Count-adapted source cutoffs replace a dimension-dependent square-function estimate. Fixed-amplitude height cutoffs and complete column integrability avoid an unavailable bounded-height conclusion. Nested nonlinear comparison passes the complete signed force, while an inverse-power test excludes positive stable clusters through base dimension six. The minimizing closure in dimension eight uses a directed cone.

Draft status. This candidate proof was obtained by our team on October 09, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction

Fix $1/2 < s < 1$ and put $p = 2s \in (1, 2)$. The fractional Laplacian is normalized to have Fourier multiplier $|\xi|^p$. We consider

$$(-\Delta)^{p/2} v = v - v^3 \quad \text{in } \mathbb{R}^N, \qquad \partial_{x_N} v > 0. \tag{1}$$

For the higher-dimensional reduction, the dimension notation is

$$\begin{aligned} N &= \text{dimension of the monotone solution},\\ n = N - 1 &= \text{dimension of a stable endpoint},\\ d = n - 1 = N - 2 &= \text{sheet-base dimension}. \end{aligned} \tag{2}$$

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Thus the endpoint required when $N = 8$ lives in $\mathbb{R}^7$, and the limiting sheet equations live in $\mathbb{R}^6$. The final section obtains the lower-dimensional conclusions by normalized constant lifting.

The additional hypothesis is the following sharp variation estimate. For every bounded stable solution u in $\mathbb{R}^n$, $n \le 7$, there is a finite constant C_u such that

$$\int_{B_R(z)} |Du|\,dx \le C_u R^{n-1} \qquad (z \in \mathbb{R}^n,\ R \ge 1). \tag{3}$$

Uniformity in z is part of this hypothesis. The constant may depend on the fixed solution and on s.

The central intermediate result is the stable endpoint classification in $\mathbb{R}^7$: a bounded stable cubic solution satisfying (3) is a constant well or a translated and rotated one-dimensional layer; see theorem 12.3. Its growth constant may depend on that individual solution. Applying this classification to the two directional endpoints yields the monotone conclusion below.

The main quantitative conclusions of this paper are these. Augmented records have relative normal scale $D \ge \sqrt{R}$ in every base dimension. Actual complete root counting gives $R^2/D^{p+1} \le C$ before fine extraction and then makes the artificial record budget negligible. The amplitude-tail exponent is $2/d$, hence $1/3$ at $d = 6$. Source screening works for every $p \in (1, 2)$ in every fixed base dimension. The final scalar cluster contradiction holds for all $d \le 6$.

Theorem 1.1 (Sharp stable growth and the monotone problem)**.** *Fix $s \in (1/2, 1)$. Assume that for each $1 \le n \le 7$ every bounded stable classical solution u of*

$$(-\Delta)^s u = u - u^3 \quad \text{in } \mathbb{R}^n$$

has a finite constant C_u such that

$$\int_{B_R(z)} |Du|\,dx \le C_u R^{n-1} \qquad (z \in \mathbb{R}^n,\ R \ge 1).$$

Then, for every $1 \le N \le 8$ and every bounded classical solution v of

$$(-\Delta)^s v = v - v^3 \quad \text{in } \mathbb{R}^N, \qquad \partial_{x_N} v > 0,$$

there exist $e \in S^{N-1}$ with $e_N > 0$, $a \in \mathbb{R}$, and a strictly increasing centered one-dimensional layer q such that

$$v(x) = q(e \cdot x - a), \qquad (-\partial_t^2)^s q = q - q^3, \quad q(0) = 0, \quad \lim_{t \to \pm\infty} q(t) = \pm 1.$$

Every fractional Laplacian has Fourier symbol $|\xi|^{2s}$, and no directional limit of v is prescribed.

The proof is organized sequentially. Section 2 fixes the uniform analytic inputs. Section 3 establishes the macroscopic geometric data. Section 4 constructs augmented records, and Section 5 controls complete roots and absorbs the artificial budget. Section 6 treats multiplicity one. Section 7 gives strong count cutoffs and the source commutator estimate. Section 8 removes height truncations and proves complete column integrability. Section 9 constructs the actual complete ranks and their profiles, and Section 10 establishes the nested nonlinear comparison and selector replacement. Section 11 passes the signed force and covariance. Section 12 excludes the resulting cluster and proves stable endpoint rigidity. Section 13 gives the directed-cone minimizing closure.

For reference, the scale bookkeeping is

quantity	scale or decay
total endpoint energy	R^d
relative budget	$S = a^2R^d = R^{d-2}D^2$
relative angular energy	a^2R^{d-2}
force normalization	$R^{d-2}D$
interaction ratio	R^2/D^{p+1}
amplitude tail	$K^{-2/d}$
scalar-cluster cutoff exponent	$d - 2(5p+1)/(p+1) < 0$ for $d \le 6$

At the largest required base dimension $d = 6$, the two remaining dimensional decay exponents are

$$2 - \frac{4}{d} - \frac{4}{p+1} = \frac{4}{3} - \frac{4}{p+1} < 0, \qquad d - \frac{2(5p+1)}{p+1} = -\frac{4(p-1)}{p+1} < 0.$$

The source-screening exponent is $-(p-1)^2/(p+1) < 0$ in every fixed dimension. These strict inequalities use $1 < p < 2$; no uniform estimate as p approaches either endpoint is asserted.

2. Analytic preliminaries

Lemma 2.1 (Bounded entire cubic solutions have uniform smoothness and closed equation and stability classes)**.** *Let $n \ge 1$, $1 < p < 2$, and let $A = (-\Delta)^{p/2}$ have Fourier symbol $|\xi|^p$. Every bounded classical entire solution of $Au = u - u^3$ satisfies $|u| \le 1$ and is smooth, with*

$$\|D^k u\|_{L^\infty(\mathbb{R}^n)} \le C_{n,p,k} \qquad (k = 0, 1, 2, \ldots).$$

Every sequence of such solutions has a subsequence converging locally in every C^k topology to another entire solution. In particular this holds for arbitrary translates and rotations of one bounded solution. If every member of the sequence is stable, meaning

$$\int_{\mathbb{R}^n} |\xi|^p |\widehat{\phi}(\xi)|^2 d\xi - \int_{\mathbb{R}^n} (1 - 3u^2)\phi^2 dx \ge 0 \quad (\phi \in C_c^\infty(\mathbb{R}^n)),$$

with a unitary Fourier transform in this quadratic form, then its limit is stable.

Proof. First allow the constants to depend on $M = \|u\|_\infty$. This initial bootstrap does not use a range bound or stability. We first justify the distributional equation used in it. For $\varepsilon > 0$ let

$$A_\varepsilon w(X) = \frac{c_{n,p}}{2} \int_{|H|>\varepsilon} \frac{2w(X) - w(X+H) - w(X-H)}{|H|^{n+p}}\, dH.$$

For a compact smooth test ϕ, Fubini and translation of variables give $\int u A_\varepsilon \phi = \int \phi A_\varepsilon u$. On a compact neighborhood of $\operatorname{supp}\phi$, classical local regularity bounds the symmetric second difference of u by $C|H|^\theta$ for some $\theta > p$; for C^2_{loc} solutions one takes $\theta = 2$. Thus $A_\varepsilon u \to Au$ uniformly on the test support. Also

$$\|A_\varepsilon \phi - A\phi\|_1 \le C\varepsilon^{2-p} \|D^2\phi\|_1 \longrightarrow 0,$$

by the L^1 second-difference bound for ϕ. The boundedness of u therefore permits passage to the limit and proves $\int uA\phi = \int (Au)\phi$. The same near–far split gives the useful test estimate

$$\|A\psi\|_1 \le C_{n,p}(\|\psi\|_1 + \|D^2\psi\|_1) \qquad (\psi \in C_c^\infty(\mathbb{R}^n)).$$

Both the truncation identity and this estimate use only boundedness and local classical regularity, rather than the cubic equation; they therefore apply to the other bounded classical functions considered below.

For regularity we use Theorem 1.1(a)–(b) of Ros-Oton–Serra[15], arXiv:1412.3892v1, p. 3. Set $L = -A$, so its right side is $u^3 - u$. The constant positive spectral density satisfies that

theorem's ellipticity assumptions. Part (a), applied on all translated unit balls, gives a global C^p bound depending only on n, p, M, since $p \notin \mathbb{N}$. For noninteger $r > 0$, the Hölder product rule makes the inhomogeneous space $C^r(\mathbb{R}^n)$ an algebra and gives

$$\|u^3 - u\|_{C^r(\mathbb{R}^n)} \le C_{n,r}\big(\|u\|_{C^r(\mathbb{R}^n)} + \|u\|^3_{C^r(\mathbb{R}^n)}\big).$$

Part (b) permits every positive input exponent. Whenever $u \in C^r(\mathbb{R}^n)$ and $r + p \notin \mathbb{N}$, it therefore raises the regularity to C^{r+p}, uniformly on translated balls. Its global C^r hypothesis is satisfied at each step, and the displayed polynomial bound controls the new norm in terms of the preceding bound. These local bounds give the corresponding global norm: pairs at distance less than $1/4$ lie in one controlled ball, and larger distances are controlled by the bounded derivatives. At integer sums choose a slightly smaller input exponent; each iteration may still gain at least $p/2$. Iteration gives bounded derivatives of every fixed order, with constants depending only on n, p, k, M. Uniform bounds one order higher and Arzela–Ascoli, followed by a diagonal over balls and derivative orders, give local compactness in every C^k topology.

The equation is closed under this convergence for a sequence with a common global amplitude bound. In the symmetric second-difference formula, the near part is dominated by $C|H|^{2-n-p}$ using the common Hessian bound; its integral over $|H| < \varepsilon$ is $O(\varepsilon^{2-p})$. The far part beyond L is $O(L^{-p})$ by the amplitude bound. On the intervening compact annulus local convergence passes the integral. Thus $Au_j \to Au$ pointwise and the cubic equation passes to the limit.

Now translate a sequence of points approaching $\sup u$. The compactness just proved applies to these translates with the fixed bound M, and the limiting solution attains the original supremum at zero. If that supremum were greater than one, its fractional Laplacian there would be nonnegative by the positive kernel, whereas $u - u^3$ would be negative. This is impossible. Apply the same argument to $-u$ to get the lower bound. Hence $|u| \le 1$, and the bootstrap constants can be taken with $M = 1$, giving the universal bounds in the statement. They now give the asserted compactness for any sequence of bounded entire cubic solutions, without an additional common-bound hypothesis.

Finally fix a compact smooth real test ϕ. Its Fourier kinetic energy is independent of the solution and finite. The potential coefficient $1 - 3u_j^2$ converges uniformly on its compact support, so its stability inequality passes to the limit directly. Rotation and translation preserve the equation and this quadratic form, proving all the stated variants. □

Lemma 2.2 (Positive massive resolvents and the convolution and derivative bounds needed by the cubic method)**.** *Let $n \ge 1$, $1 < p < 2$ and $\lambda > 0$. There is a strictly positive radial kernel $G_\lambda \in L^1(\mathbb{R}^n)$, smooth away from zero, whose convolution is the inverse of $(-\Delta)^{p/2} + \lambda$ on Schwartz functions. Its mass is $1/\lambda$. With constants depending only on n, p, λ, for $r = |X|$,*

$$G_\lambda(X) \asymp \max\{1, r^{p-n}\} \quad (0 < r \le 1), \qquad G_\lambda(X) \asymp r^{-n-p} \quad (r \ge 1),$$

$$|DG_\lambda(X)| \le Cr^{p-n-1} \quad (0 < r \le 1), \qquad |DG_\lambda(X)| \le Cr^{-n-p-1} \quad (r \ge 1).$$

In particular $DG_\lambda \in L^1$. The pointwise convolution estimates, understood for $X \ne 0$, are

$$G_\lambda * G_\lambda \le CG_\lambda, \qquad |DG_\lambda| * G_\lambda \le CG_\lambda.$$

If $K(X) = c_{n,p}|X|^{-n-p}$ is the positive kernel of the fractional Laplacian with Fourier symbol $|\xi|^p$, then

$$\frac{|DG_\lambda(X)|^2}{K(X)} \le CG_\lambda(X).$$

More precisely, the ratio of the left side to $G_\lambda(X)$ is at most Cr^{2p-2} near zero and at most Cr^{-2} at infinity.

Proof. Put $\alpha = p/2 \in (1/2, 1)$. We construct the kernel and its estimates rather than appealing to an unspecified stable-density bound. The Stieltjes representation is

$$\frac{1}{\lambda + t^\alpha} = \int_0^\infty \frac{\rho_\lambda(r)}{t+r} dr, \qquad \rho_\lambda(r) = \frac{\sin(\pi\alpha)}{\pi} \frac{r^\alpha}{r^{2\alpha} + 2\lambda r^\alpha \cos(\pi\alpha) + \lambda^2}.$$

To verify it, put $F(z) = (\lambda + z^\alpha)^{-1}$, with the branch $-\pi < \arg z < \pi$. There is no zero of $\lambda + z^\alpha$ on this branch, because $|\arg z^\alpha| < \pi\alpha < \pi$. For fixed $t > 0$, apply the Cauchy integral formula to $F(z)/(z-t)$ on a keyhole contour with radii $\varepsilon < t < R$. For R sufficiently large, $|F(z)| \le 2R^{-\alpha}$ on the outer circle, so its integral is $O(R^{-\alpha})$. On the inner circle F is bounded and $|z - t| \ge t/2$ for small ε, so the integral is $O(\varepsilon)$. The cut boundary values satisfy

$$F(-r + i0) - F(-r - i0) = -\frac{2i \sin(\pi\alpha) r^\alpha}{r^{2\alpha} + 2\lambda r^\alpha \cos(\pi\alpha) + \lambda^2}.$$

Taking account of the two orientations along the cut gives

$$F(t) = \frac{1}{2\pi i} \int_0^\infty \frac{F(-r - i0) - F(-r + i0)}{t + r}\, dr,$$

which is the claimed representation. The integrand is $O(r^\alpha)$ at zero for $t > 0$ and $O(r^{-1-\alpha})$ at infinity, so these passages are absolutely convergent. Finally $\rho_\lambda \ge 0$, and letting $t \downarrow 0$ by monotone convergence proves both the formula at zero and $\int_0^\infty \rho_\lambda(r) r^{-1} dr = 1/\lambda$.

Use $(t+r)^{-1} = \int_0^\infty e^{-v(t+r)} dv$ and positivity to write

$$\frac{1}{\lambda + t^\alpha} = \int_0^\infty e^{-vt} a_\lambda(v) dv, \qquad a_\lambda(v) = \int_0^\infty \rho_\lambda(r) e^{-vr} dr > 0.$$

It suffices first to treat $\lambda = 1$. Writing $b = \cos(\pi\alpha) \in (-1, 0)$ and using $2r^\alpha \le 1 + r^{2\alpha}$, we have

$$(1 + b)(1 + r^{2\alpha}) \le 1 + 2br^\alpha + r^{2\alpha} \le 1 + r^{2\alpha}.$$

Thus the denominator defining ρ_1 is comparable to $1 + r^{2\alpha}$, with positive constants depending on p, so $\rho_1(r) \asymp r^\alpha$ for $r \le 1$ and $\rho_1(r) \asymp r^{-\alpha}$ for $r \ge 1$. It follows that

$$a_1(v) \asymp v^{\alpha - 1} \quad (0 < v \le 1), \qquad a_1(v) \asymp v^{-\alpha - 1} \quad (v \ge 1).$$

For the upper bounds use respectively $\rho_1(r) \le Cr^{-\alpha}$ and $\rho_1(r) \le Cr^\alpha$ and integrate on the full positive line. For the lower bounds restrict respectively to $1/v \le r \le 2/v$ and $1/(2v) \le r \le 1/v$. These explicit Laplace comparisons prove both estimates including their transition range.

Let $h_v(X) = (4\pi v)^{-n/2} e^{-|X|^2/(4v)}$ be the heat kernel. Define

$$G_1(X) = \int_0^\infty a_1(v) h_v(X) dv.$$

Tonelli gives positivity and $\int G_1 = \int a_1 = 1$, the latter being the Stieltjes identity at zero. The heat multiplier $e^{-v|\xi|^2}$ shows that this convolution has multiplier $(1 + |\xi|^p)^{-1}$, so it is the asserted massive inverse. Strict positivity follows from $a_1, h_v > 0$. Differentiation under this integral is valid away from zero, with uniform integrable majorants on compact annuli, giving smoothness and radiality.

For $0 < r \le 1$ and $n > p$, the small-v part of the heat integral is comparable, by $v = r^2 z$, to

$$r^{p-n} \int_0^{r^{-2}} z^{\alpha - 1 - n/2} e^{-1/(4z)} dz.$$

The integral over $(0, \infty)$ is finite because $n > p$; restricting to $0 < z < 1$, which is contained in the integration interval, gives a positive lower bound. The remaining part $v \ge 1$ is uniformly

bounded and hence at most Cr^{p-n}. This proves the two-sided near estimate when $n > p$. If $n < p$, necessarily $n = 1$; at $X = 0$ the small-v power $v^{\alpha-1-n/2}$ has exponent greater than minus one. The whole heat integral is consequently bounded above for $r \le 1$, and restriction to $1 \le v \le 2$ gives a positive uniform lower bound. Thus $G_1 \asymp 1$ in this case. There is no logarithmic borderline $n = p$, since n is an integer and $1 < p < 2$.

For $r \ge 1$, the part $v \ge 1$ is comparable to

$$r^{-n-p} \int_{r^{-2}}^{\infty} z^{-\alpha-1-n/2} e^{-1/(4z)} dz.$$

Its integral is bounded above by the integral over $(0, \infty)$, which is finite, and below by the integral over $[1, 2]$. The part $v \le 1$ is at most $Ce^{-r^2/8}$: use $e^{-r^2/(4v)} \le e^{-r^2/8} e^{-1/(8v)}$, and integrate the latter exponential against $v^{\alpha-1-n/2}$. This gives the far estimate, including the transition range.

Differentiating the Gaussian gives $|Dh_v(X)| = (r/(2v))h_v(X)$. For $r \le 1$, the same split and substitution now give

$$|DG_1(X)| \le Cr^{p-n-1} \int_0^{r^{-2}} z^{\alpha-2-n/2} e^{-1/(4z)} dz + Cr \le Cr^{p-n-1}.$$

Here the integral over the full positive line is finite for every $n \ge 1$, because $n > p - 2$; also $p - n - 1 < 0$, so the Cr term is absorbed for $r \le 1$. For $r \ge 1$, the part $v \ge 1$ is at most

$$Cr^{-n-p-1} \int_{r^{-2}}^{\infty} z^{-\alpha-2-n/2} e^{-1/(4z)} dz \le Cr^{-n-p-1},$$

while the part $v \le 1$ is $O(re^{-r^2/8})$ and satisfies the same bound. These are the derivative estimates in both dimensional regimes. Their radial integrals are finite: near zero the exponent is $p - 2 > -1$, and at infinity it is $-p - 2 < -1$. These are also the weak derivatives: $\|Dh_v\|_1 = Cv^{-1/2}$, and $\int_0^{\infty} a_1(v) v^{-1/2} dv < \infty$, since its small-$v$ exponent is $\alpha - 3/2 > -1$ and its large-v exponent is $-\alpha - 3/2 < -1$. Fubini therefore justifies differentiating the heat mixture in distributions and in L^1, including at the origin.

The exact scaling

$$G_\lambda(X) = \lambda^{n/p-1} G_1(\lambda^{1/p} X)$$

gives positivity, mass $1/\lambda$, multiplier $(\lambda + |\xi|^p)^{-1}$ and all the stated bounds for every fixed $\lambda > 0$. Changing the natural transition radius $\lambda^{-1/p}$ to one affects only constants, which are allowed to depend on λ.

Here are the convolution details, with λ fixed. If $r = |X| \ge 4$, split at $|Y| = r/2$. On the first part $G_\lambda(X - Y) \le Cr^{-n-p}$ and $|DG_\lambda(X - Y)| \le Cr^{-n-p-1}$; on the second part $G_\lambda(Y) \le Cr^{-n-p}$. The L^1 norms of the other factors therefore give

$$(G_\lambda * G_\lambda)(X) \le Cr^{-n-p},$$
$$(|DG_\lambda| * G_\lambda)(X) \le C(r^{-n-p-1} + r^{-n-p}) \le Cr^{-n-p}.$$

The lower bound for $G_\lambda(X)$ proves both desired inequalities there.

For $n < p$, the proved kernel is globally bounded, so $G_\lambda * G_\lambda \le \|G_\lambda\|_\infty \|G_\lambda\|_1$ and $|DG_\lambda| * G_\lambda \le \|G_\lambda\|_\infty \|DG_\lambda\|_1$. Near zero these constants are dominated by CG_λ because that kernel has a positive lower bound. For $n > p$ and $0 < r \le 1/4$, split into $|Y| \le 2r$, $2r < |Y| \le 1$, and $|Y| > 1$. Writing $e = X/r$, the substitution $Y = rZ$ bounds these near parts respectively by

$$Cr^{2p-n} \int_{|Z| \le 2} |e - Z|^{p-n} |Z|^{p-n} dZ, \qquad Cr^{2p-n-1} \int_{|Z| \le 2} |e - Z|^{p-n-1} |Z|^{p-n} dZ.$$

The singularities at 0 and e are separated by one. Each integral is finite, uniformly over $|e| = 1$, because their local powers exceed $-n$: this uses $p > 0$ for the kernel factors and $p > 1$ for the derivative factor. The middle parts are bounded respectively by

$$C \int_{2r}^{1} s^{2p-n-1} ds, \qquad C \int_{2r}^{1} s^{2p-n-2} ds.$$

Each is a power, a logarithm, or a bounded term. All these terms are bounded by Cr^{p-n}, since $n > p$ and the improvements in the two possibly singular powers are respectively $p > 0$ and $p-1 > 0$. The far parts are uniformly bounded by the kernel tails. On the remaining compact annulus the two integrable singularities occur at distinct points with a fixed separation; their integrals are uniformly bounded by the same estimates, while G_λ has a positive lower bound there. This proves both pointwise convolution inequalities.

Finally use $K(X) = c_{n,p} r^{-n-p}$ and the two-sided bounds for G_λ. When $n > p$, near zero,

$$\frac{|DG_\lambda(X)|^2}{K(X)G_\lambda(X)} \le Cr^{2(p-n-1)+(n+p)-(p-n)} = Cr^{2p-2}.$$

When $n < p$, necessarily $n = 1$, the lower bound $G_\lambda \ge c > 0$ instead gives the near-zero upper bound $Cr^{2(p-2)+1+p} = Cr^{3p-3} \le Cr^{2p-2}$. At infinity the same calculation with the far powers gives Cr^{-2}. They are bounded in the required ranges, and positivity/continuity controls the intermediate annulus. This proves the last assertion and completes the construction. □

Lemma 2.3 (The positive massive resolvent represents bounded data and compares bounded distributional subsolutions)**.** *Let $n \ge 1$, $1 < p < 2$, $\lambda > 0$, and let $A = (-\Delta)^{p/2}$ have Fourier symbol $|\xi|^p$. Let G_λ be the kernel of lemma 2.2. For every real $f \in L^\infty(\mathbb{R}^n)$,*

$$u = G_\lambda * f$$

is the unique bounded distributional solution of $(A+\lambda)u = f$. Here the distributional equation means

$$\int_{\mathbb{R}^n} u(A+\lambda)\phi = \int_{\mathbb{R}^n} f\phi \qquad (\phi \in C_c^\infty(\mathbb{R}^n)).$$

For such tests $A\phi \in L^1$, so this convention is meaningful for bounded functions. More generally, if $v \in L^\infty$ satisfies $(A+\lambda)v \le f$ in distributions, then $v \le G_\lambda * f$ almost everywhere, and everywhere when v is continuous. The resolvent is order preserving and

$$\|G_\lambda * f\|_\infty \le \lambda^{-1} \|f\|_\infty.$$

Its continuous representative is uniformly continuous. Moreover,

$$D(G_\lambda * f) = DG_\lambda * f$$

as a weak derivative, with a bounded continuous right side. If $f \in W^{k,\infty}$, then for every multi-index α with $|\alpha| \le k$,

$$D^\alpha(G_\lambda * f) = G_\lambda * D^\alpha f, \qquad \|D^\alpha(G_\lambda * f)\|_\infty \le \lambda^{-1} \|D^\alpha f\|_\infty.$$

These weak derivatives have continuous representatives. In particular every bounded classical solution has the stated convolution representation, and differentiating that representation is valid under the displayed hypotheses.

Proof. By lemma 2.2 the even kernel G_λ is positive, belongs to L^1, has mass $1/\lambda$, and has weak derivative $DG_\lambda \in L^1$. Its Fourier convolution multiplier is $(\lambda+|\xi|^p)^{-1}$. Hence $u = G_\lambda * f$ is well defined and bounded with the claimed norm, and positivity proves order preservation. Translation continuity in L^1 gives

$$\sup_X |u(X+H) - u(X)| \le \|f\|_\infty \|G_\lambda(\cdot + H) - G_\lambda\|_1 \longrightarrow 0.$$

This supplies its uniformly continuous representative.

We extend the inverse identity to bounded data by duality. The second-difference estimate in the proof of lemma 2.1 gives, for every $\phi \in C_c^\infty$,

$$\|A\phi\|_1 \le C(\|\phi\|_1 + \|D^2\phi\|_1) < \infty.$$

Fubini in that same L^1 second-difference formula gives $\widehat{A\phi}(\xi) = |\xi|^p \widehat{\phi}(\xi)$. Put $g = (A+\lambda)\phi \in L^1$. Since $G_\lambda * g \in L^1$, its ordinary Fourier transform is

$$\widehat{G_\lambda * g}(\xi) = \frac{\widehat{g}(\xi)}{\lambda + |\xi|^p} = \widehat{\phi}(\xi).$$

These are Fourier transforms of integrable functions; the identity also holds at $\xi = 0$, since $\lambda > 0$. Uniqueness of the Fourier transform on L^1 therefore yields

$$G_\lambda * ((A+\lambda)\phi) = \phi \quad \text{almost everywhere.}$$

Evenness and Fubini, justified by $\|f\|_\infty \|G_\lambda\|_1 \|(A+\lambda)\phi\|_1 < \infty$, now give

$$\int (G_\lambda * f)(A+\lambda)\phi = \int f\,(G_\lambda * ((A+\lambda)\phi)) = \int f\phi.$$

Thus u is a bounded distributional solution.

Here is the maximum principle in the precise bounded distributional class. Suppose $z \in L^\infty$ and $(A+\lambda)z \le 0$ in distributions. Let ρ_ε be a nonnegative compactly supported smooth mollifier of mass one and put $z_\varepsilon = \rho_\varepsilon * z$. It is smooth and bounded, and at each fixed ε every derivative has a global bound depending only on that mollifier and $\|z\|_\infty$. Its classical fractional Laplacian is

$$(A+\lambda)z_\varepsilon(X) = \int z(Y)(A+\lambda)\rho_\varepsilon(X-Y)\,dY \le 0.$$

For the equality, use the symmetric-difference formula and $\int \|\delta_H^2 \rho_\varepsilon\|_1 |H|^{-n-p} dH < \infty$ to justify Fubini; the inequality is exactly the distributional subsolution tested with the nonnegative compact function $\rho_\varepsilon(X - \cdot)$.

If $M_\varepsilon = \sup z_\varepsilon > 0$, translate a sequence of points approaching that supremum. The uniform fixed-ε derivative bounds and a diagonal Arzela–Ascoli argument give a local C^2 limit z_*, with $z_*(0) = M_\varepsilon$ and $z_* \le M_\varepsilon$ globally. The subsolution inequality passes pointwise to this limit. Indeed the common Hessian bound makes the fractional integral over displacements of length below r be $O(r^{2-p})$, boundedness makes its tail beyond L be $O(L^{-p})$, and local convergence passes the intermediate annulus. At its global maximum $Az_*(0) \ge 0$ by the positive kernel. Consequently

$$(A+\lambda)z_*(0) \ge \lambda M_\varepsilon > 0,$$

a contradiction. Therefore $z_\varepsilon \le 0$ for every ε. Letting $\varepsilon \downarrow 0$ at almost every Lebesgue point gives $z \le 0$ almost everywhere. Applied to the difference between a bounded subsolution v and the already constructed u, this proves the comparison assertion. Applying it to both differences between two solutions proves uniqueness. For continuous representatives, an almost-everywhere inequality holds everywhere.

For a bounded classical solution w, the classical-to-distributional truncation argument at the beginning of the proof of lemma 2.1 applies independently of the nonlinearity. Precisely, on a compact neighborhood of each test support it uses only a bound $|2w(X) - w(X+H) - w(X-H)| \le C|H|^\theta$ with $\theta > p$, together with global boundedness for the far tail. This holds for the C^2_{loc} classical convention, or the local $C^{p+\alpha}$ convention with $\alpha > 0$. That argument gives $\int wA\phi = \int (Aw)\phi$. Thus every bounded classical solution of $(A+\lambda)w = f$ is

a distributional solution, and the uniqueness just proved gives $w = G_\lambda * f$. Continuity makes the equality pointwise.

Finally choose $G_{\lambda,N} \in C_c^\infty(\mathbb{R}^n)$ converging to G_λ in $W^{1,1}$; this is possible because $G_\lambda, DG_\lambda \in L^1$. For each such compact smooth kernel, Fubini with a compact test yields

$$D(G_{\lambda,N} * f) = DG_{\lambda,N} * f$$

in distributions. The two convolutions converge uniformly to $G_\lambda * f$ and $DG_\lambda * f$, respectively, since f is bounded. Passing to the limit proves the asserted weak derivative identity. Translation continuity of DG_λ in L^1 makes the right side bounded and uniformly continuous.

If $f \in W^{k,\infty}$, weak integration by parts against the compact kernel gives, for every $|\alpha| \le k$,

$$D^\alpha(G_{\lambda,N} * f) = G_{\lambda,N} * D^\alpha f.$$

Since $G_{\lambda,N} * f \to u$ uniformly and $G_{\lambda,N} * D^\alpha f \to G_\lambda * D^\alpha f$ uniformly, passage in distributions gives $D^\alpha u = G_\lambda * D^\alpha f$. Positivity and mass $1/\lambda$ give the displayed derivative norm bounds. Translation continuity of G_λ in L^1 gives continuous, in fact uniformly continuous, representatives of all these weak derivatives. The compact-kernel approximation justifies the integrations by parts without requiring decay or integrability of f or its derivatives. □

Lemma 2.4 (The normalized cubic layer has its exact algebraic tail)**.** *Let $1 < p < 2$, let $A = (-\partial_t^2)^{p/2}$ have Fourier symbol $|\tau|^p$, and let q be a bounded classical strictly increasing solution of*

$$Aq = q - q^3, \qquad q(0) = 0, \qquad \lim_{t\to-\infty} q(t) = -1, \quad \lim_{t\to+\infty} q(t) = 1.$$

Then q is smooth with bounded derivatives of every fixed order, $h = q' > 0$, $\int h = 2$, and

$$h(t) = c_{1,p}|t|^{-1-p} + o(|t|^{-1-p}) \quad (|t| \to \infty),$$

$$1 - q(t) = \frac{c_{1,p}}{p} t^{-p} + o(t^{-p}) \quad (t \to +\infty), \qquad 1 + q(t) = \frac{c_{1,p}}{p}|t|^{-p} + o(|t|^{-p}) \quad (t \to -\infty),$$

where

$$c_{1,p} = \frac{4^{p/2}\Gamma((1+p)/2)}{\sqrt{\pi}\,|\Gamma(-p/2)|}$$

is the positive kernel constant of A. In particular $h(t) \le C(1+|t|)^{-1-p}$ and $1 - |q(t)| \le C(1+|t|)^{-p}$. Each asymptotic is uniform under a bounded translation of t.

Proof. The bounded cubic bootstrap in lemma 2.1 gives the stated smoothness and uniform bounds on every fixed derivative. Monotonicity and the endpoint limits give $-1 < q < 1$, $h = q' \ge 0$, $h \in L^1 \cap L^\infty$, and $\int h = 2$. Differentiating the equation, justified by bounded higher derivatives and the symmetric second-difference formula, gives

$$(A + 2)h = \beta h, \qquad \beta = 3(1 - q^2) \ge 0, \qquad \beta(t) \to 0.$$

Let G_λ be the one-dimensional positive massive kernel of lemma 2.2, and set $f = \beta h$. The established derivative bounds and $h \in L^1 \cap L^\infty$ show that f is nonnegative, smooth, bounded and integrable. Apply lemma 2.3 with $n = 1$, $\lambda = 2$, the bounded classical solution h, and bounded right side f. It gives the pointwise representation

$$h = G_2 * f.$$

Substitution $r = q(t)$, or the ordinary chain rule and its endpoint values, gives the exact mass

$$\int_{\mathbb{R}} f(t)dt = \int 3(1 - q^2)q'\,dt = 3\int_{-1}^{1}(1 - r^2)dr = 4.$$

In particular f is nonzero, so strict positivity of G_2 in this representation proves $h > 0$ everywhere.

We first prove a pointwise polynomial bound on h. Let C_0 satisfy $G_2 * G_2 \le C_0 G_2$ as in lemma 2.2. Choose $0 < \varepsilon < 1$ with $\varepsilon C_0 < 1/2$ and a smooth compact cutoff $0 \le \chi \le 1$, equal to one on a sufficiently large interval, so $(1-\chi)\beta \le \varepsilon$. With $f_0 = \chi\beta h$ the representation gives

$$h \le G_2 * f_0 + \varepsilon G_2 * h.$$

The compact support and boundedness of f_0, together with the kernel's positive lower bound on compact intervals and its two-sided far bounds, give $G_2 * f_0 \le C_\chi G_2$ pointwise. Iterating the last inequality N times yields

$$h \le C_\chi G_2 \sum_{j=0}^{N-1} (\varepsilon C_0)^j + \varepsilon^N G_2^{*N} * h.$$

The last term is bounded by $(\varepsilon/2)^N \|h\|_\infty$ because $\int G_2 = 1/2$. Let $N \to \infty$. We obtain $h \le C G_2 \le C(1+|t|)^{-1-p}$. It follows that

$$f(t) = \beta(t)h(t) = o((1+|t|)^{-1-p}).$$

We compute the exact far coefficient from the Stieltjes density and heat mixture already established in lemma 2.2. Specializing those formulas to one dimension, with $\alpha = p/2 \in (1/2, 1)$, gives

$$G_\lambda(t) = \frac{1}{\sqrt{4\pi}} \int_0^\infty a_\lambda(v) v^{-1/2} e^{-t^2/(4v)} dv,$$
$$a_\lambda(v) = \int_0^\infty \rho_\lambda(r) e^{-vr} dr, \qquad \rho_\lambda(r) = \frac{\sin(\pi\alpha)}{\pi} \frac{r^\alpha}{r^{2\alpha} + 2\lambda r^\alpha \cos(\pi\alpha) + \lambda^2}.$$

The denominator in ρ_λ is bounded below by a positive multiple of $\lambda^2 + r^{2\alpha}$. Consequently $\rho_\lambda(r) \le C r^\alpha$ for all $r > 0$, and

$$\rho_\lambda(r) \sim \frac{\sin(\pi\alpha)}{\pi\lambda^2} r^\alpha \quad (r \downarrow 0).$$

After the substitution $w = vr$, dominated convergence with majorant $C w^\alpha e^{-w}$ gives

$$v^{\alpha+1} a_\lambda(v) \longrightarrow A_\lambda := \frac{\sin(\pi\alpha)\Gamma(\alpha+1)}{\pi\lambda^2}, \qquad a_\lambda(v) \le C v^{-\alpha-1}.$$

Substitute $v = |t|^2 z$ in the Gaussian representation. The majorant $C z^{-\alpha-3/2} e^{-1/(4z)}$ is integrable at both ends, so another dominated convergence gives

$$\lim_{|t|\to\infty} |t|^{1+p} G_\lambda(t) = \frac{A_\lambda}{\sqrt{4\pi}} \int_0^\infty z^{-\alpha-3/2} e^{-1/(4z)} dz$$
$$= \frac{A_\lambda 4^\alpha \Gamma(\alpha+1/2)}{\sqrt{\pi}} = \frac{c_{1,p}}{\lambda^2}.$$

The last equality uses $\Gamma(\alpha+1)\sin(\pi\alpha)/\pi = 1/|\Gamma(-\alpha)|$. This is exactly the normalization constant of the stated operator: integration by parts in the elementary Laplace integral gives

$$|\tau|^{2\alpha} = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - e^{-v\tau^2}) v^{-1-\alpha} dv.$$

Insert the one-dimensional Gaussian kernel in this identity. Its off-diagonal positive singular kernel has coefficient $4^\alpha\Gamma(\alpha+1/2)/(\sqrt{\pi}|\Gamma(-\alpha)|)$ by the same Gaussian substitution just evaluated. Thus the coefficient computed for G_λ agrees with the positive fractional-kernel constant for Fourier symbol $|\tau|^p$.

Return to $h = G_2 * f$. In the integral over $|s| \le |t|/2$ the quantity $|t|^{1+p} G_2(t-s)$ is uniformly bounded, tends for each fixed s to $c_{1,p}/4$, and is multiplied by the integrable function $f(s)$. Dominated convergence makes this part tend to $(c_{1,p}/4)\int f = c_{1,p}$. On $|s| > |t|/2$, the established little-oh bound on f gives

$$\sup_{|s|>|t|/2} f(s) = o(|t|^{-1-p}).$$

Its convolution against G_2 is thus $o(|t|^{-1-p})$, because the kernel has mass $1/2$. This proves the stated exact asymptotic for h at both ends. Finally

$$1 - q(t) = \int_t^\infty h(r)dr, \qquad 1 + q(t) = \int_{-\infty}^t h(r)dr.$$

To justify integration of the remainder, set $\epsilon_T = \sup_{|r|\ge T} \big||r|^{1+p} h(r) - c_{1,p}\big|$, which tends to zero. For $t > 0$,

$$\left| \int_t^\infty h(r)dr - \frac{c_{1,p}}{p} t^{-p} \right| \le \frac{\epsilon_t}{p} t^{-p},$$

and the same bound holds on the negative half-line after reflection. This proves the two well tails; integration of the already proved global bound on h gives their global upper bounds. For clarity, the translation assertion means that for every $B < \infty$,

$$\sup_{|b|\le B} \Big| |t|^{1+p} h(t+b) - c_{1,p} \Big| \longrightarrow 0 \qquad (|t| \to \infty),$$

and analogously for the two well tails with power p and coefficient $c_{1,p}/p$. Indeed $|t+b|/|t| \to 1$ uniformly on $|b| \le B$, while the remainders in the established one-variable limits are uniformly small at every sufficiently large absolute argument. No translation parameter enters the limit exchanges used above. □

3. Macroscopic geometry under sharp growth

The sharp growth bound is used before the relative-scale construction. The integral-varifold argument retains square-curvature control and excludes classical junctions. Euclidean area growth is retained when applying the stable Bernstein theorem. This supplies planar quantized macroscopic limits through sheet dimension six; it does not by itself produce the interaction-scale finite-rank force equations.

Lemma 3.1 (Bounded square curvature and vanishing cut boundary exclude classical junctions in every sheet dimension)**.** *Let $d \ge 2$, and let $M_j \subset \Omega \subset \mathbb{R}^{d+1}$ be oriented embedded smooth hypersurfaces, properly embedded relative to their cut domains, possibly cut along piecewise smooth boundaries. More explicitly, their closure inside Ω adds only the declared cut boundary, so no undeclared interior endpoint or missing sheet is permitted. Suppose on every compact subset of Ω that their areas and $\int_{M_j} |A_j|^2$ are uniformly bounded, that $\mathcal{H}^{d-1}(\partial M_j)$ tends to zero, and that $|M_j|$ converges as a varifold to an integral stationary varifold V. A cut does not change the underlying smooth hypersurface and no rounding of its boundary corners is made. Then there is no neighborhood of a singular point in which the support of V is a finite union of embedded $C^{1,\alpha}$ hypersurfaces-with-boundary meeting only along a common $C^{1,\alpha}$ boundary, for any $0 < \alpha < 1$.*

Proof. We first establish a special crossing obstruction. Suppose on a cylinder the limit is a finite integer-weighted collection of complete distinct hyperplanes with common $(d-1)$-dimensional axis, while the approximants have curvature integral $\int |A_j| \to 0$ and boundary measure tending to zero. Write the ambient coordinates as $(z, x, t) \in \mathbb{R}^{d-1} \times \mathbb{R}^2$, choosing the last direction so that all the planes have equations $t = bx$ with finite slopes b. Choose

$L > 4 + 4\max|b|$. After shrinking the cylinder and making a fixed uniform dilation, we may assume that it contains a product of a z ball with $[-2,2] \times [-L,L]$, together with a small outer margin. A fixed dilation preserves the convergence to zero of the curvature and boundary integrals. Restrict to a smaller z ball and a short positive x interval $I_+ \Subset (0,1)$ on which the finitely many plane branches are separated.

Write $\pi(z,x,t) = (z,x)$, and take a smaller connected base cylinder $Q \Subset \mathbb{R}^{d-1} \times I_+$. For a branch of slope b, choose a smooth nonnegative tube cutoff $\chi_b(t-bx)$, equal to one near that branch and supported away from every other branch above a neighborhood of $\overline{Q}$. The tube is compactly contained in the ambient cylinder in the t direction. Choose a smooth odd function ζ, with the sign of its argument, equal to that sign near each of the finitely many nonzero limiting components $\nu \cdot e_t$, and zero near zero. Project the weighted current $\chi_b \zeta(\nu_j \cdot e_t)[M_j]$ by π, using tests compactly supported above Q. Its top-dimensional density $\rho_{j,b}$ satisfies, for almost every base point,

$$\rho_{j,b}(z,x) = \sum_{X \in M_j \cap \pi^{-1}(z,x)} \chi_b(t-bx)|\zeta(\nu_j(X) \cdot e_t)| \geq 0.$$

Indeed the oriented projection factor is $\nu_j \cdot e_t$, while the coarea Jacobian is $|\nu_j \cdot e_t|$. The product of the odd weight with the orientation sign is $|\zeta|$. The sum is locally integrable by the area formula and the local area bound. Reversing an orientation changes both signs, so all weak-limit expressions depend only on the unoriented tangent plane.

The boundary formula for this weighted current bounds the distributional variation on each smaller base cylinder by

$$|D\rho_{j,b}| \leq C\mathcal{H}^{d-1}(\partial M_j) + C\int_{M_j} |A_j| + C\int_{M_j} |\chi_b'(t-bx)|\,|P_{TM_j}\nu_b|,$$

where all terms on the right are restricted to a fixed compact ambient receiver and ν_b is the fixed branch normal. Differentiating the smooth normal weight costs $C|A_j|$; differentiating χ_b gives a fixed multiple of $\chi_b' P_{TM_j}\nu_b$. No boundary of the auxiliary cylinder is present, since the tests are compactly supported there. The first two terms tend to zero by hypothesis. The last does also: its integrand is a continuous function of position and unoriented tangent plane, and on the limiting planes either the tube cutoff is zero or $P\nu_b = 0$. Varifold convergence makes $\rho_{j,b}$ converge weakly to the constant positive integer multiplicity q_b of that branch. BV Poincare, together with convergence of the means on smaller base cylinders, now gives $\rho_{j,b} \to q_b$ strongly in local L^1.

Choose position and tangent tolerances tending to zero sufficiently slowly. Varifold convergence and the area formula make the integrated number of projected incidences in each selected tube outside those tolerances tend to zero. Strong convergence to $q_b > 0$ therefore ensures a correctly positioned fiber point with tangent plane close to the branch plane outside a base set of vanishing measure. This holds simultaneously for the finite list of branches.

Slice by z. Sard's theorem gives, for almost every z, a smooth embedded curve away from the declared cut endpoints. Its slicing Jacobian is $J = |P_{\mathbb{R}^2_{x,t}}\nu_j|$. If v is its unit tangent, then its normal in the slice plane is $P_{\mathbb{R}^2}\nu_j/J$, so the scalar second fundamental form satisfies $A_j(v,v) = \kappa J$, up to the chosen signs. Thus $|\kappa|J \leq |A_j|$, and coarea gives

$$\int dz \int_{(M_j)_z} |\kappa| \leq \int_{M_j} |A_j| \longrightarrow 0$$

on a fixed compact cylinder containing the rectangle used below. The image under projection to z of the cut boundary has $(d-1)$-dimensional measure at most $\mathcal{H}^{d-1}(\partial M_j)$, since that

projection is 1-Lipschitz. Outside this vanishing exceptional set there is no interior cut endpoint; properness excludes any undeclared endpoint or accumulation. Fubini and Markov's inequality applied to the occupation error and to the total turning select a regular slice with no such endpoints, total turning tending to zero, and a common $x \in I_+$ at which two distinct branches have correctly positioned and directed points.

Use the rectangle $[-2,2] \times [-L,L]$ chosen above. Small total turning keeps the horizontal tangent component on each chosen arc bounded away from zero. Orient it positively. Each arc is then a graph with slope $b + o(1)$, cannot close, and cannot terminate in the rectangle. The margin in L prevents escape through a horizontal side before reaching $x = -1$ and $x = 1$. The two arcs consequently contain graphs on $[-1,1]$ with heights $b_1 x + o(1)$ and $b_2 x + o(1)$. Their height difference has opposite signs at the two endpoints. They intersect, contrary to embeddedness of the slice. This proves the special obstruction.

Suppose now that V has the classical junction in the statement, with common edge L. Shrink its neighborhood so the interiors $\Sigma_i \setminus L$ of the finitely many incident pieces are connected and disjoint. The restriction of V near any interior point is an integral stationary varifold supported on one $C^{1,\alpha}$ embedded hypersurface. The first constancy lemma, Bellettini–Wickramasekera, *Stable CMC integral varifolds of codimension 1* [3], Appendix A, Lemma A.1, arXiv:1802.00377v1, p. 85, applies: its support is connected and C^1, and its generalized mean curvature is zero, hence locally in L^q for every $q > d$. Thus V has a fixed positive integer multiplicity q_i on each connected sheet interior. The common edge has zero $\mathcal{H}^d$ measure, so there is no additional mass there. Stationarity on an interior graph is now the unweighted weak minimal-graph equation. Uniform ellipticity on compact graph charts and quasilinear Schauder regularity make each interior graph smooth. These facts are obtained before any edge conormal formula is used.

At an edge point the $C^{1,\alpha}$ parametrizations and their first derivatives converge under dilation to the corresponding linear half-hyperplane parametrizations. The area formula therefore gives, with the just-fixed weights q_i, the unique tangent varifold as a finite weighted union of those half-hyperplanes. In particular $\|V\|(B_r(z)) \le C_z r^d$ for small r. On a compact part of the edge the constants are uniform. This supplies the bounded rescaled area needed below, rather than inferring it just from finite unscaled mass.

Pass to a subsequence so the measures $|A_j|^2 \mathcal{H}^d \llcorner M_j$ converge weakly to a locally finite measure λ. At $\mathcal{H}^{d-1}$ almost every $z \in L$,

$$r^{2-d} \lambda(B_r(z)) \longrightarrow 0.$$

For $d > 2$, the set where this fails has sigma-finite $\mathcal{H}^{d-2}$ measure: restrict to compact sets and to points with upper $(d-2)$-density above $1/k$, then use arbitrarily small balls satisfying $\lambda(B_r) > r^{d-2}/k$ and the Vitali covering theorem. For $d = 2$ the exceptional set consists of atoms of λ and is countable. In either case the exceptional set has zero $\mathcal{H}^{d-1}$ measure.

Choose such an edge point z. For every fixed K, the density limit gives $r^{2-d}\lambda(B_{Kr}(z)) \to 0$, by applying that limit at radius Kr. Choose $r_i \downarrow 0$ successively so that the corresponding bounds hold with error tending to zero for every integer $K \le i$. After fixing r_i, choose an approximating index $j(i)$ sufficiently late. Weak convergence of the curvature measures, tested with a cutoff supported in $B_{2Kr_i}(z)$ and equal to one on $B_{Kr_i}(z)$, controls the rescaled curvature-square integral; the rescaling factor is r_i^{2-d}. At the same fixed radius the cut-boundary measure, whose factor is r_i^{1-d}, tends to zero as the approximating index increases. Varifold convergence similarly supplies the area bound with factor r_i^{-d}. Choose $j(i)$ to enforce these bounds and convergence against the first i members of a countable determining family

of compact varifold tests. Thus the rescaled hypersurfaces

$$M_i^* = r_i^{-1}(M_{j(i)} - z)$$

converge to the unique tangent junction cone and, for every fixed compact receiver, have bounded area, vanishing cut-boundary measure, and curvature-square integral tending to zero. Their curvature integrals tend to zero by Cauchy–Schwarz. All receivers remain within the original open set for large i.

Using the projector formulation of curvature[7], we record the boundary identity needed to identify this cone and later to pass square curvature to the limit. On a smooth hypersurface let $P = P_{TM}$ and $\mathsf{A}_{ikl} = P_{iq}\partial_q P_{kl}$, with tangential differentiation understood through any smooth ambient extension of P. For each fixed i and every compactly supported C^1 test $\Psi(X, P)$,

$$\int_M \left(P_{ij}\partial_{X_j}\Psi + \mathsf{A}_{ikl}\partial_{P_{kl}}\Psi + \mathsf{A}_{jij}\Psi \right) d\mathcal{H}^d = \int_{\partial M} \Psi \eta_i \, d\mathcal{H}^{d-1}. \tag{4}$$

Here η is the outward unit conormal, repeated indices are summed, and the matrix test is extended smoothly to a neighborhood of the orthogonal projectors. The identity is the tangential divergence theorem for $\Psi(X, P)Pe_i$: the two chain-rule terms are the first two terms displayed, and $\mathrm{div}_M(Pe_i) = \mathsf{A}_{jij}$. It applies face by face to the declared piecewise smooth boundary; its corners have zero $\mathcal{H}^{d-1}$ measure.

Take $\Psi = \chi(P)Y_i(X)$ and sum in i, where Y is compactly supported and χ is smooth. Both terms containing A tend to zero along the diagonal, since $|\mathsf{A}| = \sqrt{2}|A|$ and the curvature integral tends to zero. The boundary term tends to zero by the rescaled boundary bound. Varifold convergence therefore gives

$$\int \chi(P) \, \mathrm{div}_P Y \, dC = 0.$$

There are finitely many distinct unoriented planes among the tangent planes of the cone's half-hyperplanes. Choose χ equal to one on one such plane and zero on all the others. The displayed equation says that the weighted half-hyperplanes in that plane are stationary by themselves. The two possible half-hyperplanes with the given axis have opposite conormals, so their total integer weights must agree. Every occupied plane therefore occurs as a complete plane. If at least two distinct planes occurred, the special crossing obstruction proved above would apply to the rescaled embedded approximants. Hence all these planes coincide.

This tangent-plane conclusion holds at $\mathcal{H}^{d-1}$-almost every point of the edge. Such points are dense in every smaller edge patch, and each incident piece has a continuous tangent field up to that edge. The finitely many incident tangent planes consequently coincide at every edge point. At any such point the unique tangent varifold is the weighted union of the corresponding linear half-hyperplanes, as already established from the $C^{1,\alpha}$ parametrizations. It is stationary, because dilation preserves stationarity and the first-variation integrand passes under varifold convergence. All its half-hyperplanes now lie in a single plane. Stationarity forces equal positive total weight on its two sides, so at least one incident piece lies on either side.

Fix an edge point and project the nearby pieces onto their common tangent plane there. After shrinking the neighborhood this projection gives $C^{1,\alpha}$ graphs on the two sides of one common $C^{1,\alpha}$ projected edge. Choose one graph from either side. Their values and first derivatives agree along the edge, so their union is a $C^{1,\alpha}$ graph g over a full base ball B_{base}. It remains to check the equation across the edge; tangency alone would not supply that equation.

On each open side the scalar flux $Dg/\sqrt{1 + |Dg|^2}$ is smooth, divergence-free, and extends continuously to the projected edge. Its two traces agree. To justify integration by parts at

the edge, write that edge locally as $x_d = h(x')$, with $h \in C^{1,\alpha}$, and truncate the two domains at $x_d = h(x') \pm \varepsilon$. The flux is smooth on these truncated domains; the divergence theorem applies, and the flux, boundary graph area factors, and conormals converge uniformly to their continuous traces as $\varepsilon \downarrow 0$. With a compactly supported test there is no outer boundary contribution. The limiting edge terms from the two sides cancel because the base conormals are opposite and the flux traces agree. Thus

$$\int \frac{Dg}{\sqrt{1+|Dg|^2}} \cdot D\phi \, dx = 0 \qquad (\phi \in C_c^\infty(B_{\text{base}})).$$

The glued graph is a weak solution of the minimal-graph equation on the full ball. Its gradient is locally bounded, so the equation is uniformly elliptic there. Interior quasilinear Schauder regularity gives $C^{2,\alpha}$ and then smoothness by bootstrapping. Analytic elliptic regularity gives real analyticity, since the coefficients of this elliptic equation are real analytic functions of the gradient.

Pair one fixed piece on one side with any two pieces on the other side. The resulting analytic minimal graphs agree on the open side occupied by the fixed piece, hence agree on the connected base ball by analytic continuation. Repeating with the two sides interchanged shows that every incident piece belongs to the same smooth graph. The support near the selected edge point is therefore one smooth embedded minimal hypersurface. Stationarity gives constant integer multiplicity there by the same constancy theorem. The point is regular, contrary to the proposed classical junction at a singular point. This proves the lemma. □

Proposition 3.2 (Stable macroscopic limits with square curvature and Euclidean mass are finite parallel planes through sheet dimension six)**.** *Let $2 \le d \le 6$. Suppose V is an integral stationary d-varifold in $\mathbb{R}^{d+1}$ with $\|V\|(B_r(z)) \le Cr^d$ for every z and $r > 0$, with the approximating hypersurfaces of lemma 3.1. Suppose on its regular support the intrinsic scalar stability inequality holds for every compact test supported away from the singular support. Then V is a finite sum of parallel affine hyperplanes with constant positive integer multiplicities.*

Proof. The zero varifold gives the empty sum, so suppose that $V \neq 0$. Fix $\alpha = 1/4$. By lemma 3.1, V satisfies the structural hypothesis (S3) in Section 3 of Wickramasekera[14]. To apply his theorem, fix $a \in \operatorname{spt}\|V\|$ and $r > 0$, restrict V to $B_{2r}(a)$, and translate and dilate to $B_2(0)$. The resulting varifold is integral, has finite mass by the assumed growth bound, and has the origin in its support. Its stationarity is (S1). For $2 \le d \le 6$, (S2) asks for the unweighted intrinsic scalar stability inequality only on open balls disjoint from the singular support; the hypothesis supplies exactly those tests. In particular this application requires no preliminary size bound for the singular set and no extension of test functions across it. Theorem 3.1 applied to this varifold gives empty singular set in the interior. Since a and r were arbitrary, the entire support $S = \operatorname{spt}\|V\|$ is a smooth embedded minimal hypersurface, possibly disconnected.

We check the global hypotheses needed for the Bernstein theorem component by component. Let Σ be a connected component of S. It is closed in $\mathbb{R}^{d+1}$: if $x_j \in \Sigma$ converges to x, then $x \in S$, and a sufficiently small support neighborhood of x is a single connected smooth sheet. That neighborhood contains x_j for large j, so it connects x to Σ. The inclusion $\Sigma \hookrightarrow \mathbb{R}^{d+1}$ is therefore proper. It is also complete in its intrinsic metric: an intrinsic Cauchy sequence is ambient Cauchy, and a smooth local graph about its ambient limit gives intrinsic convergence.

Corollary 3.2 of Wickramasekera gives orientability on every ball restriction used above. This yields orientability of Σ without an unstated compatibility assumption in the exhaustion. Indeed any closed curve in Σ is contained in a sufficiently large ball, whose regular support is orientable by that corollary. Thus transport of a local orientation around any closed

curve preserves it. An orientation chosen at one point extends along paths independently of the path, giving a global orientation on Σ. Because the ambient space is oriented and the codimension is one, this supplies a global continuous unit normal, which is smooth by the smoothness of Σ; hence Σ is two-sided.

The constancy theorem on the connected smooth support and stationarity give

$$V \restriction \Sigma = q_\Sigma |\Sigma|, \qquad q_\Sigma \in \mathbb{N}.$$

Stationarity also gives zero classical mean curvature there. Since the singular support is now empty, every compactly supported C^1 function on Σ is among the stability tests in the hypothesis; if that inequality is written with varifold weight, divide by the constant q_Σ. Thus Σ is stable for compactly supported normal variations. Finally

$$\mathcal{H}^d(\Sigma \cap B_R(0)) \le q_\Sigma^{-1} \|V\|(B_R(0)) \le CR^d \qquad (R > 0),$$

so it has Euclidean area growth.

Theorem 1 of Bellettini[2] applies to every smooth properly immersed, two-sided, stable minimal hypersurface in $\mathbb{R}^{d+1}$ of dimension $d \in \{2, 3, 4, 5, 6\}$ with this area-growth bound. All of these hypotheses have just been checked for Σ. Its conclusion makes Σ, being connected, one affine hyperplane. Distinct components are disjoint, so their affine hyperplanes are parallel.

It remains to prove finiteness, since local regularity alone would not give it. Choose any finite collection of components $\Sigma_1, \dots, \Sigma_k$, with their positive integer multiplicities $q_1, \dots, q_k$. Each affine plane satisfies $\mathcal{H}^d(\Sigma_i \cap B_R(0))/(\omega_d R^d) \to 1$ as $R \to \infty$. The mass bound consequently gives

$$\sum_{i=1}^{k} q_i \le \liminf_{R\to\infty} \frac{\|V\|(B_R(0))}{\omega_d R^d} \le C/\omega_d.$$

Since this holds for every finite collection and each $q_i \ge 1$, there can be only finitely many components and their total integer multiplicity is finite. This proves the assertion. □

Lemma 3.3 (Parallel-plane blowdowns force an all-scale integer density and one linear plane)**.** *Let $d \ge 1$ and let ρ be a positive Radon measure on $\mathbb{R}^{d+1}$, absolutely continuous with respect to Lebesgue measure, with $\rho(B_R) \le CR^d$ for $R \ge 1$. Assume every diverging sequence R_j has a subsequence for which $R_j^{-d}(X \mapsto X/R_j)_\# \rho$ converges to a finite integer-weighted collection of parallel affine d-planes, with total multiplicity bounded by a fixed integer M. Then $\rho(B_R)/(\omega_d R^d)$ converges to an integer $m \in \{0, \dots, M\}$; every blowdown consists of a single linear plane with multiplicity m, with the zero varifold understood when $m = 0$. If a fixed point has a positive uniform d-dimensional large-ball lower mass bound, then $m \ge 1$.*

Proof. Write

$$b(t) = \frac{\rho(B_{e^t})}{\omega_d e^{dt}}, \qquad t \in \mathbb{R}.$$

This function is continuous: absolute continuity of ρ makes every sphere ρ-null, so dominated convergence applies to the indicators of balls on each bounded radius interval. It is bounded for $t \ge 0$ by the assumed upper mass bound.

We first identify precisely the limits of its diverging translates. Let $t_j \to \infty$ and $R_j = e^{t_j}$. By hypothesis, after passing to a subsequence,

$$\mu_j := R_j^{-d}(X \mapsto X/R_j)_\# \rho \rightharpoonup \mu = \sum_{l=1}^{N} m_l \mathcal{H}^d \llcorner P_l, \qquad m_l \in \mathbb{N}, \quad \sum_l m_l \le M,$$

where the P_l are parallel affine planes; an empty collection is allowed. Set $a_l = \operatorname{dist}(0, P_l)$. Every sphere has zero μ-measure, including a sphere tangent to one of the planes, because $d \geq 1$. Consequently, for every $r > 0$,

$$\mu_j(B_r) \longrightarrow \mu(B_r) = \omega_d \sum_l m_l (r^2 - a_l^2)_+^{d/2}.$$

This convergence is uniform when r ranges in a compact subinterval of $(0, \infty)$. Indeed, both ball-mass functions are nondecreasing, and the limiting function is continuous. Choose a finite partition on which its increments are arbitrarily small and sandwich each value of $\mu_j(B_r)$ between the values at the neighboring partition points. Pointwise convergence at those finitely many points proves uniform convergence. Multiplying by the continuous factor $(\omega_d r^d)^{-1}$, and then putting $r = e^s$, gives

$$b(t_j + s) \longrightarrow f(s) := \sum_l m_l (1 - a_l^2 e^{-2s})_+^{d/2} \quad \text{locally uniformly for } s \in \mathbb{R}.$$

Each such profile is nondecreasing. Moreover, if $f(0)$ is not an integer, at least one unweighted factor $(1 - a_l^2)_+^{d/2}$ lies strictly between zero and one, since otherwise every term at zero is an integer. For that index $0 < a_l < 1$, so its weighted term strictly increases on $[0, 1]$; hence

$$f(1) > f(0) \qquad \text{whenever } f(0) \notin \mathbb{Z}.$$

This argument concerns one actual subsequential limit and requires no compactness assertion for a parameter space of plane collections.

Suppose now that $\liminf_{t\to\infty} b(t) < \limsup_{t\to\infty} b(t)$. Choose a noninteger q strictly between these two numbers and $\epsilon > 0$ such that $\liminf b < q - \epsilon < q < \limsup b$. There are arbitrarily late times s_j with $b(s_j) > q$, followed by times when $b < q - \epsilon$. Let $\tau_j > s_j$ be the first intervening time at which $b(\tau_j) = q - \epsilon$, and let t_j be the last time in $[s_j, \tau_j]$ at which $b(t_j) = q$. Such first and last times exist by continuity on the corresponding compact interval. Then $t_j \to \infty$ and

$$b(t) < q \qquad (t_j < t \leq \tau_j).$$

Extract the translated profile just established along these t_j. It satisfies $f(0) = q$, $f \geq q$ on $[0, 1]$, and $f(1) > q$. Local uniform convergence first shows that $\tau_j - t_j > 1$ for all sufficiently large j: otherwise $b(\tau_j) = q - \epsilon$ would contradict $b(t_j + s) \geq q - o(1)$ uniformly for $0 \leq s \leq 1$. But then $b(t_j + 1) < q$, contradicting its convergence to $f(1) > q$. Thus $b(t)$ has a limit L.

For every plane blowdown extracted above, the same translated-profile identity now gives $f(s) = L$ for all $s \in \mathbb{R}$. No term with $a_l > 0$ is compatible with this identity: it is strictly increasing once $e^s > a_l$, while every other term is nondecreasing. Therefore all $a_l = 0$, and

$$L = \sum_l m_l =: m \in \{0, \ldots, M\}.$$

Parallel planes through the origin coincide, so this blowdown is one linear plane with multiplicity m, or the zero measure if $m = 0$. This also describes every possible blowdown: if a sequence of rescaled measures already converges, the hypothesis supplies a further subsequence with a plane limit, and uniqueness of the limit along that convergent sequence identifies the two. Thus no additional blowdown can arise from a loss of planes or from planes escaping to infinity.

Finally, suppose $\rho(B_R(z)) \geq cR^d$ for all sufficiently large R, at one fixed z, with $c > 0$. The inclusion $B_{R-|z|}(z) \subset B_R(0)$ gives

$$\liminf_{R\to\infty} \frac{\rho(B_R(0))}{\omega_d R^d} \geq \frac{c}{\omega_d} > 0.$$

Hence $m \geq 1$, as asserted. □

Proposition 3.4 (Sharp stable gradient growth gives the dimension-free detector and full extension data)**.** *Let $d \geq 2$, $1 < p < 2$, and let $u : \mathbb{R}^{d+1} \to [-1, 1]$ be a bounded smooth stable solution of $(-\Delta)^{p/2}u = u - u^3$. Suppose*

$$\int_{B_R(z)} |Du| \leq C_0 R^d \qquad (z \in \mathbb{R}^{d+1},\ R \geq 1).$$

Let U be its Poisson extension and normalize $d\mu = d_p y^{1-p} dX dy$ so the extension represents the Fourier multiplier $|\xi|^p$. Then:

1. *The complete extension energy and the well deficit satisfy*
$$\int_{B_R(z)} \left[W(u) + 1 - u^2 + \int_0^\infty |DU|^2 d_p y^{1-p} dy \right] \leq CR^d, \qquad W(t) = (1 - t^2)^2/4.$$
2. *Put $J_e = P_{e^\perp} D_X U$ for any unit e. For every fixed $L, \epsilon > 0$, call a point failed if its trace is not $C^2(B_L)$ within ϵ of a constant well or a translated rotated one-dimensional layer. Any disjoint family of unit balls centered at failed points in $B_R(z)$ has cardinality*
$$N \leq CR^{-2} \int_{B_{2R}(z)\times(0,\infty)} |J_e|^2 d\mu.$$
The constant is uniform in e.
3. *With $g = |D_X U|$, $\nu = D_X U/g$, the full angular inequality is*
$$\int g^2 |D\nu|^2 \eta^2 d\mu \leq \int g^2 |D\eta|^2 d\mu.$$
Consequently its left side is $O(R^{d-2})$ in a ball of radius R for a fixed-scale normalized cutoff. The derivative D here includes height.
4. *There are the uniform height tails*
$$\int_{B_R\times(0,h)} |DU|^2 d\mu \leq CR^d(h^{2-p} + h), \qquad \int_{B_R\times(T,\infty)} |DU|^2 d\mu \leq CR^d T^{1-p}.$$
For each receiver phase deficit $0 < \delta < 1/4$,
$$\int_{B_R \cap \{1-|u|\leq\delta\}} \left[W(u) + \int_0^\infty |DU|^2 d_p y^{1-p} dy \right] \leq C\delta^{(p-1)/p} R^d.$$

Proof. All bounded-data massive-resolvent representations and distributional Kato comparisons below are justified by lemma 2.3. Their right sides are bounded: the cubic field and its derivatives are uniformly bounded, and the squared-increment source is bounded by the global gradient bound near the origin of its kernel and by boundedness in its tail. The derivative convolution identities in that lemma apply whenever they are used below.

Write $n = d + 1$, $A = (-\Delta)^{p/2}$ and $f(t) = t - t^3$. The mass-one Poisson kernel and the Dirichlet-to-Neumann coefficient are

$$P_y(H) = \kappa_{n,p} \frac{y^p}{(|H|^2 + y^2)^{(n+p)/2}},$$
$$\kappa_{n,p} = \frac{\Gamma((n+p)/2)}{\pi^{n/2}\Gamma(p/2)}, \qquad d_p = \frac{2^{p-1}\Gamma(p/2)}{\Gamma(1 - p/2)}.$$

Thus $\int P_y = 1$ and $U = P_y * u$. The identity $\Gamma(1-p/2) = (p/2)|\Gamma(-p/2)|$ gives $c_{n,p} = p d_p \kappa_{n,p}$. Consequently the symmetric second-difference formula for $\partial_y P_y * u$ yields $-d_p \lim_{y\downarrow 0} y^{1-p} U_y = Au = f(u)$, with the same normalization of A. The separate constant $c_{n,p}$ below denotes the

singular-kernel coefficient of A, as in Section 2. Constants in estimates may depend on n, p, C_0 and the fixed tested accuracies, but not on center, radius or unit direction.

Uniform global derivative bounds, compactness of translated or rotated cubic solutions, and closure of the equation and stability follow from lemma 2.1. We use those established conclusions throughout. Local convergence of their Poisson extensions, with every derivative on compact positive-height sets, follows by splitting each integrable derivative kernel into a compact part and a uniformly small far tail.

The all-center variation estimate and positivity give

$$\int_{B_R(z)} |D_X U(\cdot, y)| \le C_0 R^d$$

for every y. Poisson scaling gives $\partial_y P_y = -y^{-1} \operatorname{div}_H(HP_y)$. Integration by parts against $u(X-H)$ therefore yields

$$U_y(X,y) = -y^{-1} \int HP_y(H) \cdot Du(X-H) dH.$$

Its first moment is $y \int |H| P_1(H) dH < \infty$, because $p > 1$. Finite-radius cutoffs justify the integration by parts, with vanishing boundary terms. It follows that $\int_{B_R(z)} |U_y(\cdot, y)| \le CR^d$ as well. Uniform trace regularity gives $|D_X U| \le C$. For $y \le 1$, use $\int \partial_y P_y = 0$ and evenness to write $U_y = \frac{1}{2} \int \partial_y P_y(H)[u(X+H) + u(X-H) - 2u(X)] dH$. Here

$$|\partial_y P_y(H)| \le C y^{p-1} (|H|^2 + y^2)^{-(n+p)/2},$$

and the second difference is at most $C \min(|H|^2, 1)$. Its radial integrals over $(0, y)$, $(y, 1)$ and $(1, \infty)$ are bounded respectively by Cy, $Cy^{p-1} \int_y^1 r^{1-p} dr$, and $Cy^{p-1} \int_1^\infty r^{-1-p} dr$. Thus $|U_y| \le Cy^{p-1}$. For $y \ge 1$, differentiation of the Poisson kernel gives $|DU| \le C/y$. Multiply one of these pointwise factors by the proved L^1 factor and integrate with weight y^{1-p}. The two needed integrals are $\int_0^1 y^{1-p} dy$ and $\int_1^\infty y^{-p} dy$. This proves the complete gradient-energy bound and

$$\int_{B_R \times (0,h)} |DU|^2 d\mu \le CR^d (h^{2-p} + h), \qquad \int_{B_R \times (T,\infty)} |DU|^2 d\mu \le CR^d T^{1-p}.$$

The displayed small/large-height formulas first follow for $h \le 1$, $T \ge 1$; their remaining ranges follow from the full-energy bound.

We record the normalized trace-energy identity that makes extension stability precise. For $\phi \in C_c^\infty(\mathbb{R}^n)$, with the unitary Fourier transform,

$$\int_{\mathbb{R}^n \times (0,\infty)} |D(P_y * \phi)|^2 d\mu = \int_{\mathbb{R}^n} |\xi|^p |\widehat{\phi}(\xi)|^2 d\xi. \tag{5}$$

Indeed the convolution multiplier of P_y, obtained by the Gaussian integral representation of its denominator, is $\psi_p(y|\xi|)$, where

$$\psi_p(t) = \frac{1}{\Gamma(p/2)} \int_0^\infty s^{p/2-1} e^{-s - t^2/(4s)} ds.$$

Differentiation gives $(t^{1-p}\psi_p')' = t^{1-p}\psi_p$, $\psi_p(0) = 1$, and $-\lim_{t \downarrow 0} t^{1-p}\psi_p'(t) = d_p^{-1}$. For the last limit, substitute $s = t^2/(4r)$ in the derivative integral and apply dominated convergence to $r^{-p/2} e^{-r - t^2/(4r)}$. The same representation gives exponential decay of ψ_p and ψ_p' at infinity, since $s + t^2/(4s) \ge t$, with part of the exponential retained to integrate the powers of s. Multiplying the mode equation by ψ_p, integrating on (ε, T), and passing to the endpoints therefore gives $\int_0^\infty t^{1-p}[(\psi_p')^2 + \psi_p^2] dt = d_p^{-1}$. Plancherel, Tonelli and the substitution $t = y|\xi|$ prove (5).

If Φ is smooth and compactly supported in the closed upper half-space with trace ϕ, its energy is at least the left side of (5). To justify this without a compact-support assumption on $P_y * \phi$, set $v_\xi(y) = \widehat{\Phi}(\xi, y) - \psi_p(y|\xi|)\widehat{\phi}(\xi)$. For each $\xi \neq 0$, integration of the scalar mode equation against v_ξ gives zero cross term: its bottom boundary value is zero and $y^{1-p}\partial_y\psi_p(y|\xi|)$ has a finite limit there; at infinity $\widehat{\Phi} = 0$ and the mode decays exponentially. All cross terms are integrable after integration in ξ by Cauchy–Schwarz and the finite energies. The energy decomposition is consequently

$$\int |D\Phi|^2 d\mu = \int |D(P_y * \phi)|^2 d\mu + \int |D(\Phi - P_y * \phi)|^2 d\mu.$$

Trace stability and (5) now imply

$$\int |D\Phi|^2 d\mu - \int f'(u)\Phi(X,0)^2 dX \geq 0.$$

The same inequality applies to the compact tests formed from J below. To see the approximation at the bottom, apply the preceding Poisson estimates to each smooth trace derivative: J has bounded tangential derivatives and $J_y = O(y^{p-1})$. Flatten its height dependence to its trace in $0 < y < \delta$, with a smooth interpolation on $(\delta, 2\delta)$, and then mollify. The change in weighted gradient energy is bounded by $C(\delta^{2-p} + \delta^p)$, tending to zero, and the trace converges in L^2. For fixed norm regularization this also holds after composition with $(|J|^2+\epsilon^2)^{1/2}$. Thus these tests follow by continuity from smooth compact tests.

Every physical derivative is a Jacobi field with Robin coefficient $f'(u)$. The following norm calculation applies both to $J = D_X U$ and to $J = P_{e^\perp} D_X U$. First let η be a test compact in position and height, allowing a trace at zero, and put $g_\epsilon = (|J|^2 + \epsilon^2)^{1/2}$. Testing the derivative equations by $J_i\eta^2$ and summing gives

$$\int (\eta^2 |DJ|^2 + 2\eta \textstyle\sum_i J_i DJ_i \cdot D\eta) d\mu = \int f'(u)|J(X,0)|^2 \eta(X,0)^2 dX.$$

Subtract this identity from stability for $g_\epsilon\eta$. The mixed terms cancel exactly because $g_\epsilon Dg_\epsilon = \sum_i J_i DJ_i$, leaving

$$\int \eta^2(|DJ|^2 - |Dg_\epsilon|^2) d\mu \leq \int g_\epsilon^2 |D\eta|^2 d\mu - \epsilon^2 \int f'(u)\eta(X,0)^2 dX.$$

All ϵ^2 terms are finite at this stage because the test is compact and y^{1-p} is integrable at zero. Send $\epsilon \downarrow 0$, obtaining

$$\int \eta^2(|DJ|^2 - |D|J||^2) d\mu \leq \int |J|^2 |D\eta|^2 d\mu.$$

Define the nonnegative angular integrand to be zero almost everywhere on $\{J = 0\}$: Sobolev locality gives $DJ = 0$ there almost everywhere, so this agrees with the difference-of-squares expression. On $J \neq 0$ it is $|J|^2|D(J/|J|)|^2$. Now replace the test by $\eta(X)\chi_T(y)$, with $\chi_T = 1$ on $[0,T]$, zero above $2T$, and $|\chi_T'| \leq C/T$, and let $T \to \infty$. Its height error is at most $CT^{-2}\int_{\operatorname{supp}\eta\times(T,2T)} |J|^2 d\mu$, which tends to zero by the full-height bound already proved. Fatou gives the left side, while dominated convergence gives the physical-gradient term. Thus for every compact physical test η, the displayed angular inequality holds at full height. For $J = D_X U$ this proves assertion 3; its right side is $O(R^{d-2})$ for a physical cutoff with $|D\eta| \leq C/R$ supported in $B_{2R}(z)$.

Let $\mathcal{A}_e = |DJ_e|^2 - |D|J_e||^2$, with the zero-set convention just established. We need a local form of zero-charge rigidity. If $\mathcal{A}_e$ vanishes on a nonempty open positive-height box, then either J_e vanishes there or, on some smaller connected open ball, $J_e/|J_e|$ is a fixed unit vector $v \in e^\perp$. These local linear relations extend globally because the Poisson derivatives

are real analytic for $y > 0$. This analyticity follows directly from the kernel: near (X_0, y_0), complexify X, y by less than cy_0. The denominator $(X - H) \cdot (X - H) + y^2$ has positive real part bounded below by $c'(|X_0 - H|^2 + y_0^2)$; the kernel, with its local analytic branches, is therefore holomorphic and dominated by an integrable multiple of $(|X_0 - H|^2 + y_0^2)^{-(n+p)/2}$. Integration against bounded u preserves analyticity. The identity principle on the connected upper half-space now extends each component relation. If J_e is zero on the box, it is zero everywhere; otherwise $w \cdot D_X U = 0$ for every $w \perp e, v$. Passing to the trace shows that u depends on at most the two coordinates in $\operatorname{span}\{e, v\}$, or on the single e coordinate in the zero case.

We justify the stability descent used here explicitly. Write such a trace as $v(x)$, $x \in \mathbb{R}^2$, independent of $z \in \mathbb{R}^{n-2}$. Test original stability with $\phi(x)\zeta(z/L)$ and divide by $L^{n-2}\|\zeta\|_2^2$. After the change of Fourier variable $\eta \mapsto \eta/L$, its kinetic part is

$$\|\zeta\|_2^{-2} \iint (|\xi|^2 + |\eta|^2/L^2)^{p/2} |\widehat{\phi}(\xi)|^2 |\widehat{\zeta}(\eta)|^2 d\xi d\eta.$$

It tends to $\int |\xi|^p |\widehat{\phi}|^2$. Dominated convergence follows from $(a + b)^{p/2} \le a^{p/2} + b^{p/2}$ and the smooth compact tests. The potential term is exactly the two-dimensional one. The descended equation has the same Fourier multiplier, by integrating the normalized physical kernel in the invariant coordinates. The two-dimensional Poisson kernel is its corresponding marginal, so trace stability implies extension stability by the same Dirichlet-minimization argument above. This descent also covers a trace depending on only one coordinate by regarding it as a two-variable function independent of the second coordinate.

Cabré–Sire, *Nonlinear equations for fractional Laplacians II*[6], Theorem 2.12, arXiv:1111.0796v1, p. 11, therefore applies with $s = p/2$: a bounded two-variable solution with stable extension is one-dimensional strictly monotone or constant. Its nonlinearity condition is $f \in C^{1,\gamma}$, $\gamma > \max\{0, 1 - p\} = 0$. The cubic has this regularity on its range; if a global norm is required, replace it outside $[-2, 2]$ by a smooth function with bounded derivatives, without affecting the solution or its stability. In their boundary convention $a = 1 - p$ and the boundary nonlinearity is $f_{\mathrm{CS}} = (2 - p)f/d_p$, as explained in their Section 2: dividing our extension inequality by d_p gives precisely their stability coefficient $f'_{\mathrm{CS}}/(1 + a) = f'/d_p$. Thus no dilation or change of the trace equation is being made. A constant limit must be $-1, 0$ or 1 by the equation, and zero is unstable: a test dilated to scale L has kinetic energy $O(L^{2-p})$ and negative potential term of order L^2. For a nonconstant monotone profile, translate to either end. Its limits are constant solutions, and stability passes to those limits by the compactness proved above, so the end values are the two stable equilibria -1 and 1. Their Theorem 2.4 and Remark 2.5 give existence and uniqueness, up to translation, of the increasing cubic layer. Fix its translation by $q(0) = 0$. The reflected function $-q(-t)$ is another centered increasing solution, so uniqueness makes q odd. Thus every increasing detector layer is a translate of this same q, and reversing the unit normal includes the decreasing orientation. Constant stable limits are the wells.

Fix the box $\mathcal{Q} = B_{1/4}(0) \times (1, 2)$. There is $c_{L,\epsilon} > 0$ such that every failed center X_0 and every unit e satisfy

$$\int_{\mathcal{Q}+(X_0,0)} \mathcal{A}_e \, d\mu \ge c_{L,\epsilon}.$$

Otherwise translate failed centers to zero, extract a converging sequence of directions, and use lemma 2.1 for local smooth convergence of the stable fields. Their Poisson derivatives converge on this box. Wherever the limiting $J_e \neq 0$, the angular integrands converge pointwise; on its zero set the limiting integrand is zero almost everywhere and all the approximating integrands

are nonnegative. Fatou therefore gives zero limiting charge on $\mathcal{Q}$. The local rigidity and two-variable classification just proved make that limiting trace a constant well or a translated rotated layer. Local C^2 convergence on a neighborhood of $\overline{B}_L$ contradicts the fixed failure accuracy. This proves the positive lower bound uniformly in e.

For disjoint unit balls centered in $B_R(z)$, the corresponding radius-$1/4$ boxes are disjoint and lie in $B_{3R/2}(z) \times (1,2)$ when $R \geq 1$. Choose a physical cutoff equal to one on $B_{3R/2}(z)$, supported in $B_{2R}(z)$, with derivative at most C/R. Summing the preceding charges and using the full projected angular inequality gives

$$c_{L,\epsilon} N \leq \int \eta_R^2 \mathcal{A}_e \, d\mu \leq C R^{-2} \int_{B_{2R}(z)\times(0,\infty)} |J_e|^2 d\mu.$$

This proves assertion 2 for every $R \geq 1$.

By the complete gradient-energy bound, assertion 2 gives $N \leq C R^{d-2}$ for any fixed detector accuracy. Take $\theta = 1/\sqrt{2}$, a larger $\theta' \in (\theta, 1)$, and $L = 1$. For the normalized increasing cubic layer q, strict positivity of q', already given by lemma 2.4, gives $m = \min_{\{|q| \leq \theta'\}} q' > 0$; the set of parameters in this minimum is compact by the well limits. Choose detector accuracy below $\min\{1 - \theta, \theta' - \theta, m/2\}$. At a nonfailed center with $|u| \leq \theta$, a constant-well comparison is impossible, and a layer comparison has $|Du| \geq m/2$, uniformly in its rotation and translation. Choose a maximal disjoint family of failed unit balls with centers in $B_R(z)$. Balls with the same centers and radius two cover all failed centers in $B_R(z)$. Their union has volume at most $CN \leq C R^{d-2}$; on the rest of the central band the gradient floor bounds its volume by $(2/m) \int_{B_R(z)} |Du| \leq C R^d$. Consequently

$$|B_R(z) \cap \{|u| \leq 1/\sqrt{2}\}| \leq C R^d.$$

Put $z_u = 1 - u^2$ and

$$\Gamma_u(X) = c_{n,p} \int (u(X) - u(Y))^2 |X - Y|^{-n-p} dY.$$

The exact product identity $A(u^2) = 2uAu - \Gamma_u$ gives

$$(A + 1) z_u = (2z_u - 1) z_u + \Gamma_u \leq \mathbf{1}_{\{z_u \geq 1/2\}} + \Gamma_u,$$

since $0 \leq z_u \leq 1$. The segment formula, the all-center variation bound and $|u| + |Du| \leq C$ give $\int_{B_R(z)} |u(X + H) - u(X)| dX \leq C_0 R^d |H|$ and $\|u(\cdot + H) - u\|_\infty \leq C \min\{|H|, 1\}$. Multiplying yields, for every translation H,

$$\int_{B_R(z)} |u(X + H) - u(X)|^2 dX \leq C R^d |H| \min(|H|, 1).$$

Integrating the kernel gives $\int_{B_R} \Gamma_u \leq C R^d$, since $\int_0^1 r^{1-p} dr$ and $\int_1^\infty r^{-p} dr$ are finite. The central-volume bound applies to $\{z_u \geq 1/2\}$. Bounded-subsolution comparison in lemma 2.3 gives $z_u \leq G_1 * (\mathbf{1}_{\{z_u \geq 1/2\}} + \Gamma_u)$. Tonelli and $\int G_1 = 1$ preserve the all-center bounds: for this nonnegative source F,

$$\int_{B_R(z)} G_1 * F = \int G_1(H) \int_{B_R(z-H)} F \, dX \, dH \leq C R^d.$$

Thus $\int_{B_R} z_u \leq C R^d$, and $W(u) = z_u^2/4 \leq z_u/4$ proves assertion 1.

We finish with the near-well estimate, retaining its dependence on the receiver. Let G_3 be the positive kernel of $(A + 3)^{-1}$. Its pointwise and derivative estimates, integrability and convolution bound $|DG_3| * G_3 \leq C G_3$ are already established in lemma 2.2; its action on bounded data is lemma 2.3.

Each nonnegative deficit $w = 1 - u$ or $1 + u$ satisfies

$$(A + 3)w = a_w w, \qquad a_w = 1 + 3w - w^2, \qquad 1 \le a_w \le 13/4 \quad (0 \le w \le 2),$$

and hence $w = G_3 * (a_w w)$. Therefore $G_3 * w \le w \le (13/4)G_3 * w$. Differentiation gives $|Dw| \le (13/4)|DG_3| * w \le (13/4)^2(|DG_3| * G_3) * w \le CG_3 * w \le Cw$. Since $D(a_w w) = (1 + 6w - 3w^2)Dw$, the first derivative bound gives $|D(a_w w)| \le Cw$. Differentiating once more and using the same convolution comparison gives

$$|D^2 w| \le C|DG_3| * w \le C(|DG_3| * G_3) * w \le CG_3 * w \le Cw.$$

These are justified by the uniform smoothness and $DG_3 \in L^1$. A nontrivial deficit is everywhere positive by the positive resolvent; an identically zero deficit is a constant well and the desired estimate is immediate. Gronwall now gives $w(X - H) \le Cw(X)$ for $|H| \le 1$. Also

$$\int \min\{1, |H|^{-n-p}\}\, w(X - H)dH \le Cw(X),$$

because that positive weight is bounded by $CG_3(H)$.

For $0 < y \le 1$, split the tangential formula $D_X U = \mp P_y * Dw$ at $|H| = 1$. The near part uses the local ratio and the kernel mass; the far part uses $P_y(H) \le Cy^p|H|^{-n-p}$ and the last weighted receiver bound. Thus $|D_X U(X, y)| \le Cw(X)$. For the height derivative symmetrize $\partial_y P_y * w$. The local second difference is at most $C|H|^2 w(X)$ by the relative Hessian bound and local ratios. Its integral is at most $Cy^{p-1}w(X)$ after splitting at $|H| = y$ and one; the far part uses $|\partial_y P_y(H)| \le Cy^{p-1}|H|^{-n-p}$ and the same weighted receiver bound. Thus $|U_y(X, y)| \le Cy^{p-1}w(X)$. For $y \ge 1$, direct differentiation gives

$$|D_X P_y(H)| + |\partial_y P_y(H)| \le Cy^{p-1} \min\{1, |H|^{-n-p}\}.$$

For the tangential derivative at $|H| \ge 1$, this follows by setting $r = |H|$ and bounding $yr^{n+p+1}/(r^2 + y^2)^{(n+p+2)/2}$; the other cases are immediate. Apply the weighted receiver bound to the derivative kernels, rather than to P_y itself. It gives $|DU(X, y)| \le Cy^{p-1}w(X)$ for all $y \ge 1$.

Both global deficit functions $w_+ = 1 - u$ and $w_- = 1 + u$ satisfy these estimates, with the same constants. At a given receiver X, take the smaller of the two resulting upper bounds; this gives the displayed estimates with receiver value $1 - |u(X)|$, without differentiating that absolute value. Put $E_\delta = B_R \cap \{1 - |u| \le \delta\}$. For $0 < y \le 1$, the relative estimates and the previously proved L^1 bounds give

$$\int_{E_\delta} |DU|^2 dX \le C\delta R^d(1 + y^{p-1});$$

for $1 \le y \le H$, they give $\int_{E_\delta} |DU|^2 dX \le C\delta y^{p-1}R^d$. Multiplying by $d_p y^{1-p}$ and integrating therefore yields $C\delta H R^d$ up to $H \ge 1$. The universal height tail contributes $CH^{1-p}R^d$ above it. Finally, $z_u \le 2\delta$ on this receiver, so $W(u) = z_u^2/4 \le (\delta/2)z_u$ and its integral is $O(\delta R^d)$. Hence the left side of the near-well estimate is bounded by $CR^d(\delta H + H^{1-p})$. Taking $H = \delta^{-1/p}$ gives $C\delta^{(p-1)/p}R^d$, as claimed. All assertions are proved. □

Proposition 3.5 (The stable sharp-growth endpoint has planar quantized energy blowdowns through dimension seven)**.** *Under the hypotheses of proposition 3.4 with $2 \le d \le 6$, put*

$$e(X) = \frac{d_p}{2} \int_0^\infty y^{1-p}|DU|^2 dy + W(u), \quad M(X) = d_p \int_0^\infty y^{1-p} D_X U \otimes D_X U\, dy.$$

Let $\sigma > 0$ be the complete energy of the centered one-dimensional cubic layer. There is an integer $m \ge 0$, independent of subsequence, such that every $R_j \to \infty$ has a subsequence and

a unit vector e_0 with

$$R_j^{-d}(X \mapsto X/R_j)_{\#} e \rightharpoonup \sigma m \mathcal{H}^d \lfloor e_0^{\perp}, \qquad R_j^{-d}(X \mapsto X/R_j)_{\#} M \rightharpoonup \sigma m (e_0 \otimes e_0) \mathcal{H}^d \lfloor e_0^{\perp}.$$

In particular, for

$$\mathcal{E}(R) = \operatorname{tr} \int_{B_R} M - \lambda_{\max}\left(\int_{B_R} M \right),$$

one has $\mathcal{E}(R) = o(R^d)$. For every fixed detector, the number of disjoint failed unit balls in B_R is consequently $o(R^{d-2})$.

Along that same subsequence the complete trace levels have this explicit common-varifold limit: for every compact interval $I \Subset (-1,1)$ and every compactly supported C^1 position/normal test F which is even in its unit-normal argument,

$$\int_I \left| R_j^{-d} \int_{\{u=c\}} F(X/R_j, Du/|Du|) d\mathcal{H}^d - m \int_{e_0^{\perp}} F(Z, e_0) d\mathcal{H}^d(Z) \right| dc \longrightarrow 0.$$

The level integrals are defined for almost every phase, which suffices for this formula; all actual trace-level pieces are included.

Proof. Use the centered increasing cubic layer q fixed in the proof of proposition 3.4, and let $Q = P_y^{(1)} * q$. Its uniqueness, oddness, and the hypotheses of the layer theorem were verified there. Its smoothness and exact well tails are those of lemmas 2.1 and 2.4. Thus every detector layer is a translate and rotation of this same profile, with the decreasing orientation obtained by reversing the normal. The normal marginal of the mass-one Poisson kernel in $\mathbb{R}^{d+1}$ is exactly the mass-one one-dimensional Poisson kernel: its Fourier convolution multiplier on functions of one coordinate is the one-dimensional extension multiplier. Hence the corresponding extensions are precisely translates and rotations of Q, with no change of scale. All detector comparisons below use its calibrated functions g_y, r_y.

We justify the finiteness and normalization of the one-layer energy before using it. By lemma 2.4, $q' > 0$ and $\int_{\mathbb{R}} q' = 2$. Hence $\int_{\mathbb{R}} |Q_t(t,y)| dt = 2$. The Poisson identity

$$Q_y(t,y) = -y^{-1} \int_{\mathbb{R}} h P_y^{(1)}(h) q'(t-h) dh$$

and the finite first moment of $P_y^{(1)}$, which uses $p > 1$, give $\int_{\mathbb{R}} |Q_y(t,y)| dt \le C_p$. Uniform trace regularity gives $|Q_t| \le C$ and $|Q_y| \le C y^{p-1}$ below height one, while differentiated Poisson kernels give $|DQ| \le C/y$ above it. Multiplying these pointwise bounds by the two L^1 bounds and integrating in y shows

$$\int_{\mathbb{R} \times (0,\infty)} y^{1-p} (|Q_t|^2 + |Q_y|^2) dt\, dy < \infty.$$

This argument does not assume finiteness of the potential integral.

The Hamiltonian equality in Cabré–Sire[6], Theorem 1.1, arXiv:1111.0796v1, printed pp. 2–3, applies to this bounded one-variable extension layer. In their boundary convention use $a = 1-p$, $f_{\mathrm{CS}} = (2-p) f/d_p$, and its potential $G_{\mathrm{CS}} = (2-p) W/d_p$, where $f = -W'$. Their coefficient $1 + a = 2 - p$ then converts the equality exactly to

$$\frac{d_p}{2} \int_0^{\infty} y^{1-p} (Q_t^2 - Q_y^2)(t,y) dy = W(q(t)).$$

The regularity condition on f_{CS} was checked in the detector proof; $q' > 0$ and the limits ± 1 supply the layer hypotheses, and $G_{\mathrm{CS}}(1) = 0$. The complete squared-gradient integrals were

proved finite above, so integration of this equality is justified and also proves $W(q) \in L^1(\mathbb{R})$. If

$$\sigma_h = d_p \int y^{1-p} Q_t^2 dt\, dy, \qquad \sigma_v = d_p \int y^{1-p} Q_y^2 dt\, dy, \qquad \sigma_W = \int_{\mathbb{R}} W(q) dt,$$

then $\sigma_h - \sigma_v = 2\sigma_W$, and the complete layer energy is

$$0 < \sigma := \tfrac{1}{2}(\sigma_h + \sigma_v) + \sigma_W = \sigma_h < \infty.$$

Positivity follows from $Q_t = P_y^{(1)} * q' > 0$. This supplies the same finite calibrated constant to every detector layer.

The complete energy bound in proposition 3.4 gives local weak compactness after the prescribed scaling. Fix a compact phase interval $I \Subset (-1, 1)$, a finite extension height T, and a detector accuracy $\eta > 0$. The detector removes at most $C_{I,T,\eta,b} R^{d-2}$ fixed balls from B_{bR}. At the remaining compact-phase points, comparison with the layer extension $Q(t, y)$ gives a gradient floor and the calibration

$$|D_X U| = g_y(U) + O(\eta), \qquad U_y = r_y(U) + O(\eta),$$

where $Q(t_y(c), y) = c$, $g_y(c) = Q_t(t_y(c), y) > 0$, and $r_y(c) = Q_y(t_y(c), y)$. We spell out the Poisson comparison needed also at height zero. Let u be C^2 within ε of a translated rotated layer q_e on $B_L(X_0)$, with $L \geq 4(T + 1)$, and test $X \in B_{L/4}(X_0)$, $0 < y \leq T$. The uniform cubic regularity of lemma 2.1 applies in dimensions $d+1$ and one to u and the layer; rotations and translations preserve those bounds. Thus both traces have a common global C^2 bound. Write $w = u - q_e$. Positivity and the tangential derivative representation give

$$|P_y * w| + |P_y * D_X w| \leq C_T(\varepsilon + L^{-p}).$$

On $|H| < L/2$ use the local value/gradient bounds, and on its complement use the global boundedness and the Poisson tail $C_T L^{-p}$. For the normal derivative use the symmetric second difference, whose local part is bounded by $C\varepsilon \min(|H|^2, 1)$. Since $\int \partial_y P_y = 0$, this gives

$$|\partial_y P_y * w| \leq C_T(\varepsilon + L^{-p}) y^{p-1}.$$

Indeed split its local radial integral at y and one, using $p < 2$, and use $|\partial_y P_y(H)| \leq C_T y^{p-1} |H|^{-n-p}$ on $|H| > L/2$. The latter tail also controls the subtraction of $2w(X)$. These estimates are uniform down to $y = 0$. Repeating the tangential argument with $D_X^2 w$ gives the needed horizontal C^2 convergence. On a fixed compact phase-height rectangle, $g_y(c)$ has a positive minimum; moreover $\partial_c r_y(c) = O_T(y^{p-1})$, by the same normal derivative estimate for q'. The inverse phase coordinate therefore converts the preceding comparison to the displayed calibration, with the sharper error $O_T((\varepsilon + L^{-p}) y^{p-1})$ for $U_y - r_y(U)$. The value at $y = 0$ is zero.

To obtain such a comparison for an extension-central point, first choose a fixed trace ball large enough that absence of compact central trace values on this connected ball would force the trace there into one well side, and then put the extension near that well by the positive Poisson tail. Thus a central trace point lies a bounded distance away. Enlarge the detector window to contain the receiver and the height range, and enlarge the deleted physical balls by that fixed distance. The detector-good trace comparison about this central trace point supplies the estimate just proved at the extension point. All constants are independent of R. Choose the window L large enough to make its tail small before choosing its trace accuracy, and only afterward take the macroscopic limit.

We specify exhausting cut domains, so no regularity outside a detector-covered receiver is being assumed. At a fixed refinement stage choose a finite list of phase-height rectangles and position receivers, and cover a containing physical ball by the detector construction just described, with a further buffer. Choose integers $B_j \to \infty$ sufficiently slowly and apply those

detectors on $B_{(B_j+3)R_j}$. After fixing that finite detector family, select a scaled outer radius $b_j \in (B_j, B_j+1)$ and put

$$D_j = B_{b_j R_j} \setminus \bigcup_l \overline{B}_{r_{j,l}}(X_{j,l}).$$

The radii $r_{j,l}$ lie in fixed intervals of enlargement factors at each fixed stage. Choose them and $b_j \in (B_j, B_j+1)$ generically so the spherical faces meet piecewise smoothly and almost every tested (c, y) gives transverse level intersections with every face. Sard's theorem for the smooth maps $(|X - X_{j,l}|, U(X, y), y)$, followed by Fubini in the chosen radius, gives this simultaneous conclusion for the finite list; the outer sphere is treated by the same argument. At height zero use the analogous trace maps. Enlarging the detector balls by a fixed extra margin places every retained face away from the failed points. Thus the detector-good gradient floor holds on the retained closure for each tested compact parameter range. Every such level is an oriented smooth hypersurface, properly embedded relative to this exhausting cut domain; its only boundary pieces are the declared spherical cuts. For any fixed compact scaled position receiver, the outer sphere is absent for all large j. All subsequent cut-area and cut-flux estimates are local on such receivers, so this outer boundary contributes nothing to them. At a fixed stage the number of detector balls is $O(R_j^{d-2})$. The complete-energy bound on each fixed-radius ball and the uniform trace gradient bound show that their deleted energy and physical cut area on a fixed receiver are $O(R_j^{d-2})$. Boundary coarea gives the precise averaged estimate

$$\int_I \int_h^T \mathcal{H}^{d-1}(\partial[R_j^{-1}(\{U(\cdot, y) = c\} \cap D_j)] \cap K) dy\, dc \le C_{I,h,T,K} R_j^{-1}$$

for every fixed compact receiver K; the same estimate holds with $h = 0$, since $D_X U$ is uniformly bounded. At each refinement stage ℓ, first fix its finite detector list, windows, ball-enlargement radii, outer receiver radius B_ℓ, and tested parameter rectangles and position receivers. Absorb all their constants, including every power of B_ℓ, into one finite C_ℓ. Choose the stage $\ell(j) \to \infty$ slowly enough that $C_{\ell(j)}/R_j \to 0$, while its detector accuracies tend to zero and its rectangles and receivers exhaust the required ranges. Thus all geometric choices at a stage precede its macroscopic limit; subsequent subsequences preserve these bounds. Deleted energy after multiplication by R_j^{-d}, all cut fluxes below, and the scaled boundary averages therefore still tend to zero. This order of choices keeps the gradient floors for each fixed rectangle independent of the macroscopic radius.

For a smooth compact position/plane test F, represented as an even function of the normal, fix $b < \infty$ so its position support lies in B_b, and define

$$L_j(c, y; F) = R_j^{-d} \int_{\{U(\cdot,y)=c\} \cap D_j} F(X/R_j, \nu)\, d\mathcal{H}^d.$$

The normal phase velocity is $v_c = \nu/g$ and the normal height velocity is $v_y = -U_y \nu/g$, where $g = |D_X U|$. The material derivatives of the unit normal are exactly

$$\partial_c^{\mathrm{mat}} \nu = g^{-1} D_X \nu\, \nu, \qquad \partial_y^{\mathrm{mat}} \nu = \nu_y - (U_y/g) D_X \nu\, \nu.$$

The tangential divergences of the two velocities are respectively $\mathrm{div}_{\{U=c\}} \nu/g$ and $-U_y \,\mathrm{div}_{\{U=c\}} \nu/g$, because the derivative of a scalar coefficient has zero pairing with the normal in tangential divergence. On each compact parameter rectangle, g^{-1} and $|U_y|$ are uniformly bounded on the retained domain. Thus no separate derivative of g is needed. Differentiation of the test and area element, followed by normal coarea, bounds the interior

parameter variation by

$$CR_j^{-d}\int_{(D_j\cap B_{bR_j})\times(0,T)\cap\{U\in I\}}(R_j^{-1}+|D_X\nu|+|\nu_y|)dXdy=o(1).$$

Indeed central volume is $O(R_j^d)$, the angular budget is $O(R_j^{d-2})$, and the inverse weight y^{p-1} is integrable down to height zero. Cauchy–Schwarz gives the dimension-independent factor R_j^{-1}. The cut flux must be retained at critical values as well: for $\mathcal{T}_j(X,y)=(U(X,y),y)$ on $\partial D_j\times(0,T)$, its two parameter derivatives are the signed pushforwards

$$-R_j^{-d}(\mathcal{T}_j)_\#[F(\nu\cdot n)d\mathcal{H}^d dy],\qquad R_j^{-d}(\mathcal{T}_j)_\#[FU_y(\nu\cdot n)d\mathcal{H}^d dy].$$

Their total variations are $o(1)$ by the cut-area estimate and the compact-parameter bounds. These identities follow by differentiating $\delta(U-c)gF$ and integrating its spatial divergence. They include singular critical-value contributions; no division by a vanishing boundary gradient is made. BV Poincare on parameter rectangles makes each $L_j(\cdot,\cdot;F)$ differ from its parameter mean by $o(1)$ in L^1. These means are locally bounded: the coarea identity gives

$$\int_I\int_h^T R_j^{-d}\mathcal{H}^d(\{U(\cdot,y)=c\}\cap D_j\cap B_{bR_j})dy\,dc\le R_j^{-d}\int_h^T\int_{B_{bR_j}}|D_XU|dXdy\le C_{b,h,T}.$$

Extract the means for a countable dense family of compact position/plane tests. Positivity and these bounds define by the Riesz representation theorem a limit varifold V, and parameter Poincare gives L^1 convergence of the tests to $V(F)$. Overlapping rectangles and receivers identify the same V.

The trace levels at $y=0$ require a separate boundary argument. For each fixed j, $U(\cdot,y)\to u$ in local horizontal C^2 as $y\downarrow 0$, by the quantitative Poisson estimates just proved. The central gradient floor holds up to height zero outside the deleted balls. For almost every c the trace level is regular and transverse to the finitely many smooth faces of the cut boundary on the compact position receiver. Choose local implicit-function charts for these trace levels and their cut faces. On every such chart the positive-height level parametrizations, normals and area elements converge to their trace counterparts. Intersections with the fixed cut domain have convergent indicators almost everywhere on the level: its boundary intersection has zero $\mathcal{H}^d$ measure at these regular transverse values. For completeness a majorant uniform in the tested phase can be obtained without a transversality constant uniform in phase. On the retained closure of a fixed compact position receiver, for phases in a compact buffer I' and y sufficiently near zero, the horizontal gradient floor is positive. Cover this compact set by finitely many small boxes on each of which one spatial component of D_XU is bounded away from zero. Each level in such a box is a graph in that fixed coordinate, with a common bound on its slope; its projected domain lies in a fixed bounded box. The sum of these graph-area bounds is a finite majorant independent of $c\in I$ and small y, at this fixed index. Restriction to the cut domain only decreases those graph areas. Dominated convergence first on the charts and then in phase proves that the BV boundary trace of $L_j(c,y;F)$ is precisely

$$L_j(c,0;F)=R_j^{-d}\int_{\{u=c\}\cap D_j}F(X/R_j,Du/|Du|)d\mathcal{H}^d$$

in L^1 of every tested compact phase interval. The constants in this fixed-index trace identification need not be uniform in j.

They are not used for the limiting estimate. If $\overline{L}_j(F)$ is the interior parameter mean on $I\times(0,T)$, the elementary BV slicing trace inequality is

$$\int_I|L_j(c,0;F)-\overline{L}_j(F)|dc\le T^{-1}\int_I\int_0^T|L_j(c,y;F)-\overline{L}_j(F)|dydc+|D_yL_j|(I\times(0,T)).$$

It follows first from the one-dimensional BV fundamental theorem on almost every vertical slice and then by integration. Both terms tend to zero by parameter Poincare and the complete variation/cut-flux estimates above. Since $\overline{L}_j(F) \to V(F)$, the actual trace-level integrals converge in phase L^1 to this same varifold. No boundary value is inferred merely from interior convergence. Buffer the phase interval by I' when applying the argument, so its side boundaries introduce no extra term.

We make the parameter extractions used for both integrality and stability explicit. Let $Q_\ell = I_\ell \times (h_\ell, T_\ell)$ be a countable exhaustion of $(-1,1)\times(0,\infty)$ by bounded rectangles, with $h_\ell > 0$, and let K_ℓ be compact position receivers increasing to the whole space. For almost every (c,y) define

$$M_{j,c,y} = R_j^{-1}(\{U(\cdot,y) = c\} \cap D_j).$$

Coarea, the gradient floor, and the angular budget give, on each fixed Q_ℓ and K_q, bounded integrals over parameters of the scaled area and of $\int_{M_{j,c,y}\cap K_q} |A_{j,c,y}|^2$. Boundary coarea gives

$$\int_{Q_\ell} \mathcal{H}^{d-1}(\partial M_{j,c,y} \cap K_q)\, dc\, dy \longrightarrow 0.$$

The varifold test integrals already converge to those of V in $L^1(Q_\ell)$. Include in their countable determining family smooth position cutoffs equal to one near each K_q. Pass to one common subsequence on which the L^1 errors for every such test and every boundary integral above are summable: at its rth index make the sum of the first r errors less than 2^{-r}. Fubini then gives, for almost every parameter simultaneously on this exhaustion, varifold convergence $|M_{j,c,y}| \rightharpoonup V$, local area bounds, and

$$\mathcal{H}^{d-1}(\partial M_{j,c,y} \cap K_q) \longrightarrow 0 \quad \text{for every } q.$$

Sard and transversality exclude only a null parameter set at each index, so every member of these sequences is a smooth properly cut level for almost every parameter.

To obtain one sequence of levels with curvature bounded on all compact receivers, work first on a positive-measure rectangle Q_1. Choose positive weights w_q so the sum of the bounds for the parameter integrals of $w_q \int_{M_{j,c,y}\cap K_q} |A_{j,c,y}|^2$ is finite, uniformly in j. Fatou gives a parameter in the preceding full-measure set at which the liminf of this weighted sum is finite. Along a subsequence attaining that finite liminf, every compact curvature integral is bounded. The local area and vanishing-boundary statements persist. The first variations are locally bounded, since the interior mean curvature satisfies $|H| \le \sqrt{d}|A|$ and the boundary contribution has mass at most $\mathcal{H}^{d-1}(\partial M_{j,c,y})$ on a containing receiver. Allard's integral-varifold compactness theorem[1], in the formulation and proof of Simon[16], Chapter 8, Theorem 5.8 and Remark 5.9, printed pp. 218–219, therefore makes V integral: the approximating varifolds have integer multiplicity one, and both their masses and first variations are bounded on every compact receiver. The same level sequence supplies the smooth approximation hypotheses of lemma 3.1. This selection of one parameter is used only for integrality and that approximation; no almost-everywhere curvature bound along the entire sequence has been asserted.

We now pass the established plane-projector boundary identity (4) to the limit. On a smooth scaled level use the same notation $P = P_{TM}$ and $\mathsf{A}_{ikl} = P_{iq}\partial_q P_{kl}$. For a compactly supported C^1 test $\Psi(X,P)$ with position support in K_q, the absolute value of its boundary term in that identity is at most $\|\Psi\|_\infty \mathcal{H}^{d-1}(\partial M \cap K_q)$. It tends to zero along the selected levels; their exhausting outer boundary lies outside this fixed receiver for all large indices. The identity applies face by face to their declared spherical cuts, as verified in lemma 3.1.

Suppose on any local sequence of these levels that the curvature-square integral is bounded on a receiver containing the test support. Since

$$\sum_{i,k,l} |\mathsf{A}_{ikl}|^2 = 2|A_M|^2,$$

weak compactness of the vector measures $\mathsf{A}^j|M_j|$, with their L^2 bounds, gives a limit tensor $\mathsf{A} \in L^2(V)$ there. Indeed testing against continuous functions and applying Cauchy–Schwarz identifies the limiting measures as absolutely continuous with respect to V; the variational characterization of the L^2 norm gives its weighted lower semicontinuity. The other coefficients in (4) are continuous functions of (X, P), so varifold convergence, weak tensor convergence, and the vanishing boundary error pass that identity to V.

On a smooth limit patch Σ let P_Σ be its smooth projector, extended smoothly to a neighborhood. Fix indices a, b and use $\Psi(X, P) = \zeta(X)[P_{ab} - (P_\Sigma(X))_{ab}]$, with a harmless matrix cutoff equal to one near all orthogonal projectors. On V the test itself is zero, its position derivative is $-\zeta\, \partial_{X_j}(P_\Sigma)_{ab}$, and its matrix derivative selects A_{iab}. The limiting identity therefore yields

$$\int \zeta \left[\mathsf{A}_{iab} - (P_\Sigma)_{ij}\partial_{X_j}(P_\Sigma)_{ab}\right] d\|V\| = 0.$$

Thus A equals the classical tangential derivative of P_Σ on that patch. Both its squared norm and the approximating squared norms have the same factor two relative to scalar square curvature. Consequently, for every nonnegative compact continuous weight ζ supported on smooth patches,

$$\int \zeta |A_V|^2 d\|V\| \le \liminf_j \int_{M_j} \zeta |A_{M_j}|^2 d\mathcal{H}^d.$$

This keeps the sharp coefficient. Neither smooth convergence of the levels nor a constancy assertion for multiplicity is needed for this identification.

We next identify the complete energy. The height tails of proposition 3.4 remove y near zero and infinity. For $h \le y \le T$, differentiating the mass-one Poisson kernel gives $|D_{X,y}P_y| \le Cy^{-1}P_y$. Applied separately to the two nonnegative traces $1 \pm u$, this proves $|DU| \le C_h(1 - |U|)$. Also $\int_{B_R}(1 - U^2) \le C_T R^d$. Indeed

$$1 - U^2 = P_y(1 - u^2) + \mathrm{Var}_{P_y}(u), \qquad \mathrm{Var}_{P_y}(u)(X) \le \int P_y(H)|u(X - H) - u(X)|^2 dH.$$

The first term has integral at most CR^d by the all-center deficit bound. The translation estimate in proposition 3.4 bounds the integrated second term by

$$CR^d \int P_y(H)|H| \min\{|H|, 1\} dH \le CR^d \int P_y(H)|H| dH \le C_T R^d,$$

using the finite first moment and $y \le T$. Hence the compact-height energy in $|U| > 1 - \delta$ is $O_{h,T}(\delta R^d)$. The trace potential satisfies the same estimate using $W(u) \le C\delta(1 - u^2)$. Deleted-ball energy is negligible. On a central trace interval the coarea identity is

$$R_j^{-d} \int W(u)\phi(X/R_j) dX = \int_I W(c) R_j^{-d} \int_{\{u=c\}} \frac{\phi(X/R_j)}{|Du|} d\mathcal{H}^d dc,$$

with the integral restricted to that central interval and the deleted domain on both sides. There $|Du| = g_0(c) + o(1)$ uniformly; the boundary-trace convergence just proved identifies its limit as $\int_I W(c)/g_0(c) dc$ times the limiting area measure. Exhaust phases with the established well-tail estimate. Bulk coarea uses the positive-height parameter convergence in exactly the same way. Thus coarea and the calibration identify

$$e_{R_j} \rightharpoonup \sigma\|V\|, \qquad M_{R_j} \rightharpoonup \sigma(\nu \otimes \nu)\|V\|.$$

The scalar coefficients in these two limits agree because the finite-energy Hamiltonian calculation above gives $\sigma = \sigma_h$. Thus the coefficient identification uses neither a presumed potential-energy integrability nor the conclusion of the present proposition.

We verify the center-uniform mass hypothesis before using the geometric closure. For arbitrary fixed $z \in \mathbb{R}^{d+1}$ and $r > 0$, the translated complete-energy bound of proposition 3.4 applies to $B_{rR_j}(R_j z)$ once $rR_j \geq 1$. Thus

$$e_{R_j}(B_r(z)) = R_j^{-d} \int_{B_{rR_j}(R_j z)} e(X)dX \leq Cr^d,$$

with a constant independent of z, r, j. Vague convergence and the open-set lower bound, obtained equivalently by increasing compact continuous tests inside the ball, give $\sigma\|V\|(B_r(z)) \leq \liminf_j e_{R_j}(B_r(z)) \leq Cr^d$. Since z, r were arbitrary and the same constant and convergent sequence were used, this is the asserted mass bound for every center and radius.

The exact horizontal stress $eI - M$ has distributional divergence zero. To record its boundary signs, let $e_{h,T}$ and $M_{h,T}$ be the same integrals restricted to $h < y < T$, retaining $W(u)$ in $e_{h,T}$. The weighted extension equation gives

$$\partial_{X_i}(e_{h,T}\delta_{ij} - (M_{h,T})_{ij}) = d_p[y^{1-p}U_y U_{X_j}]_{y=h}^{y=T} + W'(u)u_{X_j}.$$

At the bottom, $d_p y^{1-p}U_y \to W'(u)$ and $U_{X_j} \to u_{X_j}$, so its subtracted contribution cancels the displayed potential derivative. At the top the pointwise bound $|DU| \leq C/T$ makes the boundary product $O(T^{-1-p})$ on compact position sets. The full-height energy bounds permit passage of the truncated tensors in distributions, first $h \downarrow 0$ and then $T \to \infty$. Its limit is $\sigma P_{TV} V$, so V is stationary.

Finally choose an ambient smooth compact test ϕ whose support is disjoint from the singular support of V, and apply the full angular inequality with physical test $\phi(X/R_j)$, divided by R_j^{d-2}. This support restriction is made before taking the limit. Use the common parameter subsequence constructed above. At any parameter in its full-measure convergence set, put

$$C_j^{\phi}(c, y) = \int_{M_{j,c,y}} |A_{j,c,y}|^2 \phi^2 d\mathcal{H}^d.$$

If $\liminf_j C_j^{\phi} = \infty$, the desired lower bound is automatic. Otherwise take a subsequence, depending on this parameter, attaining that finite liminf. On every compact subset of the open set $\{\phi \neq 0\}$ the curvature-square integral is bounded, because $|\phi|$ has a positive lower bound there. Local area bounds, varifold convergence to the same V, and vanishing cut boundary persist. Apply the preceding curvature identity and tensor identification on a countable exhaustion of that open set. Its weighted lower semicontinuity, followed by monotone exhaustion of the weight ϕ^2, gives

$$\int_{\operatorname{reg} V} |A_V|^2 \phi^2 d\|V\| \leq \liminf_j C_j^{\phi}(c, y)$$

for almost every parameter. The zero set of ϕ contributes nothing to either weighted integral. Thus no fixed-parameter curvature bound along the entire common sequence is needed.

On a compact phase-height rectangle Q, coarea and the uniform calibration bound the angular left side below by

$$d_p \int_Q y^{1-p} g_y(c) \int_{M_{j,c,y}} |A_{j,c,y}|^2 \phi^2 d\mathcal{H}^d \, dc \, dy - o(1).$$

The calibration error tends to zero because its uniform error multiplies a bounded parameter-averaged scaled curvature integral on Q. Fatou and the preceding almost-everywhere liminf

lower bound apply to this nonnegative integral. Exhausting phase and height makes its scalar coefficient

$$d_p \int_0^\infty \int_{-1}^1 y^{1-p} g_y(c)\, dc\, dy = d_p \int_0^\infty \int_{\mathbb{R}} y^{1-p} Q_t(t,y)^2 dt\, dy = \sigma.$$

The angular right side has limit $\sigma \int |D\phi|^2 d\|V\|$ by the complete covariance convergence. Cancelling this common positive coefficient gives on the regular limit

$$\int_{\operatorname{reg} V} |A_V|^2 \phi^2 d\|V\| \le \int_{\operatorname{reg} V} |D\phi|^2 d\|V\|.$$

The right side before this restriction is $\int |D\phi|^2 d\|V\|$ over all of V. Here ϕ and its gradient vanish on a neighborhood of the singular support, so this is exactly the displayed integral over $\operatorname{reg} V$; no assertion about the mass of the singular support is needed. Exhausting phases and heights gives the common coefficient σ, which cancels. Every compactly supported intrinsic test on a regular patch has an ambient extension constant along its normals on the patch and supported away from the singular support. Its ambient gradient on the patch is the intrinsic gradient. Remove the locally constant integer multiplicity on each regular component, using stationarity on its smooth interior. This supplies precisely the intrinsic stability away from the singular support required in proposition 3.2. That proposition and lemma 3.1 show that every subsequential limit is a finite collection of parallel affine planes. Their total integer multiplicity is uniformly bounded by the energy bound. Apply lemma 3.3 to $\rho = e\, dX/\sigma$. Every blowdown is a single linear plane with the same integer multiplicity. We also justify the covariance integral over the unit ball. Every approximating matrix-valued measure M_{R_j} is positive semidefinite; the total variation of any one entry is bounded by its nonnegative trace measure, since $|v_\alpha v_\beta| \le (v_\alpha^2 + v_\beta^2)/2$. The trace measures converge weakly to $\sigma m \mathcal{H}^d \llcorner e_0^\perp$, which gives ∂B_1 zero mass. Approximate $\mathbf{1}_{B_1}$ from inside and outside by smooth cutoffs. The error in each matrix entry is bounded by the trace mass on the shrinking spherical annulus; the limsup of that error tends to zero by the same weak convergence and the zero boundary mass. Consequently

$$R_j^{-d} \int_{B_{R_j}} M \longrightarrow \sigma m \omega_d\, e_0 \otimes e_0.$$

Continuity of the trace and of the largest eigenvalue gives $\mathcal{E}(R_j)/R_j^d \to 0$ on this subsequence, also when $m = 0$. This holds after extraction from every diverging sequence. If the full nonnegative ratio did not tend to zero, a sequence bounded below by a fixed positive number would contradict that subsequence conclusion. Thus $\mathcal{E}(R) = o(R^d)$ for the full radius limit. The detector estimate in proposition 3.4 with a maximizing covariance direction at radius $2R$ yields $N \le C\mathcal{E}(2R)/R^2 = o(R^{d-2})$.

We record the complete trace-level conclusion, including restoration of the deleted pieces. The boundary-trace argument above gives convergence in phase L^1 of the deleted-domain level integrals to $V(F)$. On a compact phase interval the removed portion has error at most

$$R_j^{-d} \|F\|_\infty \int_{\{u \in I\} \cap (B_{bR_j} \setminus D_j)} |Du| dX = o(1),$$

by normal coarea, the uniform physical derivative bound and the $C_j R_j^{d-2}$ detector volume; the slow diagonal was chosen with $C_j/R_j^2 \to 0$. Thus the convergence holds for the entire actual trace level. At this point the plane closure and integer-density argument have identified $V = m\mathcal{H}^d \llcorner e_0^\perp$. The limit is consequently exactly the phase-independent expression in the statement. A countable dense family of even compact C^1 tests and compact phase intervals

can be used in the original subsequence choice. The bound

$$R_j^{-d}\int_I \mathcal{H}^d(\{u=c\}\cap B_{bR_j})dc = R_j^{-d}\int_{\{u\in I\}\cap B_{bR_j}} |Du|dX \le C_b$$

extends convergence to every such test by uniform approximation, and a countable exhaustion extends it to every compact phase interval. This proves the stated complete-level interface and finishes the proof. □

Lemma 3.6 (The finite-level recurrence gives sharp noncollapse in every sheet dimension)**.** *Under the hypotheses of proposition 3.4, for each fixed $0<\theta<1$ there are $c_\theta, R_\theta>0$ such that*

$$\int_{B_R(z)} |Du| \ge c_\theta R^d, \qquad \int_{B_R(z)} W(u) \ge c_\theta R^d$$

for every $|u(z)|\le\theta$ and $R\ge R_\theta$. In particular a nonconstant solution in proposition 3.5 has $m\ge 1$.

Proof. The bounded resolvent representations, their derivative identities, and comparison for the bounded Kato subsolution used below follow from lemma 2.3. The forcing terms are bounded by the uniform cubic derivative bounds; comparison requires no global L^2 or decay hypothesis.

Use the hypotheses and extension conventions of proposition 3.4. Fix $0<\theta<1$; constants may depend on this fixed central interval. Write $n=d+1$, $A=(-\Delta)^{p/2}$, $g=|Du|$, and $m_z(R)=\int_{B_R(z)} g$.

For clarity the localized full-pair energy used below is

$$Q_B = (\mathbb{R}^n\times\mathbb{R}^n)\setminus(B^c\times B^c),$$

$$E_B(u) = \frac{c_{n,p}}{4}\iint_{Q_B}\frac{|u(X)-u(Y)|^2}{|X-Y|^{n+p}}dXdY + \int_B W(u)dX,$$

where $c_{n,p}$ is the positive kernel coefficient for the Fourier symbol $|\xi|^p$ and $W(r)=(1-r^2)^2/4$. In particular every pair with at least one endpoint in B is included. For a ball centered at z write

$$\widetilde{B}_R^+(z) = \{(X,y): y>0,\ |X-z|^2+y^2<R^2\},$$

$$J_z(R) = \frac{d_p}{2}\int_{\widetilde{B}_R^+(z)} y^{1-p}|DU|^2dXdy + \int_{B_R(z)} W(u)dX.$$

The pair energy is finite by the uniform local gradient bound and the bounded far tail. Its kinetic part is at most twice the one-receiver row with coefficient $c_{n,p}/4$, because the $B\times B$ part is counted once and the two cross parts are equal. Thus a pointwise bound of the full kinetic row plus potential by $CK*g$ gives exactly $E_B(u)\le C\int_B K*g$, as used below.

The uniform derivative bounds, compactness of translates, and closure of the equation and stability under those limits are supplied by lemma 2.1. We also use lemma 2.2 with $\lambda=1$: its positive kernel G_1 has mass one, belongs to $W^{1,1}$, satisfies the stated convolution inequalities, and, since $n\ge 3>p$, obeys

$$G_1(H)\le C\min\{|H|^{p-n},|H|^{-n-p}\}, \qquad |DG_1(H)|\le C\min\{|H|^{p-n-1},|H|^{-n-p-1}\}.$$

The bounded-data interpretations and derivative identities are precisely those of lemma 2.3.

A central point has a uniform gradient witness. There exist fixed $r_*,c_*>0$ such that $|u(x)|\le\theta$ implies $\int_{B_{r_*}(x)} g^2\ge c_*$. Otherwise choose central translates with gradient-square mass tending to zero on balls of radii tending to infinity. The compactness and limit passages of lemma 2.1 give a constant stable limiting solution whose value lies in $[-\theta,\theta]$. Its equation forces the constant to be zero. But the zero solution is unstable: the kinetic term of a

nonzero test dilated to scale L is $O(L^{n-p})$, whereas its negative potential term is $-cL^n$. This contradiction proves the witness. Bounded g and Tonelli consequently give

$$\mathbf{1}_{\{|u|\le\theta\}}(x) \le C_\theta \int_{B_{r_*}(x)} g, \qquad |B_R(z) \cap \{|u| \le \theta\}| \le C_\theta m_z(R + C_\theta).$$

We may use this argument on any fixed larger compact central interval as well.

There is a nonnegative integrable convolution kernel K controlling the full kinetic row plus potential by $K * g$, with

$$K(H) \le C|H|^{2-n-p} \quad (0 < |H| \le 1), \qquad K(H) \le C|H|^{1-n-p} \quad (|H| \ge 1).$$

For the kinetic row use the uniform Lipschitz and range bounds in

$$|u(X + H) - u(X)|^2 \le C \min(|H|, 1)|H| \int_0^1 g(X + tH)dt.$$

With $Z = tH$, its full kernel row is at most the convolution against the radial kernel

$$K_{\mathrm{kin}}(Z) = C|Z|^{1-n-p} \int_0^1 t^{p-1} \min(|Z|/t, 1)dt.$$

For $|Z| \le 1$, split this integral at $t = |Z|$; its two parts are at most $C|Z|^p$ and $C|Z| \int_{|Z|}^1 t^{p-2}dt \le C|Z|$. For $|Z| \ge 1$ the integral is bounded. These give exactly the displayed near and far upper bounds. For the potential on $|u| \le 1/2$ use the preceding compact central witness. Elsewhere $W(u) \le C|u - u^3| = C|Au|$. The differentiated resolvent equation is

$$D^2u = DG_1 * ((2 - 3u^2)Du), \qquad |D^2u| \le C|DG_1| * g.$$

It is justified by $DG_1 \in L^1$ and the bounded smooth fields. For the near-diagonal part of Au, Taylor's integral formula followed by $Z = tH$ gives a kernel bounded by

$$K_0(Z) = C|Z|^{2-n-p}\mathbf{1}_{\{|Z|<1\}}$$

acting on $|D^2u|$. Indeed the remaining parameter integral is bounded by $\int_0^1 t^{p-2}dt < \infty$. The convolution $K_0 * |DG_1|$ is bounded by $C|Z|^{1-n}$ near zero and $C|Z|^{-n-p-1}$ at infinity. For the near estimate, split into neighborhoods of the two singularities and their complement; the two local kernel orders are $2 - p > 0$ and $p - 1 > 0$, whose sum is one, and the intervening radial integral is bounded by $C \int_{|Z|}^2 r^{-n}dr \le C|Z|^{1-n}$. Compact support of K_0 gives the far bound. For the far part of Au, the first-difference segment formula gives the kernel

$$C|Z|^{1-n-p} \int_0^{\min(1,|Z|)} t^{p-1}dt,$$

which is bounded by $C|Z|^{1-n}$ below one and $C|Z|^{1-n-p}$ above one. Since $p > 1$, each near bound $|Z|^{1-n}$ is at most $|Z|^{2-n-p}$. The central witness adds a fixed compact kernel, which is absorbed after enlarging the constant. This proves the claim. Its integrability follows from $p < 2$ near zero and $p > 1$ at infinity.

Splitting its source integral at $B_{(L+1)R}(z)$ and using the all-center upper bound gives for the full-pair energy

$$E_{B_R(z)}(u) \le Cm_z((L + 1)R) + CL^{-p}R^{n-p}, \qquad L \ge 2.$$

For the exterior term, its source annuli have mass $C(2^kLR)^d$ and kernel size $C(2^kLR)^{1-n-p}$; integrate the receiver volume CR^n and sum the resulting $C2^{-kp}$ series.

Here is the precise comparison and normalization needed for the lower bound. Cabr'e–Cinti–Serra, *Stable solutions to the fractional Allen–Cahn equation in the nonlocal perimeter*

regime[4], arXiv:2111.06285v1, PDF page 32, Lemma 5.2, states for every order $p \in (0,2)$ that

$$\int_{\widetilde{B}_R^+(z)} y^{1-p}|DU|^2 dXdy \le C_{n,p} \iint_{Q_{B_{2R}(z)}} \frac{|u(X)-u(Y)|^2}{|X-Y|^{n+p}} dXdY.$$

The estimate is for the Poisson extension of the whole trace and retains all exterior partners. Their extension coefficient is exactly $d_p = 2^{p-1}\Gamma(p/2)/\Gamma(1-p/2)$, the coefficient used here. Multiplication by $d_p/2$, the fixed positive pair coefficient $c_{n,p}/4$, and the inclusion $B_R \subset B_{2R}$ therefore give

$$J_z(R) \le C_{n,p} E_{B_{2R}(z)}(u).$$

There is no additive error. Proposition 5.3 on the same page says that $R^{p-n}J_z(R)$ is nondecreasing. Its hypotheses hold: W is nonnegative and C^3, $p \in (1,2) \subset (0,2)$, and U solves precisely their weighted equation and boundary condition. The existence of the tested central point excludes the constant wells, and the constant zero solution is unstable by the dilated test used above. If the remaining trace attained either well, its positive singular-integral formula at that extremum and its equation would force it to be constant. Thus $-1 < u < 1$, and positivity of the Poisson kernel gives the same strict range for U.

To initialize this formula uniformly, put $\theta' = (1+\theta)/2$. The uniform gradient bound supplies a fixed $r_\theta > 0$ such that $|u| \le \theta'$ on $B_{r_\theta}(z)$ whenever $|u(z)| \le \theta$. Consequently

$$J_z(r_\theta) \ge |B_{r_\theta}| \min_{|t|\le\theta'} W(t) =: \kappa_\theta > 0.$$

Monotonicity yields $J_z(\rho) \ge \kappa_\theta(\rho/r_\theta)^{n-p}$ for $\rho \ge r_\theta$. Apply the preceding comparison with $\rho = R/2$ to obtain

$$E_{B_R(z)}(u) \ge C_{n,p}^{-1} J_z(R/2) \ge c_\theta R^{n-p} \qquad (R \ge 2r_\theta).$$

Combining this with the already proved upper bound, choose the fixed L so the coefficient CL^{-p} of the exterior term is below $c_\theta/2$, and absorb that term. The resulting bound is $m_z((L+1)R) \ge c'_\theta R^{n-p}$. Renaming the radius, with this fixed L, gives

$$m_z(R) \ge cR^{d+1-p}.$$

To localize the gradient mass to central phases, regularize the convex norm of the vector Jacobi field Du by $g_\varepsilon = (|Du|^2 + \varepsilon^2)^{1/2}$. The positive-kernel convexity inequality gives

$$(A+1)g_\varepsilon \le (2-3u^2)\frac{g^2}{g_\varepsilon} + \frac{\varepsilon^2}{g_\varepsilon}.$$

For each compact smooth test, $A\phi \in L^1$ by lemma 2.3. Hence $\|g_\varepsilon - g\|_\infty \le \varepsilon$, the bound $\varepsilon^2/g_\varepsilon \le \varepsilon$, and dominated convergence on the test support permit passage to the limit in distributions. Thus

$$(A+1)g \le (2-3u^2)g \le 2g\mathbf{1}_C, \qquad C = \{|u| \le \sqrt{2/3}\}.$$

Bounded-subsolution comparison in lemma 2.3 yields $g \le 2G_1 * (g\mathbf{1}_C)$. For $X \in B_R(z)$, sources in the annulus $2^{k+1}R < |Y-z| \le 2^{k+2}R$ have mass at most $C(2^kR)^d$ and kernel at most $C(2^kR)^{-n-p}$. Integrating over the receiver and summing shows that all sources outside $B_{2R}(z)$ cost at most CR^{d-p}. Since G_1 has mass one,

$$m_z(R) \le 2\int_{B_{2R}(z)\cap C} g + CR^{d-p}.$$

The preliminary bound $m_z(R) \geq cR^{d+1-p}$ makes the error at most C/R times $m_z(R)$, uniformly in the central point z. Consequently, for a fixed sufficiently large starting radius,

$$\int_{B_{2R}(z)\cap C} g \geq cm_z(R).$$

We next prove the recurrence. Define the nonnegative trace angular defect

$$a_u(X) = c_{n,p} \int \frac{g(X)g(Y) - Du(X)\cdot Du(Y)}{|X-Y|^{n+p}} dY.$$

To justify its local finiteness, put $v = Du$ and observe

$$0 \leq g(X)g(Y) - v(X)\cdot v(Y) = \frac{|v(X)-v(Y)|^2 - |g(X)-g(Y)|^2}{2} \leq C\min\{|X-Y|^2, 1\}.$$

The last bound follows from the uniform first two derivatives of u, so the defect is bounded and measurable. For a compact smooth η, each Jacobi equation can be tested with $v_i\eta^2$; the compact Lipschitz function $g\eta$ is admissible in stability by strong $H^{p/2}$ approximation. Expanding the symmetric bilinear forms and using the preceding identity gives

$$\int \eta^2 a_u \leq \frac{c_{n,p}}{2} \iint g(X)g(Y)(\eta(X)-\eta(Y))^2 |X-Y|^{-n-p} dXdY.$$

All terms are absolutely integrable: compact support handles one endpoint, bounded derivatives handle the diagonal, and the far fractional kernel is integrable. For a cutoff equal to one on $B_R(z)$, supported in $B_{2R}(z)$, and with Lipschitz constant C/R, symmetry confines one endpoint on the right to $B_{2R}(z)$. The moment

$$\sup_X \int g(Y)|X-Y|^{2-n-p} dY \leq C$$

is finite: below distance one use bounded g and $p < 2$, and on successive larger annuli use the all-center mass bound and $\sum_{k\geq 0} 2^{k(1-p)} < \infty$. This proves

$$\int_{B_R(z)} a_u \leq CR^{-2} m_z(2R).$$

On $g > 0$ let $\nu = Du/g$, $P = I - \nu\otimes\nu$, and $\mathrm{II} = PD^2uP/g$. Project the differentiated massive representation at the receiver X; with $K_p(H) = c_{n,p}|H|^{-n-p}$, Cauchy–Schwarz gives

$$\begin{aligned} |PD^2uP|^2(X) &\leq C\left(\int K_p(X-Y)g(Y)|P(X)\nu(Y)|^2 dY\right) \\ &\quad\cdot\left(\int \frac{|DG_1(X-Y)|^2}{K_p(X-Y)} g(Y) dY\right) \leq C\frac{a_u(X)}{g(X)}. \end{aligned}$$

The integrand containing $g(Y)\nu(Y)$ is set to zero where $g(Y) = 0$. Here $|P(X)\nu(Y)|^2 \leq 2(1-\nu(X)\cdot\nu(Y))$. The second factor is bounded by boundedness of g and the finite radial integrals $C\int_0^1 r^{3p-3}dr$ and $C\int_1^\infty r^{-p-3}dr$, using lemma 2.2. Multiplication by $g(X)$ and integration therefore yield

$$\int_{B_R(z)} g^3|\mathrm{II}|^2 \leq CR^{-2}m_z(2R).$$

We justify the detector-to-small-gradient-volume step uniformly over all layer parameters. Fix $J = [-\theta_1, \theta_1]$ with $\sqrt{2/3} < \theta_1 < 1$ and choose $\theta_2 \in (\theta_1, 1)$. Use the centered cubic layer q fixed in the proof of proposition 3.4; every detector layer is a translated rotated copy of it. Its derivative is strictly positive by lemma 2.4. Consequently

$$\mu_J := \min_{q(t)\in[-\theta_2,\theta_2]} q'(t) > 0,$$

because that preimage is compact. This bound is unchanged by translating or rotating the layer. Choose the detector accuracy smaller than $\min\{1-\theta_1, \theta_2-\theta_1, \mu_J/2\}$ and fix its physical comparison window. If $u(x) \in J$ and the point is detector-good, comparison to a well is impossible; comparison to a layer puts that layer's value in $[-\theta_2, \theta_2]$ and gives $g(x) \geq \mu_J/2$. Hence, for $a = \mu_J/4$, every point of $\{u \in J, g \leq a\}$ is failed. Choose a maximal 2-separated family of such points in $B_{3R}(z)$. Their unit balls are disjoint. The detector count in proposition 3.4 and its complete energy bound give at most CR^{d-2} centers. Maximality covers the small-gradient set by their radius-two balls. Therefore

$$|B_{3R}(z) \cap \{u \in J, g \leq a\}| \leq CR^{d-2}.$$

Let $0 \leq \lambda(g) \leq 1$ be a smooth squared cutoff, zero below $a/2$ and one above a. Choose $0 \leq \zeta \leq 1$, $\zeta \in C_c^\infty(B_3)$, equal to one on B_2, and set $\chi = \zeta^{d-1}$, $\psi = \zeta^d = \chi^{d/(d-1)}$; thus both cutoffs are smooth, including when $d = 2$. Write $\chi_R(X) = \chi((X-z)/R)$, and likewise for ψ_R. Put $m = m_z(R)$ and $N = m_z(A_0R)$, where the fixed $A_0 \geq 8$ contains every ball enlargement below once R is sufficiently large. The normalized level integral is

$$F_R(t) = m^{-1} \int_{\{u=t\}} \lambda(g)\psi_R \, d\mathcal{H}^d.$$

The preliminary lower bound makes $m > 0$. Define $V_R = \lambda(g)\nu\psi_R$ on $g > 0$, and extend it by zero on $g \leq a/2$. This is a compactly supported C^1 vector field. For $\varphi \in C_c^\infty(J)$, coarea and integration by parts give

$$\int_J F_R(t)\varphi'(t)dt = -m^{-1} \int_{\{u \in J\}} \varphi(u) \operatorname{div} V_R \, dX.$$

Consequently, in distributions on J,

$$F_R'(t) = m^{-1} \int_{\{u=t\}} \frac{\operatorname{div} V_R}{g} \, d\mathcal{H}^d.$$

The numerator vanishes on $g \leq a/2$. Coarea therefore makes this an L^1 derivative, rather than just a variation measure.

Here are its bounds. Set $\Omega_R = \{u \in J\} \cap B_{3R}(z)$; central-volume control gives $|\Omega_R| \leq CN$. Differentiating λ is supported in $a/2 < g < a$, and its contribution is at most CR^{d-2} by the small-gradient covering and the uniform Hessian bound. Differentiating ψ_R costs at most $C|\Omega_R|/R \leq CN/R$. For $H = \operatorname{div} \nu$, the pointwise inequality $|H| \leq \sqrt{d}|\mathrm{II}|$ and Cauchy–Schwarz give

$$\begin{aligned}\int_{\Omega_R \cap \{g > a/2\}} |H| \, dX &\leq C \left(\int_{B_{3R}(z)} g^3 |\mathrm{II}|^2 \right)^{1/2} \left(\int_{\Omega_R \cap \{g > a/2\}} g^{-3} \right)^{1/2} \\ &\leq CN/R.\end{aligned}$$

Thus $F_R \in W^{1,1}(J)$ and

$$\int_J |F_R'| \leq C \left[\frac{N}{Rm} + \frac{R^{d-2}}{m} \right].$$

Normal coarea, the lower central-source bound on $C \subset \{u \in J\}$, and the small-gradient-volume estimate also give

$$\int_J F_R \geq m^{-1} \int_{B_{2R}(z) \cap C} g - C\frac{R^{d-2}}{m} \geq c - C\frac{R^{d-2}}{m}.$$

If $N/R \geq \delta m$ for a fixed small $\delta > 0$, the preliminary lower bound ensures $m \geq 1$ for large R, and directly gives $m^{(d-1)/d} \leq m \leq \delta^{-1} N/R$. Otherwise choose δ small enough and then R large enough. Uniformly in the central point,

$$\frac{R^{d-2}}{m} \leq CR^{p-3} \longrightarrow 0.$$

Thus $\int_J F_R \geq c/2$, whereas its total variation can be made at most $c/(4|J|)$ by those choices. Its absolutely continuous representative satisfies

$$F_R(t) \geq |J|^{-1} \int_J F_R - \int_J |F_R'| \geq \frac{c}{4|J|} > 0 \qquad (t \in J).$$

Since $0 \leq \lambda \leq 1$, the complete level area weighted by ψ_R is at least cm for almost every $t \in J$.

For every regular value t, the full level $M_t = \{u = t\}$ is a smooth closed embedded hypersurface, and χ_R has compact support on it. The Michael–Simon hypersurface Sobolev inequality[8] therefore gives

$$\left(\int_{M_t} \psi_R \, d\mathcal{H}^d \right)^{(d-1)/d} \leq C \int_{M_t} (|\nabla_{M_t} \chi_R| + |H| \chi_R) \, d\mathcal{H}^d.$$

Integrate over $t \in J$. The left side is at least $cm^{(d-1)/d}$. By coarea the cutoff-gradient term is at most CN/R. On $g > a$, the mean-curvature term is bounded by

$$\int_{\Omega_R \cap \{g > a\}} g|H| \, dX \leq C \left(\int_{B_{3R}(z)} g^3 |\mathrm{II}|^2 \right)^{1/2} \left(\int_{\Omega_R \cap \{g > a\}} g^{-1} \right)^{1/2} \leq CN/R.$$

On $0 < g \leq a$ it costs CR^{d-2}, because $g|H| \leq \sqrt{d}|D^2 u| \leq C$ and that portion of Ω_R has the small-gradient volume bound. Coarea discards $g = 0$, while Sard excludes only a null set of phases. This proves

$$m_z(R)^{(d-1)/d} \leq C \left[m_z(A_0 R)/R + R^{d-2} \right].$$

The preliminary bound absorbs the last term, since

$$\frac{(d+1-p)(d-1)}{d} - (d-2) = \frac{2d - 1 - p(d-1)}{d} > 0.$$

This exponent is positive even when $d = 2$. The exponent-one Michael–Simon inequality used above is valid in that dimension, where its target exponent is two. Therefore $m_z(A_0 R) \geq cRm_z(R)^{(d-1)/d}$ for every central point and every $R \geq R_*$, with one fixed sufficiently large R_*. Set $b_k = m_z(A_0^k R_*)/(A_0^k R_*)^d$. Then

$$b_{k+1} \geq \kappa b_k^{(d-1)/d}, \qquad \kappa = cA_0^{-d} > 0, \qquad b_0 \geq cR_*^{1-p}.$$

The center-independent number $b_* = \min\{cR_*^{1-p}, \kappa^d\} > 0$ is an invariant lower bound: if $b_k \geq b_*$, then $\kappa b_k^{(d-1)/d} \geq b_*$. If $A_0^k R_* \leq R < A_0^{k+1} R_*$, monotonicity of m_z gives $m_z(R) \geq b_* A_0^{-d} R^d$. This proves the desired bound at every sufficiently large radius, uniformly in the central point.

Finally $\int_{B_{2R}(z) \cap C} g \geq cm_z(R) \geq cR^d$ and bounded g give $|B_{2R}(z) \cap C| \geq cR^d$. On C the potential is at least $1/36$. Renaming the radius proves the potential lower bound. A nonconstant continuous solution with values in $[-1, 1]$ has a point strictly between the wells, since otherwise its image in the connected ambient space would be a single well. The potential lower bound at this point excludes zero multiplicity in proposition 3.5. □

4. Augmented excess and escaping relative scales

Write $E(R) = \mathcal{E}(R)$ for the optimized covariance excess of proposition 3.5. That proposition gives $E(R) = o(R^d)$. The entries of the covariance density are locally integrable, so their ball integrals are continuous in the radius; continuity of the largest eigenvalue then makes E continuous. Even a quadratic lower bound would not force $\sqrt{E(R)/R^{d-2}}$ to diverge when $d \geq 4$, as the model $E(R) = R^2$ shows. We therefore postpone exact self-normalization and use the upper budget $S(R) = E(R) + R^{d-1}$. The first lemma constructs escaping scales from this augmented budget alone. The final proposition is a conditional absorption criterion; Section 5 verifies its complete-root hypotheses for the actual solution. Thus the artificial summand is removed only after the complete short-gap argument, without assuming the desired scale separation beforehand.

Lemma 4.1 (Dimension-independent escaping scales from an augmented excess)**.** *Let $d \geq 2$ be an integer, and let $E : [1, \infty) \to [0, \infty)$ be continuous with $E(r) = o(r^d)$. Define*

$$S(r) = E(r) + r^{d-1}, \qquad \alpha = d - \frac{3}{2}.$$

There are $R_j \to \infty$ such that, writing

$$S_j = S(R_j) = a_j^2 R_j^d = R_j^{d-2} D_j^2,$$

one has

$$S(r) \leq S_j (r/R_j)^d \quad (r \geq R_j), \qquad S(r) \leq S_j (r/R_j)^\alpha \quad (1 \leq r \leq R_j).$$

Moreover

$$a_j \to 0, \quad D_j = a_j R_j \geq \sqrt{R_j} \to \infty, \quad D_j a_j^{1/4} \geq R_j^{3/8} \to \infty.$$

For $D_j \leq r \leq R_j$, set $w_j(r) = D_j (r/R_j)^{1/4}$. Then

$$\sup_{D_j \leq r \leq R_j} \frac{w_j(r)}{r} = a_j^{1/4} \to 0, \qquad \inf_{D_j \leq r \leq R_j} w_j(r) = D_j a_j^{1/4} \to \infty.$$

The sum of $w_j(r)$ over any dyadic grid in $[D_j, R_j]$ is at most CD_j, and the number of grid radii is $o(D_j)$.

Proof. Put $F(r) = S(r)/r^\alpha$ and $G(r) = S(r)/r^d$. Both are positive and continuous, $F(r) \geq r^{1/2} \to \infty$, and $G(r) \to 0$. Fix $R_* > 1$ such that $G(r) < 1$ for every $r \geq R_*$. Choose strictly increasing numbers $L_j \to \infty$ with

$$L_j > \max_{1 \leq r \leq R_*} F(r).$$

Let t_j be the first radius at which $F(t_j) = L_j$. Continuity and divergence ensure that this first radius exists. The choice of L_j gives $t_j > R_*$, and boundedness of F on each compact interval gives $t_j \to \infty$. By the first-hit property,

$$F(r) \leq F(t_j) \qquad (1 \leq r \leq t_j).$$

The function G attains its maximum on $[t_j, \infty)$. Indeed $G(t_j) > 0$, and beyond a sufficiently large finite radius its values are less than $G(t_j)/2$; the maximum on the remaining compact interval is therefore a maximum on the whole tail. Choose such a maximizing point $R_j \geq t_j$. Then

$$G(r) \leq G(R_j) \qquad (r \geq t_j).$$

For $t_j \leq r \leq R_j$, use $d - \alpha = 3/2 > 0$ to obtain

$$F(r) = G(r) r^{3/2} \leq G(R_j) R_j^{3/2} = F(R_j).$$

For $1 \le r \le t_j$, the first-hit property instead gives

$$F(r) \le F(t_j) = G(t_j)t_j^{3/2} \le G(R_j)R_j^{3/2} = F(R_j).$$

Thus $G(r) \le G(R_j)$ for $r \ge R_j$, while $F(r) \le F(R_j)$ for $1 \le r \le R_j$. Multiplying by the respective powers of r gives exactly the simultaneous forward and backward record inequalities. Since $R_j \ge t_j \to \infty$, pass to a strictly increasing subsequence if necessary and relabel it.

Define the positive quantities $a_j = \sqrt{S(R_j)/R_j^d}$ and $D_j = a_j R_j$. Then

$$S_j = a_j^2 R_j^d = R_j^{d-2} D_j^2, \qquad a_j^2 = G(R_j) \longrightarrow 0.$$

The initial choice $R_j > R_*$ ensures $0 < a_j < 1$, hence $D_j < R_j$, at every retained index. Also $S_j \ge R_j^{d-1}$, so

$$D_j^2 \ge R_j, \qquad a_j \ge R_j^{-1/2}, \qquad D_j a_j^{1/4} \ge R_j^{1/2} R_j^{-1/8} = R_j^{3/8}.$$

In particular $[D_j, R_j]$ is nonempty for every j. The function $w_j(r) = D_j R_j^{-1/4} r^{1/4}$ increases with r, and $w_j(r)/r = D_j R_j^{-1/4} r^{-3/4}$ decreases. Their values at $r = D_j$ give the asserted exact infimum and supremum.

Finally, consider any dyadic grid $\{2^k r_*\} \cap [D_j, R_j]$, with arbitrary $r_* > 0$, or any subcollection of that grid. If it is nonempty, let $r_{\max} \le R_j$ be its largest radius. Positivity and a geometric series give

$$\sum_r w_j(r) \le w_j(r_{\max}) \sum_{\ell=0}^{\infty} 2^{-\ell/4} \le \frac{D_j}{1 - 2^{-1/4}}.$$

Its number N_j of radii satisfies

$$N_j \le 1 + \log_2(R_j/D_j) \le 1 + \log_2 R_j, \qquad \frac{N_j}{D_j} \le \frac{1 + \log_2 R_j}{\sqrt{R_j}} \longrightarrow 0.$$

For an empty grid both conclusions are immediate. If a dyadic chain is taken with either endpoint of $[D_j, R_j]$ additionally inserted, each extra endpoint contributes at most D_j to the sum and one to the count, so the same assertions hold with a larger universal constant. This proves all claims. □

Lemma 4.2 (Intermediate covariance alignment uses only the augmented upper budget)**.** *Let $d \ge 2$ and $n \ge 2$ be integers. For $r \ge 1$, let $M(r)$ be positive semidefinite symmetric $n \times n$ matrices, nondecreasing in the positive semidefinite order, and suppose that for all sufficiently large r*

$$\operatorname{tr} M(r) \le Cr^d, \qquad \lambda_1(M(r)) - \lambda_2(M(r)) \ge cr^d.$$

Let $E(r) = \operatorname{tr} M(r) - \lambda_1(M(r))$ be continuous and $o(r^d)$, and choose augmented records R_j, S_j, D_j as in lemma 4.1. Let e_r be a unit top eigenvector of $M(r)$, considered as an unoriented axis. Uniformly for $D_j \le r \le R_j$, one has

$$\angle(e_r, e_{R_j}) \le C\frac{w_j(r)}{r}, \qquad \operatorname{tr} M(r) - e_{R_j}^T M(r) e_{R_j} \le Cr^{d-2} w_j(r)^2,$$

where $w_j(r) = D_j(r/R_j)^{1/4}$ and the angle lies in $[0, \pi/2]$. The constants are independent of j, r.

Proof. Order the eigenvalues decreasingly, and choose $r_0 \ge 1$ so that the trace and gap hypotheses hold for every $r \ge r_0$. We first work with j large enough that $D_j \ge r_0$, which is possible by lemma 4.1. At these radii the positive gap makes the top eigenspace a unique unoriented axis.

Let $r \le t \le 2r$ and write $e_t = ae_r + v$, with $v \perp e_r$. For $\vartheta = \angle(e_r, e_t)$, $|v| = \sin\vartheta$, and the spectral theorem gives

$$\begin{aligned}\lambda_1(M(r)) - e_t^T M(r) e_t &= \lambda_1(M(r))|v|^2 - v^T M(r) v \\ &\ge [\lambda_1(M(r)) - \lambda_2(M(r))]\sin^2\vartheta.\end{aligned}$$

Also $E(r) \ge 0$. The functional $B \mapsto \operatorname{tr} B - e_t^T B e_t$ is nonnegative on positive semidefinite matrices: it is the sum of their quadratic forms along an orthonormal basis of $e_t^\perp$. Applying it to $M(t) - M(r) \ge 0$ proves

$$cr^d \sin^2\vartheta \le \operatorname{tr} M(r) - e_t^T M(r) e_t \le \operatorname{tr} M(t) - e_t^T M(t) e_t = E(t).$$

Put $\alpha = d - 3/2 > 0$. If also $t \le R_j$, the backward record estimate gives $E(t) \le S_j (t/R_j)^\alpha \le 2^\alpha S_j (r/R_j)^\alpha$. Since $S_j = R_j^{d-2} D_j^2$, we obtain, uniformly for every ratio $1 \le t/r \le 2$,

$$\begin{aligned}\sin\angle(e_r, e_t) &\le C r^{-d/2} S_j^{1/2} (r/R_j)^{d/2 - 3/4} \\ &= C D_j R_j^{-1/4} r^{-3/4} = C \frac{w_j(r)}{r}.\end{aligned}$$

The unoriented angle is the quotient of the spherical distance by the identification $e \sim -e$, so it satisfies the triangle inequality. Also $\vartheta \le (\pi/2)\sin\vartheta$ on $[0, \pi/2]$.

For $r < R_j$, let K be the largest integer with $r_K = 2^K r \le R_j$, and put $r_k = 2^k r$. Use the preceding estimate on each pair (r_k, r_{k+1}), $0 \le k < K$, and, when $r_K < R_j$, on the final pair (r_K, R_j), whose ratio lies in $(1, 2)$. Then

$$\angle(e_r, e_{R_j}) \le C \sum_{k=0}^{K} \frac{w_j(r_k)}{r_k} = C \frac{w_j(r)}{r} \sum_{k=0}^{K} 2^{-3k/4} \le C \frac{w_j(r)}{r}.$$

If $r = R_j$, the same conclusion is immediate. This proves the angle estimate, including endpoints and non-dyadic final ratios.

Now write $e_{R_j} = ae_r + v$, $v \perp e_r$, and use positive semidefiniteness in the opposite direction:

$$0 \le \lambda_1(M(r)) - e_{R_j}^T M(r) e_{R_j} = \lambda_1(M(r))|v|^2 - v^T M(r) v \le C r^d \sin^2\angle(e_r, e_{R_j}).$$

Moreover the backward record bound gives exactly

$$E(r) \le S_j (r/R_j)^{d-3/2} = D_j^2 R_j^{-1/2} r^{d-3/2} = r^{d-2} w_j(r)^2.$$

Adding the last two estimates proves the same-frame excess bound for all sufficiently large j.

Only finitely many remaining indices have $D_j < r_0$. For each nonempty interval $[D_j, R_j]$, positive semidefinite monotonicity gives $0 \le \operatorname{tr} M(r) - e_{R_j}^T M(r) e_{R_j} \le \operatorname{tr} M(R_j) < \infty$, and every choice of top eigenvectors has unoriented angle at most $\pi/2$. Here $D_j \ge \sqrt{R_j} \ge 1$, and the continuous functions $w_j(r)/r$ and $r^{d-2} w_j(r)^2$ are strictly positive on that compact interval. Their positive minima therefore absorb these two finite bounds into one constant. Maximizing over the finitely many initial intervals gives constants independent of j, r, without requiring a spectral gap or a continuous choice of eigenvector there. Empty intervals impose no condition. □

Proposition 4.3 (The short-gap estimate eventually absorbs the artificial record budget)**.** *Let $1 < p < 2$ and use the sequences of lemma 4.1. For each j, let (Y_j, μ_j) be a measure space equipped with a measurable finite ordered root list*

$$t_{j,1}(y) < \cdots < t_{j,N_j(y)}(y),$$

where $N_j(y)$ is a measurable nonnegative integer. Suppose there are fixed $c_0, C_0, C_1, H_0 > 0$ and measurable sets $A_j \subset Y_j$ such that:

1. *$\mu_j(A_j) \ge c_0 R_j^d$, and $N_j(y) \ge 2$ on A_j.*
2. *The first moments satisfy*
$$\int_{A_j} \sum_{k=1}^{N_j(y)} |t_{j,k}(y)|\, d\mu_j(y) \le C_0 D_j R_j^d.$$
3. *For every $H_0 \le H \le R_j$, the measure of points whose root list has a consecutive pair of separation at most H satisfies*
$$\mu_j\{y \in Y_j : \exists k < N_j(y),\ t_{j,k+1}(y) - t_{j,k}(y) \le H\} \le C_1 R_j^{d-1} H^{(p+1)/2}.$$

Then there exist $C = C(p, c_0, C_0, C_1)$ and an index j_0, with j_0 allowed to depend on H_0 and the sequences, such that for every $j \ge j_0$,
$$\frac{R_j^2}{D_j^{p+1}} \le C, \qquad 0 \le \frac{R_j^{d-1}}{S_j} \le C R_j^{1-4/(p+1)}.$$
Consequently
$$\frac{R_j^{d-1}}{S_j} \longrightarrow 0, \qquad \frac{E(R_j)}{S_j} \longrightarrow 1.$$
These conclusions do not assume beforehand that $E(R_j)$ equals the normalization S_j. No all-index bound with the stated parameter-only constant is asserted.

Proof. By lemma 4.1, $D_j \to \infty$ and $D_j/R_j = a_j \to 0$. Interpret each root coordinate as a measurable function on $\{N_j \ge k\}$, and extend its absolute value by zero elsewhere. Then
$$M_j(y) = \sum_{k \ge 1} \mathbf{1}_{\{N_j(y) \ge k\}} |t_{j,k}(y)|$$
is measurable; the sum is finite at every point. Likewise the short-gap event is the countable union of its measurable events on $\{N_j \ge k+1\}$. Set $K = \max\{1, 2C_0/c_0\}$. Markov's inequality gives
$$\mu_j\{y \in A_j : M_j(y) > K D_j\} \le (C_0/K) R_j^d \le (c_0/2) R_j^d.$$
Thus the measurable set $A_j \cap \{M_j \le K D_j\}$ has measure at least $(c_0/2)R_j^d$. Its first two roots already give a consecutive pair, and
$$0 < t_{j,2} - t_{j,1} \le |t_{j,1}| + |t_{j,2}| \le M_j \le K D_j \le 2K D_j.$$
This uses no choice of a pair from the varying finite lists.

Choose j_0 so that $H_0 \le 2K D_j \le R_j$ for all $j \ge j_0$. The first inequality is possible because $D_j \to \infty$, and the second because $2K a_j \to 0$. This choice, unlike the constant below, may depend on H_0 and the sequences. Apply the short-gap bound with $H = 2K D_j$ only at these indices:
$$(c_0/2) R_j^d \le C_1 R_j^{d-1} (2K D_j)^{(p+1)/2}.$$
Divide by R_j^{d-1} and square to obtain
$$R_j^2 \le C_\lambda D_j^{p+1}, \qquad C_\lambda = (2C_1/c_0)^2 (2K)^{p+1}.$$
The constant depends only on p, c_0, C_0, C_1. It gives $D_j^2 \ge C_\lambda^{-2/(p+1)} R_j^{4/(p+1)}$ on the same tail. Since the normalization in lemma 4.1 is $S_j = R_j^{d-2} D_j^2$, we get
$$0 \le \frac{R_j^{d-1}}{S_j} = \frac{R_j}{D_j^2} \le C_\lambda^{2/(p+1)} R_j^{1-4/(p+1)}.$$

Enlarge a single constant to cover both tail inequalities. The exponent $1 - 4/(p+1)$ is negative for $1 < p < 2$, so the displayed ratio tends to zero. Finally the defining identity $S_j = E(R_j) + R_j^{d-1}$ from lemma 4.1 gives $E(R_j)/S_j = 1 - R_j^{d-1}/S_j \to 1$. Every quantitative estimate was applied only for $j \geq j_0$, and the finite initial segment has no effect on either limit. □

5. Complete roots, coarse moments, and interaction control

The following estimates concern actual roots of the stable solution. They use the augmented budget only as an upper bound. Complete prefix counts retain count-preserving replacements at different heights and all boundary incidences of the projected detector columns.

Lemma 5.1 (Sharp growth and noncollapse bound the complete short-gap count in every dimension)**.** *Let $2 \leq d \leq 6$, $1 < p < 2$, and let u satisfy the hypotheses of proposition 3.4. Fix a compact interval $I \Subset (-1, 1)$ of positive length. In any orthonormal coordinates $(x, t) \in \mathbb{R}^d \times \mathbb{R}$, for almost every $(x, c) \in B_r^d \times I$ list the complete roots of $u(x, t) = c$ in $(-r, r)$ in increasing order. Let $N_{r,H}(x, c)$ be the number of consecutive pairs separated by at most H. There are C, H_0 independent of the orthonormal frame such that*

$$\int_{B_r^d \times I} N_{r,H}(x, c)\, dxdc \leq C r^{d-1} H^{(p+1)/2} \qquad (r \geq H \geq H_0).$$

Every measurable subcollection of these consecutive pairs obeys the same bound.

Proof. Use the stable cubic field and sharp all-center growth in proposition 3.4.

Choose $\theta_I < 1$ with $I \subset [-\theta_I, \theta_I]$, and choose $H_0 \geq 1$ at least the threshold in lemma 3.6 for this θ_I. Its noncollapse constant may depend on the distance of I from the wells.

Write $n = d+1$, $h = u_t$, and $K(X, Y) = |X - Y|^{-n-p}$. The normalized stability form used here is

$$Q_u(\phi) = \frac{c_{n,p}}{2} \iint |\phi(X) - \phi(Y)|^2 K(X, Y) dX dY + \int (3u^2 - 1)\phi^2 dX,$$

where $c_{n,p}$ is the kernel constant for Fourier symbol $|\xi|^p$. Stability means $Q_u(\phi) \geq 0$ for compact smooth ϕ. The uniform regularity in lemma 2.1 gives $|Du| + |D^2 u| \leq C_{n,p}$ globally. Thus $(A + 3u^2 - 1)h = 0$, where $A = (-\Delta)^{p/2}$, and h, Dh are bounded independently of the orthonormal frame. The weak equation can be tested with $h\eta^2$: this test is compact and smooth, near-diagonal products of differences are $O(|X - Y|^2)$, and boundedness handles all exterior partners. For $0 \leq \eta \in C_c^\infty$, subtracting this tested identity from $Q_u(h\eta)$ uses the pointwise algebra

$$(h_X \eta_X - h_Y \eta_Y)^2 - (h_X - h_Y)(h_X \eta_X^2 - h_Y \eta_Y^2) = h_X h_Y (\eta_X - \eta_Y)^2,$$

and gives

$$Q_u(h\eta) = \frac{c_{n,p}}{2} \iint h(X) h(Y) (\eta(X) - \eta(Y))^2 K(X, Y),$$

$$Q_u(h\eta) - Q_u(|h|\eta) = 4 c_{n,p} \iint h_+(X) h_-(Y) \eta(X) \eta(Y) K(X, Y).$$

In the second identity the potential terms cancel; the kinetic identity $(a - b)^2 - (|a| - |b|)^2 = 2(|ab| - ab)$ and symmetry between X, Y give exactly the factor $4c_{n,p}$. Compactly supported Lipschitz tests such as $|h|\eta$ are admissible by fixed-support H^1, hence $H^{p/2}$, approximation. Opposite signs imply $h_+(X), h_-(Y) \leq |h(X) - h(Y)| \leq C|X - Y|$, so their product is $O(|X - Y|^2)$ near the diagonal. Both sides of the second identity are therefore absolutely integrable for $p < 2$.

Choose $0 \le \eta \le 1$, equal to one on $Q_r = B_r^d \times (-r, r)$, supported in B_{3r}^n, and with Lipschitz constant C/r. For $r \ge 1$, sharp all-center growth gives

$$\sup_X \int |h(Y)| \min\{|X-Y|^2/r^2, 1\} K(X,Y) dY \le Cr^{-2}.$$

For distances below one, bounded h gives $Cr^{-2} \int_0^1 \rho^{1-p} d\rho$. On a dyadic shell of radius $1 \le \rho \le r$, its $|h|$-mass is at most $C\rho^d$; multiplying by $C\rho^{2-n-p}/r^2$ gives $Cr^{-2}\rho^{1-p}$, since $n = d+1$. The sum over these shells is $O(r^{-2})$, because $p > 1$. The shells with $\rho \ge r$ contribute a sum bounded by $Cr^{-1-p} \le Cr^{-2}$. This proves the displayed uniform row estimate. Since $(\eta(X) - \eta(Y))^2 \le C \min\{|X-Y|^2/r^2, 1\}$ and at least one endpoint lies in B_{3r}^n when this difference is nonzero, symmetry bounds the absolute value of the first quadratic form by

$$Cr^{-2} \int_{B_{3r}^n} |h| \le Cr^{d-2}.$$

Stability gives $Q_u(|h|\eta) \ge 0$. Nonnegativity of η, its value one on Q_r, and the second identity consequently imply

$$\iint_{Q_r^2} h_+(X) h_-(Y) K(X,Y) \le Cr^{d-2}.$$

We specify the complete lists and their measurability before counting. For each fixed x, the critical values of the smooth map $t \mapsto u(x,t)$ on $[-r, r]$ have measure zero by one-dimensional Sard; discard also the two cap values $u(x, \pm r)$. These exceptional pairs (x, c) form a measurable set: the critical-value incidence set is the continuous image under $(x,t) \mapsto (x, u(x,t))$ of the compact critical set in $\overline{B}_r^d \times [-r, r]$, restricted to $B_r^d \times I$, and the cap graphs are measurable. Fubini shows that the discarded set has $(d+1)$-dimensional measure zero. For every remaining pair, the roots are regular and finite. Indeed their set is closed in $[-r, r]$, avoids its endpoints, and consists of isolated points; an infinite set would have a nonregular accumulation root. Countably many local implicit-function charts cover the regular root set. Their root graphs, ordered by their height, give measurable lists $t_1(x,c) < \cdots < t_N(x,c)$ and measurable adjacent-gap and crossing-sign tests. We set all counts to zero on the discarded null set.

A positive root means a regular crossing with $h = u_t > 0$, and a negative root means one with $h < 0$; these words do not refer to the sign of the phase. Consecutive regular roots on a fixed fiber have opposite crossing orientations. To see this, $u - c$ is nonzero with constant sign on the open interval between them. The first root crosses into that sign and the second crosses out of it, so their nonzero derivatives have opposite signs. Thus every counted consecutive pair has exactly one positive and one negative root. Each root belongs to at most the two neighboring consecutive pairs. These facts justify assigning every pair to its positive endpoint and charging both endpoint counts with multiplicity at most two.

Partition $\mathbb{R}^n$ into half-open product cubes of side H in the chosen orthonormal frame, and assign each counted pair to the unique cube containing its positive root. If $C_k = Q_k^d \times [m_k H, (m_k+1)H)$, put $C_k^* = Q_k^d \times [(m_k-2)H, (m_k+3)H)$. This is the union of C_k and its two nearest neighbors in each vertical direction. Since a counted gap is at most H, its

negative endpoint lies in C_k^*. Let $p_k(x,c)$ count the pairs assigned to C_k, and put

$$A_k = \int_{C_k \cap Q_r} \mathbf{1}_{\{u \in I\}} h_+,$$
$$B_k = \int_{C_k^* \cap Q_r} \mathbf{1}_{\{u \in I\}} h_-,$$
$$P_k = \int_{B_r^d \times I} p_k(x,c)\, dx dc.$$

The half-open convention and the measurable root lists make p_k measurable, and $\sum_k p_k = N_{r,H}$ almost everywhere. The one-dimensional area formula, integrated in x, identifies A_k with the phase-integrated number of positive roots in $C_k \cap Q_r$, and B_k with the corresponding number of negative roots in $C_k^* \cap Q_r$. Each endpoint occurs in at most two consecutive pairs, so $P_k \le 2A_k$, $P_k \le 2B_k$, and $P_k^2 \le 4A_kB_k$. Every $(X,Y) \in C_k \times C_k^*$ has $|X - Y| \le C_n H$, and hence $1 \le CH^{n+p}K(X,Y)$ away from the measure-zero diagonal. The source cubes are disjoint; therefore, by the nonnegative interaction bound above,

$$\sum_k A_k B_k \le CH^{d+1+p} r^{d-2}.$$

If $P_k > 0$, choose a positive endpoint $Z_k \in C_k \cap Q_r$ of an assigned pair, so $u(Z_k) \in I$. By the chosen H_0 and lemma 3.6, $\int_{B_H(Z_k)} |Du| \ge cH^d$. These balls have overlap at most C_n: for a point X belonging to such a ball, the distinct cube C_k intersects $B_H(X)$, and only a dimensionally bounded number of side-H grid cubes do so. Moreover $Z_k \in Q_r \subset B^n_{\sqrt{2}r}$ and $H \le r$, so all balls lie in B^n_{3r}. Summing noncollapse and using sharp upper growth gives

$$cH^d \#\{k : P_k > 0\} \le \sum_{P_k>0} \int_{B_H(Z_k)} |Du| \le C_n \int_{B^n_{3r}} |Du| \le Cr^d.$$

Thus there are at most $C(r/H)^d$ active cubes. Cauchy–Schwarz now gives

$$\left(\sum_k P_k\right)^2 \le C(r/H)^d \sum_k P_k^2 \le Cr^{2d-2}H^{p+1}.$$

By Tonelli, $\sum_k P_k = \int_{B_r^d \times I} N_{r,H}$; taking square roots proves the claim. All constants used only rotationally invariant equation, stability, growth and noncollapse bounds and the geometry of a unit product grid, so they are uniform over orthonormal frames. Restricting to any measurable subfamily only decreases the nonnegative count. □

Proposition 5.2 (Actual complete ranks have a first moment controlled by the augmented upper budget)**.** *Let $2 \le d \le 6$, $1 < p < 2$, and let u satisfy the hypotheses of proposition 3.4. Suppose the integer macroscopic multiplicity supplied by proposition 3.5 is $m \ge 1$. Let*

$$M(r) = \int_{B_r \times (0,\infty)} D_X U \otimes D_X U \, d\mu, \qquad E(r) = \operatorname{tr} M(r) - \lambda_1(M(r)),$$

and take the augmented records of lemma 4.1:

$$S_j = E(R_j) + R_j^{d-1} = a_j^2 R_j^d = R_j^{d-2} D_j^2.$$

Use a top covariance axis at R_j as the t direction. Fix a compact interval $I \Subset (-1,1)$ of positive length. There are fixed $b_0, b_1, C > 0$ such that, for $\mathcal{Q}_j = B^d_{b_0 R_j} \times (-b_1 R_j, b_1 R_j)$,

$$\int_{\mathcal{Q}_j \cap \{u \in I\}} |t|\, |u_t|\, dX \le C D_j R_j^d.$$

Let $N_j(x,c)$ count all roots in $(-b_1R_j, b_1R_j)$ of $u(x,t)=c$. Then

$$R_j^{-d}\int_{B^d_{b_0R_j}\times I}|N_j-m|\,dxdc\longrightarrow 0.$$

Consequently, on a subset of base–phase measure at least cR_j^d, the complete root list has exactly m members and its total absolute height has integral at most $CD_jR_j^d$.

Completeness throughout refers to all roots in the specified finite normal window. More generally, for every fixed compact buffered phase interval $J\Subset(-1,1)$ and every fixed enlarged product receiver $B^d_{BR_j}\times(-TR_j,TR_j)$, all its central roots lie strictly between common macroscopic normal caps inside that window, with $|t|=o(R_j)$ uniformly on the central set. Those caps avoid J, and their values lie in fixed components of $[-1,1]\setminus J$ across the connected base receiver. If J contains zero, the sign at either cap is consequently constant across that receiver. A subsequence makes the two cap signs independent of j. These conclusions hold also in each selected constant small shear, using the corresponding common caps.

Proof. The uniform physical derivative bounds and translation compactness used below follow from lemma 2.1, applied to the bounded cubic field. In particular all fixed-order derivative bounds are independent of the record radius, direction, and center.

We give the estimates before choosing a generic direction, and never assume $E(R_j)=S_j$. For readability suppress j, put $R=R_j$, $D=D_j$, $a=a_j$, $S=S_j=a^2R^d$, and $w_r=D(r/R)^{1/4}$.

Choose compact intervals $I_1, I'\Subset(-1,1)$ with $I\subset\operatorname{int}I_1$ and $I_1\subset\operatorname{int}I'$. We carry out the count and BV estimates on I_1; this leaves room for the later phase cutoff to equal one on I. Choose the detector and gradient floor on I'. All constants may depend on these fixed intervals and their distance from the wells.

The covariance matrices are positive semidefinite and nondecreasing in that order directly from their integral definition. Their entries are continuous in r by dominated convergence, since the complete extension energy is finite over every bounded physical ball by proposition 3.4. Continuity of the largest eigenvalue then makes E continuous, and proposition 3.5 gives $E(r)=o(r^d)$. Its complete covariance limits and the positivity of m also give, uniformly at all sufficiently large radii,

$$\lambda_1(M(r))-\lambda_2(M(r))\ge cr^d,\qquad \operatorname{tr}M(r)\le Cr^d.$$

Otherwise a contradicting radius sequence would have a quantized covariance subsequence. Its limit is the rank-one matrix with positive coefficient $\sigma m\omega_d$. Thus lemma 4.2 gives, for $D\le r\le R$ in the one fixed record frame,

$$\int_{B_{Cr}\times(0,\infty)}|V|^2d\mu\le Cr^{d-2}w_r^2,\qquad V=D_xU.$$

To justify a fixed enlargement with $Cr>R$, put $s=Cr$, so r is comparable to R. The spectral gap at R, positive semidefinite monotonicity, and the forward record inequality give

$$cR^d\sin^2\angle(e_R,e_s)\le\operatorname{tr}M(R)-e_s^TM(R)e_s\le E(s)\le a^2s^d.$$

Hence the angle is at most C_Ca, and $\operatorname{tr}M(s)-e_R^TM(s)e_R\le E(s)+Cs^d\sin^2\angle(e_R,e_s)\le C_CS$. Since r is comparable to R, this is the asserted $Cr^{d-2}w_r^2$ bound. When $Cr\le R$, apply lemma 4.2 directly at Cr. Set $a_r=w_r/r$. By lemma 4.1, $\sup a_r\to 0$, $\inf w_r\to\infty$, $\sum_{r\text{ dyadic}}w_r\le CD$, and the number of dyadic radii is $o(D)$.

Whole-central-set confinement is available uniformly at these radii. Indeed, if a point with u in a fixed compact central interval stayed a positive scaled distance from the limiting plane, lemma 3.6 would put positive potential energy in a ball about that point disjoint from the

plane, contradicting the energy limit in proposition 3.5. Use a slightly larger compact central interval if necessary; its constants follow by the same proof, whose only dependence is on its distance from the wells. Covariance alignment consequently puts every central point in the top receiver in

$$|t| \le \epsilon_j(|X| + D), \qquad \epsilon_j \longrightarrow 0.$$

Here at a point of size $\rho \ge D$ use the confinement in its own covariance axis and the angle bound $Cw_\rho/\rho \to 0$; at fixed multiples beyond R, use the enlargement angle estimate above. At size less than D, use radius D. Uniformity follows by contradiction: any sequence of failing intermediate radii is at least $D_j \to \infty$, so the same quantized-subsequence and central noncollapse argument applies to that sequence of radii.

Take a compact interval containing both the tested phases and zero. Confinement places its whole central set in $|t| \le \epsilon_j R_j$ on a fixed enlarged receiver, with $\epsilon_j \to 0$. Caps at fixed heights $\pm cR_j$ therefore avoid that interval. Each cap over a connected base receiver is connected, and its continuous trace cannot move between the two components of the interval's complement. This proves the asserted constant cap signs, after a finite-valued subsequence. A bounded constant shear preserves the normal coordinate and changes the physical norm by at most $C\kappa|t|$. Inserting this bound in the confinement inequality adds $C\epsilon_j\kappa|t|$ to its right side, which is absorbed for all sufficiently large j, uniformly for the fixed bounded shear family. Thus the same confinement, caps and sign argument are valid in those frames.

Fix a small $\kappa > 0$ and consider the actual sheared fields $u_b(x,t) = u(x+\kappa bt, t)$, $b \in B_1^d$. The determinant is one and the normal coordinate is unchanged. Absorb $\epsilon_j\kappa|t|$ in the preceding inequality. Every central root in the fixed enlarged top normal window with auxiliary base in B_r, $D \le r \le R$, then has $|t| < c_0 r$, where $c_0 > 0$ can be arbitrarily small for sufficiently large j, uniformly in r and b. Choose $0 < C_0 < T$ fixed. The tested roots of the one complete top list therefore all lie in $(-C_0 r, C_0 r)$ at every intermediate base, and these intervals have phase-free end caps. Thus restricting the base and replacing the normal window by this local interval gives exactly the same roots and ranks.

Write $v = D_x u$, $h = u_t$, $Z_r = (V, a_r U_t)$, and $\mathcal{A}_r = |DZ_r|^2 - |D|Z_r||^2$, with D including height. Every component of Z_r is a constant multiple of a physical Jacobi field and has the common Robin coefficient $1 - 3u^2$. Thus the norm–stability calculation in the proof of proposition 3.4 applies to Z_r: no derivative falls on the constant a_r. For every compact physical-and-height cutoff ζ, it gives

$$\int \zeta^2 \mathcal{A}_r \, d\mu \le \int |Z_r|^2 |D\zeta|^2 d\mu.$$

As in that calculation, remove the regularization $(|Z_r|^2+\varepsilon^2)^{1/2}$ at fixed compact height before removing the height cutoff. Sobolev locality defines the angular integrand to be zero almost everywhere on $\{Z_r = 0\}$. The constant in this inequality is one, independently of a_r.

Now choose $\zeta(X,y) = \chi_r(X)\rho_T(y)$, where $\chi_r = 1$ on B_{Cr}, is supported in $B_{C'r}$ and has $|D\chi_r| \le C/r$, while $\rho_T = 1$ below T, vanishes above $2T$, and has $|\rho_T'| \le C/T$. The preceding same-frame covariance estimate and the complete extension bound in proposition 3.4 give

$$\int_{B_{C'r}\times(0,\infty)} |Z_r|^2 d\mu \le Cr^{d-2} w_r^2 + Ca_r^2 r^d \le Ca_r^2 r^d,$$

since $w_r = a_r r$. The height-cutoff term is at most CT^{-2} times this finite full-height integral and tends to zero as $T \to \infty$ at each fixed record and radius. The norm regularization has already been removed, so no constant ε^2 is integrated over an infinite height range. Fatou

now gives

$$\int_{B_{Cr}\times(0,\infty)} \mathcal{A}_r \, d\mu \le Cr^{-2} \int_{B_{C'r}\times(0,\infty)} (|V|^2 + a_r^2 U_t^2) d\mu \le C a_r^2 r^{d-2}.$$

The constant is independent of a_r, and the factor a_r^2 comes from the two displayed complete energy bounds, without identifying the augmented budget with the true excess. For later use, a local Robin estimate also gives $\int_{B_{Cr}} |v|^2 \le Cr^{d-2} w_r^2$. In fact, testing the Jacobi equation of a component of V by that component times a fixed unit-scale squared cutoff yields a Caccioppoli inequality with a bounded bottom coefficient. The weighted trace inequality $\int_{y=0} f^2 \le \varepsilon \int y^{1-p} |Df|^2 + C_\varepsilon \int y^{1-p} f^2$, on nested unit half-balls, absorbs that coefficient for sufficiently small ε. A second application bounds the trace norm by the extension $L^2(d\mu)$ norm on a slightly larger half-ball. Summing a bounded-overlap cover and using the preceding same-frame estimate proves the assertion.

On a trace chart where $|v| \ge c > 0$, choose one unscaled tangential Jacobi component J with $|J| \ge c/\sqrt{d}$ at the chart center. The bounded trace derivatives and the Poisson kernel's finite first moment give a uniform modulus of continuity for this component up to height zero. Thus, on a half-ball of radius depending only on c, n, p, J has one sign and $|J| \ge c/(2\sqrt{d})$. These choices do not involve a_r. Put $W = Z_r/J$, so one component of W is identically one. Subtracting the common Robin identities gives

$$\operatorname{div}(y^{1-p} J^2 DW) = 0$$

with homogeneous weak Neumann bottom condition. Both J^2 and its physical first derivatives are uniformly bounded, and J^2 has a positive lower bound on this half-ball. For each physical coordinate X_α, differentiation in the weak sense gives

$$\operatorname{div}(y^{1-p} J^2 D\partial_\alpha W) = -\operatorname{div}(y^{1-p} \partial_\alpha(J^2) DW),$$

with the corresponding homogeneous weak conormal identity. This also holds for the tangential difference quotients: their bottom flux is the difference quotient of the zero flux, so no Robin boundary term remains. Testing these physical difference quotients with a squared cutoff and absorbing the mixed terms therefore yields, on nested fixed half-balls $H' \Subset H$,

$$\int_{H'} y^{1-p} (|D_X W|^2 + |D D_X W|^2) \le C_c \int_H y^{1-p} |DW|^2.$$

The weighted trace inequality applied to $D_X W$ bounds its squared trace norm by this left side. Finally the exact identity and its two elementary consequences are

$$\mathcal{A}_r = J^2 \left(|DW|^2 - \frac{|W \cdot DW|^2}{|W|^2} \right),$$

$$|DW|^2 \le C_c \mathcal{A}_r,$$

$$|D_X(v, a_r h)|^2 - |D_X|(v, a_r h)||^2 \le C_c |D_X W(\cdot, 0)|^2.$$

For the middle inequality use the component $W_i = 1$, which gives $|W \cdot DW|^2 \le (|W|^2 - 1)|DW|^2$; all quotient values are uniformly bounded. These estimates give

$$\int_{B_{Cr} \cap \{|v| \ge c\}} (|D_X(v, a_r h)|^2 - |D_X|(v, a_r h)||^2) \le C_c a_r^2 r^{d-2}.$$

Here a bounded-overlap fixed-radius cover sums the chart estimates and uses the extension angular bound on a slightly enlarged receiver. The chart radii, coefficients and quotient bounds are independent of $0 < a_r \le 1$, since J was chosen from the unscaled tangential components.

Use the centered increasing cubic layer q already fixed in propositions 3.4 and 3.5. Its normalization and uniqueness up to translation and reversal were checked there, and lemma 2.4 gives $q' > 0$. Put $\delta_I = \operatorname{dist}(I', \{-1, 1\}) > 0$.

The compact set

$$K_I = q^{-1}([-1 + \delta_I/2,\, 1 - \delta_I/2])$$

is bounded, and $c_q := \min_{K_I} q' > 0$. Choose the detector's fixed C^2 accuracy smaller than $\min\{\delta_I/2, c_q/2\}$. A point with $u \in I'$ cannot be approximated by a constant well at that accuracy. For a comparison layer its value belongs to the interval defining K_I, and C^1 closeness then gives $|Du| \geq c_q/2 =: c_I$ at the point, uniformly in the translation and unit direction. The C^2 comparison supplies this C^1 closeness as well.

Choose once for this record a maximal disjoint family of unit balls centered at failed points with $u \in I'$ in a fixed enlarged top receiver. Their fixed enlargements cover every failed central point in the receivers under consideration. This one finite family is used for every intermediate radius and every b; in particular the masked root ranks below do not change when the radius changes. The restriction of the centers to $\{u \in I'\}$ is needed for their confinement. Under the inverse shear, project each enlarged ball into the auxiliary base and enclose its projection in a base ball of one common fixed radius. Call the union of these base balls B_b^{det}. Outside its columns the full-gradient floor holds throughout the tested central band.

If such a projected ball meets B_r, its physical center (x_*, t_*) satisfies $|x_* - \kappa b t_*| \leq r + C$. Applying the central confinement estimate to this center gives $|t_*| \leq C\epsilon_j(r + D + |t_*|)$. Absorption, $r \geq D \to \infty$, and the same estimate for x_* place the center in B_{Cr}. The disjoint-ball detector estimate in proposition 3.4 and the same-frame covariance bound therefore bound the number of contributing balls by $Ca_r^2 r^{d-2}$. Let $P_{\mathrm{det}}(r, b)$ be the phase-integrated complete root incidence on the boundaries of these projected balls, summed before cancellation. On each such fixed sphere, the one-dimensional area formula and bounded h_b give at most Cr: indeed $\int_{I_1} N(x, c)dc = \int \mathbf{1}_{\{u_b \in I_1\}} |h_b| dt \leq Cr$ on its complete normal window. Hence

$$P_{\mathrm{det}}(r, b) \leq Ca_r^2 r^{d-1}.$$

The boundary count used here is the actual BV trace, even for these fixed spheres. Indeed, for each fixed sphere base point, one-dimensional Sard excludes critical phases and cap values for almost every phase. At each remaining base–phase point all complete roots are finite and simple, so their implicit-function graphs make the count locally constant; both BV traces therefore equal that actual count. For a prefix count additionally exclude its endpoint phase $u_b(x, z) = c$; Fubini in z justifies the integrated prefix boundary term. Unions have no greater boundary incidence than the sum of those spheres.

For generic b, let $P_x(r, b)$ and $P_c(r, b)$ be the complete positive critical-point quantities, outside B_b^{det},

$$P_x = 2\int_{Q_{Cr} \cap \{u_b \in I_1\}} \delta(h_b) |\partial_t h_b|\, |v_b|, \qquad P_c = 2\int_{Q_{Cr} \cap \{u_b \in I_1\}} \delta(h_b) |\partial_t h_b|,$$

where $h_b = \partial_t u_b$, $v_b = D_x u_b$. They are defined by positive coarea in the direction parameter and then generic physical regular-level coarea. In the original variables, $h_b = h + \kappa b \cdot v$. On $h_b = 0$, the functional $(\kappa b, 1/a_r)$ annihilates $(v, a_r h)$, so

$$|(\partial_t + \kappa b \cdot D_x) h_b| \leq Ca_r^{-1} \sqrt{|D_X(v, a_r h)|^2 - |D_X|(v, a_r h)||^2}.$$

Also parameter coarea gives

$$\int_{B_1^d} \delta(h + \kappa b \cdot v) db \leq C(\kappa |v|)^{-1} \mathbf{1}_{\{|h| \leq \kappa |v|\}}.$$

On this central critical set outside detector columns the full gradient floor gives $|v| \geq c_I/2$. Pulling back by the determinant-one shear and using Cauchy–Schwarz with the last two trace estimates proves

$$\int_{B_1^d} (P_x(r,b) + P_c(r,b))db \leq \frac{C}{a_r}\sqrt{a_r^2 r^{d-2}}\sqrt{r^{d-2}w_r^2} \leq Cr^{d-2}w_r.$$

No denominator involving the number of roots enters this estimate.

Here are the cumulative count identities that retain all changes of occupation. On the complement of the projected detector columns put

$$N_z(x,c) = \int_{-C_0 r}^{z} \delta(u_b(x,t) - c)|h_b(x,t)|dt, \qquad N = N_{C_0 r},$$

and set them zero on those columns. In the sense of distributions, differentiation of the prefix count gives a regular endpoint term plus a critical term. Before imposing the detector characteristic, their exact formulas are

$$D_x N_z = \delta(u_b(x,z) - c)\,\mathrm{sgn}(h_b)v_b(x,z) - 2\int_{-C_0 r}^{z} \delta(u_b - c)\delta(h_b)(\partial_t h_b)v_b\,dt,$$

$$D_c N_z = -\delta(u_b(x,z) - c)\,\mathrm{sgn}(h_b) + 2\int_{-C_0 r}^{z} \delta(u_b - c)\delta(h_b)\partial_t h_b\,dt.$$

The lower cap terms vanish. Smooth sign and phase regularizations, followed by positive coarea and exhaustion, justify these identities as BV measure identities. The detector characteristic adds only its already counted boundary incidence. For the generic-parameter assertion, use independent original variables (b, X) and the function $H(b, X) = h(X) + \kappa b \cdot v(X)$. On its tested zero set outside the detector columns, $v \neq 0$, so $D_b H = \kappa v \neq 0$. Parametric Sard therefore makes $H(b,\cdot)^{-1}(0)$ a regular hypersurface for almost every b in that region. The physical shear is a diffeomorphism for each fixed b, so the same conclusion holds for h_b.

Let F_k be the kth complete root when $N = m$, and zero otherwise, $1 \leq k \leq m$. The same F_k is used at every radius. For integer arguments let $\Phi_k(n) = \mathbf{1}_{\{n<k\}}$ and $\Psi_m(n) = \mathbf{1}_{\{n=m\}}$, choosing bounded one-Lipschitz extensions. Then exactly

$$F_k = \Psi_m(N)\left[-C_0 r + \int_{-C_0 r}^{C_0 r} \Phi_k(N_z)dz\right].$$

BV composition, the prefix identities, and integration in z give

$$\int_{B_r \times I_1} |D_x F_k| \leq C\int_{Q_{Cr}\cap\{u_b \in I_1\}} |v_b| + Cr(P_x + P_{\mathrm{det}}),$$

$$\int_{B_r \times I_1} |D_c F_k| \leq C|Q_{Cr} \cap \{u_b \in I_1\}| + CrP_c.$$

The central volume is $O(r^d)$ by the deficit bound in proposition 3.4. Cauchy–Schwarz and the transverse trace bound make the first bulk term $Cr^{d-1}w_r$. The shear does not change v or volume. Rescale only the base variable and apply BV Poincare on $B_1^d \times I_1$. Its mean oscillation is at most

$$C\left[w_r + 1 + \frac{P_x + P_{\mathrm{det}}}{r^{d-2}} + \frac{P_c}{r^{d-1}}\right].$$

Summing over dyadic radii from D to R and averaging over b bounds this entire positive sum by CD. Indeed $\sum w_r \leq CD$, the sum of unit terms is $o(D)$, the detector term is $a_r w_r$, and the averaged critical terms are at most $Cw_r + Cw_r/r$. Choose one generic b realizing this sum bound. Means of the same F_k on consecutive products differ by at most a fixed multiple of these oscillations, because consecutive product volumes have bounded ratio. At

the bottom scale $|F_k| \le CD$ by confinement. Telescoping the means and applying top-scale Poincare thus proves

$$\int_{B_R \times I_1} |F_k| \le CDR^d.$$

The complete top count is $m + O(a)$ in L^1 after scaling. To see its rate in the same selected direction, first let N denote the complete count with the detector columns still set to zero, and put $\widetilde{N}(\xi, c) = N(R\xi, c)$. The full-count prefix identities have no cap terms. They give

$$|D_\xi \widetilde{N}| \le CR^{1-d}(P_x + P_{\mathrm{det}}), \qquad |D_c \widetilde{N}| \le CR^{-d} P_c,$$

where the variations are on a fixed base–phase product. The selected positive dyadic sum, at its top radius, gives

$$\frac{P_x + P_{\mathrm{det}}}{R^{d-2}} + \frac{P_c}{R^{d-1}} \le CD.$$

Hence both variations are at most $CD/R = Ca$. BV Poincare bounds the L^1 distance to the mean by Ca. Choosing an integer nearest that mean changes this bound by at most a factor two, because $\widetilde{N}$ is integer-valued almost everywhere. Thus $\widetilde{N}$ is $O(a)$ in L^1 from an integer k_j.

The complete count on the detector columns has phase-integrated mass at most Ca^2R^{d-1}: each projected ball has bounded base volume, each complete normal window contributes at most CR by the one-dimensional area formula and bounded h_b, and there are at most Ca^2R^{d-2} such balls at the top scale. Thus deleting these columns changes the normalized count mean by $O(a^2/R) = o(1)$.

We identify the mean using the complete trace-level conclusion of proposition 3.5. Normal coarea expresses the original-axis count against a smooth base test as a complete level integral with amplitude $|\nu \cdot e_t|$. This continuous even amplitude is obtained from the stated smooth normal tests by uniform smooth approximation; the error is bounded by the approximation error times $R^{-d} \int_{\{u \in I_1\} \cap B_{CR}} |Du| \le C$. In the auxiliary frame the amplitude changes by at most $\kappa|v|$ before coarea, whose phase-integrated cost is at most CaR^d. The smooth base test changes by $o(1)$ uniformly in b, since its displacement is $\kappa bt/R = o(1)$ on the whole tested central set. The covariance axis agrees with the normal of the limiting plane, so that normal coarea amplitude equals one in the limit.

These conclusions hold for the selected directions $b = b_j$ and the varying record frames along the entire record sequence. Indeed, from any subsequence extract convergent orthogonal frames and a convergent b_j, and then the common energy, covariance and trace-level subsequence in proposition 3.5. The positive rank-one covariance limit identifies the limiting optimizing axis with the plane normal. Uniform approximation of the resulting moving position–normal tests, using the preceding bounded trace-area mass, gives the same count mean limit m on this further subsequence. Hence a subsequence contradicting that mean limit cannot exist. Indicators of fixed inner base balls may be used by approximation, since their boundaries have zero limiting plane measure. It follows that $k_j \to m$, and integer-valuedness gives $k_j = m$ for all sufficiently large j. Restoring the detector columns in the already established L^1 estimate therefore yields

$$\int_{B_R \times I_1} |N^{\mathrm{full}} - m| \le CaR^d.$$

The rank functions F_k were set to zero on projected detector columns, even when the restored complete count there happens to equal m. Those fibers must be charged separately. The preceding restoration estimate bounds their complete phase-integrated root count by Ca^2R^{d-1}. Confinement bounds the absolute height of every such root by CR. Their total first moment

is therefore at most $Ca^2R^d \le CDR^d$, since $D = aR$ and eventually $a \le 1 \le R$. Outside these columns, the m rank functions account for every root on a good-count fiber. On the remaining count-failure fibers, $N^{\mathrm{full}} \le (m+1)|N^{\mathrm{full}} - m|$ and the complete first moment is at most CR times their count mass. Hence this second contribution is at most $CaR^{d+1} = CDR^d$. Adding these two contributions to the rank estimate proves the auxiliary first-moment bound for the complete root lists. Return to the optimizing frame on fixed nested receivers: the shear preserves t and volume, and $|h| \le |h_b| + \kappa|v|$. Confinement and the trace transverse bound give

$$\int_{\{u\in I_1\}} |t||v| \le o(R)\, CaR^d = o(DR^d).$$

This proves the claimed moment for $|h|$. We justify strong complete counts in the original optimizing axis separately; weak coarea convergence alone would not suffice. Use fixed nested base–phase products, with the target product strictly inside its buffer. On the buffer, outside the projected detector columns, the auxiliary functions F_k are the actual kth root when the complete auxiliary count is m, and zero otherwise; on the detector columns they retain their zero value. These are the BV functions constructed from the masked prefix counts above. Put

$$G_{j,k}(\xi, c) = R_j^{-1} F_k(R_j\xi, c).$$

The already established auxiliary first moment gives $\|G_{j,k}\|_{L^1} \le Ca_j$. The displayed BV derivative bounds and the selected positive dyadic oscillation sum give, on a fixed buffered product,

$$|D_\xi G_{j,k}| \le Ca_j, \qquad |D_c G_{j,k}| \le C(a_j + R_j^{-1}) \le Ca_j.$$

Here every quantity is the total variation on that product. For the second estimate the top term P_c/R_j^{d-1} is at most CD_j because it is one nonnegative summand of the selected dyadic sum. Division by R_j gives the asserted phase-variation bound. For the first, the top term $(P_x + P_{\mathrm{det}})/R_j^{d-2}$ is at most CD_j for the same reason. Multiplying by a fixed smooth base–phase cutoff supported in this buffered product, equal to one on a smaller product whose phase interval contains I, and zero near the boundary changes the L^1 norm and total variation by at most Ca_j. Extend the resulting functions by zero to $\mathbb{R}^{d+1}$ in the joint variables (ξ, c). Denote them again by $G_{j,k}$. Whole-central-set confinement gives $|G_{j,k}| \le \epsilon_j \to 0$ on their support: every nonzero value is a cutoff multiple of an actual complete central root height divided by R_j.

We recall the elementary BV Lipschitz truncation needed here. If $G \in BV(\mathbb{R}^{d+1})$ and $\mu = |DG|$, then at its Lebesgue points

$$|G(z) - G(w)| \le C|z - w|[\mathcal{M}\mu(z) + \mathcal{M}\mu(w)],$$

where $\mathcal{M}\mu$ is the centered maximal density of the measure. Indeed telescope averages over concentric balls from radius $|z-w|$ to zero, using BV Poincare on each ball; compare the two largest-ball averages using one containing ball of comparable radius. This proves the displayed pointwise estimate. The disjoint-ball covering argument gives $|\{\mathcal{M}\mu > \lambda\}| \le C\mu(\mathbb{R}^{d+1})/\lambda$. These assertions apply to singular as well as absolutely continuous derivative measures.

Apply this to every $G_{j,k}$ at threshold $\sqrt{a_j}$. Outside a joint set of measure at most $C_m\sqrt{a_j}$, all m functions have Lipschitz constant $C\sqrt{a_j}$ on their common good set. Their scalar McShane extensions $g_{j,k}$ to the whole joint space have that same Lipschitz constant and agree with $G_{j,k}$ on the good set. Clipping each extension to $[-\epsilon_j, \epsilon_j]$ retains these properties. On the smaller buffered product where the cutoff is one, also remove the auxiliary count-failure products, the projected detector columns, and the Lebesgue null sets. The first set has scaled measure $O(a_j)$ by integer-valuedness and the complete count estimate; the detector

set has scaled base measure at most Ca_j^2/R_j^2 by its fixed-ball count. Thus the common good set A_j has complement of measure tending to zero on that product, and on A_j all $g_{j,k}$ are the m distinct actual auxiliary root heights divided by R_j.

For each k define on the whole joint space

$$\Phi_{j,k}(\xi, c) = (\xi + \kappa b\, g_{j,k}(\xi, c), c).$$

This is the base change from the auxiliary field $u_b(x,t) = u(x + \kappa bt, t)$ to the original field at that root. It is a global bi-Lipschitz map for all large j: its perturbation from the identity has Lipschitz constant $C\kappa\sqrt{a_j} < 1/2$. Existence of its inverse follows as well by the contraction mapping theorem applied to the identity plus that Lipschitz perturbation. Its map and inverse have uniform Lipschitz constants, and its displacement is at most $\kappa\epsilon_j \to 0$. Therefore the images $\Phi_{j,k}(A_j)$ cover each fixed inner base–phase product outside a set of measure tending to zero. To check this explicitly, the inverse of any point of that inner product lies in the smaller buffer for large j; points whose inverse is not in A_j are in the image of a set of vanishing measure, whose measure is at most the uniform Lipschitz Jacobian bound times that measure. Intersect these m image sets; their complement still has vanishing measure.

Every point in this intersection has at least m distinct original-axis roots. The kth transported point is an actual auxiliary root, and hence an original root by the displayed base-change formula. If two such transported roots had the same height, their original base and height would give the same auxiliary base $\xi = (x - \kappa bt)/R_j$, contradicting the strict ordering of the auxiliary ranks at that base. The common macroscopic normal window contains these roots by confinement. They are simple original roots for almost every joint base–phase point: on the critical set $u_t = 0$, the map $(x,t) \mapsto (x, u(x,t))$ has zero Jacobian, so its image has zero $(d+1)$-dimensional measure by the area formula. Consequently the complete original-axis simple-root count satisfies $N_j^o \geq m$ outside a joint set of measure tending to zero.

The normal coarea and trace-varifold argument already given identifies the weak total original-axis count mass on every fixed inner product with m times its volume. Its boundary has zero limiting plane/phase measure, so one may approximate its indicator from inside and outside by compact tests. Since the counts are nonnegative integers,

$$\int |N_j^o - m| = \int (N_j^o - m) + 2\int (m - N_j^o)_+ \longrightarrow 0.$$

The first term tends to zero by that weak mass convergence, and the second is bounded by $2m$ times the vanishing exceptional measure. This proves the claimed original-axis local base–phase L^1 count convergence. Together with the already proved original-axis complete first moment, it supplies a good-count product of measure at least cR_j^d with the asserted height integral. The argument was carried out for sufficiently large j. The first moment over the compact receiver is finite at each omitted index, by boundedness of u_t, so enlarging C covers that finite initial segment as well. □

Proposition 5.3 (The actual stable solution absorbs the augmented budget before fine rank extraction)**.** *Under the hypotheses and notation of proposition 5.2, assume additionally that the integer multiplicity in proposition 3.5 is $m \geq 2$. Explicitly, U is the Poisson extension, $d\mu = d_p y^{1-p} dX dy$,*

$$M(r) = \int_{B_r \times (0,\infty)} D_X U \otimes D_X U \, d\mu, \qquad E(r) = \operatorname{tr} M(r) - \lambda_{\max}(M(r)),$$

and the augmented records are

$$S_j = E(R_j) + R_j^{d-1} = a_j^2 R_j^d = R_j^{d-2} D_j^2, \qquad D_j = a_j R_j.$$

Choose the normal axis e_j to be a unit top eigenvector of $M(R_j)$, and write $V_j = D_x U$ in that orthonormal record frame. After discarding finitely many indices if necessary,

$$\frac{R_j^2}{D_j^{p+1}} \le C, \qquad \frac{R_j^{d-1}}{S_j} \longrightarrow 0, \qquad \frac{E(R_j)}{S_j} \longrightarrow 1.$$

Moreover there is a constant C, independent of j and $L \ge 2$, such that

$$\int_{B_{LR_j}\times(0,\infty)} |V_j|^2 d\mu \le C S_j L^d (1+\log L)^2.$$

The same outward estimate, with a changed uniform constant, holds for the actual tangential derivatives in every determinant-one constant shear $S_k(x,t) = (x+kt,t)$ with $|k| \le 1$. Thus the augmented upper normalization becomes the true excess normalization, and supplies the outward same-frame bound, before any fine rank-current, profile, or force argument.

Proof. Fix the smaller product receiver of proposition 5.2, with base $B^d_{b_0R_j}$, normal interval $(-b_1R_j, b_1R_j)$, and one fixed compact phase interval $I \Subset (-1,1)$ of positive length. Here $b_0, b_1 > 0$ are fixed. Use

$$Y_j = B^d_{b_0R_j} \times I, \qquad d\mu_j = dx\,dc,$$

and list every actual root in that normal interval in the orthonormal record frame. These lists are measurable and finite almost everywhere, as established in lemma 5.1; define them to be empty on the exceptional null set. The complete-count conclusion of the anchoring proposition gives a measurable set $A_j = \{N_j = m\}$ with $\mu_j(A_j) \ge cR_j^d$ for all sufficiently large j. One-dimensional coarea and its first-moment estimate give

$$\int_{A_j} \sum_k |t_{j,k}(x,c)|\,dx\,dc \le \int_{Y_j} \sum_k |t_{j,k}(x,c)|\,dx\,dc = \int_{\substack{|x|<b_0R_j,\ |t|<b_1R_j \\ u(x,t)\in I}} |t|\,|u_t|\,dx\,dt \le CD_jR_j^d.$$

Choose a fixed $b \ge \max\{1, b_0, b_1\}$. The smaller normal interval cuts out a contiguous block of the complete list in $(-bR_j, bR_j)$. Thus every consecutive pair of a smaller list is still consecutive in that enlarged complete list. In particular, for $H_0 \le H \le R_j \le bR_j$,

$$\mu_j\{\text{a smaller list has a consecutive gap at most } H\}$$
$$\le \int_{B^d_{bR_j}\times I} N_{bR_j,H}(x,c)\,dx\,dc \le CR_j^{d-1}H^{(p+1)/2}.$$

The fixed enlargement changes only the constant in lemma 5.1, which is uniform in the orthonormal frame. All root-list, positive-measure, first-moment, and short-gap hypotheses of proposition 4.3 are now satisfied, with constants independent of j. Its eventual conclusion gives the three displayed interaction and absorption assertions. Remove the finite initial segment permitted there.

We supply the outward estimate using only the augmented upper records. As established in the proof of proposition 5.2, complete covariance convergence and $m \ge 2$ give $r_0, c, C > 0$ with

$$\operatorname{tr} M(r) \le Cr^d, \qquad \lambda_1(M(r)) - \lambda_2(M(r)) \ge cr^d \qquad (r \ge r_0).$$

Their positive semidefinite monotonicity, continuity, and $E(r) = o(r^d)$ are also those already checked for this covariance integral.

Use the two-scale matrix inequality proved in lemma 4.2: if $r_0 \le \rho \le \tau \le 2\rho$ and e_r denotes a unit top eigenvector of $M(r)$, then

$$c\rho^d \sin^2 \angle(e_\rho, e_\tau) \le \operatorname{tr} M(\rho) - e_\tau^T M(\rho) e_\tau \le \operatorname{tr} M(\tau) - e_\tau^T M(\tau) e_\tau = E(\tau).$$

This algebraic inequality holds independently of the location of the record radius. The positive gap makes e_r a unique unoriented axis for $r \geq r_0$; its angle lies in $[0, \pi/2]$.

Use the forward record inequality in lemma 4.1. For every $\tau \geq R_j$,

$$E(\tau) \leq S(\tau) \leq S_j(\tau/R_j)^d = a_j^2 \tau^d.$$

Choose j large enough that $R_j \geq r_0$. The two-scale inequality then gives $\angle(e_\rho, e_\tau) \leq Ca_j$ whenever $R_j \leq \rho \leq \tau \leq 2\rho$. Joining R_j to $r \geq R_j$ by adjacent dyadic radii and a final ratio at most two, and using the triangle inequality for unoriented-axis angles, yields

$$\angle(e_{R_j}, e_r) \leq Ca_j(1 + \log(r/R_j)).$$

This upper bound is valid even when its right-hand side exceeds one. For the fixed original optimizing axis $e_j = e_{R_j}$, expand its Rayleigh quotient in a top eigenbasis at radius r. Positive semidefiniteness gives

$$\begin{aligned}\int_{B_r \times (0,\infty)} |V_j|^2 d\mu &= \operatorname{tr} M(r) - e_j^T M(r) e_j \\ &\leq E(r) + \lambda_1(M(r)) \sin^2 \angle(e_j, e_r) \\ &\leq Ca_j^2 r^d (1 + \log(r/R_j))^2.\end{aligned}$$

Setting $r = LR_j$ proves the claimed uniform outward bound.

Finally write U_j for U in the jth orthonormal record frame and put $U_j^k(x,t,y) = U_j(x + kt, t, y)$. The chain rule gives $D_x U_j^k = V_j \circ S_k$ exactly, since t is independent of x in this shear. Its determinant is one, it leaves y unchanged, and $S_k(B_{LR_j}) \subset B_{2LR_j}$ for $|k| \leq 1$. Therefore

$$\int_{B_{LR_j} \times (0,\infty)} |D_x U_j^k|^2 d\mu \leq \int_{B_{2LR_j} \times (0,\infty)} |V_j|^2 d\mu \leq CS_j L^d (1 + \log L)^2,$$

where the factor 2^d and $(1 + \log(2L))/(1 + \log L)$ are absorbed into a uniform constant. This is a change-of-variables identity for the actual extension. Discarding the already specified finite initial segment leaves constants independent of j, L, k. □

6. Multiplicity-one rigidity

Lemma 6.1 (Stable zero-free balls have a uniform well point)**.** *Let $n \geq 1$ and $1 < p < 2$. Every nonnegative bounded stable entire solution of $(-\Delta)^{p/2} u = u - u^3$ with $|u| \leq 1$ is identically one; every nonpositive such solution is identically minus one. For every bounded stable entire solution with $|u| \leq 1$, there is a constant $C_{n,p}$ independent of the solution such that*

$$1 - |u(X)| \leq C_{n,p}(1 + \operatorname{dist}(X, \{u = 0\}))^{-p}.$$

In particular every sufficiently large ball contained in a fixed sign phase contains a point in its concentric half-ball arbitrarily close to the corresponding well, with radius threshold depending only on n, p and the desired closeness.

Proof. Write $A = (-\Delta)^{p/2}$. We use uniform smoothness, compactness of translates and closure of the equation and stability from lemma 2.1. All its constants are uniform over the solutions considered here, since $|u| \leq 1$.

Suppose first that $u \geq 0$, and put $m = \inf u \in [0, 1]$. Translate points whose values tend to m; a subsequence converges locally smoothly to a stable entire solution v with $v \geq m$ and

$v(0) = m$. If $m = 0$, then $Dv(0) = 0$, and the bounded Hessian gives $v(Y) = O(|Y|^2)$ near zero. Thus the singular integral at the minimum is absolutely convergent, and

$$0 = Av(0) = -c_{n,p} \int_{\mathbb{R}^n} \frac{v(Y)}{|Y|^{n+p}} dY.$$

The integrand is nonnegative, so continuity forces $v \equiv 0$. This contradicts stability. Indeed, for a nonzero real $\phi \in C_c^\infty$, $\phi_R(X) = \phi(X/R)$ gives

$$Q_0(\phi_R) = R^{n-p} \int |\xi|^p |\widehat{\phi}(\xi)|^2 d\xi - R^n \int \phi^2 < 0$$

for large R, in every $n \geq 1$. If instead $0 < m < 1$, the same limiting solution has $Av(0) \leq 0$ at its global minimum, contradicting $Av(0) = m - m^3 > 0$. Hence $m = 1$, and the range bound gives $u \equiv 1$. The oddness of the cubic and the invariance of its stability potential under $u \mapsto -u$ prove the nonpositive assertion.

For every $\varepsilon \in (0, 1)$ there is a radius R_ε, depending only on n, p, ε, such that

$$u > 0 \text{ on } B_{R_\varepsilon}(X) \quad \Longrightarrow \quad u(X) \geq 1 - \varepsilon.$$

Otherwise there would be solutions positive on $B_j(X_j)$ with $u_j(X_j) < 1 - \varepsilon$. Their translates have a locally smooth stable limit $v \geq 0$ on all of $\mathbb{R}^n$, while $v(0) \leq 1 - \varepsilon$, contrary to the one-sign assertion. Applying the same statement to $-u$ handles negative balls with the same radius.

Let $Z = \{u = 0\}$. We use $\operatorname{dist}(X, \varnothing) = +\infty$ and $(1 + \infty)^{-p} = 0$. If Z is empty, continuity and connectedness give one strict sign on $\mathbb{R}^n$, hence a constant well by the assertion just proved; the requested bound is then $0 \leq 0$.

Suppose $Z \neq \varnothing$, and put $D = \operatorname{dist}(X, Z)$. First assume $D \geq D_0 := \max\{2, 2R_{1/4}\}$. The open ball $B_D(X)$ is connected and zero-free, so $\varsigma u > 0$ there for one $\varsigma \in \{-1, 1\}$. Every $Y \in B_{D/2}(X)$ has $B_{R_{1/4}}(Y) \subset B_D(X)$. The preceding compactness assertion therefore gives $\varsigma u(Y) \geq 3/4$ throughout this smaller ball. The globally defined deficit $w = 1 - \varsigma u$ satisfies $0 \leq w \leq 2$ and the exact equation

$$(A + 1)w = F(w), \qquad F(w) = -w + 3w^2 - w^3.$$

For $0 \leq w \leq 1/4$, $F(w) = w(-1 + 3w - w^2) \leq 0$, while $|F(w)| \leq C$ on $[0, 2]$ with an absolute constant.

The bounded-data representation in lemma 2.3 gives $w = G_1 * F(w)$; it requires no integrability of either bounded function. Since $G_1 > 0$, the nonpositive source in $B_{D/2}(X)$ can be dropped in an upper bound. Using the far estimate in lemma 2.2 and $D/2 \geq 1$, we obtain

$$\begin{aligned} 1 - |u(X)| = w(X) &\leq C \int_{|H| \geq D/2} G_1(H) dH \\ &\leq C_{n,p} \int_{D/2}^{\infty} r^{-1-p} dr \leq C_{n,p} D^{-p}. \end{aligned}$$

For $D < D_0$, use $1 - |u(X)| \leq 1$; enlarging the constant by a quantity depending only on D_0, n, p proves the asserted $(1 + D)^{-p}$ estimate at every point. Because D_0 is already uniform over solutions, so is this constant.

Finally, if $B_R(X_0) \subset \{\varsigma u > 0\}$, then every $Y \in B_{R/2}(X_0)$ has distance at least $R/2$ from Z. The estimate gives

$$1 - \varsigma u(Y) \leq C_{n,p}(1 + R/2)^{-p} \quad \text{throughout } B_{R/2}(X_0).$$

This tends to zero with a radius threshold depending only on n, p and the desired closeness, and in particular supplies the claimed well point. □

Lemma 6.2 (Multiplicity one gives the sharp upper energy uniformly at each fixed geometric depth)**.** *Let u satisfy the hypotheses of proposition 3.4 with $2 \le d \le 6$, and suppose its multiplicity in proposition 3.5 is one. Translate a zero to the origin. Put*

$$\lambda_u(A) = \int_A \left[W(u) + \frac{d_p}{2} \int_0^\infty y^{1-p} |DU|^2 dy \right] dX.$$

For a unit ν write $C_\nu(l,\theta) = \{X : |P_{\nu^\perp} X| < l,\ |X \cdot \nu| < \theta\}$ and $A_{\nu,r} = C_\nu(r,r) \times (0,r)$. Suppose $l_k \to \infty$, $\theta_k / l_k \to 0$, and the zeros in $C_{\nu_k}(l_k, l_k)$ are trapped in $C_{\nu_k}(l_k, \theta_k)$. On a subsequence let $\nu_k \to \nu$. Let r_k / l_k range in a compact subinterval of $(0,\infty)$, let z_k / l_k range in a fixed compact subset of $\nu^\perp$ up to an error tending to zero, and let $\nu'_k \to \nu$. Then

$$J(U, z_k + A_{\nu'_k, r_k}) \le \sigma \omega_d r_k^d + o(l_k^d),$$

where J is the extension energy including its trace potential, and σ is the complete one-layer energy in proposition 3.5. More precisely, the remainder is uniform for $r_k / l_k \in [\rho_-, \rho_+]$, $\operatorname{dist}(z_k / l_k, K) \le \epsilon_k$, and $|\nu'_k - \nu| \le \epsilon_k$, for every fixed $0 < \rho_- \le \rho_+ < \infty$, compact $K \subset \nu^\perp$, and prescribed $\epsilon_k \downarrow 0$. In particular the conclusion holds at every fixed finite depth of a geometric iteration with $r_k = c_j l_k$, where $c_j > 0$ is fixed before $k \to \infty$.

Proof. We first establish complete zero-set coverage for an arbitrary quantized blowdown, before using trapping. Given any $R_j \to \infty$, take a subsequence in proposition 3.5 with

$$\Lambda_j(A) := R_j^{-d} \lambda_u(R_j A) \rightharpoonup \sigma \mathcal{H}^d \restriction \Pi,$$

where Π is a linear d-plane. For every fixed $0 < \vartheta < 1$ and $0 < L < \infty$,

$$\sup_{\substack{|X| \le L R_j \\ |u(X)| \le \vartheta}} \operatorname{dist}(X / R_j, \Pi) \longrightarrow 0.$$

Indeed, otherwise a subsequence of the normalized central points converges to a point off Π. A ball of radius cR_j about each such point lies in a fixed compact receiver disjoint from Π. The potential lower bound in lemma 3.6 puts at least $c' R_j^d$ energy in that ball, contradicting weak energy convergence there. This is the same noncollapse argument used for central confinement in proposition 5.2; it applies here at arbitrary blowdown scales. Compactness of normalized point receivers makes the assertion uniform on each fixed compact receiver. In particular it confines every zero.

For the converse, suppose points $Z_j \in \Pi \cap \overline{B}_L$ remain a fixed distance $\delta > 0$ from $R_j^{-1}\{u = 0\}$. Pass to $Z_j \to Z \in \Pi$. Then, for large j, $B_{\delta R_j / 2}(R_j Z)$ is zero-free. Clearing in lemma 6.1, applied to every point of its concentric half-ball, gives

$$1 - |u| \le C_\delta R_j^{-p} \quad \text{on } B_{\delta R_j / 4}(R_j Z).$$

The complete near-well energy estimate in proposition 3.4, with deficit $C_\delta R_j^{-p} < 1/4$, consequently yields

$$R_j^{-d} \lambda_u(B_{\delta R_j / 4}(R_j Z)) \le C_\delta R_j^{1-p} \longrightarrow 0.$$

The limiting plane assigns positive mass to $B_{\delta/4}(Z)$, a contradiction. Together with forward confinement this proves local Hausdorff convergence of the complete zero sets to Π; reverse approximation on a fixed ball may use zeros in a slightly larger ball. Both inclusions are uniform on fixed compact receivers.

Apply this fact to the scales l_k. If Π is any subsequential energy plane, every point of $\Pi \cap C_\nu(1,1)$ is approximated by scaled zeros. A point strictly inside that cylinder and its approximating zeros are inside $C_{\nu_k}(1,1)$ for large k; trapping then makes the limiting ν coordinate zero. Thus $\Pi \cap C_\nu(1,1) \subset \nu^\perp$. Since this intersection contains an open neighborhood

of zero in the linear plane Π, it follows that $\Pi = \nu^\perp$. Every subsequence has such a further energy subsequence with this same limit. Therefore the whole sequence satisfies

$$l_k^{-d}\lambda_u(l_k\,\cdot) \rightharpoonup \sigma\mathcal{H}^d \restriction \nu^\perp.$$

Write $Z_k = z_k/l_k$, $\rho_k = r_k/l_k$, and pass to a subsequence with $Z_k \to Z \in \nu^\perp$, $\rho_k \to \rho > 0$, and $\nu_k' \to \nu$. The scaled cylinders converge to $Z + C_\nu(\rho,\rho)$. Its top and bottom faces do not meet $\nu^\perp$, while its lateral boundary meets that plane in a sphere of dimension $d-1$, which has zero $\mathcal{H}^d$ measure. For every small fixed $\epsilon > 0$, continuity of the center, axis, and radius parameters gives the inclusions

$$Z + C_\nu(\rho-\epsilon,\rho-\epsilon) \subset Z_k + C_{\nu_k'}(\rho_k,\rho_k) \subset Z + C_\nu(\rho+\epsilon,\rho+\epsilon)$$

for all sufficiently large k. The fixed inner and outer cylinders are continuity sets for the limiting measure. Weak convergence and then $\epsilon \downarrow 0$ show that their middle masses tend to $\sigma\omega_d\rho^d$. Since $\rho_k^d \to \rho^d$, the subsequence criterion gives

$$\lambda_u(z_k + C_{\nu_k'}(r_k,r_k)) = \sigma\omega_d r_k^d + o(l_k^d).$$

If the stated uniformity for fixed K, $[\rho_-,\rho_+]$, and ϵ_k failed, choose offending parameters. Their normalized centers and radii have convergent subsequences, and the common ϵ_k bounds force the limiting center onto K and the limiting axis to be ν. The preceding cylinder argument gives a contradiction. Finally, nonnegativity of the energy density gives exactly

$$\begin{aligned} J(U, z_k + A_{\nu_k',r_k}) &= \int_{z_k + C_{\nu_k'}(r_k,r_k)} \left[W(u) + \frac{d_p}{2}\int_0^{r_k} y^{1-p}|DU|^2 dy \right] dX \\ &\le \lambda_u(z_k + C_{\nu_k'}(r_k,r_k)). \end{aligned}$$

This proves the claimed upper bound without losing any distant trace partners or requiring a separate height-tail estimate. At each fixed finite iteration depth $\rho_k = c_j > 0$ is fixed, so the same argument applies. □

Proposition 6.3 (A stable sharp-growth endpoint of multiplicity one is one-dimensional)**.** *Let $2 \le d \le 6$, $1 < p < 2$, and let u satisfy the hypotheses of proposition 3.4. If its integer multiplicity in proposition 3.5 is one, then u is a translate and rotation of the increasing one-dimensional cubic layer.*

Proof. The uniform derivative bounds and translate compactness used in the contact comparisons follow from lemma 2.1, for the actual field and the one-dimensional layer.

All solution, stability, extension, and growth conventions are those of proposition 3.4. Translate so $u(0) = 0$. Such a zero exists: a one-sign stable solution is a well by lemma 6.1, and a well has multiplicity zero. Use the notation $C_\nu(l,\theta)$ and J of lemma 6.2. The centered extended increasing layer is denoted by $Q(t,y)$; its complete energy equals σ, including the Hamiltonian identity

$$\sigma = d_p \int_{\mathbb{R}\times(0,\infty)} y^{1-p} Q_t^2 \, dt\, dy.$$

This identity, the finiteness and positivity of σ, and the normalization of Q were proved in proposition 3.5; no new layer-energy calculation is needed.

We use the complete zero-set convergence established in the proof of lemma 6.2. Along any quantized blowdown, the entire scaled zero set converges locally in Hausdorff distance to its limiting plane. The same forward confinement holds for every fixed compact central phase interval, by the noncollapse argument in that proof. These conclusions concern every zero and every central point in a fixed enlarged receiver.

We use two external equation and geometric facts with their exact roles. O. Savin, *Rigidity of minimizers in nonlocal phase transitions*[11], arXiv:1610.09295v1, Section 4, Lemma 4.5, Remark 4.4, and Propositions 4.6–4.7 are comparison estimates for solutions of the extension equation, with specified signed tangent balls and a well point in a stated inner ball. They do not assume minimizing energy. To match the conormal normalization, divide J by d_p and use potential W/d_p. The simplifying unit convention in Section 2 of that paper, uniform convexity outside $[g(-1), g(1)]$, can also be met: fix $L_0 > 1$ with $q(L_0) > 1/\sqrt{3}$ and $q(-L_0) < -1/\sqrt{3}$, and apply its estimates to $\widetilde{U}(X, y) = U(L_0X, L_0y)$. Its boundary potential is $\widetilde{W} = L_0^p W/d_p$, and its layer is $\widetilde{q}(t) = q(L_0t)$; since $W''(r) = 3r^2 - 1$, the stated convexity holds outside $[\widetilde{q}(-1), \widetilde{q}(1)]$. Returning to the original coordinates changes radii and comparison constants only by fixed factors depending on p. We absorb those factors below; the footprint and auxiliary-ball comparisons are unaffected, since their scale ratios tend to zero. The geometric weak Harnack results are Propositions 6.2 and 6.4, with Remark 6.3, in O. Savin, *Some remarks on the classification of global solutions with asymptotically flat level sets*[10], arXiv:1610.03448v1. Their hypotheses concern a closed set and its contact property (P): at a supporting paraboloid of opening a in the permitted interval, the set is locally a C^2 graph with Hessian bounded by Λa. They are valid in every finite base dimension. We verify the signed-ball, well-point, contact-regularity, and energy inputs below; a minimizer classification is not invoked for u.

We will use the following elementary geometric transfer, for which we give the quantitative details. Work in coordinates (x, t) where the old cylinder is $C(l, l)$, all its zeros have $|t| \le \theta$, and all radius-q balls tangent to the bottom cap $t = -\theta$ over B_l are negative, where

$$T = l^2/\theta, \qquad q = T^{1-\tau/2}, \qquad \theta/l \to 0, \quad T \to \infty.$$

The region in this cylinder below $-\theta$ is negative: it is connected, has no zeros, and intersects each of those negative balls. The upper assertion follows by reflection.

Fix a buffered base ball of radius bl, compactly inside B_l, with $b > 0$ fixed. Let ϕ be a C^2 lower barrier there, with

$$|\phi| \le C\theta, \qquad |D\phi| \le C\theta/l, \qquad \|D^2\phi\| \le C\theta/l^2,$$

such that every zero in this base ball and $|t| < l/2$ lies above its graph. The region below this graph and above $-l/2$ is connected and has no zeros, and connects vertically to the negative region below $-\theta$; hence it is negative. Consider a ball of radius $r \le \rho q$, $0 < \rho < 1$ fixed, tangent below ϕ at $P_0 = (x_0, \phi(x_0))$, with x_0 in the concentric half-buffer. Its center is

$$C_r = P_0 - r\nu, \qquad \nu = (-D\phi(x_0), 1)/\sqrt{1 + |D\phi(x_0)|^2}.$$

For any point $P = (x, t)$ of that ball the squared-distance inequality gives

$$t \le \phi(x_0) + D\phi(x_0) \cdot (x - x_0) - \frac{|x - x_0|^2}{2r}.$$

Since $r\|D^2\phi\| \le Cq/T \to 0$, Taylor's lower bound for ϕ shows $t < \phi(x)$ whenever its horizontal projection stays in the buffered base. This calculation allows an indefinite Hessian.

The horizontal center displacement satisfies

$$|(C_r)_x - x_0| \le Cr\theta/l = o(l), \qquad r|D\phi(x_0)|^2 = o(\theta), \qquad \frac{q\theta}{l^2} = T^{-\tau/2} \to 0.$$

Choose the old radius-q negative ball with the same horizontal center and center $A = ((C_r)_x, -\theta - q)$; its horizontal center is inside B_l. If $q \le bl/16$, the new ball stays wholly

in the buffered base and in $|t| < l/2$ for large indices, so the preceding local argument already makes it negative. If $q > bl/16$, then $\theta/q \to 0$. Write

$$L = (C_r)_t - A_t = q - r + \delta, \qquad \delta = \phi(x_0) + \theta + r\left(1 - \frac{1}{\sqrt{1 + |D\phi(x_0)|^2}}\right), \qquad |\delta| \le C\theta.$$

Thus $L \ge (1-\rho)q/2$ for large indices. For a point of the new ball outside the old ball, put $v = P - C_r$. Comparing $|v|^2 \le r^2$ with $|v + Le_t|^2 \ge q^2$ gives

$$v_t \ge r - \frac{q\delta + \delta^2/2}{L} \ge r - C_\rho\theta.$$

Consequently such points have

$$|P_x - (C_r)_x| \le C_\rho\sqrt{r\theta + \theta^2} = o(l), \qquad |t| \le C_\rho\theta.$$

For the vertical bound, when $r > C_\rho\theta$ use the displayed top-cap estimate; when $r \le C_\rho\theta$ use the radius bound directly. These exceptional points remain in the buffered local negative subgraph; all other points are in the old negative ball. This proves that the entire new ball is negative. Constants may depend on the fixed buffer and barrier bounds, and all conclusions are uniform in their contact points. The reflected statement holds for upper barriers. It applies also to affine caps after a translation of height $O(\theta)$ and a rotation of slope $O(\theta/l)$, provided the cap lies on the corresponding side of all local zeros. In the applications below that last condition will be supplied explicitly by the improved local zero slab.

Fix a lower height $\theta_* > 0$. Let $\tau \in (0, 1/2)$ be the small barrier exponent from Section 4 of the fractional paper. Consider a sequence of admissible flat cylinders with

$$\theta_k \ge \theta_*, \qquad \theta_k/l_k \to 0, \qquad \{u = 0\} \cap C_{\nu_k}(l_k, l_k) \subset C_{\nu_k}(l_k, \theta_k),$$

and assume the balls of radius

$$q_k = (l_k^2/\theta_k)^{1-\tau/2}$$

tangent below and above the flat cylinder belong, respectively, to the negative and positive phases. These are the signed-ball hypotheses in Savin's Theorem 5.1. All balls used to supply its well point have radii tending to infinity. Lemma lemma 6.1 therefore supplies those well points at their actual centers, including remote centers outside the macroscopic receiver.

We specify the two geometric applications, including their domains and the constants, before using their energy consequence. In this argument every cylinder is centered at an actual zero; initially this is $u(0) = 0$, and the recentered cylinders below will also be centered at zeros. Suppress k temporarily and use its orthogonal cylinder coordinates. Put $\varepsilon = \theta/l$ and define the relatively closed sets in $\{|z| < 1\}$ by

$$\Gamma^- = \{(z,h) : u(lz, lh - \theta) = 0, \quad 0 \le h \le 2\varepsilon\},$$
$$\Gamma^+ = \{(z,h) : u(lz, \theta - lh) = 0, \quad 0 \le h \le 2\varepsilon\}.$$

They contain $(0, \varepsilon)$ and lie in $h \ge 0$. The slab assumption means that these are the entire zero sets within the old cylinder, shifted and, for Γ^+, reflected; no zero in that cylinder has been omitted. For either set and $A > 0$, let $\mathcal{D}_A$ be its contacts from below by paraboloids

$$h = -\frac{A}{2}|z - y|^2 + b, \qquad |y| \le 1,$$

which lie below the whole set in $|z| < 1$ and touch at a point with $|z| \le 3/4$. Let D_A be their projection onto the z base. This is precisely Definition 6.1 in the geometric paper. At a given projected base a lower supporting contact has a unique height; the restricted closed contact graphs therefore provide measurable contact heights. Upper contacts of the physical zero set are lower contacts of Γ^+.

Here we verify property (P) on exactly these contact and vertex domains. Fix any positive constants c_-, c_+ and consider $c_-\varepsilon \le A \le c_+\varepsilon$. The corresponding physical opening is $a = A/l$. Its vertex base is $ly \in \overline{B}_l$, its contact base lies in $\overline{B}_{3l/4}$, and its contact height lies in $[-\theta, \theta]$. On a fixed base buffer about that contact, its physical graph ϕ is below all zeros in the old cylinder and satisfies

$$|\phi| \le C\theta, \qquad |D\phi| \le C\theta/l, \qquad |D^2\phi| = a \le c_+\theta/l^2.$$

The bounds follow directly from $|x_0 - ly| \le 7l/4$ and the value of ϕ at contact. They do not require the paraboloid to exit the lateral boundary through the bottom. The local lower subgraph is negative by its vertical connection to the old negative region. Choose $T = c/a$, with a fixed small universal $c > 0$, so the sphere tangent below ϕ has curvature larger in absolute value and stays below it on that buffer. The two quantitative comparisons are, uniformly for these fixed opening constants,

$$q_k/T^{1-\tau} \longrightarrow \infty, \qquad T^{1/2-\tau}/l_k \le C_{\theta_*, c_-, c_+} l_k^{-2\tau} \longrightarrow 0.$$

Thus hypothesis (a) of Proposition 4.6 in the fractional paper holds on its whole footprint. The local upper graph of this sphere has the same three barrier bounds. Apply the quantitative ball-transfer argument already proved to every radius-$T^{1-\tau}/4$ ball tangent below that sphere on the required footprint. Its entire ball is negative, including outside the contact neighborhood. The radius-$T^{1-\tau}/2$ ball at the central contact is negative by the same transfer, and its center differs from $-(T^{1-\tau}/2)e_t$ by $o(T^{1-\tau})$ in contact-centered coordinates. Clearing therefore supplies the well point in the exact ball specified by hypothesis (c). This checks all of Proposition 4.6, including its auxiliary-ball hypothesis (b). Its curvature bound C/T and the contact slope $o(1)$ give a local graph with Hessian at most Λa. After the isotropic scaling by l, its Hessian is at most ΛA. Here Λ can be fixed using the universal constant of Proposition 4.6 and c before c_-, c_+ are chosen: these opening constants affect only how far along the sequence the two displayed comparisons hold. Reflection gives the identical result for Γ^+. Hence both sets satisfy the geometric paper's property (P) with this one fixed Λ on every needed fixed interval of openings.

Apply Proposition 6.4 of the geometric paper first to Γ^+, with its height parameter equal to ε and opening 8ε. Its hypotheses are $(0, \varepsilon) \in \Gamma^+ \subset \{h \ge 0\}$ and the just-verified property (P). Its conclusion is a contact family

$$\mathcal{U} \subset \mathcal{D}^+_{8\varepsilon} \cap (\overline{B}_{1/2} \times [0, 3\varepsilon/2])$$

whose projection U has measure at least $c(d, \Lambda) > 0$. All its vertices have $|y| \le 1$, as in the contact definition; its contact projection is inside $B_{1/2}$, as in that proposition's conclusion. Set

$$\mu_0 = \min\{1/2, c(d, \Lambda)/|B^d_{1/2}|\} > 0, \qquad \mu = \mu_0/2.$$

Choose $M = M(\mu, d, \Lambda) \ge 1$ from Proposition 6.2 and fix

$$\omega = \frac{1}{32M}.$$

These choices are independent of k. In physical variables the family $\mathcal{U}$ consists of upper contacts over $lU \subset B_{l/2}$ at heights $t \ge -\theta/2$, with physical opening $8\theta/l^2$ and slopes $O(\theta/l)$.

Suppose there is a zero in the old cylinder with base in $B_{l/2}$ and $t \le -(1-\omega)\theta$. This says exactly that Γ^- meets $B_{1/2} \times [0, \omega\varepsilon]$. Apply Proposition 6.2 to Γ^- with height parameter $\omega\varepsilon$, measure parameter μ, and opening interval

$$[\omega\varepsilon, M\omega\varepsilon].$$

All its hypotheses have been checked, including (P) on this exact interval, all vertex bases $|y| \le 1$, and contact bases $|z| \le 3/4$. It gives

$$|D^-_{M\omega\varepsilon} \cap B_{1/2}| \ge (1-\mu)|B^d_{1/2}|.$$

Remark 6.3 of that paper places every such contact at $h \le 8M\omega\varepsilon = \varepsilon/4$. Let $\mathcal{L}$ be these contacts restricted to the projected base $L = D^-_{M\omega\varepsilon} \cap B_{1/2}$. In physical variables they are lower contacts over $lL \subset B_{l/2}$, all with $t \le -3\theta/4$, opening $M\omega\theta/l^2$, and slopes $O(\theta/l)$. Thus both contact projections are inside the same physical base $B_{l/2}$, their measures are at least $(1-\mu)|B^d_{l/2}|$ and $\mu_0|B^d_{l/2}|$, and their contact heights are separated by at least $\theta/4$ wherever the projections overlap. If instead an upper bad zero in that old cylinder has base in $B_{l/2}$ and $t \ge (1-\omega)\theta$, apply this entire argument after $(u(x,t),t) \mapsto (-u(x,-t),-t)$. It has the same constants and produces the reflected pair of separated families on the same base.

We now show that either failure contradicts sharp energy. This gives the Harnack exclusion for the old cylinder's zeros on the full buffered base $B_{l_k/2}$, rather than merely assuming an exclusion on that base from a quarter-base output. Restore the index k and use the two families just constructed. Their measures have sum at least $(1+\mu_0/2)|B^d_{l_k/2}|$.

At each contact there is a genuine signed tangent ball of radius cl_k for a fixed small $c > 0$. Its curvature is larger than that of the supporting paraboloid for large k, so it lies on its signed side; it stays inside the cylinder and connects there to the prescribed signed cap. Clearing gives its well point. Lemma 4.5 and Remark 4.4 of the fractional paper approximate the extension on every fixed normal half-disk of radius $3K$ by the same oriented translated layer, with error

$$O_K(l_k^{-1}) + O_K(\theta_k/l_k) = o(1).$$

The orientation is increasing for both contact families: the lower family has negative phase below it and the upper family has positive phase above it. This estimate is uniform in the contact point. The columns used below are the fixed vertical (t,y) half-disks above each projected contact base. For a contact height $s = s(x)$, write their normal energy as

$$\mathcal{J}_{K,x,s} = \frac{d_p}{2} \int_{(t-s)^2+y^2<K^2,\ y>0} y^{1-p}(|U_t(x,t,y)|^2 + |U_y(x,t,y)|^2)\,dt\,dy + \int_{s-K}^{s+K} W(u(x,t))\,dt.$$

Dropping the nonnegative horizontal derivatives from the ambient energy and then using Fubini bounds the integral of any disjoint such columns by that ambient energy. Because the contact slope is $O(\theta_k/l_k) = o(1)$, the tilted layer comparison on radius $3K$ restricts to these fixed columns with an additional $o(1)$ error for every fixed K. It follows that $\mathcal{J}_{K,x,s} \ge \sigma_K - o(1)$ uniformly over the contacts, where $\sigma_K = J(Q, B^+_K) \uparrow \sigma$. To justify this implication, suppose a sequence of offending contacts existed and translate each fixed column to its center. If its normal energies are unbounded there is nothing to prove; otherwise weighted weak H^1 compactness applies on the half-disk. Its uniform limit is Q, the kinetic energy is weakly lower semicontinuous, and the trace potential converges uniformly. This contradicts the offending bound. Thus projection measures can be added without a change-of-variables Jacobian.

When a base has both contacts, their radius-K normal half-disks are disjoint for large k. Otherwise their center separation would lie in $[\theta_*/4, 2K]$ along a subsequence. Evaluate the lower contact's radius-$3K$ profile comparison at the upper contact, where the trace is zero. The comparison gives $0 = q(s_+ - s_-) + o(1)$, whereas strict increase and centering of q give $q(s_+ - s_-) \ge q(\theta_*/4) > 0$, a contradiction. Thus the two families of contact columns can be added without double counting. For fixed K, all these columns lie inside $A_{\nu_k, l_k/2}$ for large k, since their bases lie in $B_{l_k/2}$, their centers have $|s| \le \theta_k = o(l_k)$, and their heights are at

most K. Choose K first so

$$(1+\mu_0/2)\sigma_K > (1+3\mu_0/8)\sigma,$$

and then let k tend to infinity. Adding the measures of both contact projections gives

$$J(U, A_{\nu_k, l_k/2}) \geq (1+\mu_0/4)\sigma |B^d_{l_k/2}|.$$

This contradicts lemma 6.2. Because the preceding normalized construction starts from any lower bad zero of the old cylinder over $B_{l_k/2}$, and its reflected construction starts from any upper bad zero there, neither kind can occur for sufficiently large k. Hence every zero of the old cylinder over this full buffered base has $|t| < (1-\omega)\theta_k$; in particular this holds for every zero in $C_{\nu_k}(l_k/2, l_k/2)$. The output cylinder uses base $l_k/4$, leaving the base $l_k/2$ as a fixed buffer. Its new cap planes are lower and upper barriers for every zero in that buffer. Their new radius is

$$q'_k = q_k[16(1-\omega)]^{-(1-\tau/2)} = \rho q_k, \qquad 0 < \rho < 1.$$

Apply the proved ball-transfer argument to these affine barriers, with this fixed ρ and that buffer. It puts every lower new ball in the negative phase and every upper new ball in the positive phase. This proves the full Harnack conclusion, including its signed balls, rather than inferring it from the smaller radius alone.

The translated cylinders needed to obtain a graph limit are admissible as well. For a base in $B_{l_k/2}$, center one at any zero on that fiber, take base radius $l_k/4$, and use height $2\theta_k$ to contain the entire old zero slab. All its cap contact bases then lie in $B_{3l_k/4}$ with a fixed margin inside the old base. Its cap heights differ from the old center by $O(\theta_k)$. On a fixed smaller buffered base these cap planes lie below and above every old zero. The associated new tangent radius is a fixed factor less than q_k; for example base $l_k/4$ and height $2\theta_k$ give $q'_k = 32^{-(1-\tau/2)}q_k$. The ball-transfer argument proves the signed-ball hypotheses for this recentering. Keep this chosen center fixed for its subsequent Harnack steps: the base is multiplied by $1/4$ and the height by $1-\omega$ at each step, with no further height doubling. At each further fixed Harnack depth the same buffer construction applies, with the radius and height equal to $c_j l_k, d_j\theta_k$ for fixed positive factors. The physical lower height is at least $d_j\theta_* > 0$, and lemma 6.2 supplies the sharp energy upper bound uniformly in all normalized centers at this fixed depth. Consequently, for every fixed integer $j \geq 0$, for all sufficiently large k, any two old zeros $(x_0, t_0), (x, t)$ with $|x_0| \leq l_k/2$ and $|x - x_0| < 4^{-(j+1)}l_k$ satisfy

$$\frac{|t-t_0|}{\theta_k} \leq 2(1-\omega)^j.$$

Indeed the iterated cylinder has that base radius and that half-height about (x_0, t_0); at a fixed depth its vertical radius is comparable to l_k, so every old zero over its base is inside its vertical window once k is large. The threshold is uniform over the chosen centers by the uniform energy bound and the preceding Harnack argument. First taking $k \to \infty$, and then $j \to \infty$, gives a modulus of continuity with exponent $\beta = -\log(1-\omega)/\log 4 > 0$ for every limit of the complete zero sets under $(x, t) \mapsto (x/l_k, t/\theta_k)$. Opposite cap signs guarantee a zero on every interior vertical fiber, and the displayed estimate makes each limiting fiber a singleton. Compactness of the closed sets in fixed inner cylinders therefore yields a subsequence converging uniformly on interior bases to the graph of a continuous function f.

We verify the harmonic equation without assuming a local quadratic is globally supporting. Suppose a quadratic test from below has positive Laplacian at an interior point of the graph limit. Subtract a sufficiently small positive multiple of the squared distance from that point, so its Hessian B still has $\operatorname{tr} B > \delta > 0$ and the lower contact is strict. Translate the base

point to zero and write this polynomial as $P(z)$, adding a constant so $P(0) = f(0)$. Choose a fixed small $b > 0$ so $P < f$ on the boundary of B_b and $P \le f$ on $\overline{B}_b$. For small fixed δ the coefficients may be bounded by δ^{-1} by decreasing δ if necessary.

In the actual rescaled zero sets, minimize $z_n - P(z)$ over all zero points with $z \in \overline{B}_b$ in the physical normal window $|t| \le 2\theta_k$, equivalently $|z_n| \le 2$; this window contains every zero of the old cylinder over that base. Every fiber there is nonempty and the set is compact; uniform convergence to the graph and the strict boundary gap put a minimizing contact in $B_{b/2}$, converging to the touching point, after reducing b if needed. Translate P vertically by that minimum. Its physical graph Γ_k is below every actual zero on a fixed buffered base and touches an actual zero. Its height, slope and Hessian on that buffer obey

$$|\Gamma_k| \le C_b\theta_k, \qquad |D\Gamma_k| \le C_b\theta_k/l_k, \qquad |D^2\Gamma_k| \le C_b\theta_k/l_k^2.$$

Here, and in applying the next proposition, a translation of size $O(l_k)$ in the base and $O(\theta_k)$ in height is allowed within the old buffered cylinder. No definiteness is imposed on B. This is the local-contact localization used in Savin's Theorem 6.1; the explicit transfer argument above handles its portions outside that neighborhood.

Let C_S be the constant in Proposition 4.7 and choose

$$T_k^{-1} = \frac{\delta}{2C_S}\frac{\theta_k}{l_k^2}, \qquad \varepsilon_0 = \delta^2.$$

After a fixed tangential rotation diagonalizing B, the slope and coefficients satisfy

$$|D\Gamma_k(0)| \le \varepsilon_0, \quad |D^2\Gamma_k| \le \varepsilon_0^{-2}T_k^{-1}, \quad \varepsilon_0 \ge T_k^{-\tau/2}$$

for small fixed δ and large k. Its required footprint has radius $T_k^{1/2-\tau} = o(bl_k)$, so it is inside the buffered support. At every point of that footprint, apply the ball-transfer argument to Γ_k with $r_k = T_k^{1-\tau}/2$. It applies uniformly because $r_k/q_k \to 0$, $r_k\theta_k/l_k^2 \to 0$, and the contact points stay inside the half-buffer. Thus all the auxiliary tangent balls, including their portions outside the local contact neighborhood, are in the actual negative phase. The ball at the central contact has its center within $o(r_k)$ of $-r_ke_t$; clearing at that latter point gives the well point in $B_{r_k/2}(-r_ke_t)$ required by hypothesis (c). The global extension solves the equation throughout the required domain. All hypotheses of Proposition 4.7 now hold, and it gives

$$\frac{\theta_k}{l_k^2}\operatorname{tr} B \le C_S T_k^{-1} = \frac{\delta}{2}\frac{\theta_k}{l_k^2},$$

a contradiction. Reflection excludes upper tests with negative trace. The limiting graph is therefore harmonic in the viscosity, hence classical, sense.

Harmonic interior estimates give $f(0) = 0$ and $|f(z) - Df(0)\cdot z| \le C|z|^2$ on a fixed interior base, with a universal bound on $|Df(0)|$. Choose a fixed $\eta > 0$ so small that this error on $B_{4\eta}$ is less than $\eta^{3/2}/8$. Uniform graph convergence then puts all actual zeros on that buffered base within $\theta'/4$ of the tangent plane for large k, where $\theta' = \eta^{3/2}\theta_k$. Rotate about the fixed zero to take this plane as horizontal; its rotation slope is $O(\theta_k/l_k)$. The rotation changes vertical distances by a factor tending to one, and maps the new radius-ηl_k receiver into the old base $B_{2\eta l_k}$ and the old cylinder. The zeros in it therefore lie in height at most $\theta'/2$, leaving strict room below and above the output caps at height $\pm\theta'$.

For every point on a new cap over the new base $B_{\eta l_k}$, express its cap as an affine graph in the old coordinates. Its slope is $O(\theta_k/l_k)$ and its height is $O(\theta_k)$. The old-coordinate contact bases lie in $B_{(3/2)\eta l_k}$ for large k, while the zero-slab comparison holds on $B_{4\eta l_k}$; hence they have a fixed buffer. These lower and upper cap graphs lie respectively below and above all

zeros in that buffer. Their required new ball radius obeys

$$q'_k = \left((\eta l_k)^2/\theta'\right)^{1-\tau/2} = \eta^{(1-\tau/2)/2} q_k = \rho q_k, \qquad 0 < \rho < 1.$$

Apply the quantitative ball transfer to each affine cap with this fixed ρ. It proves that all new lower balls are negative and all new upper balls positive, including after the stated rotation and centering. Thus the output cylinder and signed balls satisfy the complete improvement conclusion.

If no single small-flatness threshold gave this conclusion for the fixed field and fixed lower height θ_*, a failure sequence with $\theta/l \to 0$ and $\theta \geq \theta_*$ would have $l \to \infty$ and would contradict the preceding compactness argument. Hence there is one threshold $\varepsilon_1(\theta_*, u) > 0$ valid for every radius and orientation of a cylinder centered at the fixed zero 0, with $\theta \geq \theta_*$ and $\theta/l \leq \varepsilon_1$. It is independent of the later iteration depth.

It remains to initialize the fixed-field iteration with complete zero-set coverage and opposite signs. Take the augmented records and their top covariance axes ν_j from proposition 5.2. Pass to a quantized subsequence. Its positive rank-one covariance limit makes ν_j converge, up to sign, to a normal of the limiting plane. The complete zero-set and central-set confinement recalled above applies on every fixed enlarged ball B_{LR_j} in these axes.

Fix $I = [-1/4, 1/4]$. Take a fixed smaller product receiver from proposition 5.2, and denote its fixed positive base and normal radius factors by b_0, h_0. Thus its base contains a ball $B_{b_0 R_j}$ and its complete normal lists are taken in a normal window containing $(-h_0 R_j, h_0 R_j)$, after reducing these factors if necessary. Choose $0 < \gamma < \min\{b_0, h_0\}/4$ and restrict the count convergence to the base $B_{\gamma R_j}$. A complete root list means every solution of $u(x,t) = c$, $c \in I$, in that finite normal window. Its almost-everywhere finiteness, simplicity, and measurability are those proved in lemma 5.1; the phase-free caps are supplied by the anchoring proposition. The word central means exactly $u \in I$ here.

Choose a fixed outer box large enough to contain this whole product receiver and B_{16R_j}. Increase its fixed size once more if necessary, and choose a fixed $c > 0$ so the levels $t = \pm cR_j$ and the base $B_{\gamma R_j}$ are in its interior. The direct central-set confinement proved above applies on a containing ball for this entire box: every I point in it has $|t| = o(R_j)$. Hence, for large j, every root in the original complete window is in $(-\gamma R_j, \gamma R_j)$. There are also no further I roots between this inner window and the chosen levels $\pm cR_j$, since every point in those intervening regions is covered by that same confinement. Consequently the root count between $\pm cR_j$ agrees with the count from the anchoring proposition on $B_{\gamma R_j} \times I$, and tends strongly in scaled L^1 to one. The two outer regions of the box above and below the strip $|t| = o(R_j)$ are connected and avoid I. Their values therefore belong to constant components of $[-1, 1] \setminus I$, and each corresponding cap has constant sign.

If the two signs were the same, their values would be in the same component of $[-1, 1] \setminus I$. For almost every base and $c \in I$ the entire finite simple-root list would then have even count, by successive sign changes of $u - c$. Hence its distance from the odd integer one would be at least one on the whole fixed base–phase receiver. This contradicts the strong L^1 count convergence supplied by proposition 5.2. The signs are therefore opposite; orient ν_j with the negative phase below and positive above.

Choose $0 < \varepsilon < \varepsilon_1(\theta_*, u)$ and take $l_0 = R_j/16$. The complete zero-set confinement on B_{16R_j}, proved directly above for every fixed ball, gives for large j

$$\{u = 0\} \cap B_{4l_0} \subset \{|X \cdot \nu_j| < \varepsilon l_0/2\}.$$

The upper and lower parts of this ball outside that strip have the signs just determined, since they connect to the corresponding common cap regions in the larger box without crossing a

zero. Set $\theta_0 = \varepsilon l_0$. The initial required radius satisfies

$$q_0 = (l_0/\varepsilon)^{1-\tau/2} = o(l_0).$$

Every ball of that radius tangent to the lower or upper cap of $C_{\nu_j}(l_0, \theta_0)$ stays in B_{4l_0} for large j, and lies respectively below $-\theta_0$ or above θ_0. It is therefore in the corresponding actual sign phase. This supplies all admissible initial cylinders. The improvement can be iterated about the fixed zero until the first height less than θ_*. With $l_j = \eta^j l_0$ and $\theta_j = \eta^{3j/2}\varepsilon l_0$, that stopping index obeys

$$\eta^{3/2}\theta_* \le \theta_j < \theta_*, \qquad l_j = l_0^{1/3}(\theta_j/\varepsilon)^{2/3} \ge \eta l_0^{1/3}(\theta_*/\varepsilon)^{2/3} \longrightarrow \infty.$$

At every previous step the height is at least θ_* and the flatness is at most ε_1, so the same fixed-field threshold applies. Take a limiting final normal. For this θ_* the entire zero set lies in a slab of half-width θ_*. Now send θ_* to zero and extract normals again; the zero set is contained in a linear hyperplane. Its two complementary half-spaces have constant signs. If their signs were the same, continuity on the separating hyperplane would make u globally nonnegative or globally nonpositive. The one-sign classification in lemma 6.1 would then make it a well, contradicting $u(0) = 0$. Their signs are therefore opposite. Continuity from both sides then gives zero at every point of that hyperplane, so it is exactly the zero set.

The negative half-space has an interior tangent ball of every radius T at the origin. The clearing lemma supplies the required well point. Lemma 4.5 and Remark 4.4 of the fractional paper give, for each fixed $K > 0$, $|U - Q(X\cdot\nu, y)| \le C_K/T$ throughout B_K^+, for all sufficiently large T. Let $T \to \infty$ with K fixed. Equality follows on every such half-ball, and then throughout the upper half-space by exhaustion. Taking traces proves the claimed planar-layer representation. □

7. Strong count screens and positive source commutators

A generic bounded H^1 cutoff does not control the source square function in six base dimensions for the full range of p. The cutoff must depend on the complete source count. Its square root is the natural Sobolev variable. The construction below bounds the excluded-source commutator by count differences before integrating the positive Poisson kernel.

Lemma 7.1 (A slow logarithmic angular cutoff makes the complete source count strongly Sobolev convergent)**.** *Let $\Omega \subset \mathbb{R}^d$ be bounded and Lipschitz, let I be a bounded interval, and let $a_j \downarrow 0$. Let $N_j \ge 0$ be measurable complete root counts on $\Omega \times I$ with $N_j \to m$ in L^1, where $m \ge 0$ is a constant integer. Suppose nonnegative root-amplitude square norms $Q_j \in L^2(I; H^1(\Omega))$ can be formed with a cutoff that is one at physical angle at most κ_j^2, zero at angle at least κ_j, and logarithmic between, and that the complete-root entropy and count restoration give*

$$0 \le Q_j^2 \le N_j, \quad \int_{\Omega\times I}(N_j - Q_j^2) \le C(a_j^2\kappa_j^{-4} + a_j^2), \quad \int_{\Omega\times I}|D_\xi Q_j|^2 \le C(|\log\kappa_j|^{-2} + a_j^2).$$

The constant C is independent of j for the chosen cutoffs. Choose $0 < \kappa_j < 1$, $\kappa_j \downarrow 0$, slowly enough that $a_j/\kappa_j^2 \to 0$. Then

$$Q_j \to \sqrt{m} \text{ in } L^2(I; H^1(\Omega)), \qquad q_j(\xi) := \left(\int_I Q_j(\xi, c)^2 dc\right)^{1/2} \to \sqrt{m|I|} \text{ in } H^1(\Omega).$$

Consequently a fixed cutoff of q_j above any level larger than $\sqrt{m|I|}$ tends to one strongly in H^1. For the quasicontinuous representatives, the set on which it differs from one has relative H^1 capacity tending to zero on every fixed compact subset of Ω.

Proof. Since $a_j^2\kappa_j^{-4} = (a_j/\kappa_j^2)^2 \to 0$, the restoration error tends to zero. In fact,

$$\|Q_j^2 - m\|_{L^1(\Omega\times I)} \le \|N_j - m\|_{L^1(\Omega\times I)} + \int_{\Omega\times I}(N_j - Q_j^2) \longrightarrow 0.$$

For nonnegative numbers, $|\sqrt{x}-\sqrt{y}|^2 \le |x-y|$, so $Q_j \to \sqrt{m}$ in $L^2(\Omega\times I)$. Also $|\log\kappa_j| \to \infty$, and the assumed entropy bound makes $D_\xi Q_j \to 0$ in L^2. This proves the first convergence, including $m = 0$. These are implications of the displayed quantitative hypotheses for the chosen cutoffs.

We record the Hilbert-norm estimate used here and below. By Fubini, any $F \in L^2(I; H^1(\Omega))$ is an H^1 map from Ω into the Hilbert space $L^2(I)$. Its norm $q_F(\xi) = \|F(\xi,\cdot)\|_{L^2(I)}$ satisfies

$$|Dq_F|^2 \le \int_I |D_\xi F|^2 dc \quad \text{almost everywhere,} \qquad |q_F - q_G| \le \|F(\xi,\cdot) - G(\xi,\cdot)\|_{L^2(I)}.$$

For the first inequality, project F onto the first N vectors of an orthonormal basis of $L^2(I)$. The ordinary finite-dimensional norm chain rule gives that inequality for each projected norm. These norms converge in $L^2(\Omega)$ to q_F; their squared gradients are bounded by $\int_I |D_\xi F|^2 dc$. Weak H^1 compactness and lower semicontinuity, localized by any nonnegative compact smooth weight, give the displayed almost-everywhere inequality. The second inequality is the Hilbert-space triangle inequality. Apply both with $F = Q_j$ and $G = \sqrt{m}$ to obtain $q_j \to q_* := \sqrt{m|I|}$ strongly in $H^1(\Omega)$.

Choose $L > q_*$ and a fixed smooth cutoff ψ equal to one on $[0, L]$ and zero above a larger fixed level. The Lipschitz composition and chain rules give

$$\|\psi(q_j) - 1\|_2 \le \|\psi'\|_\infty \|q_j - q_*\|_2, \qquad \|D\psi(q_j)\|_2 \le \|\psi'\|_\infty \|Dq_j\|_2,$$

proving the cutoff convergence. For the capacity assertion use the scalar quasicontinuous representatives of q_j, and their quasicontinuous Sobolev compositions. For a compact $K \Subset \Omega$, choose $0 \le \zeta \le 1$, $\zeta \in C_c^\infty(\Omega)$, equal to one near K. For every fixed threshold $T > q_*$, the function

$$v_j = \zeta \min\{1, (q_j - q_*)_+/(T - q_*)\}$$

is in $H_0^1(\Omega)$ and is one quasi-everywhere on $K \cap \{q_j \ge T\}$. The truncation and product rules give the quantitative bound

$$\operatorname{cap}_2(K \cap \{q_j \ge T\}; \Omega) \le \|v_j\|_{H^1(\Omega)}^2 \le \frac{C_\zeta}{(T - q_*)^2}\|q_j - q_*\|_{H^1(\Omega)}^2.$$

Here relative capacity uses $H_0^1(\Omega)$ tests with energy $\int_\Omega(|Dv|^2 + v^2)$ and quasicontinuous value at least one quasi-everywhere on the set. The estimate tends to zero. Taking $T = L$ and using $\{\psi(q_j) \ne 1\} \subset \{q_j > L\}$ proves the stated conclusion for the screen. □

Lemma 7.2 (A logarithmically weighted fractional seminorm is controlled by H1)**.** *For every integer $d \ge 1$ and $0 < \alpha < 2$, define*

$$\mathcal{J}_\alpha(f) = \iint_{\mathbb{R}^d\times\mathbb{R}^d} \frac{|f(\xi) - f(\eta)|^2}{|\xi - \eta|^{d+\alpha}}(1 + \log_+|\xi - \eta|)\, d\xi\, d\eta.$$

There is $C_{d,\alpha}$ such that $\mathcal{J}_\alpha(f) \le C_{d,\alpha}\|f\|_{H^1(\mathbb{R}^d)}^2$ for every $f \in H^1(\mathbb{R}^d)$.

Proof. Choose a finite measurable representative of f; changing it on a Lebesgue null set changes the integrand only on a product null set. Define the diagonal value to be zero.

Thus the nonnegative double integral is unambiguous, initially allowing the value $+\infty$. For a smooth function the fundamental theorem of calculus and Cauchy–Schwarz give

$$|f(\xi+z)-f(\xi)|^2 \le |z|^2 \int_0^1 |Df(\xi+tz)|^2 dt.$$

Integrating in ξ yields the gradient bound below. Approximation in H^1, and continuity of translations in L^2, extend it to every $f \in H^1$. Translation invariance and the triangle inequality give the other bound:

$$\|f(\cdot+z)-f\|_2^2 \le \min\{|z|^2\|Df\|_2^2, 4\|f\|_2^2\}.$$

Tonelli and the change of variables $\xi = \eta + z$ give exactly

$$\mathcal{J}_\alpha(f) = \int_{\mathbb{R}^d} \frac{1+\log_+|z|}{|z|^{d+\alpha}} \|f(\cdot+z)-f\|_2^2 dz.$$

Use the gradient estimate for $|z|<1$ and the L^2 estimate for $|z|\ge 1$. Polar coordinates then give

$$\mathcal{J}_\alpha(f) \le |\mathbb{S}^{d-1}| \left[\frac{\|Df\|_2^2}{2-\alpha} + 4\left(\frac{1}{\alpha}+\frac{1}{\alpha^2}\right)\|f\|_2^2 \right].$$

Indeed the radial integrals are $\int_0^1 r^{1-\alpha}dr = (2-\alpha)^{-1}$ and $\int_1^\infty r^{-1-\alpha}(1+\log r)dr = \alpha^{-1}+\alpha^{-2}$. Both are finite precisely in the stated range, proving the bound in every $d \ge 1$. □

Lemma 7.3 (A square-root count cutoff controls the excluded-source commutator in every dimension)**.** *Let $d \ge 1$, $0 < \alpha < 2$, and $L > 0$. Let $q \in H^1(\mathbb{R}^d)$ be nonnegative, and let $\chi \in H^1(\mathbb{R}^d)$ have compact support and satisfy $0 \le \chi \le 1$. Fix a Lipschitz function $\psi : [0,\infty) \to [0,1]$ equal to one on $[0,1]$ and zero on $[2,\infty)$, and set*

$$\theta(\xi) = \chi(\xi)\psi(q(\xi)/L).$$

For every measurable $E \subset \{\theta = 0\}$, up to Lebesgue null sets,

$$\int_E q(\eta)^2 \int_{\mathbb{R}^d} \frac{|\theta(\xi)-\theta(\eta)|^2}{|\xi-\eta|^{d+\alpha}} (1+\log_+|\xi-\eta|)\, d\xi\, d\eta \le C_{d,\alpha,\psi}(L^2\|\chi\|_{H^1}^2 + \|q\|_{H^1}^2).$$

Proof. Take finite measurable representatives with $q \ge 0$ and $0 \le \chi \le 1$; modifying their null exceptional sets does not affect any integral below. Extend ψ to negative arguments by its value at zero. The Lipschitz map $s \mapsto \psi(s/L) - 1$ vanishes at zero, so Sobolev composition puts $\psi(q/L) - 1$ in $H^1 \cap L^\infty$, with gradient bounded by $L^{-1}\operatorname{Lip}(\psi)|Dq|$. In particular this argument does not require $\psi(q/L)$ itself to be in $L^2(\mathbb{R}^d)$. The bounded Sobolev product rule applied to $\theta = \chi + \chi[\psi(q/L)-1]$ gives $\theta \in H^1$, supported in the compact support of χ, and

$$\|\theta\|_2 \le \|\chi\|_2, \qquad |D\theta| \le |D\chi| + L^{-1}\operatorname{Lip}(\psi)|Dq| \quad \text{almost everywhere.}$$

Consequently

$$\|\theta\|_{H^1}^2 \le C_\psi(\|\chi\|_{H^1}^2 + L^{-2}\|Dq\|_2^2).$$

Remove from E the null set where its assumed relation $\theta(\eta) = 0$ fails. Such a removal changes neither side of the claim, by Tonelli. All sets and integrals here use Lebesgue representatives; no pointwise assertion outside a set of zero Sobolev capacity is needed.

Split E into the sources with $q(\eta) < 2L$ and those with $q(\eta) \ge 2L$. On the first part $q(\eta)^2 \le 4L^2$, so its contribution is at most $4L^2\mathcal{J}_\alpha(\theta)$, with the seminorm of lemma 7.2.

For a source in the second part, put $r = q(\eta) \ge 2L$ and $t = q(\xi) \ge 0$. If $t \ge 2L$, then $\theta(\xi) = 0$. If $t < 2L$ and $r \le 4L$, the Lipschitz bound and $\psi(2) = 0$ give

$$r|\theta(\xi)| \le r\psi(t/L) \le \frac{r}{L}\operatorname{Lip}(\psi)(2L-t) \le 4\operatorname{Lip}(\psi)|r-t|.$$

If $r > 4L$ and $t < 2L$, then $r \le 2(r-t)$ and $|\theta(\xi)| \le 1$. Thus all cases give the pointwise bound

$$q(\eta)^2|\theta(\xi)|^2 \le C_\psi|q(\xi)-q(\eta)|^2.$$

Because $\theta(\eta) = 0$ on E, Tonelli bounds the high-source part by $C_\psi\mathcal{J}_\alpha(q)$. The preceding lemma and the displayed H^1 bound therefore give for the whole commutator

$$\begin{aligned}4L^2\mathcal{J}_\alpha(\theta) + C_\psi\mathcal{J}_\alpha(q) &\le C_{d,\alpha,\psi}(L^2\|\chi\|_{H^1}^2 + \|Dq\|_2^2 + \|q\|_{H^1}^2)\\ &\le C_{d,\alpha,\psi}(L^2\|\chi\|_{H^1}^2 + \|q\|_{H^1}^2).\end{aligned}$$

The factor L^2 has cancelled the inverse factor in the cutoff derivative, so the constant is independent of L, q, χ, E. □

Proposition 7.4 (Strong count entropy supplies a negligible extra count screen)**.** *Let $\Omega \Subset \Omega' \subset \mathbb{R}^d$ be bounded open sets, with Ω' Lipschitz. Let I be a finite interval of positive length, let $m \ge 0$, and let nonnegative $Q_j \in L^2(I; H^1(\Omega'))$ satisfy*

$$\int_{\Omega'\times I} |Q_j - \sqrt{m}|^2 + |D_\xi Q_j|^2 \longrightarrow 0.$$

Define $q_j(\xi) = \left(\int_I Q_j(\xi,c)^2 dc\right)^{1/2}$ and $q_* = \sqrt{m|I|}$. Then $q_j \to q_*$ in $H^1(\Omega')$. For any fixed $L > q_*$ and any Lipschitz $\psi : [0,\infty) \to [0,1]$ equal to one on $[0,1]$ and zero on $[2,\infty)$,

$$\psi(q_j/L) \longrightarrow 1 \quad \text{in } H^1(\Omega').$$

For the quasicontinuous representatives of q_j, the relative Sobolev capacities also satisfy

$$\operatorname{cap}_2(\{q_j \ge 2L\} \cap \Omega; \Omega') \le \frac{C_{\Omega,\Omega'}}{(2L-q_*)^2}\|q_j - q_*\|_{H^1(\Omega')}^2 \longrightarrow 0,$$

where $\operatorname{cap}_2(A;\Omega')$ is the infimum of $\int_{\Omega'}(|Dv|^2 + v^2)$ over $v \in H_0^1(\Omega')$ whose quasicontinuous representative is at least one quasi-everywhere on A.

Proof. The general Hilbert-norm estimate established in the proof of lemma 7.1 applies to Q_j on Ω'. It uses only $L^2(I;H^1)$ membership and gives

$$\|q_j - q_*\|_{L^2(\Omega')}^2 \le \|Q_j - \sqrt{m}\|_{L^2(\Omega'\times I)}^2, \qquad \|Dq_j\|_{L^2(\Omega')}^2 \le \|D_\xi Q_j\|_{L^2(\Omega'\times I)}^2.$$

Thus $q_j \to q_*$ in $H^1(\Omega')$. Since $L > q_*$, $\psi(q_*/L) = 1$; Lipschitz composition gives

$$\|\psi(q_j/L) - 1\|_2 \le \operatorname{Lip}(\psi)L^{-1}\|q_j - q_*\|_2, \qquad \|D\psi(q_j/L)\|_2 \le \operatorname{Lip}(\psi)L^{-1}\|Dq_j\|_2.$$

This proves the second convergence.

Apply the general capacitary truncation estimate in the same proof with ambient domain Ω', compact set $\overline{\Omega}$, and threshold $T = 2L$. Since $\Omega \Subset \Omega'$, the cutoff equal to one near $\overline{\Omega}$ has a fixed H^1 product constant depending only on these domains. Restricting the resulting superlevel set from $\overline{\Omega}$ to Ω gives exactly the stated denominator $(2L-q_*)^2$ and capacity bound. The superlevel sets here and the composed screens use the scalar quasicontinuous representative of q_j. □

Proposition 7.5 (Positive Poisson source screening from count entropy in every base dimension)**.** *Let $d \ge 1$, $n = d+1$, $1 < p < 2$, and $R \ge 1$. Write $X = (x,t) \in \mathbb{R}^d \times \mathbb{R}$. For $y > 0$ put*

$$P_y(X) = \kappa_{n,p}\frac{y^p}{(|X|^2+y^2)^{(n+p)/2}}, \qquad \mathcal{Q}_y(X) = \frac{|X|}{y}P_y(X),$$

where $\kappa_{n,p} > 0$ normalizes P_y to have integral one. Suppose a nonnegative measurable source m satisfies

$$P_y * m \le \frac{C_1}{1+y}, \qquad \mathcal{Q}_y * m \le \frac{C_1}{1+y}, \qquad \int_{\mathbb{R}} m(R\eta, t)dt \le C_0 q(\eta)^2$$

almost everywhere, for a nonnegative $q \in H^1(\mathbb{R}^d)$. Let χ, L, ψ, θ, E be as in lemma 7.3, and set $m_E(x,t) = \mathbf{1}_E(x/R)m(x,t)$. Then for either $\mathcal{P}_y = P_y$ or $\mathcal{P}_y = \mathcal{Q}_y$,

$$\int_{\mathbb{R}^n}\int_0^\infty y^{1-p}|\theta(x/R)(\mathcal{P}_y * m_E)(X)|^2 dy\, dX \le CR^{d+1-p}(1+\log R)(L^2\|\chi\|_{H^1}^2 + \|q\|_{H^1}^2),$$

with C depending only on d, p, C_0, C_1, ψ. Consequently, if p is fixed, these Sobolev norms and L remain bounded, $D > 0$, $R \to \infty$, and $R^2/D^{p+1} \le \Lambda < \infty$, the displayed energy is $o(R^{d-2}D^2)$.

Proof. The column bound and Tonelli give $\|m\|_{L^1(\mathbb{R}^n)} \le C_0 R^d \|q\|_2^2 < \infty$. Both kernels are nonnegative and integrable at each fixed height: P_y has mass one, while the change of variables $X = yZ$ shows that $\int \mathcal{Q}_y = \int |Z|P_1(Z)dZ < \infty$, because $p > 1$. We use finite measurable representatives of the Sobolev functions; all identities and estimates below are unchanged by null-set modifications. Since θ vanishes almost everywhere on E, for almost every receiver the convolution equals

$$\theta(x/R)(\mathcal{P}_y * m_E)(X) = \int [\theta(x/R) - \theta(z/R)]\mathcal{P}_y(X-Z)m_E(Z)dZ, \qquad Z = (z,s).$$

Cauchy–Schwarz with the nonnegative measure $\mathcal{P}_y(X-Z)m_E(Z)dZ$ and the assumed row bound gives

$$|\theta(x/R)\mathcal{P}_y * m_E|^2 \le \frac{C_1}{1+y}\int |\theta(x/R) - \theta(z/R)|^2 \mathcal{P}_y(X-Z)m_E(Z)dZ.$$

All subsequent integrations are justified by Tonelli, allowing infinite integrals before the final finite bound.

At physical distance $r = |X - Z| > 0$, the height-integrated radial kernels in this estimate are

$$K_P(r) = \kappa_{n,p}\int_0^\infty \frac{y\,dy}{(1+y)(r^2+y^2)^{(n+p)/2}},$$
$$K_Q(r) = \kappa_{n,p} r\int_0^\infty \frac{dy}{(1+y)(r^2+y^2)^{(n+p)/2}}.$$

For $0 < r \le 1$, discard $(1+y)^{-1} \le 1$ and set $y = rs$. The remaining integrals in s have numerators s and 1, respectively, and are finite since $n + p > 3$. Thus both kernels are at most Cr^{2-n-p}. For $r \ge 1$, use $y/(1+y) \le 1$ to obtain $K_P(r) \le Cr^{1-n-p}$. For the second kernel, split its integral at $y = r$:

$$\begin{aligned} K_Q(r) &\le Cr^{1-n-p}\int_0^r \frac{dy}{1+y} + Cr\int_r^\infty y^{-1-n-p}dy \\ &\le Cr^{1-n-p}(1+\log r). \end{aligned}$$

Consequently, for either kernel,

$$K(r) \le C\begin{cases} r^{2-n-p}, & 0 < r \le 1, \\ r^{1-n-p}(1+\log r), & r \ge 1. \end{cases}$$

We integrate out the receiver normal coordinate by putting $a = |x - z| > 0$ and $\tau = t - s$. If $a \geq 1$, the substitution $\tau = av$ in the far bound gives

$$\int_{\mathbb{R}} K(\sqrt{a^2 + \tau^2}) d\tau \leq C a^{2-n-p}(1 + \log a).$$

The integrals in v are finite, including the factor $\log(1 + v^2)$, since $n + p > 2$. If $0 < a < 1$, the part with $a^2 + \tau^2 < 1$ is bounded by

$$C \int_{\mathbb{R}} (a^2 + \tau^2)^{(2-n-p)/2} d\tau = C a^{3-n-p},$$

where $n+p > 3$ is used. On the remaining part with $|\tau| \leq 2$ the kernel is uniformly bounded; on $|\tau| > 2$ its far bound is integrable, uniformly in $a < 1$. Therefore this tail is $O(1)$, and both the near bound and this tail are at most Ca^{2-n-p}, since $0 < a < 1$ and $n+p > 3$. Since $n = d + 1$, both cases give the single base-kernel estimate

$$\int_{\mathbb{R}} K(\sqrt{a^2 + \tau^2}) d\tau \leq C a^{-d-(p-1)}(1 + \log_+ a).$$

The exceptional base diagonal $x = z$ is a Lebesgue null set, so it does not affect the nonnegative double integral.

Integrate the source normal coordinate using its column bound, and set $x = R\xi$, $z = R\eta$. The two base Jacobians and the base kernel produce R^{d+1-p}. Also

$$1 + \log_+(R|\xi - \eta|) \leq (1 + \log R)(1 + \log_+ |\xi - \eta|).$$

The result is bounded by the integral in lemma 7.3 with $\alpha = p - 1$, times $CC_0C_1R^{d+1-p}(1 + \log R)$, proving the energy estimate.

Finally the bounded interaction ratio gives $D^2 \geq \Lambda^{-2/(p+1)} R^{4/(p+1)}$. The ratio of the claimed energy scale to $R^{d-2}D^2$ is at most a constant times

$$R^{3-p-4/(p+1)}(1 + \log R) = R^{-(p-1)^2/(p+1)}(1 + \log R) \longrightarrow 0.$$

The exponent and its sign do not depend on d. □

8. Fixed-amplitude heights and complete column integrability

In dimension six, $H^1 \cap W^{2,1}$ regularity and the rank metric-Hessian energy do not imply local boundedness of the limiting heights. The test construction therefore keeps the amplitude cutoff M fixed while taking the record limit. Complete column uniform integrability then removes this cutoff. The weak force is integrated by parts as a whole test, so derivatives of the height cutoff are never individually discarded.

Lemma 8.1 (Anisotropic surface Sobolev gives a nonzero amplitude-tail exponent in every finite dimension)**.** *Let $d > 2$, let (T, ω) be a finite measure space, and for each parameter $\tau \in T$ let Σ_τ be a smooth d-dimensional hypersurface, $\ell_\tau \geq 0$ a smooth function, and $0 \leq \chi_\tau \leq 1$ a compactly supported Lipschitz function. Let $B : T \to [0, \infty)$ be measurable, and suppose all the displayed surface integrals depend measurably on τ. Suppose*

$$\int_{\Sigma_\tau \cap \{\chi_\tau = 1\}} \ell_\tau \leq C_0, \qquad \left(\int_{\Sigma_\tau} \chi_\tau^{2q} \ell_\tau^q \right)^{1/q} \leq C_1 B(\tau), \quad q = \frac{d}{d-2}, \quad \int_T B \, d\omega \leq C_2.$$

Then for $K \geq 1$,

$$\int_T \int_{\Sigma_\tau \cap \{\chi_\tau = 1,\ \ell_\tau > K\}} \ell_\tau \, d\mathcal{H}^d d\omega \leq C K^{-2/d}.$$

One may take $C = C_0^{2/d} C_1 C_2$. In particular, the exponent is $1/3$ when $d = 6$.

For $R > 0$ and $0 < a \le 1$, write $X = (x,t) \in \mathbb{R}^d \times \mathbb{R}$, stretch to $z = (x,s)$ with $t = as$, and consider the level $F_y(z) = U(x, as, y) = c$, scaled by R^{-1}. Set $Z = (D_x U, aU_t)$, $\gamma = |Z|$, $\ell = \gamma/a$, and $\mathcal{A}_Z = |DZ|^2 - |D\gamma|^2$. Here $D = (D_x, \partial_t, \partial_y)$ uses the physical variables and includes height. The fields are pulled back by the stretch when differentiated with respect to z.

Let $0 \le \chi = \chi_{c,y} \le 1$ be an ambient C^1 cutoff, or a locally Lipschitz ambient cutoff whose pullback is differentiable at $\mathcal{H}^d$-almost every point of the tested level. Require its restriction to the level to have compact support in its regular interior, away from any declared surface boundary. In particular $D_z\chi = (D_x\chi, a\partial_t\chi)$ denotes the derivative of its pullback, not a height derivative. The surface Sobolev hypothesis above follows with

$$B(c,y) = \frac{1}{a^2 R^{d-2}} \int \delta(U-c)[\chi^2 \mathcal{A}_Z + \gamma^2 |D_z\chi|^2] dX.$$

Thus a bound Ca^2R^{d-2} for the integral of its numerator in phase and height, together with the almost-everywhere uniform central level-energy bound

$$\int_{\{\chi=1\}} \delta(U-c)\gamma^2 dX \le Ca^2R^d,$$

implies the stated tail after normal coarea, with normalization a^2R^d. A bound on the larger region $\{\chi > 0\}$, or on a physical receiver containing that region, is sufficient as well. Here U is smooth, all cutoff and level integrals are measurable, and the phase-height measure is $dc\,d\nu(y)$ on fixed phase and height ranges with finite ν. The notation $\delta(U-c)dX$ means the coarea level measure; it is used only where the regular level and the displayed weights are defined. No differential equation for U is required for this geometric implication.

Proof. For almost every parameter, write $E_K(\tau)$ for the tail integral. On $\{\ell_\tau > K\}$, one has $\ell_\tau \le K^{1-q}\ell_\tau^q$, so

$$E_K(\tau) \le \min\{C_0, K^{1-q}(C_1 B(\tau))^q\}.$$

The inequality $\min(A, x^q) \le A^{1-1/q}x$ for $A, x \ge 0$, applied with $x = C_1 K^{-(q-1)/q} B(\tau)$, gives

$$E_K(\tau) \le C_0^{1-1/q} C_1 K^{-2/d} B(\tau).$$

Integrating proves the assertion using only $B \in L^1(T,\omega)$, rather than an L^q bound for B.

We check the geometric implication for $0 < a \le 1$. At a fixed extension height write $z = (x,s)$, $t = as$, and $F(z) = U(x,as,y)$. Then $D_z F = Z$, so $D_z Z$ is the symmetric Hessian of F. Put $n = Z/\gamma$ on $\{\gamma > 0\}$. The two angular tensors admit the sum-of-squares expressions

$$\mathcal{A}_z := |D_z Z|^2 - |D_z\gamma|^2 = \sum_{i=1}^d |P_{n^\perp}\partial_{x_i} Z|^2 + a^2 |P_{n^\perp}\partial_t Z|^2,$$

$$\mathcal{A}_Z = \sum_{i=1}^d |P_{n^\perp}\partial_{x_i} Z|^2 + |P_{n^\perp}\partial_t Z|^2 + |P_{n^\perp}\partial_y Z|^2.$$

Consequently $\mathcal{A}_z \le \mathcal{A}_Z$, precisely because $a \le 1$ and the height summand is nonnegative. No Jacobi-field identity is used. On a regular level $\Sigma_z = \{F = c\}$, decompose the Hessian into tangent-tangent, tangent-normal, and normal-normal blocks. The tangent-tangent block is γA_{Σ_z}, the tangent-normal block is $D_{\Sigma_z}\gamma$, and subtracting $|D_z\gamma|^2$ removes the normal-normal square and one of the two mixed-block squares. Therefore

$$|D_{\Sigma_z}\gamma|^2 + \gamma^2 |A_{\Sigma_z}|^2 = \mathcal{A}_z.$$

Let $\widehat{\Sigma} = R^{-1}\Sigma_z$. We pull ℓ and χ to $\widehat{\Sigma}$, so their values at z/R are their unscaled values at z. Since $dz = a^{-1}dX$, coarea gives, for each compactly supported nonnegative test h,

$$\int_{\widehat{\Sigma}} h\, d\mathcal{H}^d = \frac{1}{aR^d} \int \delta(U-c)\gamma\, h((x,t/a)/R)dX.$$

In particular

$$\int_{\widehat{\Sigma}\cap\{\chi=1\}} \ell\, d\mathcal{H}^d = \frac{1}{a^2R^d} \int_{\{\chi=1\}} \delta(U-c)\gamma^2 dX.$$

These are level-integral identities, rather than pointwise identities between ambient and surface volume forms. Sard's theorem and coarea allow omission of exceptional phases at each fixed height; Fubini then applies to the finite product measure. The argument is entirely on the regular interior and asserts no removability across a critical set or a cut boundary.

For clarity we state the precise Sobolev input. The first-order Michael–Simon inequality[8] on any smooth d-dimensional hypersurface M is

$$\left(\int_M v^{d/(d-1)}\right)^{(d-1)/d} \le C_d \int_M (|D_M v| + |H_M| v) \qquad (0 \le v \in C^1_c(M)).$$

Here H_M is the trace mean curvature. If M has boundary, the support must be compact in its interior; no completeness or minimality is required. Apply this to $v = f^{2(d-1)/(d-2)}$, for nonnegative compactly supported f. Cauchy–Schwarz bounds its right side by

$$C_d \left(\int_M (|D_M f|^2 + |H_M|^2 f^2)\right)^{1/2} \left(\int_M f^{2d/(d-2)}\right)^{1/2}.$$

If the last integral is zero the desired inequality is immediate; otherwise divide by its square root and square the result. This gives the dimension-d, exponent-two form. Taking $M = \widehat{\Sigma}$ and $f = \chi\sqrt{\ell}$, with $q = d/(d-2)$, yields

$$\left(\int_{\widehat{\Sigma}} \chi^{2q}\ell^q\right)^{1/q} \le C_d \int_{\widehat{\Sigma}} (|D_{\widehat{\Sigma}}(\chi\sqrt{\ell})|^2 + |H_{\widehat{\Sigma}}|^2\chi^2\ell).$$

The support is compact in the regular surface, where $\gamma > 0$, so $\sqrt{\ell}$ is smooth on a neighborhood of it. Compactly supported Lipschitz cutoffs are admissible by chartwise smooth approximation both uniformly and in $W^{1,2}$, with support in one fixed compact subset of the regular interior; there the mean curvature is bounded and the approximation also converges in the relevant finite-exponent norms. The scaled tangential derivative and curvature are R times their unscaled counterparts. Using the preceding area formula and $|H_{\Sigma_z}|^2 \le d|A_{\Sigma_z}|^2$, the right side is at most

$$\frac{C_d}{a^2R^{d-2}} \int \delta(U-c)\left[\gamma^2|D_z\chi|^2 + \chi^2(|D_{\Sigma_z}\gamma|^2 + \gamma^2|A_{\Sigma_z}|^2)\right] dX \le C_d B(c,y).$$

Indeed $D_z\sqrt{\ell} = (2\sqrt{a\gamma})^{-1}D_z\gamma$, which, multiplied by the area factor $\gamma/(aR^d)$ and the derivative factor R^2, yields exactly $1/(4a^2R^{d-2})$. The curvature term has the same scaling. The cutoff is compactly supported away from every removed surface boundary, so the Sobolev inequality produces no boundary term. The central level-energy assumption supplies C_0 in the abstract tail bound, and the integrated angular/cutoff assumption supplies the bound for $\int B$. Finally,

$$\int_{\widehat{\Sigma}\cap\{\chi=1,\ell>K\}} \ell = \frac{1}{a^2R^d} \int \delta(U-c)\gamma^2 \mathbf{1}_{\{\chi=1,\gamma/a>K\}} dX.$$

Integrating in the fixed finite phase-height measure and applying the abstract estimate proves the normalized $CK^{-2/d}$ tail. In particular the exponent is $1/3$ for $d = 6$. □

Lemma 8.2 (The failed phase-track mass is negligible through six base dimensions)**.** *Let $3 \le d \le 6$, $1 < p < 2$, $R_j \to \infty$, and let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Suppose q_j is bounded in $H^1(\Omega)$, $m_j \ge 0$ is an integrable physical source with*

$$\int_{\mathbb{R}} m_j(R_j\xi, t)dt \le Cq_j(\xi)^2,$$

and a measurable set $E_j \subset R_j\Omega$ has $|E_j| \le CR_j^{d-2}$. Suppose also $S_j \ge cR_j^{d-2+4/(p+1)}$. Then

$$\int_{E_j\times\mathbb{R}} m_j \le CR_j^{d-4/d} = o(S_j).$$

The strict inequality needed for the last conclusion is

$$2 - \frac{4}{d} - \frac{4}{p+1} < 0.$$

Proof. Put $A_j = R_j^{-1}E_j \subset \Omega$. Its measure is at most CR_j^{-2}. Sobolev embedding on the fixed bounded Lipschitz domain, valid since $d > 2$, gives

$$\sup_j \|q_j\|_{L^{2d/(d-2)}(\Omega)} \le C_\Omega \sup_j \|q_j\|_{H^1(\Omega)} < \infty.$$

Tonelli, the change of variables $x = R_j\xi$, the source-column bound and Hölder's inequality now yield

$$\begin{aligned}\int_{E_j\times\mathbb{R}} m_j &= R_j^d \int_{A_j}\int_{\mathbb{R}} m_j(R_j\xi, t)dt\, d\xi \\ &\le CR_j^d \int_{A_j} q_j^2 \le CR_j^d \|q_j\|^2_{L^{2d/(d-2)}(\Omega)} |A_j|^{2/d} \\ &\le CR_j^{d-4/d}.\end{aligned}$$

The constant is independent of j. Divide by $S_j \ge cR_j^{d-2+4/(p+1)}$. The resulting ratio is at most $CR_j^{2-4/d-4/(p+1)}$, and

$$2 - \frac{4}{d} - \frac{4}{p+1} \le \frac{4}{3} - \frac{4}{p+1} < 0$$

for $d \le 6$ and $p < 2$. Since $R_j \to \infty$, this proves the claimed little-o bound. □

Lemma 8.3 (Strong clipped lifts give finite-record bounded-height supports)**.** *Let $\Omega \subset \mathbb{R}^d$ be bounded and open, let $m \ge 1$ be an integer, and let $f_k \in H^1(\Omega)$ for $1 \le k \le m$. Suppose $0 \le \theta_j \le 1$, $\theta_j \in H^1(\Omega)$, and $\theta_j \to 1$ strongly in H^1. On a measurable set O_j containing $\{\theta_j > 0\}$ up to null sets, let $t_{j,k}$ be finite measurable functions. For $L > 0$ let T_L clip real numbers to $[-L, L]$, and suppose the zero-extended functions*

$$F_{j,k,L} = \theta_j T_L(t_{j,k}) \quad \text{on } O_j$$

belong to $H^1(\Omega)$ and converge strongly to $T_L(f_k)$ there for each fixed L.

Fix a smooth $A : [0,1] \to [0,1]$ vanishing on $[0, 3/4]$ and equal to one on $[7/8, 1]$. For each $M > 0$, fix smooth $h_M : \mathbb{R} \to [0,1]$ equal to one on $[-M, M]$, zero outside $(-2M, 2M)$, with $|h_M'| \le C/M$. Define

$$\beta_{j,M} = A(\theta_j) \prod_{k=1}^m h_M(F_{j,k,4M}), \qquad \beta_M = \prod_{k=1}^m h_M(f_k).$$

Then $\beta_{j,M} \to \beta_M$ strongly in $H^1(\Omega)$ for fixed M. Wherever $\beta_{j,M} > 0$, one has $\theta_j > 3/4$ and $|t_{j,k}| < 8M/3$ for every k, almost everywhere. Also $\beta_M \to 1$ strongly in $H^1(\Omega)$ as $M \to \infty$.

Proof. For fixed M, the smooth Lipschitz compositions of the strongly convergent lifts converge strongly in H^1. To see the gradient convergence explicitly, write the derivative difference as the sum of a bounded multiplier times the strongly vanishing gradient difference and $(h'_M(F_{j,k,4M}) - h'_M(T_{4M}(f_k)))DT_{4M}(f_k)$. The second term tends to zero in L^2 by bounded convergence in measure against a fixed L^2 function. The entire sequence has this property by the subsequence criterion. The limit is $h_M(f_k)$, because $h_M(T_{4M}(f_k)) = h_M(f_k)$. Similarly $A(\theta_j) \to 1$ strongly in H^1, since $|DA(\theta_j)| \le \|A'\|_\infty |D\theta_j| \to 0$ and its bounded values converge in measure to one. Finite products of these bounded strongly H^1 convergent factors converge strongly in H^1, by the product rule and the same fixed-gradient dominated-convergence argument.

If $\beta_{j,M} > 0$, then $\theta_j > 3/4$, so this point lies in O_j outside a fixed null set. The defining equality $F_{j,k,4M} = \theta_j T_{4M}(t_{j,k})$ therefore applies there for every k. Since every h_M factor is nonzero, $|F_{j,k,4M}| < 2M$, and hence

$$|T_{4M}(t_{j,k})| = \frac{|F_{j,k,4M}|}{\theta_j} < \frac{8M}{3} < 4M.$$

The clipping is inactive, which proves the asserted actual-height bound on this nonzero set. Finally $\beta_M \to 1$ almost everywhere, its values are bounded, and

$$|D\beta_M| \le (C/M) \sum_k |Df_k|.$$

Dominated convergence for its values and this gradient bound give strong H^1 convergence to one. No local boundedness of any f_k was assumed. □

Lemma 8.4 (Whole-test integration removes height truncations from weak forces)**.** *Let $f \in H^1_{\mathrm{loc}}(\Omega) \cap W^{2,1}_{\mathrm{loc}}(\Omega)$ and $F \in L^1_{\mathrm{loc}}(\Omega)$, where $\Omega \subset \mathbb{R}^d$ is open. Let $0 \le b \le 1$ belong to $H^1_{\mathrm{loc}}(\Omega)$. Suppose $0 \le \beta_M \le 1$ belong to $H^1_{\mathrm{loc}}(\Omega)$ and converge to one almost everywhere. If for every smooth compactly supported η and every M,*

$$\int Df \cdot D(\eta b^2 \beta_M^2) = -\int F \eta b^2 \beta_M^2,$$

then $\int (\Delta f - F)\eta b^2 = 0$ for every such η. In particular, if these identities hold for cutoffs b_δ with $0 \le b_\delta \le 1$ and $b_\delta \to 1$ almost everywhere as $\delta \downarrow 0$, then $\Delta f = F$ distributionally.

Proof. Fix $\eta \in C_c^\infty(\Omega)$ and M, and put $w = \eta b^2 \beta_M^2$. The bounded-factor product rule shows that $w \in H^1 \cap L^\infty$ and $\operatorname{supp} w \subset \operatorname{supp} \eta \Subset \Omega$. Choose a smooth compact cutoff equal to one near $\operatorname{supp} \eta$; multiplying by it leaves w unchanged and shows that its zero extension belongs to $H^1(\mathbb{R}^d)$. Thus no regularity of the boundary of Ω is required for this extension.

Convolve that extension with nonnegative smooth mollifiers of mass one. For sufficiently small radii, $w_\varepsilon \in C_c^\infty(\Omega)$ have support in one fixed compact subset, satisfy $\|w_\varepsilon\|_\infty \le \|w\|_\infty$, and converge strongly to w in H^1; along a subsequence they converge almost everywhere as well. Since $f \in W^{2,1}_{\mathrm{loc}}$, ordinary distributional integration by parts gives

$$\int Df \cdot Dw_\varepsilon = -\int \Delta f \, w_\varepsilon.$$

Cauchy–Schwarz for $Df \in L^2_{\mathrm{loc}}$ passes the left side to the limit. On the fixed common support, bounded almost-everywhere convergence and $\Delta f \in L^1_{\mathrm{loc}}$ pass the right side by dominated convergence. Combining the resulting identity for w with the assumed weak-force identity yields

$$\int (\Delta f - F)\eta b^2 \beta_M^2 = 0.$$

This reasoning applies separately to every fixed M; it requires no uniform mollification sequence as M varies. Send $M \to \infty$, for example along positive integers, by dominated convergence against $\|\eta\|_\infty|\Delta f - F|$ on the fixed support of η. This proves the first conclusion. If $b = b_\delta$, apply that conclusion for each fixed δ, and then send $\delta \downarrow 0$ by the same dominated-convergence argument. It gives $\int(\Delta f - F)\eta = 0$ for every compact smooth η, which is exactly $\Delta f = F$ distributionally. The integration by parts retains the entire product before either cutoff limit, and no bound on f is needed. □

Lemma 8.5 (Normal-column control and the complete ratio tail imply transverse-column integrability)**.** *Let $\Omega \subset \mathbb{R}^d$ have finite measure and let $n_j, e_j \geq 0$ be integrable. Suppose $\{n_j\}$ is uniformly integrable and, for each $K \geq 1$, there are $r_{j,K} \geq 0$ such that*

$$e_j \leq K^2 n_j + r_{j,K} \quad \text{almost everywhere,} \qquad \lim_{K\to\infty} \limsup_{j\to\infty} \int_\Omega r_{j,K} = 0.$$

Then $\{e_j\}$ is uniformly integrable. If $0 \leq \beta_{j,M}, \beta_M \leq 1$, $\beta_{j,M} \to \beta_M$ in measure for each fixed M, and $\beta_M \to 1$ almost everywhere as $M \to \infty$, then

$$\lim_{M\to\infty} \limsup_{j\to\infty} \int_\Omega (1 - \beta_{j,M}^2)(e_j + n_j) = 0.$$

Here uniform integrability includes a uniform L^1 bound and uniform absolute continuity of the integrals.

Proof. Given $\epsilon > 0$, choose fixed K so that $\int r_{j,K} < \epsilon$ for all sufficiently large j. Uniform integrability of n_j then supplies $\delta > 0$ such that for $|E| < \delta$,

$$\int_E e_j \leq K^2 \int_E n_j + \int r_{j,K} < 2\epsilon$$

for those j. Finitely many earlier integrable functions can be included by reducing δ. Fixing one sufficiently large K also bounds the L^1 norms of e_j for large j, and the finite remaining norms are harmless. This proves uniform integrability.

For fixed M, bounded convergence in measure and uniform integrability give

$$\int |\beta_{j,M}^2 - \beta_M^2|(e_j + n_j) \longrightarrow 0.$$

To verify this, split at any positive threshold for the bounded difference, use the uniform L^1 bound on the small part, and uniform absolute continuity on the complement, whose measure tends to zero. The same argument applied to $1 - \beta_M^2$, now uniformly over j, shows

$$\lim_{M\to\infty} \sup_j \int (1 - \beta_M^2)(e_j + n_j) = 0.$$

Indeed the measure of any fixed positive superlevel set tends to zero by bounded convergence on a finite measure space. Combining the two limits gives the assertion. □

Lemma 8.6 (The actual normal Wronskian gives complete base-column integrability before profile comparison)**.** *Let $d > 1$, let $\Omega \Subset \Omega' \subset \mathbb{R}^d$ be bounded connected Lipschitz domains, and let $R_j \to \infty$, $a_j \downarrow 0$. Let smooth fields H_j and $V_j \in \mathbb{R}^d$ on $R_j\Omega' \times \mathbb{R} \times (0,\infty)$ obey $D_x H_j = \partial_t V_j$. Put $Z_j = (V_j, a_j H_j)$ and $\mathcal{A}_j = |DZ_j|^2 - |D|Z_j||^2$, where derivatives include all physical directions and height. Suppose, with $d\mu = d_p y^{1-p} dx dt dy$, $1 < p < 2$,*

$$\int H_j^2 d\mu \leq C R_j^d, \quad \int |V_j|^2 d\mu \leq C a_j^2 R_j^d, \quad \int \mathcal{A}_j d\mu \leq C a_j^2 R_j^{d-2}.$$

Let $0 \leq \rho_j(t) \leq 1$ be smooth compactly supported cutoffs with $|\rho_j'| \leq C/R_j$, and assume differentiation and integration by parts in the following complete normal integrals are legitimate

by exhaustion. For the actual fields $H_j = \partial_t U_j$, $V_j = D_x U_j$, with U_j the Poisson extension of a bounded smooth trace with bounded derivatives, it also follows from the usual physical derivative bounds. In fact the energy bounds above suffice for the distributional exhaustion argument used below. Define

$$n_j(\xi) = \int \rho_j(t) H_j(R_j\xi, t, y)^2 d_p y^{1-p} dt dy.$$

Then $\|n_j\|_{L^1(\Omega')} \le C$ and $|Dn_j|(\Omega) \le Ca_j$. In particular n_j is uniformly integrable on Ω; after subtracting its mean it tends to zero in $L^{d/(d-1)}(\Omega)$. This conclusion uses no transverse normal-profile comparison.

Proof. All energy integrals are over $R_j\Omega' \times \mathbb{R} \times (0,\infty)$. Write $d\mu_{t,y} = d_p y^{1-p} dt\, dy$ and $N_j(x)$ for the unscaled column. Tonelli's theorem gives $N_j \in L^1(R_j\Omega')$ and makes the column finite for almost every base point. If $a_j = 0$, the second energy bound and smoothness give $V_j = 0$, hence $D_x H_j = 0$. Connectedness of Ω' makes N_j constant, so all the assertions for this index follow directly. We may therefore assume $a_j > 0$.

First restrict the height integral defining N_j to $\varepsilon < y < Y$, obtaining $N_j^{\varepsilon,Y}$. On compact subsets of $R_j\Omega'$, smoothness and compact t-support justify differentiation and integration by parts. Compatibility gives $D_x(H_j^2) = 2H_j\partial_t V_j$. Writing one of these two equal terms as

$$H_j\partial_t V_j = \partial_t(H_j V_j) - (\partial_t H_j)V_j$$

and integrating the total derivative in t gives

$$D_x N_j^{\varepsilon,Y} = \int_{\varepsilon<y<Y} [\rho_j \mathscr{W}_j - \rho_j' H_j V_j] d\mu_{t,y}, \qquad \mathscr{W}_j = H_j\partial_t V_j - (\partial_t H_j)V_j.$$

For any vector Z and derivative ∂_α, the wedge identity is

$$\sum_{r<s} (Z_r \partial_\alpha Z_s - Z_s \partial_\alpha Z_r)^2 = |Z|^2 |\partial_\alpha Z|^2 - (Z \cdot \partial_\alpha Z)^2.$$

The locally Lipschitz norm $|Z_j|$ has its weak derivative almost everywhere, and $\mathcal{A}_j \ge 0$ there. At $Z_j \neq 0$, divide the wedge identity by $|Z_j|^2$, take the terms joining the last component of $Z_j = (V_j, a_j H_j)$ to its first d components, and use the nonnegative contribution of every derivative other than ∂_t. This proves

$$|\mathscr{W}_j| \le a_j^{-1} |Z_j| \sqrt{\mathcal{A}_j}$$

almost everywhere; at $Z_j = 0$, both H_j and V_j vanish and thus $\mathscr{W}_j = 0$. In particular

$$\int |Z_j|^2 d\mu = \int (|V_j|^2 + a_j^2 H_j^2) d\mu \le C a_j^2 R_j^d.$$

Cauchy–Schwarz therefore makes both $\rho_j \mathscr{W}_j$ and $\rho_j' H_j V_j$ integrable on the complete product. The capped columns converge to N_j in $L^1(R_j\Omega')$. By Fubini and dominated convergence their displayed gradients converge in the same space to the vector field

$$F_j(x) = \int [\rho_j \mathscr{W}_j - \rho_j' H_j V_j] d\mu_{t,y}.$$

Testing the capped identity against any compactly supported smooth base function and passing to the limit proves $D_x N_j = F_j$ distributionally. Thus $N_j \in W^{1,1}(R_j\Omega')$. This argument requires no separate integrability of $H_j\partial_t V_j$ and $(\partial_t H_j)V_j$, and derives the required exhaustion directly from the three energies.

The same estimates, restricted to $R_j\Omega$ when necessary, give

$$\int_{R_j\Omega} |D_x N_j|\,dx \le a_j^{-1}\left(\int |Z_j|^2 d\mu\right)^{1/2}\left(\int \mathcal{A}_j d\mu\right)^{1/2} + \frac{C}{R_j}\left(\int H_j^2 d\mu\right)^{1/2}\left(\int |V_j|^2 d\mu\right)^{1/2}$$
$$\le C a_j R_j^{d-1}.$$

Since $n_j(\xi) = N_j(R_j\xi)$, the change of variables $x = R_j\xi$ yields

$$\|n_j\|_{L^1(\Omega')} = R_j^{-d}\|N_j\|_{L^1(R_j\Omega')} \le C, \qquad |Dn_j|(\Omega) = R_j^{1-d}\int_{R_j\Omega} |D_x N_j|\,dx \le C a_j.$$

Put $q = d/(d-1) > 1$. The BV Sobolev–Poincare inequality on the fixed bounded connected Lipschitz domain Ω gives

$$\|n_j - (n_j)_\Omega\|_{L^q(\Omega)} \le C_\Omega |Dn_j|(\Omega) \le C a_j \longrightarrow 0.$$

The means are bounded by the L^1 estimate. Hence $\sup_j \|n_j\|_{L^q(\Omega)} < \infty$, and Hölder's inequality gives $\sup_j \int_E n_j \le C|E|^{1/d}$ for every measurable $E \subset \Omega$. This proves uniform integrability. □

9. Actual complete ranks and their full profiles

Definition 9.1 (The complete higher-dimensional fine record)**.** Let $3 \le d \le 6$, $1 < p < 2$, and let u satisfy proposition 3.4. Write U for its Poisson extension and $d\mu = d_p y^{1-p} dX dy$. Suppose the integer multiplicity in proposition 3.5 is $m \ge 2$. Define

$$M(r) = \int_{B_r\times(0,\infty)} D_X U \otimes D_X U\, d\mu, \qquad E(r) = \operatorname{tr} M(r) - \lambda_{\max}(M(r)).$$

An optimizing orthonormal frame at R_j means that its t axis is a unit top eigenvector of $M(R_j)$ and its x axes form an orthonormal basis of the perpendicular space. A fine record consists of the radii, augmented budgets, and original optimizing frames produced by proposition 5.3, including its outward same-frame estimate, with

$$R_j \longrightarrow \infty, \quad S_j = a_j^2 R_j^d = R_j^{d-2} D_j^2, \quad D_j = a_j R_j \longrightarrow \infty, \quad a_j \longrightarrow 0,$$
$$E(R_j)/S_j \longrightarrow 1, \qquad \lambda_j = R_j^2/D_j^{p+1} \longrightarrow \lambda \in [0,\infty).$$

Set $X = (x,t) \in \mathbb{R}^d \times \mathbb{R}$, $V_j = D_x U_j$, $H_j = (U_j)_t$, and $Z_j = (V_j, a_j H_j)$, with all fields expressed in that record's frame. Fixed enlarged receivers are understood. The common force normalization is $F_j = R_j^{d-2} D_j = S_j/D_j$. A small shear $x \mapsto x + k_j t$, $k_j \to 0$, has determinant one; when used it is always the pulled-back actual equation, with coefficient matrix

$$\begin{pmatrix} I + k_j k_j^T & -k_j & 0 \\ -k_j^T & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The outward same-frame bound is $\int_{B_{LR_j}\times(0,\infty)} |V_j|^2 d\mu \le C S_j L^d (1+\log L)^2$ for $L \ge 2$, with constants independent of j, L. No isotropic equation is assigned to a sheared field.

Lemma 9.2 (The positive Kato replacement controls the entire trace angular tensor in every dimension)**.** *Under definition 9.1, set* $Z_0 = (D_x u_j, a_j (u_j)_t)$ *and* $\gamma = |Z_0|$. *On every fixed enlarged receiver,*

$$\int \left(|D_X Z_0|^2 - |D_X \gamma|^2\right) dX \le C S_j / R_j^2.$$

The bound requires no nonzero gradient floor and its constant is independent of a_j. The corresponding actual trace vector in each selected constant small shear of definition 9.1 obeys the same estimate on fixed enlarged receivers.

Proof. Work first in the original orthogonal optimizing frame and suppress j. Put $n = d+1$, $A = (-\Delta)^{p/2}$, $f(u) = u - u^3$, $\beta = 3(1-u^2)$, and $L = A + 2 - \beta = A - f'(u)$. Uniform physical smoothness is supplied by lemma 2.1. All bounded-data resolvent and derivative identities below are those of lemma 2.3, with the positive kernels of lemma 2.2; their right sides are bounded. The weighted equation, stability, and Poisson normalization are those established in proposition 3.4 and (5).

We first obtain the local trace norm. For a physical derivative w of U, let $\eta(X,y) = \chi_r(X)\rho(y)$, with $0 \le \chi_r, \rho \le 1$, $\chi_r = 1$ on B_r, supported in B_{2r}, $|D\chi_r| \le C/r$, and $\rho = 1$ near zero, supported below height two. Its common Jacobi identity, tested with $\eta^2 w$, gives

$$\int |D(\eta w)|^2 d\mu = \int (1 - 3u^2)(\eta w)(X,0)^2 dX + \int w^2 |D\eta|^2 d\mu.$$

The one-dimensional weighted trace inequality, applied for each X, says

$$\int (\eta w)(X,0)^2 dX \le \epsilon \int |D(\eta w)|^2 d\mu + C_\epsilon \int (\eta w)^2 d\mu.$$

It follows by weighted Cauchy–Schwarz on a sufficiently short vertical interval and averaging its upper endpoint; $\int_0^\ell y^{p-1} dy$ is finite and $1 - p \in (-1, 0)$. Choose ϵ to absorb the bounded Robin coefficient. As $r \ge 1$, all cutoff derivatives are bounded, and the two identities give

$$\int_{B_r} |D_x u|^2 dX \le C \int_{B_{2r}\times(0,2)} |D_x U|^2 d\mu.$$

Summing over tangential components and using $\int_{B_r} u_t^2 \le C \int_{B_r} |Du| \le C r^d$, the outward same-frame bound gives

$$\int_{B_r} \gamma^2 dX \le C a^2 r^d \big(1 + \log(r/R)\big)^2, \qquad r \ge R.$$

The containing extension receiver is B_{2r}, so the outward bound applies also at $r = R$. No normal-speed floor or inverse power of a is used.

Every component of Z_0 solves $(A+2)Z_0 = \beta Z_0$. The convex-norm regularization used in lemma 3.6, now applied to this constantly rescaled vector Jacobi field, gives $A\gamma \le f'(u)\gamma$ in distributions. Thus $\kappa = f'(u)\gamma - A\gamma$ is a nonnegative distribution and hence a locally finite Radon measure. Put

$$F = G_2 * (\beta\gamma), \qquad Q = F - \gamma \ge 0, \qquad J = F + \gamma.$$

Indeed $Z_0 = G_2 * (\beta Z_0)$ and $\beta \ge 0$ imply $F \ge \gamma$. Since $(A+2)\gamma = \beta\gamma - \kappa$ and $(A+2)F = \beta\gamma$, subtraction of the potential gives the exact distributional identity

$$LJ = -\kappa - \beta Q.$$

The functions γ, F, J are bounded and globally Lipschitz at each index: this is immediate for γ, and follows for F from $DG_2 \in L^1$ and boundedness of $\beta\gamma$.

Here J has the needed full-height norm. Vector Kato and bounded-subsolution comparison give

$$(A+1)\gamma \le (2 - 3u^2)\gamma \le 2\mathbf{1}_{\mathcal{C}}\gamma, \qquad \gamma \le 2G_1 * (\mathbf{1}_{\mathcal{C}}\gamma), \qquad \mathcal{C} = \{|u| \le \sqrt{2/3}\}.$$

The kernel bounds imply $G_2 \le CG_1$, and the proved convolution inequality gives $G_2 * G_1 \le CG_1$. Since $\beta \le 3$, it follows that

$$0 \le J \le CG_1 * (\mathbf{1}_{\mathcal{C}}\gamma).$$

The deficit estimate gives $|\mathcal{C} \cap B_r(z)| \le Cr^d$ for every center and $r \ge 1$. This bound persists after the fixed massive convolution by Tonelli. Poisson annuli, together with boundedness for $y \le 1$, give $P_y * G_1 * \mathbf{1}_{\mathcal{C}} \le C/(1+y)$. Positive Cauchy–Schwarz therefore yields

$$(P_y * J)^2 \le \frac{C}{1+y}(P_y * G_1) * (\mathbf{1}_{\mathcal{C}}\gamma^2).$$

The height-integrated positive kernel

$$\mathscr{L} = \int_0^\infty \frac{d_p y^{1-p}}{1+y}(P_y * G_1)\, dy$$

belongs to L^1, because $\int_0^\infty y^{1-p}(1+y)^{-1}dy < \infty$. Before convolution with G_1, its radial representation is a fixed constant times

$$\int_0^\infty \frac{y}{1+y}(|X|^2+y^2)^{-(n+p)/2}dy.$$

It is at most $C|X|^{2-n-p}$ for $|X| \le 1$, using $y/(1+y) \le y$, and at most $C|X|^{1-n-p}$ for $|X| \ge 1$, using $y/(1+y) \le 1$. These powers are integrable at their respective endpoints. Splitting convolution at half the distance and using the faster far tail of G_1 gives $\mathscr{L}(X) \le C|X|^{-d-p}$ for $|X| \ge 2$.

Fix a receiver B_{bR}. Sources in a fixed larger ball cost C_bS, by the mass of $\mathscr{L}$ and the trace estimate. On the exterior annulus of radius $r_k = 2^k c_b R$, the same estimate bounds the source mass by $Ca^2 r_k^d (1+k)^2$; its contribution is at most

$$C_b R^n r_k^{-d-p} a^2 r_k^d (1+k)^2 \le C_b S R^{1-p} 2^{-kp}(1+k)^2.$$

Summation proves

$$\int_{B_{bR}\times(0,\infty)} (P_y * J)^2 d\mu \le C_b S.$$

We justify the measure test before removing its height cap. Set $W = P_y * J$, choose $0 \le \eta_R \le 1$ equal to one on B_{bR}, compactly supported in a fixed enlargement, with $|D_X \eta_R| \le C_b/R$, and let $0 \le \rho_T \le 1$ equal one below height T and zero above $2T$, with $|\rho_T'| \le C/T$. Bounded Lipschitz regularity of J gives finite local weighted energy of W: its tangential derivatives are bounded, and $\int |\partial_y P_y(H)||H| dH \le C$ bounds its height derivative near zero, using $p > 1$. For $y \ge 1$, $|DW| \le C\|J\|_\infty / y$. Thus $W\eta_R\rho_T$ has finite weighted energy. Its stability inequality follows by the smooth-trace approximation below and the compact weighted-energy approximation established in proposition 3.4.

The weak Poisson Green identity also applies to the compact trace test $J\eta_R^2$. To make the measure pairing precise, convolve J with smooth nonnegative mollifiers. These traces converge uniformly and locally in H^1, and their Poisson extensions converge in weighted gradient energy on compact capped receivers, by the uniform Lipschitz bounds just given. Their conormal distributions satisfy

$$A(J * \varphi_\delta) = (f'(u)J - \beta Q) * \varphi_\delta - \kappa * \varphi_\delta.$$

The compact tests $\eta_R^2(J * \varphi_\delta)$ converge uniformly to $\eta_R^2 J$, with a common compact support. Fubini rewrites their pairing with $\kappa * \varphi_\delta$ as the pairing of κ with their reflected mollification. Those mollified tests also converge uniformly with a common compact support, so local finiteness of κ passes this term to $\int \eta_R^2 J\, d\kappa$; the other terms converge locally in L^1. This proves the Green identity for W, including its measure conormal term. Expanding the product

energy and using $LJ = -\kappa - \beta Q$ now gives

$$\begin{aligned} 0 &\le \int |D(W\eta_R\rho_T)|^2 d\mu - \int f'(u) J^2 \eta_R^2 dX \\ &= \int W^2 |D(\eta_R\rho_T)|^2 d\mu - \int \eta_R^2 J \, d\kappa - \int \eta_R^2 J\beta Q \, dX. \end{aligned}$$

The spatial and height cutoff derivatives are orthogonal. The height term is at most $CT^{-2}\int_{B_{c_b R}\times(0,\infty)} W^2 d\mu$, which tends to zero at fixed record. The full-height norm thus permits $T \to \infty$, giving, by $J, \beta, Q \ge 0$,

$$\int_{B_{bR}} J \, d\kappa \le \int W^2 |D_X \eta_R|^2 d\mu \le C_b S/R^2.$$

At $\gamma(X) > 0$ put $n(X) = Z_0(X)/\gamma(X)$. There the defect is the ordinary nonnegative density

$$\kappa(X) = c_{d+1,p} \int \gamma(Y)[1 - n(X)\cdot n(Y)]|X - Y|^{-d-1-p} dY.$$

This formula follows by subtracting $A\gamma$ from $n(X)\cdot AZ_0 = f'(u)\gamma$. It is absolutely convergent: near X, the unit vector n is smooth and $1 - n(X)\cdot n(Y) = O(|X-Y|^2)$; the far kernel is integrable. Set $\gamma(Y)n(Y) = 0$ where $\gamma(Y) = 0$. Project the differentiated representation $DZ_0 = DG_2 * (\beta Z_0)$ perpendicular to $n(X)$. With $K(H) = c_{n,p}|H|^{-n-p}$, Cauchy–Schwarz gives

$$\begin{aligned} |P_{n^\perp} DZ_0|^2 &\le \left(\int K(X-Y)\gamma(Y) |P_{n^\perp} n(Y)|^2 dY \right) \\ &\quad \cdot \left(\int \frac{|DG_2(X-Y)|^2}{K(X-Y)} \beta(Y)^2 \gamma(Y) dY \right) \\ &\le C\kappa(X)(G_2 * (\beta\gamma))(X) \le C\kappa(X) J(X). \end{aligned}$$

Here $|P_{n^\perp} n(Y)|^2 \le 2(1 - n(X)\cdot n(Y))$, $\beta^2 \le 3\beta$, and $|DG_2|^2/K \le CG_2$ is already proved in lemma 2.2. The left side is exactly $|D_X Z_0|^2 - |D_X \gamma|^2$. On the zero set of Z_0, it vanishes almost everywhere by locality of Sobolev derivatives. Integrating on $\{\gamma > 0\}$ and using the full positive-measure bound proves the assertion in the orthogonal optimizing frame, including any singular Kato contribution on the zero set.

For completeness transfer it to the selected constant shear $S_k(x,t) = (x + kt, t)$. Its actual trace derivatives obey $V^{\mathrm{sh}} = V \circ S_k$ and $H^{\mathrm{sh}} = (H + k\cdot V)\circ S_k$. Consequently

$$Z_0^{\mathrm{sh}} = T_{a,k}(Z_0 \circ S_k), \qquad T_{a,k} = \begin{pmatrix} I & 0 \\ ak^T & 1 \end{pmatrix}.$$

Both $T_{a,k}$ and its inverse are uniformly bounded, as are the source-space shear and its inverse. For a vector field Z the angular defect is exactly

$$\mathcal{A}_Z = \sum_\alpha \frac{|Z \wedge \partial_\alpha Z|^2}{|Z|^2}$$

on $Z \ne 0$, with its almost-everywhere zero-set convention supplied by Sobolev locality. Exterior powers and the lower singular-value bound for $T_{a,k}$ give $\mathcal{A}_{T_{a,k}Z} \le C\mathcal{A}_Z$. The chain rule gives $\mathcal{A}_{Z\circ S_k} \le C\mathcal{A}_Z \circ S_k$. Since the shear has determinant one and maps a tested receiver into a fixed enlarged receiver, the proved integral bound transfers with a uniform constant. Thus the stated angular estimate holds in the selected shear as well; no isotropic resolvent identity was asserted for that sheared field. □

Lemma 9.3 (Generic detector-good slices retain every positive interior root endpoint)**.** *Let $\Omega_0 \Subset \Omega_1 \subset \mathbb{R}^d$ be bounded open base sets, let $t_- < t_+$ be finite, and set*

$$\mathcal{P}_0 = \Omega_0 \times (t_-, t_+), \qquad \mathcal{P}_1 = \Omega_1 \times (t_-, t_+).$$

The latter is the buffered product and the former the tested receiver. Let I, I' be bounded open phase intervals with $\overline{I} \subset I'$ and $\overline{I'} \subset (-1, 1)$. Fix $\kappa > 0$ and $S_b(x,t) = (x + \kappa b t, t)$ for $b \in B_1^d$. Suppose u is smooth on an open neighborhood of the compact set $\bigcup_{b\in\overline{B}_1} S_b(\overline{\mathcal{P}_1})$. Put $u_b = u \circ S_b$, $v_b = D_x u_b$, $h_b = \partial_t u_b$, and $q_b = |v_b|$. Assume $u_b(x, t_\pm) \notin I'$ for every $x \in \overline{\Omega}_1$ and $b \in \overline{B}_1$; these are the two phase-free normal caps.

For each b let $E_b \subset \mathbb{R}^d$ be a closed phase-independent excluded base set. Assume that for a fixed $\gamma > 0$,

$$|Du(S_b(x,t))| \geq \gamma \quad \text{whenever } (x,t) \in \mathcal{P}_1, \quad u_b(x,t) \in I', \quad x \notin E_b.$$

Define the complete detector-good tested-phase band by

$$G_b = \mathcal{P}_1 \cap \{u_b \in I'\} \cap ((\mathbb{R}^d \setminus E_b) \times \mathbb{R}), \qquad \pi(x,t) = x.$$

It is open and contains every point of $\mathcal{P}_0$ with $u_b \in I$ outside the excluded columns. No further deletion is included in G_b.

For almost every b, the complete detector-good normal critical set $M_b = G_b \cap \{h_b = 0\}$ is a smooth d-dimensional manifold (possibly empty), with $Dh_b \neq 0$ everywhere on it. For almost every (b, s), $s > 0$, and for any fixed horizontal unit vector e, one may follow all actual regular roots $u_b = c$ on almost every phase-line slice $(c, z) \in I \times e^\perp$, along the line $x = z + re$. The null set of phase-line slices may depend on e, b, s; no assertion for every phase simultaneously is needed. Tracks are maximal inside $\mathcal{P}_1$; retain a root only while it is in G_b and $|h_b| > sq_b$. Count only their positive one-sided endpoints lying in $\mathcal{P}_0$, with multiplicities before cancellation, including replacements which preserve the total count. The boundary of the tested base Ω_0 is not a stopping boundary: the track is followed in the larger base Ω_1. Endpoints on $\partial\mathcal{P}_1$ are not counted, and the cap condition excludes normal-cap endpoints at a tested phase. On $q_b > 0$ put $\tau_b = h_b/q_b$, and define the following positive coarea quantities on the buffered receiver:

$$P_x^{\mathrm{g}} = \int_{G_b} \delta(h_b)|\partial_t h_b|\,|v_b|, \qquad P_c^{\mathrm{g}} = \int_{G_b} \delta(h_b)|\partial_t h_b|,$$

$$B_s^{\mathrm{g}} = \int_{G_b \cap \{q_b > 0\}} \delta(|\tau_b| - s)|h_b D_x \tau_b - v_b \partial_t \tau_b|.$$

Here $\delta(h_b)$ is the regular level coarea measure $d\mathcal{H}^d/|Dh_b|$ on the detector-good zero set; it makes no assertion inside a detector column. For generic (b, s), the ratio delta has the analogous meaning $d\mathcal{H}^d/|D(|\tau_b|)|$ on $\{|\tau_b| = s, q_b > 0\}$. For a fixed horizontal unit vector e, write $x = z + re$, $z \in e^\perp$. Let $N_{\mathrm{det}}(z, c; e)$ be the number of positive one-sided endpoints of retained root-track intervals on that slice whose endpoint lies in $\mathcal{P}_0$ with its base coordinate in ∂E_b. Count every incident root track separately, even when two endpoints have the same location; receiver-boundary endpoints are excluded by the compact subreceiver convention. Define, with value $+\infty$ allowed,

$$D_b(e) = \int_I \int_{e^\perp} N_{\mathrm{det}}(z, c; e)\, dz dc, \qquad D_b = \sup_{|e|=1} D_b(e).$$

Here b and the threshold s are fixed; dependence on s is suppressed in N_{det}, $D_b(e)$, and D_b. This sliced incidence is the definition for an arbitrary closed E_b; no regularity of its boundary is assumed. Then the phase-integrated number of retained positive interior endpoints, integrated over the transverse line parameters for any fixed e, is at most $C(P_x^{\mathrm{g}} + B_s^{\mathrm{g}} + D_b(e))$, hence

at most $C(P_x^{\mathrm{g}} + B_s^{\mathrm{g}} + D_b)$. The same budget includes all detector-good fold incidences of the unthresholded regular roots; a fold contributes its two positive one-sided root incidences even if ratio retention stops earlier. Moreover,

$$|\pi(\mathcal{P}_0 \cap \{u_b \in I, h_b = 0\})| \le |E_b \cap \Omega_0| + P_c^{\mathrm{g}}.$$

The exceptional critical projection $\pi\{(x,t) \in \mathcal{P}_0 \cap G_b : u_b \in I, h_b = 0, \partial_t h_b = 0\}$ has zero base volume. The projection-zero assertion is restricted to the detector-good region. Positive exhaustion covers infinitely many charts or infinite total cost.

Proof. Use the exact buffered product $\mathcal{P}_1$, tested receiver $\mathcal{P}_0$, phase intervals and open set G_b in the statement. All geometric assertions in the proof are made in this open detector-good band; endpoints are charged only when they lie in $\mathcal{P}_0$. In original coordinates $h_b = h + \kappa b \cdot v$. At a detector-good point with $h_b = 0$, $|h| \le \kappa|v|$, so the full gradient floor gives $|v| \ge \gamma/\sqrt{1+\kappa^2} > 0$. We make both genericity assertions before restricting to the possibly irregular detector columns. On the joint open set $v(X) \ne 0$, the map

$$(X, b) \longmapsto h(X) + \kappa b \cdot v(X)$$

is a submersion, because some b_i derivative is $\kappa v_i \ne 0$. Parametric transversality therefore gives, for almost every b, a regular zero set in X. The map

$$(X, b, c) \longmapsto (u(X) - c, h(X) + \kappa b \cdot v(X))$$

is also a submersion: the c derivative is $(-1, 0)$, and the same b_i derivative supplies an independent second target direction. Apply transversality on a countable exhaustion and use Fubini in the parameters. The joint shear $(x, t, b) \mapsto (S_b(x,t), b)$ is a diffeomorphism, so these conclusions hold in the sheared variables. Restriction to the open set G_b then gives, for almost every b, the smooth d-manifold M_b with $Dh_b \ne 0$, and, for almost every phase c for such a b, the smooth $(d-1)$-manifold

$$\Sigma_{b,c} = G_b \cap \{u_b = c, h_b = 0\}.$$

The exceptional parameter sets were chosen on the open set $v \ne 0$, independently of E_b; hence no measurability or regularity of the family $b \mapsto E_b$ is required for these genericity conclusions.

At a point of $\Sigma_{b,c}$ with $\partial_t h_b = 0$, the tangent equations are $v_b \cdot dx = 0$ and $D_x h_b \cdot dx = 0$. These horizontal constraints are independent, because the full defining map has rank two while both normal derivatives are zero. Hence projection to the base has rank at most $d-2$; when $d = 1$, the independence of two horizontal constraints is impossible, so this degenerate subset is empty. The area formula on a countable atlas shows that its image has zero $\mathcal{H}^{d-1}$ measure. Projection of this image onto the transverse parameter space of a fixed base-line direction also has zero $(d-1)$-measure. At other points the projected fold hypersurface has normal parallel to v_b; its line-tangent part likewise has zero transverse-parameter image. Almost every encountered detector-good fold therefore satisfies $h_b = 0$, $\partial_t h_b \ne 0$ and $\partial_e u_b \ne 0$. Indeed $\partial_e u_b \ne 0$ first gives the local zero set as $r = R(t)$. At the fold, $R'(t_0) = 0$ and $R''(t_0) = -\partial_t h_b / \partial_e u_b \ne 0$. The one-dimensional Morse lemma gives two monotone branches on one side of r_0, so exactly two regular one-sided root tracks are incident. For the strict retained condition $|h_b| > sq_b$, a root normally stops at its ratio boundary before this fold. Retaining the fold count in the upper bound also controls the unthresholded root transport when it is used.

We also verify that a retained graph cannot end by a continuation failure which is absent from the incidence list. The transformed full gradient has the uniform lower bound $\gamma_\kappa =$

$\gamma/(1+\kappa) > 0$, because $\|DS_b^{-1}\| \le 1+\kappa$ for $|b| \le 1$. On the retained region $|h_b| > sq_b$ it follows that

$$|h_b| \ge \frac{\gamma_\kappa s}{\sqrt{1+s^2}} =: c_s > 0, \qquad |D_x T_c| = q_b/|h_b| < 1/s.$$

Along a horizontal line the graph height is therefore $1/s$-Lipschitz. At a finite interior endpoint of its parameter interval it has a unique finite limiting height. The normal cap values exclude an endpoint on a normal cap, and compact subreceivers exclude outer receiver boundaries. Smoothness gives $u_b = c$ at the limiting point and preserves $|h_b| \ge c_s$. If its base belongs to E_b, it belongs to ∂E_b, since bases of the approaching retained points lie outside that closed set. This endpoint is included with its positive one-sided multiplicity in N_{det}. If its base is outside E_b, its value $u_b = c \in I$ is interior to I', and its position is interior to $\mathcal{P}_1$ because its base lies in $\Omega_0 \Subset \Omega_1$. It consequently belongs to the open band G_b. If it still has strict $|h_b| > sq_b$, the implicit function theorem continues the retained root inside G_b, contradicting maximality. Otherwise continuity of the non-strict inequality from the retained side gives exactly $|h_b| = sq_b$, a ratio-equality endpoint. A point with $q_b = 0$ and this positive lower bound for $|h_b|$ still satisfies the retention inequality strictly, so it cannot be an additional stopping mechanism. Thus no joint critical or undefined-ratio endpoint has been omitted. This continuation argument uses no regularity of ∂E_b.

At a ratio boundary $|h_b| = sq_b$ in the good band, the gradient floor gives $q_b > 0$ and $h_b \ne 0$. Put $\psi_b = |\tau_b|$ on the open set $q_b > 0, h_b \ne 0$. For each b, Sard's theorem applied to ψ_b and to (u_b, ψ_b), followed by Fubini in b, s, c, gives a regular scalar ratio level for almost every (b, s) and a transverse intersection with $u_b = c$ for almost every c. Equivalently one may use the independent target derivatives in s and (c, s) in parametric transversality. All these assertions are first made on the smooth joint open set, then restricted to G_b.

The projection of such an intersection to the base is a locally embedded $(d-1)$-manifold: since $h_b \ne 0$, the level $u_b = c$ is a normal graph, and its ratio derivative is

$$D_x[\psi_b(x, T_c(x))] = D_x\psi_b - (v_b/h_b)\partial_t\psi_b.$$

This vector is nonzero precisely by the transverse-intersection condition. Its zero component in direction e characterizes the line-tangent part; projection of that part to $e^\perp$ has rank at most $d-2$ and hence null $(d-1)$-dimensional image by the area formula. At every remaining encounter the ratio along the root crosses s with nonzero derivative. There is exactly one retained one-sided incidence locally; the upper bound by two is sufficient also when adding other endpoint types at the same location. Detector endpoints are counted directly by $D_b(e)$, without any detector-boundary transversality assertion. Every endpoint counted here is interior to the buffered receiver; its outer base boundary and its phase-free caps contribute none.

The endpoint weights follow by positive coarea. On a slice $x = z + re$, the Jacobian of (u_b, h_b) at $h_b = 0$ is $|\partial_e u_b\, \partial_t h_b|$. Integrating first on the (r, t) slices and then in $(z, c) \in e^\perp \times I$ counts the distinct fold locations with total cost

$$\int_{G_b \cap \{u_b \in I\}} \delta(h_b)|\partial_t h_b|\,|v_b \cdot e| \le P_x^{\mathrm{g}}.$$

Multiply this by two for their two positive root incidences. At a ratio boundary the corresponding determinant is

$$\big|h_b\partial_e\psi_b - (v_b \cdot e)\partial_t\psi_b\big|.$$

Since $\psi_b = |\tau_b|$ and $\tau_b \ne 0$ there, replacing ψ_b by τ_b changes only its sign. Root coarea, or the same two-function area formula, therefore gives the phase-integrated ratio-event count

$$\int_{G_b \cap \{u_b \in I, q_b > 0\}} \delta(|\tau_b| - s)\big|h_b\partial_e\tau_b - (v_b \cdot e)\partial_t\tau_b\big| \le B_s^{\mathrm{g}}.$$

All these are positive counts on the full actual root set, before any count-preserving cancellation. Add the defining $D_b(e)$, assigning simultaneous detector/ratio endpoints to either term or overcounting them. Distinct normal heights remain distinct throughout.

The sliced detector incidence is measurable in the completed Lebesgue measure: in a countable implicit-function atlas about retained endpoints, each endpoint has $|h_b| \geq c_s$, so it belongs to a genuine local root graph. The retained set on that graph is open. Enumerate components of its open line sections by the first point in a fixed enumeration of the rationals belonging to each component, and count left and right interval endpoints separately. The component endpoints are measurable functions of the remaining parameters: their defining infima and suprema can be taken over rational subintervals compactly contained in the open section. Assign an endpoint to the first chart containing it, on compact subcharts, to avoid chart duplication. The resulting endpoint counting functions are measurable, and increasing positive exhaustion gives the stated incidence integral, including value $+\infty$. No assertion about the perimeter of E_b is needed. Positive exhaustion similarly treats infinitely many fold and ratio charts; when the total budget is finite, the resulting interior-event count is finite on almost every phase-line slice.

For the single all-phase critical projection use the smooth d-manifold M_b already established for generic b. Its projection to the d-dimensional base has Jacobian $|\partial_t h_b|/|Dh_b|$. The subset $\partial_t h_b = 0$ has zero image volume by the area formula. Every other projected point is counted at least once. Thus its detector-good projection has volume at most P_c^{g}. Every point of the full tested critical set $\mathcal{P}_0 \cap \{u_b \in I, h_b = 0\}$ whose base is outside E_b lies in G_b by its exact definition. The omitted part of its projection is contained in $E_b \cap \Omega_0$, which proves the asserted sum bound. No projection-zero conclusion has been made inside the detector columns and no phase-dependent exceptions have been unioned. □

Lemma 9.4 (Bounded ranks with vanishing positively charged jumps have Sobolev limits)**.** *Let $D \subset \mathbb{R}^k$ be bounded and Lipschitz and let $F_j \in BV(D;\mathbb{R}^q)$ be uniformly bounded in value. Suppose*

$$DF_j = G_j \mathcal{L}^k + \mu_j, \qquad \sup_j \|G_j\|_2 < \infty, \qquad |\mu_j|(D) \to 0.$$

Then every subsequence has a further subsequence along which $F_j \to F$ strongly in L^2, $G_j \rightharpoonup DF$ weakly in L^2, and $F \in H^1(D)$, with $\int |DF|^2 \leq \liminf \int |G_j|^2$ along that further subsequence. If a component of G_j tends to zero in L^2, the corresponding derivative of F is zero. In the horizontal variant on $D = \Omega \times I$, assume strong L^2 convergence and the analogous decomposition only for $D_x F_j$. Its conclusion is $F \in L^2(I; H^1(\Omega))$, with weak identification and the same lower bound for the horizontal derivatives; it does not assert a phase derivative.

Proof. The displayed decomposition gives bounded total variation, so BV compactness supplies strong L^1 convergence; the common amplitude bound upgrades it to strong L^2. Take a weakly convergent L^2 subsequence of G_j. For a fixed smooth compact test field, the derivative pairing passes to the limit because the measure term is bounded by its supremum times $|\mu_j|(D)$. This identifies the weak limit as the derivative of F. Weak lower semicontinuity proves the energy estimate. In the horizontal variant, compactness in phase is an assumption: the same fixed-test argument identifies $D_x F \in L^2(\Omega \times I)$, which gives the asserted Bochner Sobolev space. The vanishing-component assertion follows by the same identification. No jump measure is paired with the derivative of a moving coefficient. □

Proposition 9.5 (Positive root events produce ordered H1 ranks with one common center in every dimension)**.** *Let $R_j \to \infty$, $D_j \to \infty$, $a_j = D_j/R_j \to 0$, and $S_j = a_j^2 R_j^d$. The actual*

trace fields u_j are smooth on buffered physical receivers exhausting the normalized base: for every bounded Lipschitz $\Omega \Subset \Omega^+ \Subset \mathbb{R}^d$, all estimates below hold on $R_j\Omega^+$ with a normal interval containing every tested central root over $R_j\Omega$ and with its two cap values outside a fixed larger phase interval. Constants may depend on the fixed receiver, but not on j. A countable nested choice of such receivers covers every compact subset of $\mathbb{R}^d$. Fix compact nested phase intervals $I \Subset I' \Subset (-1,1)$, with $|I| > 0$.

Write $v_j = D_x u_j$, $h_j = \partial_t u_j$, and let $\mathfrak{D}_j$ be the closed phase-independent projected detector deletion. Assume uniform physical derivative bounds, central volume $O(R_j^d)$, transverse trace energy $\int |v_j|^2 \le CS_j$, and $|(v_j, h_j)| \ge \gamma > 0$ on the tested central band outside $\mathfrak{D}_j$; the cap convention above is phase-free for all $c \in I'$. Choose the actual generic directions and positive ratio thresholds s_j of lemma 9.3. A regular root is called retained precisely when its base is outside $\mathfrak{D}_j$ and $|h_j| > s_j|v_j|$. A thin root is one excluded by this ratio inequality, $|h_j| \le s_j|v_j|$; no additional root deletion is made by this term.

For almost every (x, c) let

$$N_j(x,c) = \#\{t : u_j(x,t) = c\}$$

be the complete count of all simple roots in the prescribed normal interval, before detector or ratio deletion. Normal caps forbid missing endpoints. Assume $N_j(R_j\xi, c) \to m$ strongly in local base–phase L^1, for an integer $m \ge 1$. A complete-count product means the set of $(\xi, c) \in \Omega \times I$ for which $R_j\xi \notin \mathfrak{D}_j$ and the complete list has exactly m simple roots, ordered as $t_{j,1} < \cdots < t_{j,m}$. Assume

$$R_j^{-d} \int_{R_j\Omega} \int_I \sum_{u_j(x,t)=c} |t|/D_j \, dcdx \le C_\Omega.$$

The normalized detector count and first moment are, respectively,

$$R_j^{-d} \int_{R_j\Omega \cap \mathfrak{D}_j} \int_I N_j(x,c)\, dcdx, \qquad R_j^{-d} \int_{R_j\Omega \cap \mathfrak{D}_j} \int_I \sum_{u_j(x,t)=c} |t|/D_j \, dcdx;$$

assume both tend to zero. The normalized thin-root count

$$R_j^{-d} \int_{R_j\Omega} \int_I \#\{t : u_j(x,t) = c,\ |h_j| \le s_j|v_j|\}\, dcdx$$

also tends to zero.

Assume the folds of all unthresholded roots, the ratio endpoints of retained roots, and their detector-boundary incidences have the complete positive endpoint bound from lemma 9.3 with total $P_j = o(R_j^{d-2} D_j)$, uniformly for each coordinate base-line direction. In this budget fold incidences are counted even if ratio retention stops before them, and detector incidences use the positive sliced multiplicity specified in that lemma. Finally assume one phase-independent base set E_j contains $\mathfrak{D}_j$ and the projection of the full central critical set $\{u_j \in I, h_j = 0\}$, with $|E_j| = o(R_j^{d-2} D_j)$.

Then, on a subsequence, every complete ordered normalized rank has a finite common-center limit $f_k \in H^1_{\mathrm{loc}}(\mathbb{R}^d)$, independent of phase, and $f_1 \le \cdots \le f_m$. Convergence of actual normalized heights on complete-count products means convergence in measure there, with respect to $d\xi dc$. For each fixed clipping level, the modified bounded rank has a horizontal derivative decomposition as in lemma 9.4, with a uniformly bounded L^2 density and a measure part tending to zero.

On every fixed buffered receiver the successful actual root transport preserves order and obeys, for a physical base displacement z along a coordinate direction,

$$\int \sum_{\text{surviving}} |t'(x+z,c)-t(x,c)|\,dx\,dc \le C|z|a_jR_j^d.$$

Among retained roots starting on complete-count fibers, the phase-integrated base measure of failed starts, counted with their positive root multiplicities, is at most $C_m|z|P_j$. Discarded thin roots, count-defect fibers, and detector columns have total normalized root-count mass tending to zero. The same two transport bounds hold for an arbitrary vector displacement inside the buffered receiver, with constants changed by a factor depending only on d, by using a path of at most d coordinate segments. Bounds are first on compact subreceivers so that all intermediate segments stay in the buffer.

Proof. Apply lemma 9.3 on the full detector-good buffered band. For each coordinate direction e, its joint phase-line conclusion says that for almost every $(z,c) \in e^{\perp} \times I$ every encountered detector-good normal critical event is a simple transverse fold, with two positive one-sided root incidences, and every encountered ratio boundary is regular. Detector endpoints are counted by the positive sliced incidence in that lemma. These are joint null-set conclusions, so they justify the finite charged partitions below. Use the complete count, ratio retention, thin-root convention, detector moments, and exhausting buffered receivers specified in the statement. All estimates are first made on one fixed compact receiver. The actual parameters are chosen in the simultaneous full-measure generic set for the finitely many coordinate directions and the countable receiver exhaustion; no subsequent parameter change is made.

For almost every regular complete-count fiber, list its m roots increasingly as $t_{j,k}(x,c)$. Let $T_L(r) = \max(-L, \min(L,r))$. First define $F^L_{j,k}(x/R_j, c) = T_L(t_{j,k}/D_j)$ on those fibers, and zero on count defects or detector columns. These are measurable by local implicit root charts. Root lists are finite almost everywhere: a sequence of distinct roots accumulating in the compact normal interval would limit to a normal critical root, while one-dimensional Sard and normal coarea exclude such critical roots for almost every (x,c); the caps exclude boundary roots. Starting with a retained root on a complete-count fiber, follow it along a horizontal line until the first actual fold, ratio boundary, or detector endpoint. For ordered-rank comparison also stop when another root's fold changes the complete count or its order label; at most m ranks are charged to one such positive incidence. The continuation and positive incidence conclusion of lemma 9.3 ensures that no other finite endpoint has been omitted. Successful transport is injective and order-preserving, since distinct regular roots cannot meet without a fold and the complete count is constant between the stopping events. The fundamental theorem on root graphs and normal coarea cancel the inverse normal speed, giving, for a physical displacement z in a coordinate direction,

$$\int \sum_{\text{surviving}} |t'(x+z,c)-t(x,c)|dxdc \le |z| \int_{\{u_j \in I\}} |v_j| \le C|z|a_jR_j^d.$$

The last step is Cauchy–Schwarz with the stated central volume and trace energy. Every failed retained complete-count start is charged to its first positive endpoint, so their phase-integrated base measure is at most $C_m|z|P_j$. Starts already excluded by count, thin-root, or detector deletion are accounted for separately. Complete-count defects have vanishing normalized count mass and base–phase measure: on $\{N_j \neq m\}$ the integer inequalities $N_j \le (m+1)|N_j - m|$ and $1 \le |N_j - m|$ apply. The corresponding vanishing root-count errors are assumed for thin roots and detector columns.

For an arbitrary displacement use successive coordinate segments. At most d intermediate bases occur, $\sum_i |z_i| \le \sqrt{d}|z|$, and each positive failed-start incidence is charged on its first failing segment. Integrating and summing the coordinate estimates proves the displayed transport bounds for this path. Count, thin-root and detector errors are exactly the vanishing normalized root-count errors in the hypotheses. This establishes the quantitative assertions in the statement, not merely compactness of the resulting clipped ranks.

Therefore for fixed L, with $z = R_j\zeta$,

$$\int |F^L_{j,k}(\xi+\zeta,c) - F^L_{j,k}(\xi,c)|d\xi dc \le C|\zeta| + CL\varepsilon_j + CL|\zeta|R_j^{1-d}P_j, \qquad \varepsilon_j \to 0.$$

Coordinate translations suffice for a general translation by telescoping its coordinate components. Since $R_j^{1-d}P_j = o(a_j)$, these are uniform horizontal translation bounds.

Remove the single phase-independent set E_j. Outside it no phase in I has a normal critical point. A root continues regularly through every phase in I, preserving its order, and cannot reach a normal cap. Thus the complete count is constant throughout I on each such column. If that count is not m, the clipped rank is zero at every phase on the column. Normal coarea gives

$$\int_{x\notin E_j}\int_I \sum_{\text{roots}} |\partial_c t_{j,k}|dcdx = |\{(x,t) : u_j(x,t)\in I,\ x\notin E_j\}| \le CR_j^d.$$

Consequently, for almost every $c, c' \in I$,

$$\int_\Omega |F^L_{j,k}(\xi,c) - F^L_{j,k}(\xi,c')|\,d\xi \le C/D_j + 2LR_j^{-d}|E_j\cap R_j\Omega|.$$

No separate phase-slice estimate for count defects is needed, because their rank is identically zero through I outside E_j. Here

$$R_j^{-d}|E_j| = o(D_j/R_j^2) \longrightarrow 0.$$

Averaging the phase-pair estimate shows that the $L^1(\Omega\times I)$ distance from the phase average tends to zero. Those averages have the horizontal translation bounds above, so the Fréchet–Kolmogorov compactness criterion on smaller receivers gives strong local L^1 compactness. It follows that the clipped ranks themselves converge strongly in local base–phase L^1 to phase-independent limits $f^L_k(\xi)$; their uniform bound by L upgrades this to strong L^2.

Take a diagonal over positive integers L and the receiver exhaustion. For integers $L' \ge L$, the original clips satisfy $T_L(F^{L'}_{j,k}) = F^L_{j,k}$, including their common zero convention. Their limits therefore satisfy $T_L(f^{L'}_k) = f^L_k$. The absolute first moment gives $\int_\Omega |f^L_k| \le C_\Omega/|I|$, uniformly in L. Compatibility makes $|f^L_k|$ nondecreasing in L; monotone convergence shows that their pointwise limit f_k is finite almost everywhere and $f^L_k = T_L(f_k)$. On complete-count products the raw heights also converge in measure to f_k: their first moment bounds the measure of $\{|t_{j,k}|/D_j > L\}$ by C_Ω/L; on the complement and on $\{|f_k| \le L\}$, the desired difference equals that of the L-clips, which already tends to zero in measure. Let $j\to\infty$ and then $L\to\infty$. Every clip and every rank used the same raw height t/D_j; no rank-dependent recentering has entered this common-center limit.

We now justify the entire horizontal derivative, including absence of singular-continuous variation. Modify $F^L_{j,k}$ further by setting it to zero when its own root is thin. Its L^1 difference from the preceding clip is at most L times the normalized thin-root count and thus tends to zero; boundedness by L gives the same strong L^2 limit. Fix j, a coordinate direction e, and almost every phase c and transverse line parameter z. By lemma 9.3 all encountered detector-good folds are simple, all relevant ratio boundaries are regular, and the integrated positive endpoint bound is finite. Fubini therefore makes the total number of interior fold,

ratio, and detector incidences finite on this compact line slice. Multiplicities are counted before cancellation.

For precision consider any maximal open line interval on which the base is outside the detector deletion, the complete count is m, and its kth root is retained. The value of that root may be zero; the interval is defined by its retention and complete-count assignment, not by a nonzero height. The retained-graph continuation conclusion of lemma 9.3 gives the $1/s_j$-Lipschitz height, finite one-sided endpoint traces, and a charged ratio or detector endpoint for its own failure. The rank can also be set to zero, or change labels, when a different root is born or dies. This changes the complete count or the ordered label only at an actual fold, which is included with its positive multiplicity in P_j. Between all these incidences the complete count is constant: every existing root continues by the implicit function theorem; any new sequence of roots would accumulate at a critical root or a cap, and caps are excluded. Order cannot change there. In particular boundaries introduced by the convention $N_j = m$ cause no additional breaks. Count-preserving replacements at distinct heights consist of distinct already charged fold incidences. Each incidence affects at most m retained ranks.

There are consequently finitely many such active intervals on this slice, apart from intervals reaching the exterior base boundary, which are irrelevant for compact tests. On each interval the clipped root is Lipschitz; it has finite one-sided traces at its endpoints and the function is zero on the complementary detector, ratio-deleted, or count-defect pieces. Thus the slice is a finite piecewise absolutely continuous function, with distributional derivative equal to its ordinary interval derivatives plus the charged endpoint jumps. A singular-continuous derivative cannot occur in such a finite partition. This is stronger than merely exhausting local root charts. Clipping is Lipschitz and creates no derivative jump at its own clipping heights; any oscillation through height zero stays within the same root-graph interval. Coincident geometric events are still counted with their positive root incidences.

The interval derivative in normalized base variables is

$$-z_{j,k}\mathbf{1}_{|t_{j,k}|<LD_j}, \qquad z_{j,k} = v_j/(a_j h_j).$$

On $|h_j| \geq \gamma/2$, normal coarea bounds its summed squared norm by

$$R_j^{-d} a_j^{-2} \int_{\{u_j \in I,\ |h_j| \geq \gamma/2\}} |v_j|^2/|h_j|\, dX \leq C S_j^{-1} \int |v_j|^2 \leq C,$$

independently of the clip. On the detector-good complementary roots the full gradient floor gives $|v_j| \geq c_\gamma > 0$. Their normalized first absolute derivative moment is at most

$$(a_j R_j^d)^{-1} \int_{\{u_j \in I,\ x \notin \mathfrak{D}_j,\ |h_j| < \gamma/2\}} |v_j|\, dX \leq C(a_j R_j^d)^{-1} \int |v_j|^2 \leq C a_j \longrightarrow 0.$$

Place this latter absolutely continuous part into the vanishing measure remainder. The good-speed part is the uniformly bounded L^2 density. Every endpoint jump has size at most $2L$, and at most m ranks change at a charged incidence. The total normalized jump variation is therefore at most

$$C_m L R_j^{1-d} P_j = o(1).$$

The sliced BV characterization now gives the claimed horizontal derivative decomposition; its remainder is bounded by the displayed $O(a_j)$ absolutely continuous variation and the charged jump variation. Detector boundaries are handled by their sliced incidence definition, including nonsmooth boundaries. Count-defect pieces contribute no interior derivative; their only label changes have just been charged. If needed to identify retained slopes on the removed count-defect fibers, their first moment vanishes: use K times their vanishing complete count

mass on $|z| \le K$, the L^2 good-speed bound to obtain C/K on its complement, and the preceding $o(1)$ bound on bad-speed roots, taking $j \to \infty$ before $K \to \infty$.

Apply the horizontal form of lemma 9.4 to the already strongly L^2 convergent modified clips. The same limit f_k^L, being phase-independent and $|I| > 0$, has a base H^1 derivative with bound independent of L. On each connected bounded Lipschitz base receiver the Poincaré inequality in the form

$$\|f_k^L\|_{L^2(\Omega)} \le C_\Omega\big(\|D_x f_k^L\|_{L^2(\Omega)} + \|f_k^L\|_{L^1(\Omega)}\big)$$

and the common first moment give a uniform H^1 bound. Increasing L, weak H^1 compactness and the compatible convergence in measure identify its limit as $f_k \in H^1_{\mathrm{loc}}$. The original clips are ordered on complete-count products and are all zero on their common complement, so their limits satisfy $f_1^L \le \cdots \le f_m^L$. Although the thin-root modification need not preserve finite-j order, it has the same limits. Letting $L \to \infty$ gives $f_1 \le \cdots \le f_m$. The countable exhausting receiver convention yields these conclusions on all of $\mathbb{R}^d$. Phase independence was proved before the vanishing thin-root modification, so it is preserved. This proves every assertion. □

Proposition 9.6 (Fine records give all ordered Sobolev ranks, complete columns, and clipped actual supports)**.** *Every fine record in definition 9.1 has a subsequence with a finite complete ordered family $f_1 \le \cdots \le f_m$ on $\mathbb{R}^d$. Its members belong to $H^1_{\mathrm{loc}} \cap W^{2,1}_{\mathrm{loc}}$, and*

$$\int_\Omega \left(|D^2 f_k|^2 - \frac{|D^2 f_k D f_k|^2}{1 + |D f_k|^2} \right) \le C_\Omega.$$

There are actual trace phases in one fixed compact central interval, physical root centers $T_{j,k}$ forming the selected ordered m-root family on the retained regular region and agreeing with the complete roots wherever their full count is m, a common scalar center τ_j, and cutoffs with the following properties. For every fixed height bound M, smooth compact base receiver, and fixed positive separation threshold for any selected outer gaps, bounded H^1 base factors can be chosen so their nonzero support lies on genuine root graphs, every retained height satisfies $|T_{j,k} - \tau_j| \le C_M D_j$, and the specified physical gaps exceed $c\delta D_j$. The regularity cutoff is continuous, belongs to H^1, and is zero in a relative neighborhood of the complement of its regular root region; every fixed positive threshold therefore has interior regular support on a compact tested base. Their limiting factors are the corresponding bounded height and separation factors of f_k. The factors converge strongly in H^1 for fixed parameters. On these actual regular supports the physical root graphs may be chosen with $|D_x T_{j,k}| \le 1$ for all sufficiently large j. No local boundedness of f_k is asserted.

The angle cutoffs in the selected regular-root family can be constructed with one common sequence $\kappa_j \downarrow 0$, with low-pass support $|D_x u_j|/|u_{j,t}| < \kappa_j$ and low-pass plateau $|D_x u_j|/|u_{j,t}| \le \kappa_j^2$. This sequence and the detector refinements may be taken sufficiently slowly to impose any additional countable family of estimates whose errors vanish as $j \to \infty$ for each fixed positive angle and fixed refinement. Their choice precedes the generic direction and the single phase selection. All stated count, transport, height, and strong convergence conclusions are retained under this common slow choice.

For every tested compact phase interval the regular-root construction uses a smooth physical detector cutoff $0 \le \chi_{0,j} \le 1$ and the common angular low-pass $0 \le \psi_j \le 1$, where $\psi_j(r) = 1$ for $r \le \kappa_j^2$ and $\psi_j(r) = 0$ for $r \ge \kappa_j$. Its amplitude on an actual root is

$$b_{0,j}(x,t) = \chi_{0,j}(x,t)\psi_j\big(|D_x u_j(x,t)|/|u_{j,t}(x,t)|\big),$$

extended as zero across the omitted critical and detector neighborhoods. Write $O_j(c)$ for its open selected regular-root region and $\theta_j(\cdot, c)$ for the continuous regularity cutoff above.

On $O_j(c)$ every positive amplitude $b_{0,j}$ is exactly one, and there are exactly m such actual roots. They are precisely the selected ordered family $T_{j,1},\dots,T_{j,m}$ and satisfy $\chi_{0,j}=1$ and $|D_xT_{j,k}|\le\kappa_j^2$. Other actual roots on such a fiber are allowed and have amplitude zero. In particular these conclusions hold on the nonzero support of θ_j and every actual cutoff used above.

The detector cutoff has the following quantitative localization on every fixed buffered physical receiver. Its finite detector family includes one fixed full-gradient-floor detector on the tested compact central phase band, with accuracy, window, and positive floor independent of the refinement stage. There is a finite family of physical balls $B_{C_0}(X_{j,l})$, with a fixed radius C_0 independent of j and refinement stage, whose centers are failed points of one of the finitely many fixed planar detectors tested at that stage, such that

$$\#\{l\}\le C_ja_j^2R_j^{d-2},\qquad \{\chi_{0,j}<1\}\cap\mathcal{Q}_j^{\mathrm{in}}\subset\mathcal{B}_j:=\bigcup_l B_{C_0}(X_{j,l}).$$

Here $\mathcal{Q}_j^{\mathrm{in}}$ is the tested inner physical trace receiver; the failed points and all these balls are chosen in a larger receiver separated from it by a fixed positive multiple of R_j. The constants C_j are the finite constants of the common refinement stage and are chosen slowly enough that $C_ja_j\to0$ and $C_j/R_j^2\to0$. For every fixed $A<\infty$, the physical A-neighborhood of $\mathcal{B}_j$ satisfies

$$|\{X:\operatorname{dist}(X,\mathcal{B}_j)<A\}|\le C_AC_ja_j^2R_j^{d-2}=o(S_j).$$

There are no additional zeros of $\chi_{0,j}$ from root labels, critical-point omission, complete-count defects, filled projected holes, or base and normal boundary localization on this inner receiver. Those boundary factors are imposed separately on buffered receivers; normal critical roots outside the physical detector deletion are omitted by the angular factor alone. The common angular low-pass can be chosen strictly positive for $r<\kappa_j$, so its positive support is exactly that cone. These localization statements hold after the selected bounded small shear, with a changed fixed C_0.

The common scalar center may be chosen $\tau_j=0$; in particular $\tau_j/R_j\to0$. At the same selected phase sequence and on every fixed compact receiver, the regularity cutoffs satisfy $0\le\theta_j\le1$ and $\theta_j\to1$ strongly in local H^1. They vanish in a relative neighborhood of the complement of $O_j(c_j)$ as stated above. For every $L>0$ and $k\le m$, their zero-extended clipped actual height lifts satisfy

$$\mathbf{1}_{O_j(c_j)}(\xi)\theta_j(\xi,c_j)T_L\big((T_{j,k}(R_j\xi,c_j)-\tau_j)/D_j\big)\longrightarrow T_L(f_k(\xi))\quad\text{strongly in } H^1_{\mathrm{loc}}(\mathbb{R}^d),$$

where T_L is clipping to $[-L,L]$ and the left side is defined to be zero off the genuine regular region. These assertions concern the same regularity cutoff and the same actual ordered family as the other conclusions.

With one containing macroscopic normal cap ρ_R, the complete columns

$$n_j(\xi)=\int\rho_{R_j}(t)H_j(R_j\xi,t,y)^2d_py^{1-p}dtdy,$$

$$e_j(\xi)=a_j^{-2}\int\rho_{R_j}(t)|V_j(R_j\xi,t,y)|^2d_py^{1-p}dtdy$$

satisfy $n_j\to m\sigma$ strongly locally in L^1, and both column families are uniformly integrable. The complete ratio tail satisfies $\lim_{K\to\infty}\limsup_j S_j^{-1}\int_{|V_j|>Ka_j|H_j|}|V_j|^2d\mu=0$. The retained low-angle normalized tilts $D_xu_j/(a_ju_{j,t})$, evaluated at the actual roots, converge strongly in local base–phase L^2 to $-Df_k$. Every fixed compact target observable of these tilts converges strongly in L^2 to the same observable of $-Df_k$, including the phase independence.

Here the one-layer constant and the core comparisons have the following precise meaning. Let q be the centered increasing odd cubic layer with limits ±1, let $Q(t,y)$ be its Poisson extension, put $H_* = Q_t$, and set $\sigma = \int_{\mathbb{R}\times(0,\infty)} H_*^2 d_p y^{1-p} dt\, dy$. At the selected phases c_j in the fixed compact central interval, let $\epsilon_{j,k}(\xi)$ be the actual sign of $u_{j,t}$ at the selected root, and define the comparison center exactly by

$$S_{j,k}(R_j\xi) = T_{j,k}(R_j\xi) - \epsilon_{j,k}(\xi) q^{-1}(c_j).$$

Write $O_j = O_j(c_j)$. On every compact tested base Ω, $|\Omega \setminus O_j| \to 0$. For every fixed $L > 0$, the trace comparisons in physical offset variables (z,s) satisfy

$$\sup_{\xi\in O_j\cap\Omega,\ k\le m} \|u_j(R_j\xi + z, S_{j,k}(R_j\xi) + s) - \epsilon_{j,k}(\xi) q(s)\|_{C^2(B_L^{d+1})} \longrightarrow 0.$$

Define the actual normal rectangles

$$\mathcal{C}_{j,k}^L(\xi) = \{(t,y) : |t - S_{j,k}(R_j\xi)| < L,\ 0 < y < L\}.$$

On those rectangles $H_j(R_j\xi, t, y) - \epsilon_{j,k} H_*(t - S_{j,k}, y)$ tends to zero uniformly, including its trace at $y = 0$. Moreover

$$\sup_{\xi\in O_j\cap\Omega,\ k\le m} \int_{\mathcal{C}_{j,k}^L(\xi)} |\nabla_{t,y} U_j(R_j\xi, t, y) - \epsilon_{j,k} \nabla_{t,y} Q(t - S_{j,k}(R_j\xi), y)|^2\, d_p y^{1-p} dt\, dy \longrightarrow 0.$$

For every fixed L and all sufficiently large j, the m rectangles on each such base are disjoint and lie in the plateau of the containing cap. With the rectangles set empty off O_j, the complete normal-column remainder

$$\mathcal{R}_{j,L}(\xi) = n_j(\xi) - \mathbf{1}_{O_j}(\xi) \sum_{k=1}^{m} \int_{\mathcal{C}_{j,k}^L(\xi)} H_j(R_j\xi, t, y)^2 d_p y^{1-p} dt\, dy$$

is nonnegative and satisfies

$$\lim_{L\to\infty} \limsup_{j\to\infty} \int_\Omega \mathcal{R}_{j,L}(\xi) d\xi = 0.$$

These conclusions include normalized collisions of ranks and hold for the actual pulled-back fields in the selected small shear.

For every compact interval $I \Subset (-1,1)$ of positive length, the complete simple-root count $N_j(R_j\xi, c)$ tends to m strongly in local $L^1(d\xi\, dc)$, and the same common-center family $f_1, \dots, f_m$ is obtained. For each fixed clip L, the complete normalized ranks $T_L((t_{j,k} - \tau_j)/D_j)$, set to zero on the vanishing incomplete products, converge strongly in local $L^2(d\xi\, dc)$ to $T_L(f_k(\xi))$. Their retained normalized slope and compact-target convergence holds on each such phase product, with constants allowed to depend on I. One sequence diagonal over an exhaustion of $(-1,1)$ supplies these conclusions simultaneously.

The actual generic directions can be chosen with complete positive endpoint budget $P_j = o(R_j^{d-2} D_j)$ and one all-phase critical/deletion projection of volume $o(R_j^{d-2} D_j)$ on each fixed phase interval and receiver. The successful physical root transports satisfy the quantitative bounds of proposition 9.5: their phase-averaged unscaled height movement over a displacement z is at most $C|z| a_j R_j^d$, and their failed-start count measure is at most $C_m |z| P_j$, with the stated additional vanishing thin/count/detector errors. Fixed physical displacements and displacements $z = D_j\zeta$ for bounded ζ remain in any chosen larger buffer for large j. Include these nonnegative transport errors in the countable Fubini diagonal selecting a phase sequence whenever a single selected phase is required. The common macroscopic caps and their constant signs are those of proposition 5.2.

Proof. By definition 9.1 these are the actual augmented records of proposition 5.3 under the hypotheses of proposition 5.2. In particular $S_j = E(R_j) + R_j^{d-1} = a_j^2 R_j^d = R_j^{d-2} D_j^2$, the optimizing frame and the outward same-frame estimate are those of the budget proposition, and the complete counts, absolute first moment, central confinement, and common caps are those of the anchoring proposition. The underlying fields are the original bounded smooth stable cubic solution and its actual Poisson extension with the all-center sharp gradient bound in proposition 3.4. We use no independently postulated collection of asymptotic numerical parameters. The quantitative record-selection inequalities are not used separately in this rank proof; their needed consequences are the stated outward bound, $D_j \to \infty$, and $a_j \to 0$. Every bounded Jacobi resolvent representation and derivative convolution below is justified by lemma 2.3. Its bounded-subsolution comparison also applies to the corresponding vector-norm Kato inequality, since the Jacobi fields and their coefficients are bounded at each index.

The uniform physical smoothness used in this proof is supplied by lemma 2.1. The positive massive resolvent kernels, their derivative bounds, and their convolution estimates are those of lemma 2.2, in the stated Fourier normalization. These apply to the actual cubic trace and, in one dimension, to its normalized layer.

Use the actual fine record of definition 9.1. We give the construction, including the complete positive root events. The global trace tensor needed below is supplied by lemma 9.2.

We apply the construction to arbitrary compact nested phase intervals $I \Subset I' \Subset (-1,1)$. The full-phase form of proposition 5.2 supplies the complete count and the common first moment on this product. The detector and gradient floor are chosen on I', with constants depending on its distance from the wells. We verify below the positive event, detector, and count hypotheses of proposition 9.5 on I, before invoking that proposition. It will then give strong convergence of every clipped rank in base–phase L^2 and convergence of the untruncated ranks in measure. All angular-metric, count-restoration and slope arguments below use fixed compact intervals only. On overlaps the ordered complete lists coincide, so the phase-independent limits agree. A countable nested exhaustion of $(-1,1)$ and a diagonal give the same ranks and all the stated full-phase conclusions. The endpoint bounds proved below are the data P_j used by the quantitative transport conclusion of proposition 9.5, which yields exactly the transport estimates exported here.

Suppress j and write $R = R_j$, $a = a_j$, $S = a^2 R^d$, $H = U_t$, and $Z = (V, aH)$. All estimates below are on fixed enlarged physical receivers, with complete extension height; the containing finite normal windows and phase-free caps are those of proposition 5.2. The outward same-frame bound and complete local energy give

$$\int |Z|^2 d\mu = \int (|V|^2 + a^2 H^2) d\mu \le CS.$$

The norm–stability identity used in propositions 3.4 and 5.2 applies to the constant Jacobi vector Z: each component has the same Robin coefficient, and no derivative falls on a. It gives $\int \zeta^2 \mathcal{A}_Z \, d\mu \le \int |Z|^2 |D\zeta|^2 d\mu$, where $\mathcal{A}_Z = |DZ|^2 - |D|Z||^2$. As in those proofs, remove the norm regularization at fixed compact height, then remove the height cutoff using the preceding full-height norm. A physical cutoff of scale R yields

$$\int \mathcal{A}_Z \, d\mu \le CS/R^2 = Ca^2 R^{d-2}. \tag{6}$$

This holds also for the actual pulled-back fields in each selected small shear: the source shear and the target map $T_{a,k}$ of lemma 9.2 have uniformly bounded maps and inverses, and the wedge identity there transfers the angular tensor with a uniform factor. The unit-scale Robin trace estimate already proved in proposition 5.2 and lemma 9.2 gives $\int |D_x u|^2 \le CS$ on the corresponding trace receiver.

Here is the complete positive-kernel argument for the relative tails. Work in the original orthogonal record coordinates, put $A = (-\Delta)^{p/2}$, $n = d + 1$, $v = D_x u$, $\beta = 3(1 - u^2)$, and $C_u = \{|u| \le \sqrt{2/3}\}$. By lemma 2.3 and lemma 2.2,

$$v = G_2 * (\beta v), \qquad V = K_y * (\beta v), \qquad K_y = P_y * G_2.$$

Positive Cauchy–Schwarz gives $|V|^2 \le (K_y * \beta)(K_y * (\beta |v|^2))$. The deficit bound in proposition 3.4 yields $\int_{B_r(z)} \beta \le C r^d$ for $r \ge 1$, uniformly in the center. It persists after convolution with G_2. Poisson annuli then give $K_y * \beta \le C/(1 + y)$: for $y \ge 1$, the annulus of size $2^k y$ is bounded by $C y^{-n} 2^{-k(n+p)} (2^k y)^d$, and these terms sum to C/y; for $y \le 1$ use $\beta \le 3$ and the kernel mass.

For either deficit $w = 1 \mp u$, its exact equation is $(A + 3)w = a_w w$, with $1 \le a_w \le 13/4$. Bounded-data resolvent comparison therefore gives $G_3 * w \le w$. The massive kernel bounds compare $G_2 \le C G_3$, and $\beta \le 6w$, so $G_2 * \beta \le Cw$. Convolving with P_y for the two signs gives $K_y * \beta \le C(1 - |U|)$. Vector Kato and bounded-subsolution comparison also give

$$(A + 1)|v| \le (2 - 3u^2)|v| \le 2\mathbf{1}_{C_u}|v|, \qquad |v| \le 2G_1 * (\mathbf{1}_{C_u}|v|).$$

The vector Kato comparison is the one justified in lemma 9.2. Positivity and the mass $\int G_1 = 1$ imply $|v|^2 \le 4G_1 * (\mathbf{1}_{C_u}|v|^2)$. With $m_v = \mathbf{1}_{C_u}|v|^2$ and $Q_y = P_y * G_2 * G_1$, we have proved

$$|V(X, y)|^2 \le C \min\{(1 + y)^{-1}, 1 - |U(X, y)|\}(Q_y * m_v)(X). \tag{RT}$$

All sources are retained in this positive inequality.

Set $L = \int_0^\infty y^{1-p}(1 + y)^{-1} Q_y \, dy$. Before convolution with the fixed massive kernels, its kernel is a constant times

$$J(X) = \int_0^\infty \frac{y}{1 + y} (|X|^2 + y^2)^{-(n+p)/2} dy.$$

For $r = |X| \le 1$ use $y/(1 + y) \le y$ to get $J \le C r^{2-n-p}$; for $r \ge 1$ use $y/(1 + y) \le 1$ to get $J \le C r^{1-n-p}$. These are integrable at zero and infinity precisely because $p < 2$ and $p > 1$. Half-distance splitting and the faster r^{-n-p} massive tails show that their convolution preserves the far bound $L(X) \le C|X|^{-d-p}$. Also $\|L\|_1 < \infty$ by the kernel masses. Define

$$L_{>T} = \int_T^\infty \frac{y^{1-p}}{1 + y} Q_y \, dy, \qquad L_\delta = \int_0^\infty y^{1-p} \min\{\delta, (1 + y)^{-1}\} Q_y \, dy.$$

Then $0 \le L_{>T}, L_\delta \le L$ and

$$\|L_{>T}\|_1 \le C T^{1-p}, \qquad \|L_\delta\|_1 \le C \delta^{p-1}.$$

The second estimate follows by splitting the height integral at δ^{-1}. For $0 < y \le 1$, the explicit Poisson bound and the massive near and far bounds give $P_y * G_2 \le C G_2$. To check uniformity, for $|X| \le 2y$ the convolution is at most $C y^{p-n} \le C|X|^{p-n}$; otherwise split at $|Z| = |X|/2$ and $|X - Z| = |X|/2$. The receiver-near and source-near terms have the same $|X|^{p-n}$ bound below radius two and the common $|X|^{-n-p}$ bound above it; the complementary region is bounded by the same powers after radial integration. Consequently

$$L_{<\tau} := \int_0^\tau \frac{y^{1-p}}{1 + y} Q_y \, dy \le C \tau^{2-p} (G_2 * G_1),$$

with mass at most $C\tau^{2-p}$ and far bound $C\tau^{2-p}|X|^{-n-p}$.

The preceding local Robin estimate and the outward bound in definition 9.1 imply

$$\int_{B_r} m_v \le C a^2 r^d (1 + \log(r/R))^2 \qquad (r \ge R).$$

For any nonnegative kernel of mass at most ε and far bound $C|X|^{-d-p}$, split its convolution into a fixed larger source ball and exterior annuli of radii $r_k = 2^k cR$. The inner contribution on B_{bR} is at most $C_b\varepsilon S$. Each exterior contribution is at most

$$C_b R^n r_k^{-d-p}\, a^2 r_k^d (1+k)^2 \le C_b S R^{1-p} 2^{-kp}(1+k)^2.$$

Their sum is $C_b S R^{1-p}$. A kernel with far bound $C\varepsilon|X|^{-n-p}$ instead gives $C_b\varepsilon S R^{-p}$. Apply these estimates to the three kernels above and to (RT), including d_p. For $T \ge 1$, $0 < \tau \le 1$, and $0 < \delta \le 1/2$ this proves

$$S^{-1}\int_{B_{bR}\times(T,\infty)} |V|^2 d\mu \le C_b(T^{1-p} + R^{1-p}),$$

$$S^{-1}\int_{B_{bR}\times(0,\tau)} |V|^2 d\mu \le C_b \tau^{2-p}(1 + R^{-p}),$$

$$S^{-1}\int_{B_{bR}\times(0,\infty)} |V|^2 \mathbf{1}_{\{|U|>1-\delta\}} d\mu \le C_b(\delta^{p-1} + R^{1-p}).$$

These bounds are relative to S throughout; no absolute R^d estimate is divided by a smaller transverse budget.

We verify the positive-height surface hypotheses and their complete cutoff budget. Fix $I \Subset (-1,1)$ and $J = [h,T] \Subset (0,\infty)$ with positive buffers. Use the normalized layer and the positive-height detector comparison already established in propositions 3.4 and 3.5. Its normal extension Q has $Q_t = P_y * q' > 0$. Uniform convergence to the wells for bounded y shows that the set of normal parameters with $Q \in I$ and $y \in J$ is compact. Hence $|D_X Q| = Q_t$ has a positive minimum there. The trace comparison on a sufficiently large fixed window, followed by the explicit Poisson formula, gives C^1 extension comparison on J: the near part uses the fixed C^2 trace accuracy, and the far derivative kernels have arbitrarily small L^1 mass as the window is increased. Constant-well comparisons cannot have extension value in I once these errors are smaller than its distance from the wells. Choose the fixed window and accuracy in this order. We obtain $g := |D_X U| \ge c_{I,J} > 0$ on the central positive-height band outside cylinders over the failed detector balls. These choices are independent of a, R.

The detector count is $N_{\mathrm{det}} \le C_{I,J} R^{-2} \int |V|^2 d\mu \le C_{I,J} a^2 R^{d-2}$ by proposition 3.4. Take a maximal disjoint failed-ball cover and fixed enlargements. Choose a smooth cutoff χ_{det} zero on a neighborhood of the failed set, one outside the next enlargement, with $|D_X \chi_{\mathrm{det}}| \le C_{I,J}$ and transition volume at most $C_{I,J} N_{\mathrm{det}}$. Disjoint unit centers give bounded overlap of all fixed enlargements. Its nonzero support lies in the region where the preceding extension-gradient floor holds. Multiply by a macroscopic physical cutoff χ_R, equal to one on B_{bR}, supported in B_{CR}, and with derivative C/R, to form $\chi = \chi_R \chi_{\mathrm{det}}$.

At any level point in this support, the gradient floor and uniform physical Hessian bound give a fixed radius $r_0 > 0$ on which one fixed physical derivative has a fixed sign and magnitude at least $c_{I,J}/(4\sqrt{n})$. The level in that ball is one graph in that direction, of area at most $C_{I,J} r_0^d$. At these positive heights the weighted harmonic equation is uniformly elliptic on fixed smaller balls. The squared norm $\gamma^2 := |Z|^2$ is a nonnegative subsolution, because every component is weighted harmonic. Its interior mean-value/Moser inequality gives

$$\sup_{B_{r_0}(X_0)\times\{y\}} \gamma^2 \le C_{I,J} \int_{B_{2r_0}(X_0)\times(y-2r_0,y+2r_0)} \gamma^2 d\mu,$$

after reducing $r_0 < h/8$; the enlarged heights lie in a fixed compact positive buffer. Constants do not depend on a, since the same scalar inequality is applied to the norm of a vector of solutions. Partition the physical receiver into cells of diameter much smaller than r_0, and choose one level point in each occupied cell. The enlarged balls have bounded overlap.

Summing their graph-area and subsolution estimates, and using $g \ge c_{I,J}$, gives at every tested (c, y)

$$\int_{\{\chi>0\}} \delta(U-c)\gamma^2 dX = \int_{\{U=c,\,\chi>0\}} \frac{\gamma^2}{g} d\mathcal{H}^d \le C_{I,J} S.$$

Thus the uniform central level-energy hypothesis of lemma 8.1 is satisfied.

In that lemma put $z = (x, t/a)$, so $|D_z\chi| \le |D_X\chi|$ for $0 < a \le 1$. The parameter measure is $dc\, d_p y^{1-p} dy$ on $I \times J$, of finite mass. Coarea in c gives the explicit cutoff budget

$$\begin{aligned}\int_{I\times J} B(c,y)\, dc\, d_p y^{1-p} dy &= \frac{1}{a^2 R^{d-2}} \int_{\{U\in I,\, y\in J\}} [\chi^2 \mathcal{A}_Z + \gamma^2 |D_z\chi|^2] d\mu \\ &\le \frac{C_{I,J}}{a^2 R^{d-2}} [S/R^2 + R^{-2} \int \gamma^2 d\mu + N_{\mathrm{det}}] \le C_{I,J}.\end{aligned}$$

Here the detector-transition term uses the uniform physical bound on γ and its fixed positive height interval; its cost is the detector count, not an unscaled R^d error. Every surface cutoff is compactly supported in the regular part and vanishes near all deleted surface boundaries. The long stretched normal coordinate causes no new boundary term or constant in the surface inequality. The lemma now proves

$$S^{-1} \int_{B_{bR}\times J\cap\{U\in I\}} \gamma^2 \mathbf{1}_{\{\gamma/a>K\}} d\mu \le C_{I,J} K^{-2/d} + C_{I,J} R^{-2}.$$

The last term restores all failed cylinders using their volume and bounded γ. In particular the exponent is $1/3$ when $d = 6$. We next prove the separate trace amplitude estimate, which will also supply the high-gradient tail for the direction selection. It is a direct application of lemma 8.1; we verify its trace inputs. Work first in the original orthogonal record frame, put $n = d + 1$, $Z_0 = (v, ah) = (D_x u, a u_t)$ and $\gamma = |Z_0|$, and fix compact nested phase intervals $I \Subset I' \Subset (-1, 1)$ and fixed containing physical receivers. The vector resolvent identity justified above gives

$$Z_0 = G_2 * (\beta Z_0), \qquad \beta = 3(1-u^2), \quad 0 \le \beta \le 3, \qquad \gamma^2 \le C G_2 * \gamma^2.$$

The last inequality is positive Cauchy–Schwarz and $\int G_2 = 1/2$. The same Robin trace estimate gives $\int_{B_{CR}} |v|^2 \le CS$. Also $a^2 \int_{B_{CR}} h^2 \le C a^2 R^d$ by the uniform gradient bound and sharp L^1 growth. Applying the same trace argument at every $r \ge R$ and using the record's outward bound therefore gives

$$\int_{B_r} \gamma^2 \le C a^2 r^d (1 + \log(r/R))^2 \qquad (r \ge R).$$

Choose one fixed trace detector on I', sufficient to give $|Du| \ge c_I > 0$ outside its failed physical balls. At each fixed phase $c \in I$ let Γ_c be the portion of $\{u = c\}$ in the containing receiver outside those balls. Its good points have uniform physical graph charts: some coordinate derivative is at least $c_I/\sqrt{n}$, and the global Hessian bound preserves its sign and a fixed positive lower bound on a fixed ball. The implicit graphs there have uniformly bounded slopes and area densities. A fixed-grid covering consequently gives, uniformly in the phase and center,

$$\mathcal{H}^d(\Gamma_c \cap B_r(Y)) \le C r^d \quad (0 < r \le 1), \qquad \mathcal{H}^d(\Gamma_c \cap B_r(Y)) \le C r^n \quad (r \ge 1).$$

For a ball not centered on the level, select a good level point in it when it is nonempty and use its doubled ball; this gives the same small-radius estimate. The kernel bounds in lemma 2.2 now imply

$$\sup_Y \int_{\Gamma_c} G_2(X - Y) d\mathcal{H}^d(X) \le C.$$

At small radii its dyadic contributions are Cr^{p-1}, summable because $p > 1$; above one they are Cr^{-p}. Integrate the pointwise resolvent bound over Γ_c. Sources in a fixed enlarged ball cost CS by the trace norm just proved. For sources at distance $r \geq CR$, the crude area bound $\mathcal{H}^d(\Gamma_c) \leq CR^n$ and the kernel tail give a contribution at most

$$Ca^2R^n \sum_{r \text{ dyadic},\ r \geq CR} r^{-n-p} r^d (1 + \log(r/R))^2 \leq Ca^2R^{d-p} \leq CS.$$

On Γ_c, $|Du| \geq c_I$. Normal level coarea thus proves the uniform actual trace-level estimate required by the surface lemma:

$$\int_{\Gamma_c} \frac{\gamma^2}{|Du|} d\mathcal{H}^d = \int \delta(u - c) \mathbf{1}_{\Gamma_c} \gamma^2 dX \leq CS.$$

Take a smooth physical cutoff χ, equal to one on the tested inner receiver outside fixed enlargements of the failed balls, zero near the original failed balls, and compactly supported in the containing receiver. Its detector transitions lie outside the failed set. There are at most Ca^2R^{d-2} fixed balls. Bounded derivatives of u and bounded overlap therefore give

$$\int_{\{u \in I\}} \gamma^2 |D_X \chi|^2 dX \leq CS/R^2 + Ca^2R^{d-2} \leq Ca^2R^{d-2}.$$

The macroscopic transition uses the trace norm and $|D\chi| \leq C/R$; the ball transitions use bounded γ and their physical volume. The full trace angular theorem lemma 9.2 gives

$$\int \mathcal{A}_{Z_0} dX \leq CS/R^2 = Ca^2R^{d-2}.$$

Apply the geometric calculation of lemma 8.1 directly to the stretched trace $F(x, \tau) = u(x, a\tau)$, not to a positive-height approximation. Its stretched gradient is Z_0 and its stretched Hessian is symmetric. The stretched angular tensor satisfies

$$\mathcal{A}_z = \sum_i |P_{Z_0^\perp} \partial_{x_i} Z_0|^2 + a^2 |P_{Z_0^\perp} \partial_t Z_0|^2 \leq \mathcal{A}_{Z_0}.$$

Here the projections are onto the orthogonal complement of the vector Z_0 on its nonzero set. On a regular stretched level the tangent-tangent and tangent-normal Hessian decomposition gives exactly $|D_\Sigma \gamma|^2 + \gamma^2 |A_\Sigma|^2 = \mathcal{A}_z$. Thus the required curvature is controlled by the trace tensor already proved. After scaling the stretched surface by R^{-1} its level measure is

$$d\mathcal{H}^d_{\widehat{\Sigma}_c} = \frac{\gamma}{aR^d} \delta(u - c) dX, \qquad \ell = \gamma/a.$$

The uniform level estimate above is precisely the bound on $\int_{\widehat{\Sigma}_c \cap \{\chi = 1\}} \ell$. Since $D_z \chi = (D_x \chi, a\partial_t \chi)$, its squared norm is at most $|D_X \chi|^2$. The angular/cutoff estimates above give the other hypothesis, namely

$$\int_I B(c) dc \leq C, \qquad B(c) = \frac{1}{a^2 R^{d-2}} \int \delta(u - c) [\chi^2 \mathcal{A}_{Z_0} + \gamma^2 |D_z \chi|^2] dX.$$

All stretched levels are regular on this compact cutoff support, since the full physical gradient has its floor there and $a > 0$. The cutoff vanishes before every removed boundary and before the outer receiver boundary. The surface Sobolev inequality therefore has no boundary term. Its abstract tail conclusion gives $CK^{-2/d}$; restoring the fixed-ball transition regions costs at most CR^{-2} after division by S. We have proved the actual trace estimate

$$S^{-1} \int_{B_R \cap \{u \in I,\ \gamma/a > K\}} \gamma^2 dX \leq CK^{-2/d} + CR^{-2}.$$

This is a nonzero $K^{-1/3}$ tail when $d = 6$, and no lower bound for $|u_t|$ was used to prove it.

For retained low-angle roots outside the detector deletion, $|v/h| \le 1$ and the full gradient floor imply $|h| \ge c_I'$. Normal phase coarea now yields the needed normalized slope-square tail, where $\mathcal{Q}_j$ is a fixed containing physical receiver for the tested base and its macroscopic normal caps:

$$\int_{\Omega\times I} \sum_{\text{low angle}} |z_{j,k}|^2 \mathbf{1}_{\{|z_{j,k}|>K\}} d\xi dc \le \frac{C}{S_j} \int_{\mathcal{Q}_j\cap\{u_j\in I,\ \gamma_j/a_j>c_I'K\}} \gamma_j^2 dX \le CK^{-2/d} + CR_j^{-2}.$$

This verifies uniform integrability for the strong retained-slope passage later in the proof. The fixed physical high-gradient tail also follows, since $|v| \ge c_*$ forces $\gamma/a \ge c_*/a \to \infty$.

Finally this trace tail is uniform under the selected small shears. Their actual trace vector is $Z_0^{\mathrm{sh}} = T_{a,k}(Z_0 \circ S_k)$ as in lemma 9.2, with uniformly bounded matrix and inverse and determinant-one source change. Its norm is comparable to $\gamma \circ S_k$, and a tested receiver maps into a fixed enlarged receiver. Thus the displayed trace tail transfers by those norm bounds and a change of variables, without an isotropic resolvent identity in the sheared frame. The common slow angle diagonal can impose it uniformly before choosing the generic shear. Together with the bulk argument this includes every actual root and requires no normal-speed floor until the explicitly retained low-angle slope conversion.

On a detector-good compact central rectangle write $q = |D_xU|$, $h = U_t$, and $\ell = |(D_xU, ah)|/a$. The full physical gradient has a fixed floor $|(D_xU, h)| \ge c$. On $q > Ka|h|$, if $|h| \ge c/2$ then $\ell \ge cK/2$; otherwise $q \ge c/2$ and $\ell \ge c/(2a)$, which exceeds $cK/2$ for each fixed K and all large j. Thus the high-ratio set is contained in the high-amplitude set at threshold $cK/2$. Detector boxes contribute $O(S/R^2)$ by their number and uniform derivative bounds. The relative height/well tails already proved exhaust the complementary regions. This gives the asserted complete ratio tail. The same argument at trace height, using the boundary estimate, gives

$$\epsilon_j(c_*) := S_j^{-1} \int_{\{u_j\in I',\ |D_xu_j|\ge c_*\}} |D_xu_j|^2 dX \longrightarrow 0$$

for every fixed $c_* > 0$. This is the physical high-gradient tail used in the oblique parameter argument.

Write $v = D_xu$, $h = u_t$, $q = |v|$. Choose one common $\kappa_j \downarrow 0$ so slowly that $\kappa_j \ge 1/\log R_j$, $a_j/\kappa_j^8 \to 0$, the square root of each fixed physical high-gradient tail divided by κ_j tends to zero, and every finite refinement constant below satisfies $C_j a_j \to 0$. Each fixed refinement has a finite constant, so all these requirements are compatible with a countable slow diagonal. All fixed-angle tail errors and fixed detector-refinement errors vanish at each fixed parameter before its exhaustion is taken. Enumerate any additional such nonnegative errors and impose their first l requirements at the lth fixed angle/refinement stage. Keep that stage until all those errors, the finitely many angular and event errors already used here, and the explicitly required powers of a_j are below $1/l$. This gives the exported common additional-diagonal freedom, before the generic directions and phases are selected. In the oblique directions $h_b = h + \kappa_j b \cdot v$, average over $b \in B_1^d$ and a threshold $s \in [\nu_j, 2\nu_j]$, where $\nu_j = R_j^{-3}$. The following are estimates for these parameter averages before selecting a generic pair.

At a detector-good critical or small-ratio point of $h_b = h + \kappa_j b \cdot v$, the full gradient floor implies $|v| \ge c_I$. Parameter coarea has density at most $C/(\kappa_j|v|)$, and the actual directional critical derivative is bounded by $Ca_j^{-1}\sqrt{\mathcal{A}_{(v,a_jh)}}$ because its corresponding target linear functional annihilates (v, a_jh) there. Hence Cauchy–Schwarz gives

$$\int_b (P_{x,j}^{\mathrm{good}} + P_{c,j}^{\mathrm{good}}) db \le \frac{C}{\kappa_j a_j} \left(\int \mathcal{A}_{(v,a_jh)} \right)^{1/2} (\epsilon_j(c_I) S_j)^{1/2} \le CF_j \sqrt{\epsilon_j(c_I)}/\kappa_j.$$

For a ratio event, write $\tau_b = h_b/q$ and $|\tau_b| = s$. The target functional $(\kappa_j b - \tau_b v/q, 1/a_j)$ annihilates $(v, a_j h)$, so

$$|Dh_b - \tau_b Dq| \le C a_j^{-1} \sqrt{\mathcal{A}_{(v,a_j h)}}.$$

Consequently

$$|h_b D_x \tau_b - v\partial_t \tau_b| \le C a_j^{-1} \sqrt{\mathcal{A}_{(v,a_j h)}}$$

on $|\tau_b| \le 2\nu_j$. Parameter coarea for τ_b has density at most C/κ_j, and its support satisfies $|h| \le (\kappa_j + 2\nu_j)q$, where $q \ge c_I$. Cauchy–Schwarz gives the same $CF_j\sqrt{\epsilon_j(c_I)}/\kappa_j$ bound for the ratio endpoint term. All source-coordinate shears are uniformly bounded, so the derivative estimate transfers to their pulled-back physical coordinates. Choose κ_j slowly so $\sqrt{\epsilon_j(c_I)}/\kappa_j \to 0$. Both positive event averages are therefore $o(F_j)$, using the global trace angular budget CS_j/R_j^2 in lemma 9.2.

The remaining powers in that parameter selection follow directly from the same density. At a receiver the b-average of the discarded normal mass is at most

$$\frac{C}{\kappa_j q} \int_{-\nu_j q}^{\nu_j q} |r| dr \le C\nu_j^2 q/\kappa_j.$$

Here the support has $q \ge c_I$, so its physical integral is at most $C\nu_j^2 S_j/\kappa_j$. On $|h| \le 2\kappa_j q$, averaging $q^2/|h_b|$ above the retained threshold gives at most $Cq\kappa_j^{-1}\log(C\kappa_j/\nu_j)$; $q \ge c_I$ again converts its integral to $CS_j \log R_j/\kappa_j$. On the complementary cone $|h_b| \ge |h|/2$ and the full gradient floor bounds the integral by CS_j/κ_j. Since $\kappa_j \ge 1/\log R_j$, the claimed $CS_j \log^2 R_j$ bound follows. Every retained same-phase root has $|h_b| \ge c_I\nu_j$; bounded second derivatives and Rolle's theorem give at most CR_j/ν_j such roots per fiber. A fiber with any discarded root therefore has total count at most $(1+CR_j/\nu_j)$ times its discarded count. With $\nu_j = R_j^{-3}$ its integrated count divided by F_j is at most $C(\nu_j^2 + R_j\nu_j)D_j/\kappa_j \le C/(\kappa_j R_j) \to 0$.

By averaging the sum of these nonnegative normalized quantities over b and $s \in [\nu_j, 2\nu_j]$, select one generic parameter pair realizing their bounds simultaneously, with a fixed change of constants. Finite transversality and regular-value requirements remove only a null set of parameters. Thus the discarded normal mass is at most $C\nu_j^2 S_j/\kappa_j$, the retained transverse root energy is at most $CS_j \log^2 R_j$, and the complete positive critical and ratio-endpoint budget is $o(F_j)$. The same pair is used in the actual selected shear and in all subsequent complete-root arguments at this stage of the diagonal.

The projected detector balls can be filled only by their bounded complementary components. In dimension d their filled set $\mathfrak{D}_j$ has a cover with $\sum r_i^{d-1} \le CP_*$, maximum radius $CP_*^{1/(d-1)}$, and volume $CP_*^{d/(d-1)}$, where $P_* \le Ca_j^2 R_j^{d-2} L_j^{d-1}$. Absorb the finite constants of each fixed detector refinement into L_j^{d-1}, then choose the refinement diagonal slowly enough that $L_j \uparrow \infty$ and $a_j L_j^{d-1} \to 0$, together with the preceding constraints. This is possible because $a_j \to 0$ and each fixed refinement has finite constants. A selected shear maps a physical detector ball to a base projection of at most twice its radius for large j, so these estimates are uniform in the selected direction. The all-center cylinder bound is $CR_j r_i^{d-1}$. Thus complete pressure and missing-fiber baseline costs divided by F_j are bounded respectively by

$$Ca_j L_j^{d-1}, \qquad Ca_j^{(d+1)/(d-1)} R_j^{-1/(d-1)} L_j^d,$$

which tend to zero. In particular the second bound is also the detector-volume ratio, since

$$\frac{|\mathfrak{D}_j|}{F_j} \le Ca_j^{(d+1)/(d-1)} R_j^{-1/(d-1)} L_j^d = C(a_j/R_j)^{1/(d-1)} (a_j L_j^{d-1})^{d/(d-1)} \longrightarrow 0.$$

Here is a construction of the filled detector deletion and all the estimates just used. Work in one fixed buffered receiver and its containing normal window $|t| < CR_j$. The projected failed balls have original radii $\ell_i \geq 1$, with $\sum_i \ell_i^{d-1} \leq P_*$; the finite enlargement constants of a detector refinement have been absorbed in P_*. At each fixed index this family is finite. The empty family gives an empty deletion, so suppose it is nonempty. The preceding bound for P_* and the slow-diagonal condition give $\max_i \ell_i/R_j \leq P_*^{1/(d-1)}/R_j \to 0$. For $1 \leq r = o(R_j)$, a base cylinder $B_{Cr} \times (-CR_j, CR_j)$ is covered by at most CR_j/r physical balls of radius Cr. The center-uniform sharp variation bound in proposition 3.4, after the bounded determinant-one shear, gives

$$\int_{B_{Cr}\times(-CR_j,CR_j)} |Du_j|\, dX \leq CR_j r^{d-1}.$$

Normal coarea bounds the complete phase-integrated root count of this cylinder by the same quantity. First choose the boundaries of the original projected balls, before filling or making a boxing cover. Radial coarea selects $\rho_i \in [\ell_i, 2\ell_i]$ such that

$$\int_{\partial B_{\rho_i}} \int_I N_j(x,c)\, dc\, d\mathcal{H}^{d-1}(x) \leq CR_j \ell_i^{d-2} \leq CR_j \ell_i^{d-1}.$$

Indeed the integral of the left side over $\rho_i \in [\ell_i, 2\ell_i]$ is a cylindrical normal-coarea integral at most $CR_j\ell_i^{d-1}$, and the interval has length ℓ_i. We may impose at the same time the almost-everywhere regular-radius conditions for the finite family. In particular distinct sphere patches have intersections of codimension at least two, apart from redundant identical balls, which are removed. Write $A_j = \bigcup_i \overline{B}_{\rho_i}$ and obtain the closed filled set $\mathfrak{D}_j$ by adjoining all bounded components of $\mathbb{R}^d \setminus A_j$. Its boundary is a subset of the selected original spheres. Filling removes boundary components and introduces none; the finite smooth sphere-patch decomposition, whose intersections have zero $\mathcal{H}^{d-1}$ measure, gives

$$\mathrm{Per}(\mathfrak{D}_j) \leq \sum_i \mathcal{H}^{d-1}(\partial B_{\rho_i}) \leq CP_*.$$

The filled set is bounded, and the Euclidean finite-perimeter isoperimetric inequality therefore gives $|\mathfrak{D}_j| \leq CP_*^{d/(d-1)}$.

Detector incidence uses these selected original spheres. For every horizontal unit direction e, the area formula applied on each smooth cylinder to

$$(x,t) \in \partial B_{\rho_i} \times (-CR_j, CR_j) \longmapsto (\pi_{e^\perp} x, u_j(x,t))$$

has absolute Jacobian $|\nu_i(x) \cdot e|\, |h_j(x,t)|$. Every retained one-sided endpoint on the filled deletion boundary is a preimage on one of these spheres. A factor two bounds its incident-side multiplicity. The sphere intersections have null image in the joint line/phase variables by their dimension, and the critical images of the smooth cylinder map have zero measure by Sard's theorem. The area formula and normal coarea thus bound the complete positive sliced detector incidence of lemma 9.3, uniformly in e, by

$$\mathsf{D}_j(e) \leq C \sum_i \int_{\partial B_{\rho_i}} \int_I N_j(x,c)\, dc\, d\mathcal{H}^{d-1}(x) \leq CR_j P_* = o(F_j).$$

All actual roots are counted before ratio deletion, so this dominates every required retained detector endpoint and the detector boundary term in the complete-count BV identities. Its boundary radii will not be replaced by the radii of the boxing cover below.

We construct that separate cover explicitly. The portion of $\mathfrak{D}_j$ at distance at most one from A_j is covered by B_{ρ_i+1}, whose radii are at least one and whose $(d-1)$ powers have sum at most CP_*. If x lies in the remaining portion, then $B_1(x) \subset \mathfrak{D}_j$: any path to the exterior must cross $\partial\mathfrak{D}_j \subset A_j$. The density function $r \mapsto |\mathfrak{D}_j \cap B_r(x)|/(\omega_d r^d)$ is continuous, equals

one at $r = 1$, and tends to zero at infinity. Let $r_x \geq 1$ be its largest radius with density at least $1/2$. Then its density is exactly $1/2$ at r_x, and

$$r_x \leq C|\mathfrak{D}_j|^{1/d} \leq CP_*^{1/(d-1)}, \qquad \operatorname{Per}(\mathfrak{D}_j; B_{r_x}(x)) \geq cr_x^{d-1}.$$

The second inequality is the relative isoperimetric inequality on that ball, since both the occupied and unoccupied volume are half its volume. The $5r$ covering lemma selects pairwise disjoint balls $B_{r_x}(x)$ whose fivefold dilates cover this remaining portion. Their radii are bounded below by one and their centers and radii are bounded, so the disjoint selection is finite at each fixed index. Local perimeter additivity on these disjoint open balls gives

$$\sum_x r_x^{d-1} \leq C \sum_x \operatorname{Per}(\mathfrak{D}_j; B_{r_x}(x)) \leq C \operatorname{Per}(\mathfrak{D}_j) \leq CP_*.$$

Combining these fivefold balls with the B_{ρ_i+1} gives a finite boxing cover by base balls of radii $r_i \geq 1$ with

$$\sum_i r_i^{d-1} \leq CP_*, \qquad \max_i r_i \leq CP_*^{1/(d-1)}, \qquad \sum_i r_i^d \leq CP_*^{d/(d-1)}.$$

The radii of the original inflated balls obey the same maximum bound because the original family is nonempty and has $\ell_i \geq 1$. In particular all boxing radii are $o(R_j)$, since

$$\frac{P_*^{1/(d-1)}}{R_j} \leq C(a_j/R_j)^{1/(d-1)}(a_j L_j^{d-1})^{1/(d-1)} \longrightarrow 0.$$

Their cylinders therefore lie in a fixed enlarged buffer and obey the earlier cylindrical estimate. Summing it over the boxing cover proves that the whole detector-deleted complete root count is at most CR_jP_*. No boundary coarea choice is made for these covering balls.

Every tested root lies in $|t| \leq CR_j$ by the common-cap confinement in proposition 5.2. Thus the normalized detector root count and normalized absolute first moment are bounded, respectively, by

$$CR_j^{1-d}P_* \leq Ca_j^2 L_j^{d-1}/R_j \longrightarrow 0, \qquad \frac{CR_j^{2-d}P_*}{D_j} \leq C\frac{a_j}{R_j}L_j^{d-1} \longrightarrow 0.$$

The second estimate uses $D_j = a_jR_j$. These are the two separate detector hypotheses of proposition 9.5. The construction establishes them, the volume and cover bounds, and the complete positive sliced boundary budget from the actual original failed-ball family; it introduces no further record assumption.

Apply lemma 9.3 to the selected generic direction and threshold. Its transversality and area-formula conclusions are applied outside $\mathfrak{D}_j$ and include every actual count-preserving replacement there. Detector boundary events have already been included in the positive endpoint budget $P_j = o(F_j)$. Let $\mathcal{C}_j^{\mathrm{good}}$ be the projection of the central normal critical set outside $\mathfrak{D}_j$. On that critical set the full gradient floor and $h_b = 0$ give $|v_b| \geq c_I > 0$, uniformly for the selected small shears. Hence

$$|\mathcal{C}_j^{\mathrm{good}}| \leq P_{c,j}^{\mathrm{good}} \leq C_I P_{x,j}^{\mathrm{good}} = o(F_j).$$

The single phase-independent set

$$E_j = \mathfrak{D}_j \cup \mathcal{C}_j^{\mathrm{good}}$$

contains the projection of every central normal critical point, including those inside the detector deletion, and satisfies $|E_j| = o(F_j)$. This verifies the exceptional-set hypothesis of proposition 9.5 without an assertion of transversality inside detector columns.

The coarea transfer first identifies the weak complete-count limit. On a containing physical receiver $\mathcal{Q}_j$, the central-volume and transverse trace bounds give

$$\int_{\mathcal{Q}_j\cap\{u_j\in I'\}} |v_j|dX \le |\mathcal{Q}_j\cap\{u_j\in I'\}|^{1/2}\left(\int_{\mathcal{Q}_j}|v_j|^2dX\right)^{1/2} \le Ca_jR_j^d.$$

Thus, for a fixed compact base test, the small shear changes its scaled base argument by $O(|k_j|)$ on a containing macroscopic normal window, while replacing $|h|$ by $|h+k_j\cdot v|$ costs $O(|k_j|a_j)$ after division by R_j^d. The complete trace-level limit in proposition 3.5, applied with the normal projection factor $|\nu\cdot e_t|$ and a containing normal cutoff, therefore gives weak complete count m on every fixed base–phase product. Smooth approximation of this even normal factor is legitimate; the limiting normals have $|\nu\cdot e_t|=1$.

We verify the strong convergence needed below. On the complement of the phase-independent detector deletion set the full count is

$$N_j(x,c)=\int \delta(u_j(x,t)-c)|h_j(x,t)|dt.$$

The phase-free caps and the complete-count BV identities established in proposition 5.2 give

$$D_xN_j=-2\int\delta(u_j-c)\delta(h_j)(\partial_th_j)v_jdt,\qquad \partial_cN_j=2\int\delta(u_j-c)\delta(h_j)\partial_th_jdt.$$

They count every fold before cancellation, including replacements at distinct heights. Multiplying by the characteristic of the detector-good base adds only the complete detector-boundary root incidence to the base variation. Its bound is the previously established complete pressure bound, which dominates the retained-only detector incidence in lemma 9.3. Thus on a fixed scaled product

$$|D_\xi N_j|\le CR_j^{1-d}(P_{x,j}^{\mathrm{good}}+P_{\mathrm{det},j}^{\mathrm{full}})=o(a_j),\qquad |D_cN_j|\le CR_j^{-d}P_{c,j}^{\mathrm{good}}=o(D_j/R_j^2).$$

Here the displayed inequalities are bounds for the total variations on that product. Both tend to zero. BV Poincaré makes the detector-deleted integer count close in L^1 to one integer; its weak mean is m, so this integer is m for all sufficiently large j. Restoring the detector columns costs at most $CR_jP_{*,j}$ in complete phase-count mass, whose ratio to R_j^d tends to zero by the chosen detector diagonal. Thin-root and thin-fiber modifications have the already proved vanishing complete count mass. Consequently the actual complete counts converge strongly in local base–phase L^1 to m, in the chosen small shear.

The common first moment of proposition 5.2 is preserved by the same determinant-one shear. The normal coordinate remains t; on the containing window $|t|\le CR_j$, the preceding central-band estimate bounds the additional coarea term by

$$C|k_j|R_j\int_{\mathcal{Q}_j\cap\{u_j\in I'\}}|v_j|dX\le C|k_j|a_jR_j^{d+1}=C|k_j|D_jR_j^d.$$

The detector costs and thin-root counts above supply the remaining hypotheses of proposition 9.5. That proposition, using lemma 9.4, now gives all finite ordered common-center ranks in H^1_{loc}, independent of phase. In particular its proof supplies the full derivative decomposition rather than assuming a perimeter bound for selected sheet domains.

The normalized tilt is $z=v/(ah)$. Its actual angular metric on regular roots is

$$\mathcal{A}_Z=a^2h^2\sum_\alpha(\partial_{X_\alpha}z)^T(I+z\otimes z)^{-1}\partial_{X_\alpha}z,\quad D_{\xi_i}z=R(\partial_{x_i}z-az_i\partial_tz),\quad \partial_cz=\partial_tz/h.$$

We supply the derivative and label argument for these observables. Let $\gamma_I>0$ be the fixed full physical gradient floor on the buffered detector-good band. Fix a smooth compactly supported target map $\Phi:\mathbb{R}^d\to\mathbb{R}^q$, supported in $|z|\le K$. Define its kth rank observable to be

$\Phi(z_{j,k})$ on complete-count products outside the detector deletion while its own root is retained, and zero otherwise. On its nonzero target support, for large j we have $a_j|z| \le a_j K < 1$. The full gradient floor therefore gives $|h_j| \ge c_I > 0$. The metric identity and normal phase coarea give

$$\int_{\Omega\times I} |D_\xi \Phi(z_{j,k})|^2 \le C_\Phi \frac{R_j^{2-d}}{a_j^2} \int \mathcal{A}_{(v,a_j h)} dX \le C_\Phi,$$

$$\int_{\Omega\times I} |\partial_c \Phi(z_{j,k})|^2 \le C_\Phi \frac{R_j^{-d}}{a_j^2} \int \mathcal{A}_{(v,a_j h)} dX \le C_\Phi R_j^{-2}.$$

For the first inequality use $D_\xi z = R_j(D_x z - a_j z \partial_t z)$ and $|a_j z| \le 1$; for the second use $\partial_c z = \partial_t z / h_j$. The normal coarea factor is $|h_j|$, so all remaining inverse speeds are bounded on this target support. The comparison of the Euclidean target derivative with the angular metric is uniform on $|z| \le K$.

These formulas specify the absolutely continuous derivatives. To account for the entire derivative, follow the actual ordered roots on almost every coordinate base line. A change of complete count or rank label occurs only at an actual fold, including count-preserving replacements at distinct heights; at most m labels are affected by any one positive incidence. Detector endpoints are charged by the complete incidence just proved. At an own-root ratio endpoint, $|z| = 1/(a_j s_j) \to \infty$, so this compactly supported observable vanishes in a neighborhood of that endpoint for all large j. Normal caps are phase-free. The sliced conclusion of lemma 9.3 therefore gives a measure remainder of horizontal total variation at most

$$C_{m,\Phi} R_j^{1-d} P_j = o(1).$$

For a phase slice at fixed base, a label changes only at a critical value. Positive normal coarea counts these incidences by $P_{c,j}^{\mathrm{good}}$; degenerate critical values have null base image by the same lemma. The detector set is independent of phase, and own-root ratio boundaries again carry zero observable. The phase jump remainder consequently has variation at most $C_{m,\Phi} R_j^{-d} P_{c,j}^{\mathrm{good}} = o(1)$. Finite positive cost gives finitely many charged fold incidences on almost every compact slice. Any additional retained intervals have zero endpoint traces for this observable and their squared derivative sum is bounded by the displayed estimates; zero-trace one-dimensional Sobolev gluing handles their countable union. Thus no singular-continuous derivative is omitted.

Apply lemma 9.4 on the joint product. These bounded observables have a uniformly bounded L^2 derivative density, vanishing measure remainder, and phase derivative tending to zero. They converge strongly in local L^2, on a subsequence, to phase-independent limits. Do this for a countable family of compact target maps separating points. Bad-speed roots with $|h_j| < \gamma_I/2$ have $|v_j| \ge c_{\gamma_I} > 0$, so their complete normalized root-count mass is at most

$$R_j^{-d} \int_{\{x\notin\mathfrak{D}_j,\ |h_j|<\gamma_I/2,\ u_j\in I\}} |h_j| dX \le C R_j^{-d} \int |v_j|^2 dX \le C a_j^2 \to 0.$$

Here bounded h_j and the lower bound for $|v_j|$ give the first inequality. Thus no positive-measure family can escape to an infinite target through bad-speed roots. The trace amplitude tail already proved bounds the squared normalized slopes on every good-speed root $|h_j| \ge c_I$, not only the smaller low-angle family: normal coarea gives the same bound by $S_j^{-1}\gamma_j^2$. Hence these slopes are uniformly integrable and tight. Detector and incomplete products have vanishing measure. Roots with physical angle greater than one have $|v_j| \ge c_I$ and their good-speed slope-square mass tends to zero by the physical high-gradient tail. The separating-map limits therefore determine a finite phase-independent z_k; convergence in measure and uniform

integrability give strong local base–phase L^2 convergence of the retained low-angle slopes to z_k.

We identify this limit with the height gradient directly. Let $\gamma_I > 0$ denote the fixed full physical gradient floor on the buffered detector-good phase band. For a fixed clip L, set the clipped normalized kth height to zero on detector and count defects and when its own root is thin. It has the same L^2 limit $T_L(f_k)$ as in proposition 9.5. On each retained root interval its ordinary horizontal derivative is exactly $-z_{j,k}\mathbf{1}_{\{|(T_{j,k}-\tau_j)/D_j|<L\}}$. Every endpoint jump has size at most $2L$ and is charged to the positive budget, so its normalized variation is $C_{m,L}R_j^{1-d}P_j = o(1)$. On $|h_j| \geq \gamma_I/2$ the derivative density has bounded L^2 norm by normal coarea and the transverse trace bound. On its complement the full gradient floor gives $|v_j| \geq c_{\gamma_I}$, and its total absolute derivative is at most

$$\frac{1}{a_j R_j^d}\int_{\{x\notin\mathfrak{D}_j,\ |h_j|<\gamma_I/2,\ u_j\in I\}} |v_j|dX \leq \frac{C}{a_j R_j^d}\int |v_j|^2 dX \leq Ca_j \to 0.$$

Place this part in the vanishing measure remainder. On the good-speed part, the strong slope convergence and the high-gradient tail identify the derivative limit. For almost every L the set $\{|f_k| = L\}$ has zero measure, so convergence in measure of the heights and slope uniform integrability give

$$DT_L(f_k) = -z_k\mathbf{1}_{\{|f_k|<L\}}$$

distributionally. Let $L \to \infty$ through such levels; f_k is finite almost everywhere and belongs to H^1_{loc} by the rank proposition. Sobolev locality and dominated convergence give $Df_k = -z_k$. This proves both the identification and the asserted strong slope and compact-observable conclusions, including their phase independence.

For the weak Hessian put $\mathsf{M}(z) = (I+z\otimes z)^{-1}$, $r_k = \frac{1}{2}\log(1+|z_k|^2)$, and $w_k = z_k/\sqrt{1+|z_k|^2}$. The required bounded contractions can be chosen compactly supported in target space, so the preceding full BV decomposition applies to them. For $r(z) = \frac{1}{2}\log(1+|z|^2)$ choose a smooth triangular truncation $\rho_L(r)$ which equals r below L, is zero above $2L$, has $|\rho_L'| \leq C$, and obeys $0 \leq \rho_L(r) \leq Cr$. For $w(z) = z/\sqrt{1+|z|^2}$ use $\eta(r(z)/L)w(z)$, with $\eta = 1$ below one and zero above two. Their target derivatives have uniformly bounded dual-metric norm, since $|D_z r|_{\mathsf{M}^{-1}} \leq 1$, $|D_z w|_{\mathsf{M}^{-1}} \leq C$, and $|w| \leq 1$. The angular estimate gives Sobolev bounds independent of their truncation level: for each fixed level, its input support is bounded, so $a_j|z| < 1$ for all sufficiently large j and the same normal-speed floor applies. Their vanishing jump remainders are handled by the preceding full BV decomposition. Moreover $0 \leq r(z) \leq |z|$ and $|w(z)| \leq 1$. Strong convergence of $z_{j,k}$ in measure and the uniform L^2 bound therefore identify the weak Sobolev limits of the truncations with r_k and w_k, respectively. Thus $r_k, w_k \in H^1_{\mathrm{loc}}$, $e^{r_k} = \sqrt{1+|z_k|^2} \in L^2_{\mathrm{loc}}$ and $z_k = e^{r_k}w_k$. To justify the product without an unbounded Sobolev chain rule, take smooth scalar cutoffs χ_L equal to one below L, zero above $L+1$, and with uniformly bounded derivatives. The functions $e^{r_k}\chi_L(r_k)w_k$ belong to $W^{1,1}_{\mathrm{loc}}$. Their usual derivative terms converge in L^1 by Cauchy–Schwarz and dominated convergence, and the cutoff term is bounded by

$$C\left(\int_{\{r_k>L\}}(1+|z_k|^2)\right)^{1/2}\|Dr_k\|_{L^2} \longrightarrow 0.$$

Consequently $z_k \in W^{1,1}_{\mathrm{loc}}$ and $f_k \in W^{2,1}_{\mathrm{loc}}$.

We give an explicit target contraction for the full metric estimate, since ordinary Euclidean clipping alone would not justify a bound independent of the target radius. In radial coordinates $z = \sinh(s)n$, $s \geq 0$, $n \in S^{d-1}$, the metric $\mathsf{M}(z)$ has line element

$$ds^2 + \sinh^2(s)|dn|^2.$$

Choose smooth $p_L : [0,\infty) \to [0,\infty)$ equal to s on $[0,L]$, zero on $[3L,\infty)$, with $p_L \le s$ and $|p_L'| \le 2$, and put

$$\mathcal{T}_L(\sinh(s)n) = \sinh(p_L(s))n.$$

It is smooth near the origin, compactly supported as a function of its input, and its differential obeys

$$(D\mathcal{T}_L(z)\zeta)^T \mathsf{M}(\mathcal{T}_L(z)) D\mathcal{T}_L(z)\zeta \le 4\zeta^T \mathsf{M}(z)\zeta.$$

Indeed the radial factor is $(p_L')^2 \le 4$ and the tangential factor is $\sinh^2(p_L(s))/\sinh^2(s) \le 1$. On the fixed input support of $D\mathcal{T}_L$, normal coarea and $a_j|z| < 1$ give

$$\int_{\Omega\times I} \sum_i (D_{\xi_i}\mathcal{T}_L(z_{j,k}))^T \mathsf{M}(\mathcal{T}_L(z_{j,k})) D_{\xi_i}\mathcal{T}_L(z_{j,k}) \le C_\Omega,$$

where C_Ω is independent of L; the finitely many discarded initial indices may depend on L. The full derivative decomposition already established for compact target maps removes the endpoint remainders. Strong L^2 convergence of the values and weak L^2 convergence of their ordinary derivative densities, with the measure remainder tending to zero, identify the latter limit as $D\mathcal{T}_L(z_k)$. The bounded continuous square root of $\mathsf{M}(\mathcal{T}_L(z_{j,k}))$ converges strongly in every finite L^q; pairing against bounded test fields identifies the corresponding weighted weak limit. Weak lower semicontinuity therefore gives the same bound for $\mathcal{T}_L(z_k)$. This step can equivalently be obtained by testing the quadratic form against bounded smooth dual fields and taking their supremum. On $\{|z_k| < \sinh L\}$ the map is the identity; $W^{1,1}$ locality therefore identifies its derivative with Dz_k there. Increasing L and using monotone convergence yields

$$\int_\Omega \sum_i (D_i z_k)^T \mathsf{M}(z_k) D_i z_k \le C_\Omega.$$

As $z_k = -Df_k$, this is exactly the metric Hessian bound in the statement. No spatial endpoint embedding is used.

Use one fixed full-gradient-floor detector, and incorporate a countable increasing family of profile detectors as follows. Fix the compact phase interval I under consideration and put $A_I = \sup_{c\in I} |q^{-1}(c)|$. At refinement level l, prescribe a window $W_l \uparrow \infty$ and an accuracy $\eta_l \downarrow 0$. Apply the detector of proposition 3.4 to these parameters in the original orthogonal frame, using the transverse field of definition 9.1. Its disjoint failed-unit-ball count is bounded by $C_l S_j/R_j^2 = C_l a_j^2 R_j^{d-2}$. A maximal disjoint family covers all failed centers by fixed enlargements of those balls. A uniformly small shear changes the covering constants by a uniform factor. Include their union and the fixed gradient-floor failed set in the physical deletion. Choose $l = l(j) \to \infty$ sufficiently slowly that all the existing event and angular requirements hold and the resulting finite refinement constants satisfy $C_j a_j \to 0$. In particular $C_j a_j^2 \to 0$ and $C_j/R_j^2 \to 0$. This choice is made in the common slow diagonal before the generic direction and phases are selected. Here is a construction that also fixes the quantitative localization of $\chi_{0,j}$. At a fixed refinement stage combine the failed sets of its finitely many detector tests, including the fixed full-gradient-floor test. In a containing physical receiver choose a maximal disjoint family of unit balls centered in that union. Color each center by one detector test at which it fails. The subfamily of each color is disjoint, so proposition 3.4 and the outward bound of definition 9.1 give a total count at most $C_j S_j/R_j^2 = C_j a_j^2 R_j^{d-2}$. Maximality covers every failed point in the inner receiver by doubled balls. The larger physical receiver has a macroscopic buffer, so all the fixed enlargements below, and every later fixed-length normal phase track, stay inside it for large j.

For these centers take radial smooth factors $\phi_{0,l}$ equal to zero at distances at most six and one at distances at least eight, and $\phi_{1,l}$ equal to zero at distances at most four and one at

distances at least five. Put $\chi_{0,j} = \prod_l \phi_{0,l}$ and $\chi_{1,j} = \prod_l \phi_{1,l}$. Choose smooth ζ_l equal to one on the closed annulus of radii six and eight and supported in the annulus of radii $11/2$ and $17/2$, and set

$$\zeta_j = \chi_{1,j} \left(1 - \prod_l (1 - \zeta_l) \right).$$

These are finite products. They all vanish on a neighborhood of the failed set, which is covered by the doubled balls. If $\chi_{0,j} > 0$, all distances exceed six, so $\chi_{1,j} = 1$ in a neighborhood. If $0 < \chi_{0,j} < 1$, at least one $\phi_{0,l}$ is in transition, hence its $\zeta_l = 1$ and $\zeta_j = 1$. Multiplication by $\chi_{1,j}$ ensures that the annular amplitude still vanishes near failed points lying in other overlapping balls. Their derivative supports avoid the original failed set, where the full gradient floor on the tested central band applies. Every point with $\chi_{0,j} = 1$ is detector-good. Moreover $\{\chi_{0,j} < 1\}$ is contained in the union of the radius-eight balls: no critical, count, rank, base-boundary, normal-cap, or filled projected-hole deletion is multiplied into this physical detector cutoff.

The disjoint unit balls have a fixed bound on the overlap of their radius-nine enlargements. The product rules, bounded derivatives of the fixed radial factors, and this overlap bound give the following displayed derivative/volume estimate. They also give the asserted localization and its fixed-distance enlargements, since each enlarged ball has fixed volume depending only on that distance. In the selected shear, pull back these cutoffs from the original orthogonal frame. The pulled-back balls lie in balls of one larger fixed radius, and the derivative and volume estimates have uniform constants because the shear and its inverse are uniformly bounded and have determinant one. All spatial and normal boundary factors remain separate. The common angle functions are chosen with strictly positive logarithmic transition before their prescribed outer endpoint.

$$|\operatorname{supp} \zeta_j| + \int (|D\chi_{0,j}|^2 + |D\chi_{1,j}|^2 + |D\zeta_j|^2) \le C_j a_j^2 R_j^{d-2}.$$

Here volumes and derivatives are in the physical trace receiver. Such nested cutoffs are obtained by placing their transitions in fixed successive annuli about each covering ball. Every fixed refinement has a finite constant, so it can be absorbed into the slow diagonal. Use the same slowly chosen κ_j as in the parameter selection, including $a_j/\kappa_j^8 \to 0$.

On detector-good regular roots put $r = a_j|z| = |v/h|$. Choose smooth radial amplitudes ψ_j, ξ_j, ω_j as follows. The low-pass ψ_j is one for $r \le \kappa_j^2$, zero for $r \ge \kappa_j$, with logarithmic transition. The band amplitude ξ_j is one on $[\kappa_j^2, \kappa_j]$, zero below κ_j^3 and above $\kappa_j^{1/2}$, and logarithmic on its two transition intervals. The outer low-pass ω_j is one below κ_j, zero above $\kappa_j^{1/2}$, with logarithmic transition. Smooth the logarithmic junctions inside fixed proportional subintervals; retain the plateau of ξ_j on the whole closed transition set of ψ_j. All nonzero amplitudes have $r \le \kappa_j^{1/2} < 1$, hence the full gradient floor gives $|h| \ge c_I > 0$ there. Define actual root amplitudes

$$b_{0,j} = \chi_{0,j}\psi_j(r), \qquad b_{1,j} = \chi_{1,j}\xi_j(r), \qquad b_{2,j} = \zeta_j\omega_j(r).$$

Whenever $0 < b_{0,j} < 1$, at least one of $b_{1,j}, b_{2,j}$ is exactly one. Indeed a partial angle weight lies in ξ_j's plateau and has $\chi_{1,j} = 1$, while a partial detector weight lies in ζ_j's plateau and has $\omega_j = 1$ if the angle weight is already one. They all vanish in a neighborhood of a normal critical point outside the detector set, because its angle tends to infinity; they also vanish near the physical failed set by the detector cutoffs. The free caps contain no central root. Thus their symmetric sums over all actual roots are continuous in (x, c), with ordinary

implicit-root charts on every nonzero support and zero extension across the omitted critical charts. No rank labels or count-defect perimeter are used for these symmetric sums.

Set

$$Q_{0,j}^2 = \sum_{u_j=c} b_{0,j}^2, \qquad Q_{A,j}^2 = \sum_{u_j=c} (b_{1,j}^2 + b_{2,j}^2).$$

We verify their dimension-free Sobolev bounds. Write $\mathsf{M}(z) = (I + z \otimes z)^{-1}$. The derivative of a logarithmic radial amplitude $\varphi(r)$ obeys

$$(D_z\varphi)^T \mathsf{M}(z)^{-1} D_z\varphi \le C|\log \kappa_j|^{-2}.$$

Indeed its radial derivative is at most $C/(r|\log \kappa_j|)$, its dual metric radial factor is $\sqrt{a_j^2 + r^2}$, and every transition has $r \ge \kappa_j^3 \gg a_j$. For a root graph,

$$D_\xi z = R_j(D_x z - a_j z \partial_t z), \qquad \mathcal{A}_{(v,a_j h)} = a_j^2 h^2 \sum_\alpha (\partial_\alpha z)^T \mathsf{M}(z) \partial_\alpha z.$$

Since $a_j|z| \le 1$ and $|h| \ge c_I$ on all derivative supports, normal coarea and the square-root-of-sum chain rule give

$$\begin{aligned}\int_{\Omega\times I} |D_\xi Q_{0,j}|^2 + |D_\xi Q_{A,j}|^2 &\le \frac{C R_j^{2-d}}{a_j^2|\log \kappa_j|^2} \int \mathcal{A}_{(v,a_j h)} \\ &\quad + C R_j^{2-d} \int (|D\chi_{0,j}|^2 + |D\chi_{1,j}|^2 + |D\zeta_j|^2) \\ &\le C|\log \kappa_j|^{-2} + C_j a_j^2 \longrightarrow 0.\end{aligned}$$

The phase derivative has the same estimate times R_j^{-2}, using $\partial_c z = (\partial_t z)/h$; the detector part has the analogous factor. The bound above uses lemma 9.2 and is independent of the base dimension. The root-norm chain rule is legitimate by finite implicit-chart sums followed by positive exhaustion; the amplitudes vanish before every omitted critical chart, so no hidden endpoint measure appears.

The low-pass count restoration and band masses are

$$\int (N_j - Q_{0,j}^2) \le C a_j^2 \kappa_j^{-4} + C_j a_j^2, \qquad \int Q_{A,j}^2 \le C a_j^2 \kappa_j^{-6} + C_j a_j^2.$$

For the angular terms, on the detector-good set $r \ge \kappa_j^\ell$ forces $|v| \ge c_I \kappa_j^\ell$ by the full gradient floor, so bounded $|h|$, normal coarea and $\int |v|^2 \le C S_j$ give the powers $\kappa_j^{-2\ell}$. The detector terms have the displayed smaller bound from their physical volume and bounded h. The complete counts converge strongly to m by the complete-count BV argument above. Thus the square-root inequality and the derivative bounds prove

$$Q_{0,j} \to \sqrt{m}, \qquad Q_{A,j} \to 0 \quad \text{strongly in } L^2(I; H^1(\Omega)).$$

These are the actual inputs to lemma 7.1 and proposition 7.4, rather than an assumption of their conclusions for the actual fields.

Define the continuous scalar error

$$\mathfrak{e}_j = ((Q_{0,j} - \sqrt{m})^2 + Q_{A,j}^2)^{1/2}.$$

It tends to zero in $L^2(I; H^1(\Omega))$ by the norm chain rule. Choose fixed $0 < \delta_1 < \delta_0 < 1/2$, smaller than half the distance from $\sqrt{m}$ to any other square root of a nonnegative integer. On $\mathfrak{e}_j < \delta_0$, the band count is below one, so every low-pass weight is exactly zero or one. The low-pass count is therefore an integer; its square root is within δ_0 of $\sqrt{m}$, forcing it to be exactly m. Those m selected roots have $\chi_{0,j} = \psi_j = 1$, normal derivative satisfying $|u_{j,t}| \ge c_I > 0$ and physical slopes $|D_x T_{j,k}| \le \kappa_j^2 < 1$. They are a genuine ordered smooth

family on this open set. Their order cannot change while they remain regular. Its other actual roots, if any, have zero low-pass weight and are not silently included in the selected family.

Let A be a smooth cutoff equal to one for $\mathfrak{e} \le \delta_1/2$ and zero for $\mathfrak{e} \ge \delta_1$. Then $\theta_j = A(\mathfrak{e}_j)$ is continuous and converges strongly to one in $L^2(I; H^1(\Omega))$. On any compact tested base, it vanishes in a relative neighborhood of the complement of the open regular region $O_j = \{\mathfrak{e}_j < \delta_0\}$; fixed positive superlevels have closure inside that region. The selected family agrees with the complete list on $O_j(c) \cap \{N_j = m\}$. The complement has vanishing base–phase measure, and the uniform integrability of all low-angle slope squares gives

$$\int_{\Omega\times I} \mathbf{1}_{O_j(c)\cap\{N_j\neq m\}} \sum_{k=1}^{m} \left| \frac{v_j(R_j\xi, T_{j,k}(R_j\xi, c))}{a_j h_j(R_j\xi, T_{j,k}(R_j\xi, c))} \right|^2 d\xi\, dc \longrightarrow 0.$$

Thus both the value error of bounded selected heights and their extra slope contribution on count-defect fibers vanish in phase average. Here is the single-phase selection explicitly. Fix one compact interval $I_* \subset \operatorname{int} I$ of positive length. Enumerate the nonnegative phase errors on I_*: the squared H^1 error of $\theta_j - 1$ on each compact base receiver, the clipped-height L^2 errors for integer clips, the retained-slope L^2 errors, the measure of the complete-count defect and the displayed selected-slope mass on that defect, and each of the normalized transport errors required in the statement. For transport, divide total height movement by $D_j R_j^d$ and failed-start count by R_j^d. Their phase averages are bounded respectively by $C|z|/R_j$ and $C_m|z|P_j/R_j^d$, which tend to zero for fixed physical z and for $z = D_j\zeta$ with bounded ζ. Include the corresponding nonnegative displacement integrals over bounded displacement sets when needed. Thus every error in this enumeration has phase integral tending to zero. Choose integers $n_j \uparrow \infty$ slowly enough that the sum of the phase integrals of the first n_j errors tends to zero. Outside the null set of nongeneric slices there is then $c_j \in I_*$ where this finite sum is at most twice its average over I_*. Every enumerated error tends to zero at these same phases. This proves the simultaneous phase selection, rather than selecting a separate phase for each rank, clip, or receiver. Thus the regular support and its interior zero extension are actual finite-record constructions. The fixed-core comparisons proved below are uniform in the tested phase interval, so they require no additional selection based on a later profile conclusion.

At these same phases, the selected heights and slopes therefore have the same ordered limits as the complete ranks. The construction permits extra zero-weight roots on a selected fiber; their contributions are treated in the later positive source and orientation arguments.

The normalized layer and its extension are those fixed in lemma 2.4 and proposition 3.5. The finite-energy Hamiltonian calculation in the proof of the latter already establishes

$$\sigma = \frac{d_p}{2} \int y^{1-p}(Q_t^2 + Q_y^2)dt\, dy + \int W(q)dt = d_p \int y^{1-p} Q_t^2 dt\, dy > 0.$$

Thus the macroscopic energy constant is exactly the complete normal derivative constant used here.

We give the cap-localized normal-column calculation directly. Fix bounded connected Lipschitz bases $\Omega \Subset \Omega' \Subset \mathbb{R}^d$. Choose a smooth $0 \le \rho \le 1$, equal to one near zero and supported in $(-b, b)$ for a fixed b, and put $\rho_R(t) = \rho(t/R)$. Its plateau can be chosen to

contain all the tested central roots by proposition 5.2. Write

$$\mathcal{K}_j = R_j\Omega' \times (-bR_j, bR_j) \times (0,\infty),$$
$$N_j(x) = \int \rho_{R_j}(t) H_j(x,t,y)^2 d_p y^{1-p} dt\, dy,$$
$$n_j(\xi) = N_j(R_j\xi).$$

All energy estimates in the following calculation are on $\mathcal{K}_j$, with unrestricted extension height. Its physical projection is contained in B_{CR_j} for a fixed C depending only on the tested base and b. The complete local energy of proposition 3.4, the outward same-frame bound of definition 9.1, and the extension angular estimate already established above give

$$\int_{\mathcal{K}_j} H_j^2 d\mu \le CR_j^d, \qquad \int_{\mathcal{K}_j} |V_j|^2 d\mu \le CS_j, \qquad \int_{\mathcal{K}_j} \mathcal{A}_{Z_j} d\mu \le CS_j/R_j^2.$$

In particular $\int_{\mathcal{K}_j} |Z_j|^2 d\mu \le CS_j$, since $S_j = a_j^2 R_j^d$. In a selected small shear these bounds follow by the actual determinant-one pullback into a fixed enlarged physical ball: $V^{\mathrm{sh}} = V^{\mathrm{o}} \circ S_k$, $H^{\mathrm{sh}} = (H^{\mathrm{o}} + k \cdot V^{\mathrm{o}}) \circ S_k$, and the angular wedge transformation already proved above has a uniform constant. No integral over the unbounded physical normal cylinder is asserted.

Apply the compact-cap version of lemma 8.6. Its proof is local in t: in the capped identity only

$$\rho_{R_j} \mathscr{W}_j - \rho'_{R_j} H_j V_j, \qquad \mathscr{W}_j = H_j \partial_t V_j - (\partial_t H_j) V_j,$$

occurs, supported in the compact normal support of ρ_{R_j} and its derivative. Therefore its full-cylinder energy assumptions may be replaced here by the three verified bounds on $\mathcal{K}_j$. Compatibility $D_x H_j = \partial_t V_j$ gives the same distributional capped identity, and the wedge bound $|\mathscr{W}_j| \le a_j^{-1} |Z_j| \sqrt{\mathcal{A}_{Z_j}}$ gives

$$\int_{R_j\Omega} |D_x N_j| \le \frac{C}{a_j} (CS_j)^{1/2} (CS_j/R_j^2)^{1/2} + \frac{C}{R_j} (CR_j^d)^{1/2} (CS_j)^{1/2} \le C a_j R_j^{d-1}.$$

As in that lemma, height truncation and dominated convergence of these two integrable terms justify the distributional identity; no energy bound outside the cap is used. Scaling and its BV Sobolev–Poincare conclusion give

$$\|n_j\|_{L^1(\Omega')} \le C, \qquad \|n_j - (n_j)_\Omega\|_{L^{d/(d-1)}(\Omega)} \le C a_j.$$

We identify this capped mean without any unbounded-normal-domain estimate. The covariance measure limit in proposition 3.5 is tested with $\mathbf{1}_\Omega(x/R_j)\rho(t/R_j)$ and its normal-normal component. The optimizing normal axes in definition 9.1 align with its positive rank-one covariance limit: at radius R_j the limiting matrix has a simple positive top eigenvalue, so its unit top eigenvectors converge, up to sign, to the limit normal. Thus in the record coordinates the limit plane is $t = 0$. Its mass on $\partial\Omega \times \{0\}$ is zero because Ω is Lipschitz, and $\rho(0) = 1$, so approximation by smooth compact tests proves

$$\int_\Omega n_j(\xi) d\xi \longrightarrow m\sigma|\Omega|.$$

Here the equality of the two layer normalizations is the established Hamiltonian identity cited above. The same mean limit holds in the selected shear: its position change is $o(1)$ on the bounded scaled cap, and the additional normal-normal covariance has total normalized mass at most $C(|k_j|a_j + |k_j|^2 a_j^2) \to 0$. Weak covariance convergence and the zero boundary mass justify the position change, first for smooth compact tests and then for this Lipschitz base.

Consequently $n_j \to m\sigma$ strongly in local $L^{d/(d-1)}$, hence in local L^1, and the complete capped normal columns are uniformly integrable.

For the transverse columns define explicitly

$$r_{j,K}(\xi) = a_j^{-2} \int \rho_{R_j}(t)|V_j(R_j\xi, t, y)|^2 \mathbf{1}_{\{|V_j|>Ka_j|H_j|\}} d\mu_{t,y}.$$

Then $e_j \le K^2 n_j + r_{j,K}$, while the complete ratio tail already proved on the bounded physical receiver containing $\mathcal{K}_j$ gives

$$\lim_{K\to\infty} \limsup_{j\to\infty} \int_\Omega r_{j,K} = 0.$$

Indeed its integral is S_j^{-1} times that capped physical tail integral. Thus lemma 8.5 proves uniform integrability of e_j before any full transverse-profile conclusion, and its final assertion permits fixed height factors to be exhausted in these complete capped columns.

We now construct the fixed physical cores using only the preceding data. We choose the common scalar center $\tau_j = 0$. This is compatible with the phase-independent raw-height limits supplied by proposition 9.5: that proposition assumes and uses the absolute first moment of $t_{j,k}/D_j$, and makes no independent recentering of any rank. Write q for the centered increasing odd cubic layer, Q for its Poisson extension, $H_* = Q_t$, and $\sigma = \int_{\mathbb{R}\times(0,\infty)} H_*^2 d_p y^{1-p} dt\, dy$. The layer normalization, strict monotonicity, and equality of these two energy constants are already established in lemma 2.4 and proposition 3.5.

At every selected actual root $(x, T_{j,k}(x,c))$, $c \in I$, one has $\chi_{0,j} = 1$ and $|D_x T_{j,k}| \le \kappa_j^2$. Hence the refined detector gives a translated rotated layer in an increasing physical window, with C^2 error $\eta_{l(j)} \to 0$. It cannot give a constant well: c is in the fixed compact interval I, at positive distance from both wells. In the original orthogonal coordinates write the approximating layer about that point as $q(\nu\cdot(X - X_0) + b)$, with $|\nu| = 1$. The compact-phase lower bound for q' and the value at the root give $b = q^{-1}(c) + O_I(\eta_{l(j)})$. In a sheared frame the coefficients of its affine argument are $(\nu_x, \nu_t + k_j \cdot \nu_x)$. The root identity $D_x T = -v/h$, its bound κ_j^2, the physical derivative bound, and the C^1 approximation give

$$|\nu_x| \le C_I(\kappa_j^2 + \eta_{l(j)}), \qquad \nu_t = \epsilon_{j,k} + o(1), \qquad \epsilon_{j,k} = \operatorname{sign} h(x, T_{j,k}) \in \{-1, 1\}.$$

All errors here are uniform in the selected root, phase, and base on a compact receiver. The bounded shear tends to zero and does not alter these conclusions. Define the comparison center exactly by

$$S_{j,k}(x,c) = T_{j,k}(x,c) - \epsilon_{j,k} q^{-1}(c).$$

Oddness of q identifies the limiting comparison as $\epsilon_{j,k} q(t - S_{j,k})$. This phase pin is an exact definition; it is not replaced by an actual zero root. The C^2 detector comparison and convergence of its normal coefficients also give, for every fixed L,

$$\sup_{\xi\in O_j(c),\ k\le m,\ c\in I} \|u_j(R_j\xi + z, S_{j,k}(R_j\xi, c) + s) - \epsilon_{j,k} q(s)\|_{C^2(B_L^{d+1})} \longrightarrow 0,$$

where the norm is taken in the physical offset variables (z, s).

For completeness the comparison passes to the extension uniformly down to height zero. If w is the difference of the actual trace and its detector layer, its value and first two physical derivatives are bounded by η on $B_W(X_0)$ and are globally bounded by lemma 2.1. For fixed L, $W \ge 8(L + A_I + 1)$, and $0 \le y \le L$, split the Poisson convolution of $D_X w$ at radius $W/2$. The local part is at most η, and the remaining Poisson mass is at most $C_L W^{-p}$. Thus $|D_X(P_y * w)| \le C_L(\eta + W^{-p})$ on the tested core. For the height derivative use $\int \partial_y P_y = 0$ and symmetric second differences. In the local ball those differences are bounded by $C\eta \min(|H|^2, 1)$; split its radial integral at $|H| = y$ and one. Outside that ball use the uniform bound and $|\partial_y P_y(H)| \le$

$C_L y^{p-1}|H|^{-d-1-p}$. Since $1<p<2$, this gives $|\partial_y(P_y * w)| \le C_L(\eta + W^{-p})y^{p-1}$. This argument is first made in the original orthogonal frame. Pulling the resulting actual extension back by the determinant-one shear proves it for its sheared derivatives as well; no isotropic Poisson representation of a sheared trace is used. Uniform continuity of H_* under the above normal and phase perturbations consequently gives, for every fixed L,

$$\sup_{\substack{\xi\in O_j(c),\ k\le m,\ c\in I\\ |t-S_{j,k}(R_j\xi,c)|\le L,\ 0\le y\le L}} |H_j(R_j\xi,t,y) - \epsilon_{j,k}H_*(t-S_{j,k}(R_j\xi,c),y)| \le \omega_j(L), \qquad \omega_j(L)\longrightarrow 0.$$

For example the proof bounds $\omega_j(L)$ by $C_{I,L}(\eta_{l(j)} + W_{l(j)}^{-p} + \kappa_j^2 + |k_j|)$ after increasing a modulus of continuity. The same argument, and the symmetric second-difference bound for q', show that the error between $(U_j)_y$ and $\epsilon_{j,k}Q_y(t-S_{j,k},y)$ is at most $C_L\omega_j(L)y^{p-1}$. The error for the normal gradient $(H_j,(U_j)_y)$ therefore tends to zero in weighted $L^2(d_p y^{1-p}dt\,dy)$ on every such fixed rectangle, uniformly in the selected root and base: the two integrals near zero use $1-p>-1$ and $p-1>-1$, respectively. In particular no error is divided by a_j in this normal-energy comparison.

These comparisons also prove separation of the physical cores. For any fixed B, the trace comparison and the convergence of its normal coefficient to $\epsilon_{j,k}$ give $\epsilon_{j,k}\partial_t u_j > c_{I,B} > 0$ on the vertical interval $|t-T_{j,k}|\le B$, for all sufficiently large j. The lower bound follows from the positive minimum of q' on the fixed compact argument interval containing $[-B-A_I-1, B+A_I+1]$. Hence no other actual root at phase c occurs in that interval. Taking $B>2L+2A_I$ shows that the m rectangles

$$\mathcal{C}_{j,k}^L(\xi,c) = \{(t,y) : |t-S_{j,k}(R_j\xi,c)|<L,\ 0<y<L\}$$

are disjoint on $O_j(c)$. This remains true when some normalized limiting ranks coincide: physical separations tend to infinity along this regular region, although their quotient by D_j may tend to zero.

Put

$$\sigma_L = \int_{-L}^{L}\int_0^L H_*(s,y)^2 d_p y^{1-p} dy\,ds.$$

Then $\sigma_L \uparrow \sigma$, and the uniform comparison gives

$$\left| \int_{\mathcal{C}_{j,k}^L(\xi,c)} H_j(R_j\xi,t,y)^2 d_p y^{1-p} dt\,dy - \sigma_L \right| \le C_L\omega_j(L)$$

on $O_j(c)$. All these fixed physical cores lie in the plateau of the macroscopic normal cap for large j. Indeed proposition 5.2 confines every root at a phase in I to $|T_{j,k}| = o(R_j)$, uniformly over the fixed buffered base, while $|S_{j,k}-T_{j,k}|\le A_I$. We choose the containing cap to equal one on a fixed smaller macroscopic normal interval containing those roots.

For $\xi\notin O_j(c)$ set the core family empty, and define the actual nonnegative remainder

$$\mathcal{R}_{j,L}(\xi,c) = n_j(\xi) - \mathbf{1}_{O_j(c)}(\xi)\sum_{k=1}^m \int_{\mathcal{C}_{j,k}^L(\xi,c)} H_j(R_j\xi,t,y)^2 d_p y^{1-p} dt\,dy \ge 0.$$

Disjointness and the cap plateau justify this inequality. For any fixed compact base receiver Ω,

$$\int_\Omega \mathcal{R}_{j,L}(\xi,c)d\xi \le \|n_j - m\sigma\|_{L^1(\Omega)} + m(\sigma-\sigma_L)|\Omega| + m\sigma_L|\Omega\setminus O_j(c)| + C_L\omega_j(L)|\Omega|.$$

At the phase sequence selected from the preceding nonnegative errors, $\theta_j(\cdot,c_j)\to 1$ in H^1, and $\theta_j = 0$ outside $O_j(c_j)$; hence the third term tends to zero. The first term tends to zero

by the already proved complete normal-column argument. Consequently

$$\lim_{L\to\infty}\limsup_{j\to\infty}\int_\Omega \mathcal{R}_{j,L}(\xi,c_j)d\xi = 0.$$

This proves the normal-core exhaustion before any transverse full-profile conclusion. It uses positive actual core masses and the complete column, including bases with extra zero-weight roots, and requires no rate for the measure of incomplete fibers.

We give the promised finite-record height and separation factors. Work first at a fixed phase from the already chosen diagonal and on a compact receiver, writing $w_{j,k} = (T_{j,k} - \tau_j)/D_j = T_{j,k}/D_j$ on O_j and $z_{j,k} = v_j/(a_j h_j)$. For $L > 0$ let T_L denote the scalar clipping to $[-L, L]$ and define, with zero extension off O_j,

$$F_{j,k,L} = \theta_j T_L(w_{j,k}).$$

These are actual H^1 functions. Indeed replace θ_j first by $(\theta_j - \varepsilon)_+$. On a compact receiver its positive support has closure inside O_j, so the usual product rule in the smooth implicit-root charts and zero extension apply. The clipped height is bounded by L, and the derivative density is bounded by $|z_{j,k}| + L|D\theta_j|$. Thus sending $\varepsilon \downarrow 0$ in H^1 gives

$$DF_{j,k,L} = -\theta_j z_{j,k}\mathbf{1}_{\{|w_{j,k}|<L\}} + T_L(w_{j,k})D\theta_j.$$

The retained strong L^2 slope convergence, height convergence in measure, and $\theta_j \to 1$ in H^1 imply

$$F_{j,k,L} \longrightarrow T_L(f_k) \quad \text{strongly in } H^1.$$

This holds for every L, not just generic clipping levels: the indicators converge off $\{|f_k| = L\}$, and Sobolev locality gives $Df_k = 0$ almost everywhere on that level set. The second derivative term above tends to zero in L^2 because it is bounded by $L|D\theta_j|$. The bounded values give strong L^2 convergence of the lifts themselves.

Apply lemma 8.3 to these actual lifts and normalized heights $w_{j,k}$. With its fixed smooth functions A, h_M, put

$$H_{j,M} = A(\theta_j)\prod_{k=1}^m h_M(F_{j,k,4M}), \qquad H_M = \prod_{k=1}^m h_M(f_k).$$

For each fixed M, the lemma gives strong H^1 convergence and $|T_{j,k} - \tau_j| < 8MD_j/3$ on the positive support. The root heights and factors are continuous on O_j, so this pointwise conclusion and the bound $3MD_j$ on its closure hold there as well. The closure lies in a positive superlevel of θ_j and hence in the genuine regular region on a compact tested base. Its physical slopes are at most $\kappa_j^2 < 1$. The same lemma gives $H_M \to 1$ strongly in local H^1 as $M \to \infty$.

For separation factors choose a bounded smooth function r on $[0,1]$ that equals $1/s$ on $[3/4, 1]$, and put $A_{j,k} = r(\theta_j)F_{j,k,4M}$. Then $A_{j,k} \to T_{4M}(f_k)$ strongly in H^1, while $A_{j,k} = w_{j,k}$ wherever $H_{j,M} > 0$. Choose a smooth function $b : \mathbb{R} \to [0,1]$ with $b = 0$ on $(-\infty, 2]$ and $b = 1$ on $[3, \infty)$. For a specified finite set J of adjacent outer gaps, a fixed $\delta > 0$, and a smooth compact base cutoff χ, use

$$B_{j,M,\delta} = \chi H_{j,M}\prod_{i\in J} b((A_{j,i+1} - A_{j,i})/\delta).$$

Its nonzero support lies on the actual regular graphs, satisfies the preceding actual height bounds, and has $T_{j,i+1} - T_{j,i} > 2\delta D_j$ for every specified gap. Its support closure still has gaps at least $2\delta D_j > \delta D_j$ and is a compact subset of the regular root region after restricting

to the compact tested base. The bounded smooth product and chain rules give strong H^1 convergence, for these fixed parameters, to

$$\chi \prod_k h_M(f_k) \prod_{i\in J} b((f_{i+1}-f_i)/\delta).$$

To check this exact limit, where the limiting height product is positive all $|f_k| < 2M < 4M$, so the deweighted clips are the actual limiting ranks; where that product is zero both products are zero. Every factor is bounded with bounded derivative for fixed M, δ, so convergence of derivatives follows from strong L^2 convergence of the input gradients and dominated convergence against each fixed limiting gradient. No assertion of strong H^1 convergence as $\delta \downarrow 0$ is needed or made. A countable Fubini diagonal over receivers, clips, and fixed cutoff parameters retains the same phases and all preceding rank, slope, transport, and regular-support conclusions. This completes the construction. □

Proposition 9.7 (Complete actual cores give the full transverse profile and vanishing normal-extension discrepancy)**.** *Use definition 9.1 and the actual data of proposition 9.6. Let q be the centered increasing odd one-dimensional layer, let Q be its normal Poisson extension, and put $H_* = Q_t$, $\sigma = \int H_*^2 d_p y^{1-p} dt dy$. On every smaller receiver use the selected ordered actual root family $T_{j,k}$ on the genuine regular bases of proposition 9.6, at one phase sequence c_j in a fixed compact central interval. It agrees with the complete actual root family wherever the full count is m. On every genuine regular base the orientations $\epsilon_{j,k} \in \{-1, 1\}$ are the actual local orientations. Extend the comparison data measurably across the remaining bases, choosing m ordered comparison centers and signs there. The associated phase-pinned comparison centers on the regular bases are*

$$S_{j,k} = T_{j,k} - \epsilon_{j,k} q^{-1}(c_j), \qquad \epsilon_{j,k} q(T_{j,k} - S_{j,k}) = c_j,$$

chosen by nonnegative-error Fubini. The data off the regular bases are used only to define the unweighted comparison norms; they are never asserted to be actual roots or used in an actual equation test. For these extensions,

$$S_j^{-1} \int \rho_{R_j} \left| V_j - a_j \sum_{k=1}^{m} (-Df_k(x/R_j)) \epsilon_{j,k} H_*(t - S_{j,k}(x), y) \right|^2 d\mu \longrightarrow 0.$$

This estimate is independent of the bounded-sign translated-profile extensions chosen off the regular bases. Their base measure tends to zero; complete-column uniform integrability pays the actual energy there, and absolute continuity of the fixed integrable coefficient $\sum_k |Df_k|^2$ pays every extended model profile there. On all genuine regular supports the centers and signs remain the exact actual selected-root data.

The centers $S_{j,k}$ are the zeros of the corresponding comparison layers; they are defined by the displayed exact phase pinning. The orientations are the actual local orientations and no alternating pattern is required in this assertion. The estimate includes every extension height and holds in the selected small shear. Distinct ranks have physically separated fixed-radius cores in measure, including on normalized coincidence sets. In the original orthogonal coordinates let U_j^{n} be the one-dimensional Poisson extension of $t \mapsto u_j(x,t)$ at each fixed x. Then

$$\int_{Q_{R_j} \times (0,\infty)} |D_{t,y}(U_j - U_j^{\mathrm{n}})|^2 d\mu = o(S_j).$$

The same normal-discrepancy assertion holds in every selected constant small shear, with the actual pulled-back extension and the normal-only extension of its pulled-back trace. No bound on the untruncated limiting heights is required.

Proof. We use the smoothness of lemma 2.1 and the complete rank, core, column and ratio-tail conclusions of proposition 9.6. In particular its extension angular estimate (6), established before the trace-rank construction, gives on every fixed enlarged physical receiver

$$\int \mathcal{A}_{Z_j}\, d\mu \le C a_j^2 R_j^{d-2}, \qquad Z_j = (V_j, a_j H_j), \qquad \mathcal{A}_Z = |DZ|^2 - |D|Z||^2 = \sum_\alpha \frac{|Z \wedge \partial_\alpha Z|^2}{|Z|^2}.$$

The density has its almost-everywhere zero-set convention from Sobolev locality. The same estimate in the selected shear was established there by the source and target linear changes; no isotropic equation for a sheared field is needed here. These are full-extension-height estimates over bounded physical receivers. The fixed-radius core comparisons, disjointness, exact phase pinning $S_{j,k} = T_{j,k} - \epsilon_{j,k} q^{-1}(c_j)$, and exhaustion of the capped normal columns are also the preceding proposition's conclusions, including on normalized coincidence sets. We use them directly rather than reconstructing the detector family. All the nonnegative phase-averaged errors below are placed in its common countable list before the final phase selection. Thus the same sequence retains its rank, slope, column and regular-support conclusions.

We first prove the approximation on these genuine regular bases. Their complement has scaled measure tending to zero, since the regularity cutoff tends to one in measure. Choose any measurable translated-profile data on that complement; for instance choose m ordered centers and arbitrary signs. For every base and every choice of centers, Cauchy–Schwarz in the finite sum gives

$$\int \left| \sum_k (-Df_k(\xi)) \epsilon_{j,k} H_*(t - S_{j,k}, y) \right|^2 d_p y^{1-p} dt dy \le m\sigma \sum_k |Df_k(\xi)|^2.$$

The right side is a fixed integrable function on a compact receiver. Thus its integral over those exceptional bases tends to zero, independently of the extended centers and signs. The actual normalized transverse and normal energies on that complement tend to zero by uniform integrability of e_j and n_j in proposition 9.6. Consequently an approximation proved on the genuine regular bases extends to the displayed full, unweighted transverse norm. No fictitious root is introduced into a finite-index equation or a nonlinear comparison test. The same argument applies to the vanishing complete-count defect products when a proof is first carried out on the complete family.

Here an admissible compact target map means $F \in C_c^\infty(\mathbb{R}^d; \mathbb{R}^q)$ for a fixed finite q. On $H \neq 0$ put $z = V/(aH)$. The map $\mathcal{F}(n) = F(n_x/n_{d+1})$ on the two hemispheres of the unit sphere extends smoothly by zero across the equator: compact support of F makes it vanish in a neighborhood of that equator. Its derivatives are bounded by a constant depending on F. The chain rule and the angular density just defined give

$$H^2 |D_{t,y} F(z)|^2 \le C_F a^{-2} \mathcal{A}_Z.$$

On each fixed detector-good normal rectangle the normal speed is bounded above and away from zero, by fixed-core comparison with the strictly positive Q_t. Thus the derived extension angular estimate bounds the phase-averaged normal/height derivative energy of $F(z_j)$ on these cores by $C_F R_j^{d-2}$, which tends to zero after division by R_j^d. Here is the trace-anchored weighted estimate used to pass from this derivative bound to the core values. For a fixed rectangle $C = (-L, L) \times (0, L)$, an interval $J_0 \subset (-L, L)$ of length at least $l_0 > 0$, and a constant vector A, one has

$$\int_C y^{1-p} |G - A|^2 dt\, dy \le C_{L,l_0,p} \left(\int_C y^{1-p} |D_{t,y} G|^2 dt\, dy + \int_{J_0} |G(t,0) - A|^2 dt \right). \tag{TP}$$

To prove it, write

$$G(t,y)-A=[G(t,y)-G(t,v)]+[G(t,v)-G(s,v)]+[G(s,v)-G(s,0)]+[G(s,0)-A],$$

average s over J_0 and v over $(L/2,L)$, and use weighted Cauchy–Schwarz on the two vertical legs, with $\int_0^L y^{p-1}dy<\infty$. The horizontal leg is controlled by the one-dimensional fundamental theorem of calculus at height v, and averaging v bounds its contribution by the bulk horizontal derivative integral. This proves (TP) for smooth functions, and weighted-energy approximation proves its trace form. For the application, take a fixed smaller compact phase interval of positive length, and choose L large enough to contain its comparison roots and a buffer. Detector comparison and the upper and positive lower bounds for $|u_t|$ make its actual trace image such an interval J_0, with the phase and normal Lebesgue measures uniformly comparable. On complete-count products, all m local branches transported from the selected phase are present, so completeness and order identify them with the same ordered labels at the anchor phases. The incomplete products and detector/retention exceptions have vanishing base–phase measure by the preceding full-phase conclusions. Since F is bounded and the normal-phase Jacobian on these cores is bounded above and below, their contribution to the anchor integral tends to zero. Strong compact-target convergence of the actual trace ranks in proposition 9.6, integrated over that phase interval, therefore makes the anchor term in (TP) tend to zero with $A=F(-Df_k(x/R_j))$. Thus

$$R_j^{-d}\sum_k\int_{\text{core }k}H_j^2|F(z_j)-F(-Df_k(x/R_j))|^2d\mu\to 0.$$

We make explicit why this core convergence controls the entire column. Put $b_k(\xi)=-Df_k(\xi)$, choose smooth compact target truncations F_K equal to the identity on $|z|\le K$ and satisfying $|F_K(z)|\le C|z|$, and first compare with the model truncated to its own disjoint core rectangles. On a fixed core its actual normal mass is at most $\sigma+o_j(1)$ for each fixed L, uniformly in the regular base, by the preceding normal-profile comparison. Thus, after taking $j\to\infty$, replacing $F_K(b_k)$ by b_k costs at most $C\sigma\int_{\{|b_k|>K\}}|b_k|^2$, with a constant independent of L. Replacing $F_K(z_j)$ by z_j costs the complete ratio tail. On the complement of the cores the part where $|z_j|\le K$ has transverse energy at most K^2 times the complete normal-column remainder; where $H_j=0$, any nonzero transverse field belongs to the ratio tail. Finally, restoring each whole model profile costs its fixed integrable tail. More precisely, write

$$\tau_L=\int_{\mathbb{R}\times(0,\infty)\setminus((-L,L)\times(0,L))}H_*^2d_py^{1-p}dt\,dy\longrightarrow 0,$$

and let $T_j(K)$ be the normalized complete ratio-tail energy on a containing physical receiver. If E_j denotes the normalized squared error in the proposition on a compact base Ω, these estimates give

$$\limsup_j E_j\le C\Big[\limsup_j T_j(K)+K^2\limsup_j\int_\Omega\mathcal{R}_{j,L}\,d\xi + \sum_k\int_{\Omega\cap\{|b_k|>K\}}|b_k|^2d\xi+\tau_L\sum_k\int_\Omega|b_k|^2d\xi\Big].$$

The already treated exceptional-base errors have vanished here. The normal remainder $\mathcal{R}_{j,L}$ is exactly the one in proposition 9.6, with every extension height included. In obtaining this estimate the compact-target error is first sent to zero at fixed K,L; fixed-core normal comparison then replaces H_j by $\epsilon_{j,k}H_*$. Now send $L\to\infty$ with K fixed, and then $K\to\infty$. Core exhaustion, the complete ratio tail and $b_k\in L^2(\Omega)$ make the right side vanish. This proves the full transverse approximation; no product of an unbounded limiting coefficient

with an uncontrolled actual remainder is used. The fixed-M height cutoffs can be removed by lemma 8.5 and the already established uniform integrability of the columns, before asserting this untruncated profile estimate. Append all these fixed-radius profile errors, on their countable exhaustion, to the previous count, lift, slope, and transport errors before selecting the phase. Every error is nonnegative and has vanishing phase average; one summably weighted Fubini diagonal selects the common sequence c_j and retains all the actual regular supports as well as the exact definition $S_{j,k} = T_{j,k} - \epsilon_{j,k} q^{-1}(c_j)$. Genuine zero-phase roots are $o(1)$ away on the refined good bases, but the anchors and profiles use the defined comparison centers throughout. This uses no union of separately phase-selected exceptions.

The profile construction just proved is in the selected frame; no additional rank construction in the original frame is needed. For the Fourier argument write U^{o} for the actual extension in the original orthogonal frame and $U^{\mathrm{s}} = U^{\mathrm{o}} \circ S_{k_j}$, where $S_k(x,t) = (x + kt, t)$. In particular $V^{\mathrm{s}} = V^{\mathrm{o}} \circ S_{k_j}$. We use the all-center extension L^1 estimate established in the proof of proposition 3.4:

$$\int_{B_{CR}(z)} |DU^{\mathrm{o}}(X,y)| dX \le CR^d \qquad (y > 0, R \ge 1).$$

Its constants are uniform in height and center. That estimate follows there from the mass-one Poisson formula for $D_X U$ and the integrable positive height-derivative envelope $|H| P_y(H)/y$; the latter has finite height-independent mass because $p > 1$. No new resolvent identity or trace-integrability hypothesis is used here.

Choose fixed smooth cutoffs $\chi(x/R)$ and $\rho((t - \tau_j)/R)$ with values in $[0,1]$, one on a containing selected inner box and zero outside a fixed larger box; choose a height cap $\upsilon(y/R)$ equal to one below R and zero above $2R$. In original coordinates put

$$\eta_R^{\mathrm{o}}(x,t,y) = \chi((x - k_j t)/R) \rho((t - \tau_j)/R) \upsilon(y/R), \qquad w_R^{\mathrm{o}} = \chi((x - k_j t)/R) \rho((t - \tau_j)/R) u^{\mathrm{o}},$$

and let W_R^{o} be the original isotropic Poisson extension of this compact trace. The original cutoff need not be a product; its gradient is bounded by C/R uniformly for bounded k_j. Its support lies in a fixed enlargement of the record receiver. Its normal center is the same scalar τ_j as the rank heights; proposition 9.6 permits the choice $\tau_j = 0$, and in particular $\tau_j = o(R_j)$. A normal translation has no effect on any Fourier norm. Under pullback by S_{k_j} the trace cutoff is exactly the selected product $\chi(x/R)\rho((t - \tau_j)/R)$.

For the next localization calculation suppress superscripts and write $U = U^{\mathrm{o}}$, $\eta_R = \eta_R^{\mathrm{o}}$, and $U_R = W_R^{\mathrm{o}}$. Let $E_R = \eta_R U - U_R$. It has zero trace and finite weighted energy. Harmonic orthogonality, followed by testing the equation for U with $\eta_R E_R$, gives exactly

$$\int |DE_R|^2 d\mu = \int U D\eta_R \cdot DE_R d\mu - \int E_R DU \cdot D\eta_R d\mu.$$

The zero-trace approximations in these integrations follow, for example, by cutoff and the one-dimensional weighted Hardy inequality $\int y^{-1-p} |E_R|^2 \le (4/p^2) \int y^{1-p} |\partial_y E_R|^2$. Both U, E_R are uniformly bounded. Young's inequality absorbs the first term with error $CR^{-2} R^{d+1} \int_0^{2R} y^{1-p} dy = CR^{d+1-p}$. For the second term use the just-proved L^1 estimate at each height and integrate $CR^{-1} R^d \int_0^{2R} y^{1-p} dy = CR^{d+1-p}$. Consequently

$$\int |DE_R|^2 d\mu \le CR^{d+1-p}, \qquad \int |D_x U_R - \eta_R V|^2 d\mu \le CR^{d+1-p}.$$

For the second estimate use $D_x U_R - \eta_R V = U D_x \eta_R - D_x E_R$. Also $|DU| \le C/y$ by the bounded-trace Poisson kernel, so the height tail over the $O(R^{d+1})$ trace-support volume gives

$$\int_{y > R} |\eta_{R,0}^{\mathrm{o}}(X) V^{\mathrm{o}}(X,y)|^2 d\mu \le CR^{d+1-p}.$$

Thus, with the original trace cutoff $\eta^{\mathrm{o}}_{R,0}$, the original Fourier field $F^{\mathrm{o}}_j = D_x W^{\mathrm{o}}_{j,R}$ satisfies $\|F^{\mathrm{o}}_j - \eta^{\mathrm{o}}_{R,0} V^{\mathrm{o}}_j\|_2^2 \le C R^{d+1-p}$. The source-space shear has determinant one, so its pullback $F^{\mathrm{s}}_j = F^{\mathrm{o}}_j \circ S_{k_j}$ obeys the identical norm bound

$$\|F^{\mathrm{s}}_j - \chi(x/R_j)\rho((t-\tau_j)/R_j) V^{\mathrm{s}}_j\|_2^2 \le C R_j^{d+1-p}.$$

Only $W^{\mathrm{o}}_{j,R}$ is assigned an isotropic equation; this last assertion is an L^2 pullback identity. All these are full-height norms. Bounded interaction gives

$$R^{d+1-p}/S \le C R^{3-p-4/(p+1)} \to 0.$$

The exponent is $-(p-1)^2/(p+1) < 0$.

We now justify the cutoff effects before using either spectral fact. On a fixed height-retained base, every comparison center obeys $|S_{j,k} - \tau_j| \le C_M D_j = o(R_j)$. Since the whole normal profile has finite norm $\|H_*\|^2_{L^2(d_p y^{1-p} dt\, dy)} = \sigma$, its normal tail gives, uniformly for these centers,

$$\|(1 - \rho((t-\tau_j)/R_j)) H_*(t - S_{j,k}, y)\|_2^2 \le \int_{|t| > cR_j/2} H_*(t,y)^2 d_p y^{1-p} dt\, dy \longrightarrow 0.$$

Here $\rho = 1$ on $|t - \tau_j| \le cR_j$ and $C_M D_j < cR_j/2$ for large j. Multiplication by the fixed base coefficient $Df_k(x/R_j)$ and Cauchy–Schwarz in the finite sum make the resulting error $o(S_j)$. First hold M fixed; then remove it using the complete-column uniform integrability, normalized-height tightness, and absolute continuity of the fixed integrable coefficient $\sum_k |Df_k|^2$. The complement of the genuine regular bases is handled by the extension argument already given above. Enlarge the containing cap in the selected full profile estimate so it is one on the support of the selected product $\chi\rho$. Combining that profile estimate, the localization bound, and removal of ρ shows in the global weighted Hilbert norm that

$$F^{\mathrm{s}}_j = a_j \chi(x/R_j) \sum_k (-Df_k(x/R_j)) \epsilon_{j,k} H_*(t - S_{j,k}(x), y) + o_{L^2}(\sqrt{S_j}).$$

This is an approximation by whole translated profiles, with no normal cutoff remaining; it is the estimate used in Fourier space.

For a fixed physical base displacement z, the actual centers satisfy $S_{j,k}(x+z) - S_{j,k}(x) \to 0$ in measure on complete regular bases. Indeed refine fixed-radius planar comparison on a window larger than $2|z|$. Its physical tilt tends to zero and its normal derivative has a positive floor, so the implicit function theorem transports the same-phase root to $x + z$ with displacement $o(1)$ and the same orientation. Completeness and order identify its rank; count/regularity exceptions tend to zero. These nonnegative errors were included in the common phase selection. The fixed coefficient functions χDf_k are in L^2 and are continuous under base translation by z/R_j. Translation of H_* is continuous in its complete normal Hilbert norm, with a uniform bound 4σ for every shift. Dominated convergence against the fixed coefficient squares, followed by uniform integrability on the exceptional bases, therefore gives

$$S_j^{-1} \|F^{\mathrm{s}}_j(x+z,t,y) - F^{\mathrm{s}}_j(x,t,y)\|_2^2 \longrightarrow 0.$$

Average in z against a smooth even compactly supported probability density, positive on a ball. The normalized translation norms have a common finite bound, so dominated convergence applies. Its multiplier $\int \zeta(z) |e^{i\xi\cdot z} - 1|^2 dz$ has a positive lower bound on $|\xi| \ge \varepsilon$ for each $\varepsilon > 0$. Plancherel proves concentration of the normalized transverse energy at $\xi = 0$.

For the second spectral fact let Π_δ be the orthogonal projection in the normal variable onto $|\tau| \le \delta$. The base factor $\chi(x/R_j)$ commutes with this projection. For $h_* = q' \in L^1$ and

$s = p/2$, the one-dimensional extension mode identity gives

$$\|\Pi_\delta H_*\|_2^2 = d_p \int_{|\tau|\le\delta} \int_0^\infty y^{1-p} |\widehat{H}_*(\tau, y)|^2 dy\, d\tau = s \int_{|\tau|\le\delta} |\tau|^{p-2} |\widehat{h}_*(\tau)|^2 d\tau \le C\delta^{p-1}.$$

Normal translations preserve the projected norm, including centers depending measurably on the base. The whole-profile approximation and the finite-sum bound hence imply

$$\limsup_j S_j^{-1} \|\Pi_\delta F_j^{\mathrm{s}}\|_2^2 \le C\delta^{p-1} \sum_k \int |\chi D f_k|^2.$$

Sending $\delta \downarrow 0$ removes low selected normal frequencies. For any fixed L, the same argument with $\delta = L/R_j$ also gives a limit zero. Multiplication by the selected normal cutoff has introduced no unestimated frequency mixing: it was removed in the complete Hilbert norm before projection.

The unitary shear now transfers these two spectral assertions to F_j^{o}. With Fourier convention $e^{-i(x\cdot\xi + t\tau)}$, its exact relation is $\tau_{\mathrm{s}} = \tau_{\mathrm{o}} + k_j \cdot \xi$ and its tangential frequency ξ is unchanged. The physical-frequency concentration at $\xi = 0$ is therefore identical in the two frames. For $|k_j| \le k_0$,

$$\{|\tau_{\mathrm{o}}| \le \delta\} \subset \{|\xi| > \delta/(1 + k_0)\} \cup \{|\tau_{\mathrm{s}}| \le 2\delta\}.$$

Apply the corresponding weighted Plancherel projections to the same field, divide by S_j, first send $j \to \infty$, and then $\delta \downarrow 0$. The first set has vanishing mass by tangential concentration and the second by the selected low-normal-frequency bound. This proves low original normal-frequency exclusion. No claim of concentration at the finer tangential scale $R_j|\xi|$ is used, and no estimate of $|k_j|/a_j$ is required.

For a compact trace w, its full extension and its normal-only extension have the exact normal-energy difference multiplier

$$k^p - |\tau|^p - s|\xi|^2 k^{p-2}, \qquad k^2 = |\xi|^2 + \tau^2.$$

The normalized mode profile $\Phi = \psi_p$ in the proof of (5) satisfies $d_p \int y^{1-p}\Phi^2 = s$ and $d_p \int y^{1-p}\Phi'^2 = 1 - s$, by multiplying its ODE by $y^{2-p}\Phi'$ and integrating. The normal harmonic extension is orthogonal to any zero-trace normal remainder, giving the displayed difference. For $\xi \neq 0$, dividing by the transverse multiplier $s|\xi|^2 k^{p-2}$ gives

$$\frac{1 - r^s - s(1 - r)}{s(1 - r)}, \qquad r = \tau^2/k^2 \in [0, 1).$$

Concavity of r^s makes this quotient nonnegative. It is bounded on $[0, 1)$, has value $(1 - s)/s$ at zero, and tends to zero at one by Taylor expansion. At $\xi = 0$ the discrepancy multiplier itself is zero. First exclude $|\tau| \le \delta$, then $|\xi| > \epsilon$, by the two spectral facts; on the remainder send $\epsilon/\delta \to 0$. More explicitly, its limsup after division by S_j is bounded by $C\delta^{p-1} + C\omega(\epsilon/\delta)$, where $\omega(r) \to 0$ as $r \downarrow 0$; the fixed-ϵ tangential-frequency error has vanished as $j \to \infty$. Send $\epsilon \downarrow 0$ at fixed δ, and then $\delta \downarrow 0$. The normalized difference is consequently $o(1)$. Apply this original isotropic Fourier identity to $w_{j,R}^{\mathrm{o}}$, then restore the actual original trace on an inner original box where $\eta_{R,0}^{\mathrm{o}} = 1$. Choose the fixed plateau margins in the containing selected box so this original inner box is contained uniformly for all sufficiently small k_j. The full-extension error there is controlled by E_R below height R_j and by the bounded-trace derivative bound C/y above it, both with cost $O(R_j^{d+1-p})$. For the normal-only extension at an inner original base, the original cutoff is one for every normal coordinate in the inner normal interval. The excluded normal trace partners are therefore at distance at least cR_j.

The possibly nonproduct cutoff remains uniformly bounded by one, which is all the following tail estimates require. For $0 < y < R_j$ their one-dimensional Poisson estimates are

$$|\partial_t P_y^{(1)} * ((1-\eta_{R,0}^{\circ})u^{\circ})| \le C y^p R_j^{-p-1}, \qquad |\partial_y P_y^{(1)} * ((1-\eta_{R,0}^{\circ})u^{\circ})| \le C y^{p-1} R_j^{-p}.$$

Squaring, integrating with y^{1-p}, and multiplying by the inner trace volume $O(R_j^{d+1})$ gives $O(R_j^{d+1-p})$. Above height R_j use again C/y. These bounds justify the whole-height local restoration, and all these errors are $o(S_j)$ by the exponent already proved.

The normal discrepancy survives the selected constant shear by a direct estimate, independent of dimension. For a compact trace write $s = p/2$, $k = (|\xi|^2 + \tau^2)^{1/2}$ and $h(r) = 1 - s + sr^2 - r^p$. The normal energy of the difference between its full extension and its normal extension is $k^p h(|\tau|/k)$ per Fourier mode. After pulling back by $x \mapsto x + \ell t$, it is $k^p h(|\tau + \ell \cdot \xi|/k)$. The positive continuous ratio $h(r)/(r-1)^2$ is bounded above and below on $[0, \infty)$, with continuous value $p(2-p)/2$ at one; the limits at zero and infinity are positive. The triangle inequality for absolute values therefore gives

$$k^p h(|\tau + \ell \cdot \xi|/k) \le C k^p h(|\tau|/k) + C|\ell|^2 |\xi|^2 k^{p-2}.$$

Plancherel proves

$$E_n(U \circ S_\ell - N_\ell) \le C E_n(U - N) + C|\ell|^2 E_x(U).$$

Here N_ℓ is the normal-only extension of the pulled-back trace, and the Fourier estimate was applied to the original isotropic extension before pullback. Apply this global Fourier estimate to the compact-trace extension $W_{j,R}^{\circ}$ constructed above. Its transverse energy satisfies $E_x(W_{j,R}^{\circ}) \le C S_j$ by the localization estimate. Restoring the actual field and its pulled-back trace on the inner receiver adds only the already controlled $C R_j^{d+1-p} = o(S_j)$ terms; bounded shears preserve the fixed margins and these local restoration estimates. Since $\ell_j \to 0$, this proves the full-height $o(S_j)$ discrepancy in the selected shear without an isotropic equation for that sheared field. This completes the proof. □

Lemma 9.8 (Complete-column integrability transports every bounded Sobolev base test)**.** *Let $\Omega \subset \mathbb{R}^d$ be bounded and open. Suppose smooth vector fields $j_j \in L^2(\Omega)$, with no common L^2 bound required, satisfy $j_j \to j$ strongly in $L^1(\Omega)$, $j \in L^2(\Omega)$, and their divergences are uniformly integrable in $L^1(\Omega)$. Let $b_j \in H_0^1(\Omega) \cap L^\infty(\Omega)$ have a common bound in both spaces and converge in measure to b. Then*

$$\int j_j \cdot D b_j \longrightarrow \int j \cdot D b.$$

For the stated L^2 fields the finite-index left sides are ordinary integrals and satisfy $\int j_j \cdot Db_j = -\int b_j \operatorname{div} j_j$. The variant with only locally bounded smooth j_j uses the right side as the definition of the finite-index pairing; its conclusion is the same. No unproved boundary integrability is implicit in that variant.

Proof. The bounded H^1 sequence has weakly convergent subsequences; convergence in measure identifies every limit as $b \in H_0^1 \cap L^\infty$. Dunford–Pettis gives weakly convergent subsequences of $\operatorname{div} j_j$ in L^1. Strong current convergence identifies every distributional limit as $\operatorname{div} j$, hence that divergence is an L^1 function and every weakly convergent divergence subsequence has this limit. For clarity, if $|b_j|, |b| \le B$, then for each $\delta > 0$,

$$\left| \int (b_j - b) \operatorname{div} j_j \right| \le \delta \sup_j \| \operatorname{div} j_j \|_1 + 2B \int_{\{|b_j - b| > \delta\}} |\operatorname{div} j_j|.$$

The last sets have measure tending to zero, so uniform integrability makes their integrals tend to zero; afterward send $\delta \downarrow 0$. Hence $\int (b_j - b) \operatorname{div} j_j \to 0$. Weak L^1 convergence pairs with the

fixed bounded function b. Thus integration by parts gives the asserted limit. On the limiting side, bounded smooth approximation to b converging strongly in H^1 and almost everywhere justifies the pairing with $j \in L^2$ and $\operatorname{div} j \in L^1$. Here is the bounded density argument used in these integrations by parts, valid on an arbitrary open Ω. For $b \in H^1_0(\Omega) \cap L^\infty$, choose $\phi_\ell \in C^\infty_c(\Omega)$ converging strongly in H^1 and, after a subsequence, almost everywhere to b. Set $L = \|b\|_\infty + 1$ and $T_L(r) = \max\{-L, \min\{r, L\}\}$. Then $T_L(\phi_\ell) \to b$ strongly in H^1: the value convergence follows by the contraction property, while

$$DT_L(\phi_\ell) - Db = \mathbf{1}_{\{|\phi_\ell|<L\}}(D\phi_\ell - Db) - \mathbf{1}_{\{|\phi_\ell|\geq L\}} Db$$

converges to zero in L^2 by strong gradient convergence and dominated convergence, since $|b| < L$. The possible level sets cause no extra derivative by Sobolev locality. Each truncated function is compactly supported in Ω. Convolve it with a nonnegative smooth approximate identity of radius smaller than its support distance to $\partial\Omega$, choosing that radius so the H^1 approximation error is below $1/\ell$. The resulting $b_\ell \in C^\infty_c(\Omega)$ satisfy $|b_\ell| \leq L$, converge strongly in H^1, and have an almost-everywhere convergent subsequence. Thus $\int j \cdot Db_\ell \to \int j \cdot Db$ by $j \in L^2$, whereas $\int b_\ell \operatorname{div} j \to \int b \operatorname{div} j$ by dominated convergence against the L^1 divergence. The identical argument applies at every finite index in the stated L^2 case. For the locally bounded variant, the same bounded approximants instead show that the dual pairing $-\int b_j \operatorname{div} j_j$ is the limit of the compact-test pairings, independently of the approximating sequence. This proves the asserted interpretation without a boundary regularity assumption.

The subsequence criterion gives convergence of the whole sequence. □

Proposition 9.9 (One all-phase source decomposition is compatible with the strong count screen)**.** *Use definition 9.1 and proposition 9.6 on a fixed containing macroscopic trace box. Fix $I = (-\vartheta, \vartheta)$ with $\sqrt{2/3} < \vartheta < 1$, and select the actual trace phase c_j in a fixed compact subinterval of $(-1/2, 1/2)$. Write $m_c = |Du_j|\mathbf{1}_{\{|u_j|<\sqrt{2/3}\}}$. There are nonnegative sources m_*, r_* with $m_c = m_* + r_*$ in this box such that*

$$\|r_*\|_1 = o(S_j), \qquad \int m_*(R_j\xi, t)dt \leq Cq_j(\xi)^2, \quad q_j \longrightarrow \sqrt{m|I|} \quad \text{strongly in } H^1_{\mathrm{loc}}.$$

For almost every source point with $m_* > 0$, there is an actual complete c_j root at the same base within a fixed normal distance C_I, with C_I independent of j and the point. Where the regularity cutoff of proposition 9.6 is positive, that root can be chosen from its selected ordered m-root family: almost every surviving source (x,t) there satisfies $|t - T_{j,k}(x, c_j)| \leq C_I$ for some selected $k \in \{1, \dots, m\}$. The same assertion holds after imposing any of the nested regularity, height, count, and separation factors on that support; zero-weight extra actual roots do not replace a selected root in this proximity conclusion. The detector-error trace volume is $o(S_j)$ and the full central gradient has a fixed positive floor elsewhere. The construction uses a single exceptional base set for all phases in I. In the outer box it is compatible with every nested regularity, height, count and separation factor; excluded bases are paid by proposition 7.5. Sources outside a fixed larger macroscopic box have the positive cap bound $CR_j^{d+1-p} = o(S_j)$. More generally, if a tested trace corridor has volume at most $CR_j^d\ell_j$, $1 \ll \ell_j \leq CR_j$, and Euclidean distance at least $c\ell_j$ from every surviving source on the outer retained bases, its complete positive energy obeys

$$\int_{\text{corridor}} \theta_j(x/R_j)^2 \left[W(u_j) + \int_0^\infty d_p y^{1-p} |DU_j|^2 dy\right] dX \leq CR_j^d \ell_j^{1-p} + o(S_j).$$

Here $\theta_j = \chi_j \psi(q_j/L)$, where $0 \leq \chi_j \leq 1$ is a nested base product supported in one fixed buffered receiver, $\sup_j \|\chi_j\|_{H^1} < \infty$, and $L > \sqrt{m|I|}$ is fixed. The excluded normalized outer source-base set E_j must satisfy $E_j \subset \{\theta_j = 0\}$ up to null sets, as holds for the nested factors

used below. All such excluded source bases are included in the $o(S_j)$ term. The estimate holds in the selected small shear with uniformly equivalent kernels.

Proof. The bounded-data Jacobi resolvent identities, their derivative convolutions, and comparison of the bounded gradient Kato subsolution below follow from lemma 2.3. The actual cubic derivatives and the multiplying potential are bounded; subsequent source mass estimates are used only for quantitative bounds, not to justify this representation.

The uniform physical smoothness used in this proof is supplied by lemma 2.1. The positive massive resolvent kernels, their derivative bounds, and their convolution estimates are those of lemma 2.2, in the stated Fourier normalization. These apply to the actual cubic trace and, in one dimension, to its normalized layer.

Use the record of definition 9.1. The complete count and all compact-phase tilt statements used here are the full-phase conclusions of proposition 9.6, applied to the fixed interval $I = (-\vartheta, \vartheta)$. In particular the complete count tends to m on this interval, so the count-restoration estimates below use the entire interval, including its portions outside the support of m_c. Choose a fixed compact phase buffer $I \Subset I' \Subset (-1, 1)$ and fixed nested product receivers. The common-cap and whole-central-set confinement assertion of proposition 5.2 applies on every containing enlarged base receiver: all points with $u_j \in I'$ lie in $|t| = o(R_j)$, while the normal caps at fixed heights $\pm cR_j$ avoid I'. All root lists below use these full normal windows. This supplies actual phase-free normal caps, not artificial endpoints of selected tracks.

Here is also the positive gradient envelope used below. Put $g = |Du|$, $A = (-\Delta)^{p/2}$, and $f(u) = u - u^3$, so $f'(u) = 1 - 3u^2$. For $g_\varepsilon = (|Du|^2 + \varepsilon^2)^{1/2}$, convexity of the Euclidean norm gives, first with the symmetric kernel truncated away from the diagonal,

$$Ag_\varepsilon \le f'(u)|Du|^2/g_\varepsilon.$$

The truncation limit is valid because the smooth bounded derivatives supplied by lemma 2.1 control the local second differences and the fractional tail is integrable. Hence

$$(A+1)g_\varepsilon \le (2 - 3u^2)g_\varepsilon + 2\varepsilon.$$

Testing with compact nonnegative functions and sending $\varepsilon \downarrow 0$ is justified by $g_\varepsilon \to g$ uniformly and $A\phi \in L^1$, as in lemma 2.3. It gives, including the zero set of Du,

$$(A+1)g \le (2 - 3u^2)g \le 2g\mathbf{1}_{\{|u|<\sqrt{2/3}\}} = 2m_c \quad \text{in distributions.}$$

The last inequality uses the nonpositive coefficient on the complement of the stated central set. Bounded-subsolution comparison in lemma 2.3 therefore proves $g \le 2G_1 * m_c$. This proof does not divide by g at its zeros.

We record explicitly how this trace envelope controls every derivative of the actual extension. Write $n = d + 1$. At every fixed $y > 0$, differentiating the Poisson formula in the physical trace variables gives $D_X U = P_y * Du$, because Du is uniformly bounded and P_y is integrable. Thus

$$|D_X U| \le P_y * g \le 2(P_y * G_1) * m_c. \tag{EP}$$

For the height derivative use the scaling identity

$$\partial_y P_y(H) = -y^{-1} \operatorname{div}_H(HP_y(H)).$$

It follows by differentiating $P_y(H) = y^{-n}P_1(H/y)$. Since $\partial_y P_y \in L^1$ at fixed positive height and u is bounded, it first gives $U_y = (\partial_y P_y) * u$. Integrate the divergence formula by parts with a smooth cutoff equal to one on B_L and zero outside B_{2L}. The boundary error before division by y is at most $C\|u\|_\infty y^p L^{-p}$, since $|HP_y(H)| \le Cy^p|H|^{1-n-p}$ on that annulus. It tends to

zero as $L \to \infty$ for each fixed y. Moreover $\int |H| P_y(H) dH = C_p y < \infty$, exactly because $p > 1$, so bounded Du permits dominated convergence in the remaining term. Consequently

$$U_y(X,y) = -\int_{\mathbb{R}^n} \frac{H}{y} P_y(H) \cdot Du(X-H) dH, \qquad |U_y| \le \mathcal{Q}_y * g \le 2(\mathcal{Q}_y * G_1) * m_c. \quad \text{(EQ)}$$

Here $\mathcal{Q}_y(H) = |H| P_y(H)/y$ is precisely the positive kernel in proposition 7.5. Its mass is finite and independent of height. Tonelli applies to the nonnegative convolutions in (EP)–(EQ), and gives the full gradient majorant

$$|DU|^2 \le C \left[((P_y * G_1) * m_c)^2 + ((\mathcal{Q}_y * G_1) * m_c)^2 \right].$$

The complete positive norms and mixed products established below therefore control the complete actual extension energy, including its height derivative. These are original orthogonal-coordinate identities; their selected-frame versions are the constant pullbacks described next.

All isotropic resolvent, Poisson and conormal formulas are first proved in the original orthogonal record coordinates. In a selected shear $S_k(x,t) = (x + kt, t)$, pull back each convolution kernel to its composition with S_k and pull back gradients by the constant derivative matrix. Its determinant is one and its norm and inverse norm tend to one. Thus the two gradient norms, source masses, distances, and kernel annuli are uniformly equivalent. The same formulas with these pulled-back kernels give the required inequalities for the selected-frame source, with uniform constants. No isotropic equation is assigned to the sheared field.

Work with the actual fine record definition 9.1 and the actual complete roots, full ratio tail and strong low-angle slopes of proposition 9.6. Choose the fixed full-gradient-floor physical detector included in the quantitative cutoff construction of proposition 9.6. It is the detector from proposition 3.4 with window and accuracy depending only on the compact phase interval I, and gives $|Du_j| \ge c_I > 0$ on that band outside its bad balls. Its number of bad balls is at most $C_I a_j^2 R_j^{d-2}$. Write χ_j^{src} for a bounded-overlap physical cutoff, distinct from the base factors $\chi_j(\xi)$ in the statement. It is zero near those balls and one outside fixed enlargements, and has excluded trace volume and squared gradient integral at most $C_I a_j^2 R_j^{d-2} = o(S_j)$. The detector window and accuracy in this source cutoff remain fixed, so its constants and the floor c_I are independent of j.

First fix a physical angle $0 < \kappa < 1$. On the detector-good central set, $|D_x u| > \kappa |u_t|$ implies $|D_x u| \ge c_I \kappa / \sqrt{1+\kappa^2} \ge c_I \kappa/2$. Uniform physical derivative bounds give a radius $\rho_\kappa > 0$ on which the actual tangential extension derivative has norm at least $c_I\kappa/4$ at heights in $(\rho_\kappa/2, \rho_\kappa)$. All normal extension derivatives there are bounded by a fixed constant. Hence, for any fixed normalized ratio threshold K, this neighborhood lies in $|V_j| > K a_j |H_j|$ for large j. A bounded-overlap covering of the trace set by such balls, using the corresponding positive weighted bulk energy on each ball, bounds its trace volume, and thus its central source mass, by C_κ times the complete ratio-tail energy. The ratio-tail conclusion of proposition 9.6, followed first by $j \to \infty$ and then $K \to \infty$, proves that this physical high-angle mass is $o_\kappa(S_j)$. Use the additional-diagonal freedom exported by proposition 9.6 to impose these fixed-angle source estimates in the same angle sequence κ_j used by its low-pass and annular count construction. Include the physical high-angle set beginning at κ_j^2, all fixed detector refinements, and the countable receiver tests before the generic direction and phase are selected. Thus that mass is $o(S_j)$, $a_j/\kappa_j^8 \to 0$, and the source and rank low-pass cutoffs have the same positive angular support. This is one joint construction, without replacing an already fixed angle sequence. The fixed-angle estimate can be imposed uniformly before the shear is selected: if $h^{\mathrm{sh}} = h^{\mathrm{o}} + k_j \cdot v^{\mathrm{o}}$, then for $0 < \kappa \le 1$ and $|k_j| \le 1$, $|v^{\mathrm{o}}| > \kappa |h^{\mathrm{sh}}|$ implies $|v^{\mathrm{o}}| > (\kappa/2)|h^{\mathrm{o}}|$. Impose the original-frame estimates at the corresponding thresholds $\kappa_j^2/2$ in the same slow diagonal. The determinant-one source change and uniform norm equivalence

then transfer them to the selected frame. All remaining source estimates below are uniform for every c_j in the prescribed central interval, so no new phase selection is made here.

Use $\Psi_j = \psi_j$, the same common logarithmic angle cutoff exported by proposition 9.6: it is one for $|D_x u|/|u_t| \le \kappa_j^2$, strictly positive for angle below κ_j, and zero at angle at least κ_j. Put $b_j = \chi_j^{\mathrm{src}} \Psi_j$. On $b_j > 0$ the full gradient floor and $\kappa_j < 1/2$ give $|u_t| \ge c_I/2$. Thus these source tracks are genuine implicit roots throughout their positive phase intervals. Here the angular density is defined from the actual trace vector by

$$Z_0 = (D_x u_j, a_j (u_j)_t), \qquad \mathcal{A}_{Z_0} = |D_X Z_0|^2 - |D_X |Z_0||^2 \ge 0.$$

Its value on $\{Z_0 = 0\}$ is taken to be zero, which agrees almost everywhere with the derivative expression by Sobolev locality. The trace angular estimate of lemma 9.2 is $\int \mathcal{A}_{Z_0} \le C S_j / R_j^2 = C a_j^2 R_j^{d-2}$ on each containing receiver, including the selected constant small shear. With this definition, the derivative identity for the angular metric and that cited estimate give

$$\int dx \int_I \sum_{\text{tracks}} |\partial_c b_j|^2 dc = \int_{\{u \in I,\, b_j > 0\}} \frac{|\partial_t b_j|^2}{|u_t|} dX \le C a_j^{-2} \int \mathcal{A}_{Z_0} + C \int |D \chi_j^{\mathrm{src}}|^2 \le C R_j^{d-2}.$$

To check this estimate explicitly, write $v = D_x u$, $h = u_t$, $r = |v|/|h|$, $Z_0 = (v, ah)$ and $n = Z_0/|Z_0|$. On the transition $\kappa_j^2 < r < \kappa_j$, the cutoff can be chosen smooth in $\log r$ with $|r \Psi_j'(r)| \le C/|\log \kappa_j|$. Since $a_j/\kappa_j^2 \to 0$, differentiation on the target sphere gives

$$|D_X \Psi_j|^2 \le \frac{C \mathcal{A}_{Z_0}}{a_j^2 h^2 |\log \kappa_j|^2}.$$

Indeed $r = a_j |n_x|/|n_{d+1}|$ and the squared spherical gradient of $\log r$ is

$$(1 + |v/(a_j h)|^2)^2 / |v/(a_j h)|^2.$$

Dividing by $|Z_0|^2$ gives at most $C/(a_j^2 h^2)$. The normal-speed floor on $b_j > 0$ bounds every remaining power of $|h|^{-1}$. The derivative of χ_j^{src} contributes at most $C \int |D \chi_j^{\mathrm{src}}|^2$. This proves the displayed phase derivative estimate, in fact with the extra factor $|\log \kappa_j|^{-2}$ on its angular term.

A positive-amplitude track has $|h| \ge c_I/2$, so it is an implicit c-graph with $|dt/dc| \le 2/c_I$. At a finite interior endpoint its height has a unique limit; if the limiting amplitude were positive, smoothness and this floor would continue the implicit root and its amplitude, a contradiction. Its amplitude therefore tends to zero. The common caps from proposition 5.2 exclude an endpoint on a normal boundary for $c \in I$. Alternatively, enlarge an inner normal window by $2|I|/c_I + 2$ before defining the tracks: the bound on dt/dc prevents any source track from reaching this outer boundary before c_j. Spatial base boundaries cause no additional phase endpoint because this continuation is at fixed x. All tested compact bases have the prescribed larger base buffer.

If a source point of amplitude $b_j = 1$ cannot be continued along its positive track to c_j, that track has a zero-amplitude endpoint between its phase and c_j. For almost every base the preceding derivative bound makes the amplitude H^1 on each such phase interval, so the fundamental theorem of calculus and Cauchy–Schwarz charge at least $1/|I|$ to its phase derivative energy. To choose one measurable exceptional set, put

$$G_j(x) = \int_I \sum_{\text{tracks}} |\partial_c b_j(x,c)|^2 dc, \qquad E_{\mathrm{ph}} = \{x : G_j(x) \ge |I|^{-1}\},$$

including the null set on which this trackwise description fails. Equivalently, normal coarea gives $G_j(x) = \int_{\{u_j(x,t) \in I,\, b_j(x,t) > 0\}} |\partial_t b_j(x,t)|^2 / |u_{j,t}(x,t)|\, dt$, so G_j is measurable, and Chebyshev gives $|E_{\mathrm{ph}}| \le |I| \int G_j \le C R_j^{d-2}$. This set contains every failed full-amplitude track base

up to null sets and works simultaneously for all starting phases and all target phases in I; no union of separately phase-selected exceptions is taken. Every remaining full-amplitude source point reaches an actual complete c_j root, at normal distance at most $|I|/(c_I/2)$. On a positive final regular support, after the refined-detector source deletion described below, its nonzero low-pass root weight is forced to one by the annular count construction. It is therefore one of the selected m roots. Sources with an excluded regular base are handled by the count-adapted source-partner screen, rather than by an assertion of small capacity for every complete-count defect.

We derive the Sobolev count hypotheses for these particular amplitudes on all of I. Let $N_j(x,c)$ count every simple root in the complete normal window, and define

$$Q_j(\xi,c)=\left(\sum_{u_j(R_j\xi,t)=c} b_j(R_j\xi,t)^2\right)^{1/2}.$$

Only the positive-amplitude roots contribute. Their speed has a fixed positive floor; Rolle's theorem and the uniform bound for u_{tt} give a fixed minimum separation between any two such same-phase roots. Hence their number in the finite normal window is bounded at each fixed index. For $\epsilon>0$, every root with $b_j\ge\epsilon$ is simple and persists on a neighborhood in base–phase space. On a compact subproduct the corresponding root subset is compact, so finitely many implicit charts suffice. In those charts use $(b_j-\epsilon)_+$, and extend a disappearing term by zero: it vanishes at the boundary $b_j=\epsilon$, and the same continuation argument in a horizontal direction shows that an interior chart cannot end with $b_j>\epsilon$. The common caps contain no tested roots. The ordered real root heights identify overlapping charts, so the sums agree there; the zero extensions have no boundary measure. The Euclidean norm chain rule, followed by weak compactness and monotone L^2 convergence as $\epsilon\downarrow 0$, proves the weak Sobolev derivative estimate

$$|D_\xi Q_j|^2\le R_j^2\sum_{u=c}|\mathcal{T}b_j|^2,\qquad \mathcal{T}_i=\partial_{x_i}-(v_i/h)\partial_t.$$

This also explains why a disappearing track creates no uncharged singular derivative. Equivalently one may first regularize the finite Euclidean norm by $(\epsilon^2+\sum b^2)^{1/2}-\epsilon$; its derivative inequality is uniform. Normal coarea and $|v/h|\le\kappa_j$ on the positive support give

$$\int_{\Omega\times I}|D_\xi Q_j|^2d\xi dc\le R_j^{2-d}\int_{\{u\in I,b_j>0\}}|h|\,|\mathcal{T}b_j|^2dX\le C(|\log\kappa_j|^{-2}+a_j^2).$$

For the angular term insert the preceding bound for $D\Psi_j$, use the floor for $|h|$, and apply lemma 9.2, whose physical integral is $Ca_j^2R_j^{d-2}$. The detector term is bounded by $CR_j^{2-d}\int|D\chi_j^{\mathrm{src}}|^2\le Ca_j^2$. These calculations do not restrict to the support of m_c.

The local Robin trace estimate proved in lemma 9.2, applied on a fixed containing receiver, gives

$$\int_{B_{CR_j}}|v_j|^2dX\le C\int_{B_{C'R_j}\times(0,2)}|V_j|^2d\mu\le CS_j.$$

Its constants are independent of j, and the bounded shear preserves this estimate. We use it for count restoration below.

Normal coarea now gives exactly

$$\int_{\Omega\times I}(N_j-Q_j^2)d\xi dc=R_j^{-d}\int_{\{u\in I\}}(1-b_j^2)|h|dX.$$

Outside the detector enlargement its integrand is supported where $r>\kappa_j^2$. The full gradient floor gives $|v|\ge c_I\kappa_j^2/2$ there, including the case $h=0$, and bounded h then gives $|h|\le$

$C_I\kappa_j^{-4}|v|^2$. On the detector enlargement use bounded h and its volume $Ca_j^2R_j^{d-2}$. Hence

$$0 \le Q_j^2 \le N_j, \qquad \int (N_j - Q_j^2) \le C(a_j^2\kappa_j^{-4} + a_j^2R_j^{-2}) \le C(a_j^2\kappa_j^{-4} + a_j^2).$$

Every integral here is over the full phase interval I, even where $m_c = 0$. The complete counts tend to m there by proposition 9.6. The just-proved bounds therefore verify, rather than assume, all hypotheses of lemma 7.1. That lemma and the Hilbert norm chain rule yield

$$Q_j \longrightarrow \sqrt{m} \quad \text{in } L^2(I; H^1_{\mathrm{loc}}), \qquad q_j(\xi) := \left(\int_I Q_j(\xi, c)^2 dc\right)^{1/2} \longrightarrow \sqrt{m|I|} \quad \text{strongly in } H^1_{\mathrm{loc}}.$$

Normal phase coarea on $b_j > 0$ gives

$$\int b_j^2 m_c(R_j\xi, t)dt \le C \int_I Q_j(\xi, c)^2 dc = Cq_j(\xi)^2,$$

since $|Du|/|u_t| \le \sqrt{1+\kappa_j^2}$. Apply lemma 8.2 to this source and E_{ph}. Its mass is at most $CR_j^{d-4/d} = o(S_j)$; the required lower bound $S_j \ge cR_j^{d-2+4/(p+1)}$ follows from the bounded interaction ratio in definition 9.1. The strict inequality $2 - 4/d - 4/(p+1) < 0$ holds for $d \le 6$ and $p < 2$.

Apply the quantitative detector localization exported by proposition 9.6 with an inner physical receiver containing the source box and its fixed $C_I + 1$ neighborhood, and with the larger macroscopic buffer already fixed above. Let $\mathcal{B}_j$ be its union of fixed-radius balls covering $\{\chi_{0,j} < 1\}$ on the inner receiver. Its detector family includes the fixed floor detector used to define χ_j^{src} here. Take

$$\mathcal{R}_j = \{X : \mathrm{dist}(X, \mathcal{B}_j) < C_I + 1\}, \qquad C_I = 2|I|/c_I.$$

All radius and transition-annulus enlargements are contained in this fixed-radius family after increasing its fixed C_0 once. The exported localization gives

$$|\mathcal{R}_j| \le C_I' C_j a_j^2 R_j^{d-2}, \qquad \int_{\mathcal{R}_j} m_c \le C_I' C_j a_j^2 R_j^{d-2} = o(S_j).$$

The same refinement stage is chosen slowly enough that $C_j a_j \to 0$ and $C_j/R_j^2 \to 0$. A successful positive source track has $|dt/dc| \le 2/c_I$ and total phase length at most $|I|$. If its final root has $\chi_{0,j} < 1$, its source point lies in $\mathcal{R}_j$. Hence every surviving source outside $\mathcal{R}_j$ completes to a root with $\chi_{0,j} = 1$. The positive angular support is the same common ψ_j for source and root amplitudes, so this endpoint has $b_{0,j} > 0$. On a positive regular support the exported annular property makes it one of the selected m roots. This deletion changes neither the fixed source cutoff b_j nor its phase-derivative estimate, so $|E_{\mathrm{ph}}| \le CR_j^{d-2}$ retains its literal constant independent of the refinements.

Define

$$m_* = m_c \mathbf{1}_{\{b_j = 1\}} \mathbf{1}_{\{x \notin E_{\mathrm{ph}}\}} \mathbf{1}_{\{X \notin \mathcal{R}_j\}}, \qquad r_* = m_c - m_*.$$

The transition of b_j is contained in the fixed detector enlargement or the physical high-angle set beginning at κ_j^2. Those masses, the failed-track mass, and the additional refined-detector source mass have all been proved $o(S_j)$. The extra deletion only decreases the source column. This proves the asserted source decomposition and its column bound.

We prove the complete positive kernel-product estimate needed to discard this mass, including its behavior at the height endpoints. Put $n = d + 1$, $\nu = p - 1 \in (0, 1)$, and

$$\mathcal{K}_y(Z) = \frac{y^\nu}{(y + |Z|)^{n+\nu}}, \qquad \mathcal{I}(Z) = |Z|^{-\alpha}, \quad \alpha = n + p - 2 \in (0, n).$$

The positive kernels in proposition 7.5 obey

$$P_y(Z) \le C\mathcal{K}_y(Z), \qquad \mathcal{Q}_y(Z) = \frac{|Z|}{y} P_y(Z) \le C\mathcal{K}_y(Z).$$

These follow directly from their displayed formulas and the comparability of $(|Z|^2+y^2)^{1/2}$ and $|Z|+y$. The mass of $\mathcal{K}_y$ is a fixed finite constant: rescale $Z = y\zeta$ and integrate $(1+|\zeta|)^{-n-\nu}$, whose radial tail is $r^{-1-\nu}$. For $r = |Z| > 0$, every integration point X has $|X| \ge r/2$ or $|Z-X| \ge r/2$. On the first region $\mathcal{K}_y(X) \le C\mathcal{K}_y(Z)$ and the second factor has bounded mass; on the second region reverse the two factors. Thus

$$\mathcal{K}_y * \mathcal{K}_y(Z) \le C\mathcal{K}_y(Z).$$

For either $A_y, B_y \in \{P_y, \mathcal{Q}_y\}$, their radial symmetry and Tonelli's theorem now give

$$\begin{aligned} \int_0^\infty y^{1-p} \int_{\mathbb{R}^n} A_y(X-Z) B_y(X-W) dX dy &\le C \int_0^\infty (y + |Z-W|)^{-n-\nu} dy \\ &= \frac{C}{n+\nu-1} |Z-W|^{2-n-p} = C'\mathcal{I}(Z-W). \end{aligned} \tag{KP}$$

In particular, at unit source separation the height integrand is bounded near zero and is $O(y^{-n-\nu})$ at infinity; both are integrable. This verifies the finiteness that homogeneity alone would not supply. The possible singular value on $Z = W$ causes no problem after source integration because $\alpha < n$.

The same estimate holds after either kernel is convolved with any of the fixed massive resolvents G_1, G_2 from lemma 2.2. Indeed those kernels are positive, radial, have finite mass, and satisfy $G_\lambda(Z) \le C|Z|^{p-n}$ below radius one and $G_\lambda(Z) \le C|Z|^{-n-p}$ above it, for $\lambda \in \{1,2\}$. We check

$$G_\lambda * \mathcal{I}(Z) \le C\mathcal{I}(Z). \tag{KR}$$

Write $r = |Z|$. On $|Z-W| \ge r/2$, use $\mathcal{I}(Z-W) \le Cr^{-\alpha}$ and the finite massive mass. On $|Z-W| < r/2$, one has $|W| \asymp r$, while $\int_{|V|<r/2} |V|^{-\alpha} dV = Cr^{n-\alpha}$. For $r \le 1/2$ this part is at most $Cr^{p-\alpha} \le Cr^{-\alpha}$; for $r \ge 2$ it is at most $Cr^{-p-\alpha} \le Cr^{-\alpha}$. The intermediate compact annulus has the same bounded estimate because $|W|$ stays away from zero in this part. This proves (KR). Tonelli and translation invariance express the left side of (KP) with massive factors as the convolution of its original source-difference kernel with G_λ, or with $G_\lambda * G_\mu$ if both factors have a massive convolution. Apply (KR) once or twice. Consequently (KP) holds, with a fixed constant, for all the Poisson and positive conormal envelopes actually used here, including their G_1 and G_2 convolutions.

Bounded $|Du|$ and its all-center sharp BV bound give the uniform Riesz row

$$\sup_Z \int_{\mathbb{R}^n} \mathcal{I}(Z-W) m_c(W) dW \le C. \tag{KB}$$

Below distance one the radial bound is $C\int_0^1 r^{n-1-\alpha} dr = C\int_0^1 r^{1-p} dr < \infty$. On the annulus $2^k \le |Z-W| < 2^{k+1}$, $m_c \le |Du|$ has mass at most $C2^{k(n-1)}$ and the contribution is at most $C2^{k(1-p)}$, summable since $p > 1$. Thus the row is uniform in Z, j even though the complete source need not have finite global mass.

Extend the deleted source r_* by zero outside its defining containing box, so $0 \le r_* \le m_c$ globally and $\|r_*\|_1 = o(S_j)$. If $\mathcal{A}_y, \mathcal{B}_y$ are any two of the positive kernels just covered, (KP), (KB), and Tonelli imply the direct mixed estimate

$$\int_{\mathbb{R}^n} \int_0^\infty y^{1-p} (\mathcal{A}_y * r_*)(X) (\mathcal{B}_y * m_c)(X) dy dX \le C \int r_*(Z) (\mathcal{I} * m_c)(Z) dZ \le C\|r_*\|_1 = o(S_j). \tag{KD}$$

Replacing the second source by $r_* \le m_c$ proves its squared complete positive norm is $o(S_j)$ as well. Thus both deleted-source squares and their mixed products with the entire source are negligible on every receiver and for all bounded test factors. The proof does not require a finite global squared norm of the entire source, and therefore does not lose a factor by Cauchy–Schwarz against such a norm. The factor d_p in the actual weighted energy is a fixed positive constant and changes none of these bounds.

The all-center BV annuli give $P_y * m_c, \mathcal{Q}_y * m_c \le C/(1+y)$: at $y \le 1$ use bounded m_c and the finite kernel masses; at $y \ge 1$, the ball of radius y contributes C/y, while the exterior annuli contribute respectively $Cy^{-1}2^{-k(p+1)}$ and $Cy^{-1}2^{-kp}$. Positivity and the finite massive masses preserve this uniform row bound after a G_1 or G_2 convolution. For the raw Poisson and conormal kernels, the proved source-column bound therefore checks every hypothesis of proposition 7.5. On a bounded outer source receiver, choose a fixed compact base cutoff ζ equal to one there and replace q_j outside that receiver by ζq_j; this gives a global H^1 majorant with uniformly bounded norm, without changing the column bound on the source under consideration. The receiver factor is zero on every excluded outer source base. That proposition consequently bounds those excluded-source squares by $CR_j^{d+1-p}(1+\log R_j) = o(S_j)$, since

$$\frac{R_j^{d+1-p}(1+\log R_j)}{S_j} \le CR_j^{-(p-1)^2/(p+1)}(1+\log R_j) \longrightarrow 0.$$

The additional rows for massive Poisson envelopes and trace-potential envelopes are verified below. Sources outside the outer receiver have macroscopic distance from the inner tested box and will be included in the final distant-source estimate.

We justify the potential part of the corridor estimate as well. Put $n = d+1 \ge 4$, $A = (-\Delta)^{p/2}$, and $\beta = 3(1-u^2)$. The exact bounded vector resolvent identity from lemma 2.3 is $Du = G_2 * (\beta Du)$. Define K_i by its multiplier

$$\widehat{K_i}(\xi) = -i\xi_i \frac{|\xi|^{p-2}}{|\xi|^p + 2}.$$

We prove the integrable near and far bounds for this kernel, including the low-frequency cancellation. Choose a smooth annular partition $\sum_{j\in\mathbb{Z}} \varphi(2^{-j}\xi) = 1$ for $\xi \ne 0$, and let $K_{i,j}$ have multiplier $\varphi(2^{-j}\xi)\widehat{K_i}(\xi)$. On the fixed annulus $1/2 \le |\eta| \le 2$, rescaling gives

$$\widehat{K_i}(2^j\eta) = -i\frac{2^{j(p-1)}\eta_i|\eta|^{p-2}}{2 + 2^{jp}|\eta|^p} \quad (j \le 0),$$

where every η derivative is bounded by $C_N 2^{j(p-1)}$. For $j \ge 0$ factor 2^{jp} from the denominator instead; every η derivative is then bounded by $C_N 2^{-j}$, because $|\eta|^p + 2^{1-jp}$ is bounded below uniformly. Integration by parts N times in the inverse Fourier integral therefore gives

$$|K_{i,j}(X)| \le C_N \begin{cases} 2^{j(n+p-1)}(1+2^j|X|)^{-N}, & j \le 0, \\ 2^{j(n-1)}(1+2^j|X|)^{-N}, & j \ge 0. \end{cases}$$

Here N can be any fixed integer larger than $n+p+2$. This estimate uses the actual annular multiplier and its derivatives, so the odd cancellation is retained before any absolute value is taken. If $r = |X| \le 1$, the sum over $j \le 0$ is bounded. Split the remaining sum at $2^j r = 1$; its two geometric series are at most Cr^{1-n}. If $r \ge 1$, the $j \ge 0$ sum is $O(r^{-N})$. Splitting the $j \le 0$ sum at $2^j r = 1$ gives Cr^{1-n-p} on each side. Thus the series converges absolutely for $X \ne 0$ and defines an integrable kernel with

$$|K_i(X)| \le Cr^{1-n} \quad (0 < r \le 1), \qquad |K_i(X)| \le Cr^{1-n-p} \quad (r \ge 1).$$

The same estimate also gives

$$\sum_j \|K_{i,j}\|_1 \le C\left(\sum_{j\le 0} 2^{j(p-1)} + \sum_{j\ge 0} 2^{-j}\right) < \infty.$$

Thus the convergence is in L^1 as well. The Fourier multiplier of its sum is the sum of the annular multipliers, which equals the specified multiplier away from zero and also at zero by continuity, since $p-1>0$. This establishes the far estimate without applying an absolute convolution to a signed differentiated Riesz kernel.

For completeness the multiplier identity also represents the bounded actual field. Write $T_i = (-\partial_i)(-\Delta)^{(p-2)/2}$. Its odd homogeneous kernel has size $C|Z|^{1-n-p}$; applying it to a bounded C^1 function is legitimate by subtracting its value near zero, since the resulting local radial bound is Cr^{1-p}, while its tail is integrable because $p>1$. On smooth bounded u the identity $Au = \sum_i T_i \partial_i u$ follows first for compact smooth fields from the Fourier symbols and then by multiplying u by expanding smooth cutoffs. Their local second derivatives are uniformly bounded and converge locally, while boundedness controls the fractional tails; the same local first-difference and far-tail bounds apply to $T_i\partial_i u$. Next truncate the bounded forcing $\beta\partial_i u$ by such cutoffs. Its G_2 convolutions and their first derivatives converge locally, by $G_2, DG_2 \in L^1$, with common global bounds. The preceding first-difference and tail estimates therefore pass T_i to the limit. For the compact forcing the Fourier identity $T_i G_2 = K_i$ applies; on the right $K_i \in L^1$ passes the same truncation limit. Hence

$$f(u) = Au = \sum_i K_i * (\beta\partial_i u)$$

for the bounded actual solution as well.

Take absolute values only now, and use the already proved Kato envelope $|\beta Du| \le 6G_1 * m_c$. Set $\mathcal{L} = 6\sum_i |K_i| * G_1$. Then

$$|f(u)| \le \mathcal{L} * m_c, \qquad \mathcal{L}(X) \le Cr^{1-n}\ (r=|X|\le 1), \quad \mathcal{L}(X) \le Cr^{1-n-p}\ (r=|X|\ge 1).$$

These are radial-majorant bounds; $\mathcal{L}$ itself need not be radial. Here are the convolution bounds explicitly. For $r \le 1/4$, the region $|Z| \le 2r$ contributes Cr^{p+1-n}, after scaling the two integrable local singularities. On $2r < |Z| < 2$ the radial integral is bounded by $C\int_{2r}^2 \rho^{p-n}d\rho$, and the remaining region is bounded. Since $n \ge 4 > p+1$, these bounds are at most $Cr^{p+1-n} \le Cr^{1-n}$. For $r \ge 4$, split into $|Z| < r/2$, $|X-Z| < r/2$, and their complement. The first uses the mass of G_1 and the K_i far bound; the others use $G_1(Z) \le Cr^{-n-p}$ and the finite L^1 norm of K_i. The result is Cr^{1-n-p}. On the intervening compact annulus the two local singularities are separated and integrable. This proves the positive envelope with its sharp far decay. On $|u| \ge 1/2$, $W(u) \le C|f(u)|^2$. On the complementary detector-good compact phase set, the gradient floor and $|Du| \le 2G_1 * m_c$ give $W(u) \le C(G_1 * m_c)^2$. The remaining receiver set has volume $o(S)$.

For the excluded-base trace commutator retain the sharper row bounds. Both $\mathcal{L}$ and G_1 are bounded by the integrable positive kernel

$$K_0(Z) = C\begin{cases} |Z|^{1-n}, & |Z| \le 1, \\ |Z|^{1-n-p}, & |Z| \ge 1. \end{cases}$$

For G_1 this uses its sharper bounds $|Z|^{p-n}$ near zero and $|Z|^{-n-p}$ at infinity. Bounded m_c at distances below one and the all-center BV bound above one give $K_0 * m_c \le C$: the near radial integral is $C\int_0^1 dr$, and on the dyadic annulus of radius 2^k the bound is $C2^{-kp}$. Also

$\|K_0\|_1 < \infty$. This supplies the trace counterparts of the bulk deleted-source estimate (KD), as follows. For any $\mathcal{T}, \mathcal{T}' \in \{G_1, \mathcal{L}\}$, their positivity and domination by K_0 imply

$$\|\mathcal{T} * m_c\|_\infty \le C, \qquad \|\mathcal{T} * r_*\|_1 \le \|\mathcal{T}\|_1 \|r_*\|_1 \le C\|r_*\|_1.$$

Consequently, on the whole trace space,

$$\int (\mathcal{T} * r_*)(\mathcal{T}' * m_c) dX \le \|\mathcal{T}' * m_c\|_\infty \|\mathcal{T} * r_*\|_1 \le C\|r_*\|_1 = o(S_j).$$

Replacing the second source by $r_* \le m_c$ gives its squared trace norm and every cross product of the two deleted trace envelopes as $o(S_j)$ as well. This estimate is valid although m_c has infinite possible global mass. In particular, for either positive trace kernel, the difference between its squared whole-source envelope and its squared envelope with r_* removed is at most $2(\mathcal{T} * r_*)(\mathcal{T} * m_c)$; its integral is $o(S_j)$. Thus the potential bound above can discard r_* with precisely this negligible error, using $\mathcal{L}$ on $|u| \ge 1/2$ and G_1 on the detector-good complementary band. The remaining detector error has volume $o(S_j)$ and W is bounded. Bounded test factors and restriction to any corridor only decrease these positive error estimates.

For the excluded-base commutator, positive Cauchy–Schwarz likewise bounds its square by a constant times its single K_0 row with the squared cutoff difference. Its receiver-normal marginal is

$$\int_{\mathbb{R}} K_0(x - z, t - s) dt \le C \begin{cases} |x - z|^{1-d}, & |x - z| \le 1, \\ |x - z|^{-d-(p-1)}, & |x - z| \ge 1. \end{cases}$$

Indeed integrate $(r^2 + t^2)^{(1-n)/2}$ near zero and $(r^2 + t^2)^{(1-n-p)/2}$ at infinity, with $n = d + 1$. The near bound is weaker than $Cr^{-d-(p-1)}$. The pointwise cutoff identity in proposition 7.5 and the source-column bound therefore reduce the excluded-base trace norm to lemma 7.3 with $\alpha = p - 1$. After physical scaling its bound is $CR^{d+1-p}(1 + \log R) = o(S)$.

The coarser Riesz bound $C|Z|^{2-n-p}$ was sufficient for the deleted-mass mixed norms; the far receiver-normal estimate for trace screening uses the sharper K_0. We now verify the additional massive-convolution rows. Put $\mathcal{B}_y = P_y * G_1$ or $\mathcal{Q}_y * G_1$. The preceding uniform row bound and $\int G_1 = 1$ give $\mathcal{B}_y * m_c \le C/(1 + y)$.

Let $r = |x|$ and first omit G_1. Integration in the receiver normal coordinate gives

$$\int_{\mathbb{R}} P_y(x, t) dt = C \frac{y^p}{(r^2 + y^2)^{(d+p)/2}}, \qquad \int_{\mathbb{R}} \mathcal{Q}_y(x, t) dt \le C \frac{y^{p-1}}{(r^2 + y^2)^{(d+p-1)/2}}.$$

The second bound follows from $|(x, t)| \le (r^2 + t^2 + y^2)^{1/2}$ before the normal integration. Multiplying by $y^{1-p}/(1 + y)$ and integrating height, split at $y = r$ and $y = 1$. Both resulting rows are bounded by

$$J(r) = C \begin{cases} r^{2-d-p}, & 0 < r \le 1, \\ r^{1-d-p}(1 + \log r), & r \ge 1. \end{cases}$$

For example the two height integrals are bounded by constant multiples of $\int_0^\infty y[(1 + y)(r^2 + y^2)^{(d+p)/2}]^{-1} dy$ and $\int_0^\infty [(1 + y)(r^2 + y^2)^{(d+p-1)/2}]^{-1} dy$. The displayed splits give the stated powers; the integral of $(1 + y)^{-1}$ up to r accounts for the possible far logarithm. Since $1 < p < 2$, J is integrable both at zero and infinity in $\mathbb{R}^d$.

The marginal $g_1(x) = \int_{\mathbb{R}} G_1(x, t) dt$ is positive of mass one and satisfies

$$g_1(r) \le Cr^{p-d} \ (r \le 1), \qquad g_1(r) \le Cr^{-d-p} \ (r \ge 1),$$

by the bounds in lemma 2.2 and direct normal integration. Tonelli identifies the weighted height/normal row of $\mathcal{B}_y$ with g_1 convolved in $\mathbb{R}^d$ with the corresponding row without G_1. It is thus bounded by $g_1 * J$. We verify $g_1 * J \le CJ$. For $r \le 1/4$, the region $|z| \le 2r$

scales to a finite integral and contributes Cr^{2-d}; on $2r < |z| < 2$, the integrand is at most $C|z|^{2-2d}$ and contributes at most Cr^{2-d} because $d > 2$; the remaining region is bounded. These are at most $CJ(r)$. For $r \geq 4$, split into $|z| < r/2$, $|x-z| < r/2$, and their complement. The first uses the mass of g_1 and comparability of $J(x-z)$ with $J(r)$; the latter two use $g_1(z) \leq Cr^{-d-p}$ and the finite integral of J. Their bounds are again at most $CJ(r)$. On the intervening compact annulus the convolution is bounded because its two local singularities are separated. Therefore the massive row is at most

$$CJ(r) \leq Cr^{-d-(p-1)}(1 + \log_+ r).$$

All constants depend only on the fixed dimension, p, and uniform source bounds.

For excluded source bases $E \subset \{\theta = 0\}$, positivity and Cauchy–Schwarz give directly

$$|\theta(x/R)(\mathcal{B}_y * m_E)(X)|^2 \leq \frac{C}{1+y} \int |\theta(x/R) - \theta(z/R)|^2 \mathcal{B}_y(X-Z) m_E(Z) dZ.$$

Integrate the receiver normal coordinate and height using the row just proved, then the source normal coordinate using $\int m_*(R\eta, s)ds \leq Cq_j(\eta)^2$. Physical base scaling $x = R\xi$, $z = R\eta$ produces $R^{d+1-p}(1 + \log R)$ times the integral in lemma 7.3 with $\alpha = p - 1$. Its bound is uniform in j for the actual nested factors, since their H^1 norms, the fixed count threshold and $\|q_j\|_{H^1}$ are bounded; the count screen from proposition 7.4 tends to one. This proves the massive-convolution screening estimate with error $o(S_j)$ from the same strict exponent as above, without attributing a new row assertion to proposition 7.5. Under the selected uniformly bounded shear, the pulled-back kernels have uniformly comparable radial majorants and the identical estimate follows with fixed constants.

Finally retain sources at distance $c\ell$ from the corridor. Below height ℓ, BV annuli give $P_y * m \leq Cy^p \ell^{-p-1}$ and $\mathcal{Q}_y * m \leq Cy^{p-1}\ell^{-p}$; above it use C/y. Their squared weighted height integrals are at most $C\ell^{-p}$. For a massive convolution split its displacement at $c\ell/2$. The near part has the same bound; its far part has mass $C\ell^{-p}$ and the row bound $C/(1+y)$, giving the smaller squared norm $C\ell^{-2p}$. The trace potential envelopes have the latter bound directly. Multiply by the corridor volume $CR^d\ell$ and add the proved deleted-source, excluded-base and detector errors. This is the claimed complete energy estimate. A determinant-one shear tending to the identity preserves distances, kernel annuli and the actual source-column construction with uniform constants, so this argument also proves its selected-frame version. □

Lemma 9.10 (Complete displaced ranks give the exact adjacent stability in every base dimension)**.** *In definition 9.1 and proposition 9.6, suppose $\lambda > 0$. Then every adjacent gap $g_i = f_{i+1} - f_i$ is positive almost everywhere, its power g_i^{-p-1} is locally integrable, and*

$$\frac{16c_{1,p}\lambda}{m\sigma} \int \sum_{i=1}^{m-1} g_i^{-p-1}\phi^2 \leq \int |D\phi|^2 \qquad (\phi \in C_c^\infty(\mathbb{R}^d)).$$

Proof. Use the full-phase complete roots and clipped-rank convergence of proposition 9.6 in the fine record definition 9.1. We also use proposition 3.4 explicitly: the underlying actual trace is a smooth stable solution of $(-\Delta)^{p/2}u = u - u^3$ in $\mathbb{R}^n$, $n = d+1$, its Poisson extension has the stated full-height local energy bound, and $d\mu = d_p y^{1-p} dX dy$ represents the Fourier symbol $|\xi|^p$. All equation and stability arguments below are made for this actual field in orthonormal physical coordinates; the selected shear is subsequently a change of variables. Fix one positive-length compact central phase interval and a fixed bounded-height support. Apply lemma 5.1 in the selected unit oblique direction. Passing from the nonorthogonal shear base parameter to its orthogonal base changes its measure and the normal gap by factors tending to one, and fixed enlarged receivers contain both descriptions. A limiting adjacent

gap below δ is therefore approximated in measure by consecutive actual roots within $2\delta D_j$, with a harmless enlargement of this factor if necessary. Since $D_j \to \infty$ and $D_j/R_j \to 0$, the permitted range $H_0 \le H \le r$ holds for H equal to that fixed multiple of δD_j for all large j. The short-gap lemma imposes no root-slope hypothesis. Its bound gives

$$|\{g_i < \delta\} \cap \Omega| \le C_\Omega \lambda^{-1/2} \delta^{(p+1)/2}.$$

First remove the bounded-height support in measure, using finiteness of the Sobolev ranks; then send $\delta \downarrow 0$. Thus $g_i > 0$ almost everywhere.

We fix the exact stability normalization and derive the signed interaction. With a unitary Fourier transform define

$$\begin{aligned} Q_u(\varphi) &= \int_{\mathbb{R}^n} |\xi|^p |\widehat{\varphi}(\xi)|^2 d\xi + \int_{\mathbb{R}^n} (3u^2 - 1)\varphi^2 dX \\ &= \frac{c_{n,p}}{2} \iint_{\mathbb{R}^n \times \mathbb{R}^n} \frac{|\varphi(X) - \varphi(Y)|^2}{|X - Y|^{n+p}} dX dY + \int (3u^2 - 1)\varphi^2 dX, \end{aligned}$$

where

$$c_{n,p} = \frac{p 2^{p-1} \Gamma((n+p)/2)}{\pi^{n/2} \Gamma(1 - p/2)}, \qquad d_p = 2^{p-1} \frac{\Gamma(p/2)}{\Gamma(1 - p/2)}.$$

The Fourier kinetic term equals $\int |D(P_y * \varphi)|^2 d\mu$, by the normalized extension identity in proposition 3.4. Stability is $Q_u(\varphi) \ge 0$ for compact smooth tests and, by compact H^1 approximation, for the compact Lipschitz tests used below, since $p/2 < 1$.

Let e be any unit physical direction, $h = u_e$, and $H = P_y * h = U_e$. Uniform smoothness in lemma 2.1 justifies differentiating the actual cubic equation and its weak extension equation. In particular

$$(-\Delta)^{p/2} h = (1 - 3u^2)h, \quad \operatorname{div}(y^{1-p} DH) = 0, \quad -d_p \lim_{y \downarrow 0} y^{1-p} H_y = (1 - 3u^2)h,$$

where the last identity is equivalently

$$\int DH \cdot D\Psi \, d\mu = \int (1 - 3u^2) h \Psi_0 dX.$$

Take $\eta \ge 0$ compact smooth in the physical variables. For a height cutoff ρ_T, equal to one below T and zero above $2T$ with $|\rho_T'| \le C/T$, test the last identity by $H\eta^2\rho_T^2$. Product expansion gives exactly

$$\int |D(H\eta\rho_T)|^2 d\mu - \int (1 - 3u^2) h^2 \eta^2 dX = \int H^2 |D(\eta\rho_T)|^2 d\mu.$$

The harmonic extension minimizes energy among extensions of the same trace $h\eta$. Consequently $Q_u(h\eta)$ is at most this right side. The complete local full-height energy from proposition 3.4 bounds $\int H^2 d\mu$ on the compact physical support of η. Thus the height-cutoff error is at most $CT^{-2} \int_{T<y<2T} H^2 d\mu \to 0$. Sending $T \to \infty$ proves

$$Q_u(h\eta) \le \int H^2 |D_X \eta|^2 d\mu.$$

All tests at fixed T are justified by weighted energy approximation and uniform physical regularity.

The potential terms of $Q_u(h\eta)$ and $Q_u(|h|\eta)$ are identical. Their kinetic difference has pointwise numerator

$$4\eta(X)\eta(Y)[h_+(X)h_-(Y) + h_-(X)h_+(Y)].$$

The two ordered integrals agree by exchanging X, Y, and the factor $c_{n,p}/2$ therefore yields

$$Q_u(h\eta) - Q_u(|h|\eta) = 4c_{n,p} \iint h_+(X)h_-(Y)\eta(X)\eta(Y)|X - Y|^{-n-p} dX dY.$$

The identity is integrable: for opposite signs bounded Dh gives $h_+(X)h_-(Y) \le C|X-Y|^2$ near the diagonal, and $p < 2$; at infinity the compact test and bounded h suffice. Since $Q_u(|h|\eta) \ge 0$, we have established the exact inequality

$$4c_{n,p} \iint h_+(X)h_-(Y)\eta(X)\eta(Y)|X-Y|^{-n-p}dXdY \le \int H^2|D_X\eta|^2 d\mu. \tag{AS}$$

No source partner has been discarded in deriving (AS).

Fix compact phase intervals $I_1, I_2 \Subset (-1,1)$, a bounded z ball, a compact base receiver with a larger buffer, and a clipping level M. Let $F^M_{j,k}(\xi, c)$ denote the clipped normalized actual rank, zero on the vanishing incomplete products. By proposition 9.6 it tends strongly in L^2 on each containing base–phase product to $T_M(f_k(\xi))$. Translation invariance of Lebesgue measure and continuity of translations for the fixed limiting function imply

$$\int_{|z|\le L}\int_{\Omega\times I_2} |F^M_{j,k}(\xi + a_j z, c') - T_M(f_k(\xi))|^2 d\xi dc' dz \longrightarrow 0.$$

Indeed the term $F^M_{j,k} - T_M(f_k)$ after the shift is bounded by its norm on one fixed larger receiver, and the remaining term is a translation of the fixed L^2 function over displacements at most $a_jL \to 0$. The unshifted ranks on $\Omega \times I_1$ obey the same conclusion. The incomplete-product measure at either endpoint tends to zero by the complete-count conclusion and a change of variables. Remove the fixed clip on bounded-height supports, and then exhaust them in measure. Hence, jointly in ξ, z, c, c', the actual normalized separation between rank i at $(R_j\xi, c)$ and rank $i+1$ at $(R_j\xi + D_j z, c')$ tends in measure to $g_i(\xi)$. This proves the full two-source-base convergence needed for Fatou without a pointwise lower bound on normal speed along a phase path.

The cap and sign input is proposition 5.2, used explicitly. Fix one compact buffered interval $J \Subset (-1,1)$ containing I_1, I_2, zero, and the compact selected-core phase interval in its interior. On a connected enlarged base receiver that proposition places every J-phase root in $|t| \le \epsilon_j R_j$, with $\epsilon_j \to 0$, also in the selected shear. Choose a fixed small $c_* > 0$ so that the two common caps $t_\pm = \pm c_* R_j$ lie inside the containing receiver. They avoid J. The continuous value of u on either connected cap lies entirely in one component of $[-1,1] \setminus J$: below every tested phase or above every tested phase. A subsequence fixes these finitely many cap states. Choose a smooth normal cutoff equal to one on $|t| \le c_* R_j/2$, zero outside $|t| < c_* R_j$, with derivative C/R_j. Its collars contain no J-phase roots for all large j.

All following root and sign assertions are made on the almost-everywhere regular fibers used in normal coarea. For each fixed base, one-dimensional Sard excludes a null set of critical phase values; Fubini excludes a null set in the base–phase product. On its complement every root in the capped interval is regular, the list is finite, and there are no unlisted critical crossings between successive roots. If the lower cap is below J, the first root of $u = c$ has positive normal derivative; if it is above J, that derivative is negative. Indeed the sign of $u - c$ before its first root fixes the direction of that crossing. Between consecutive roots $u - c$ is nonzero with constant sign, so the crossing orientations alternate. There is therefore one common first-root sign $\epsilon_1 \in \{-1, 1\}$ across both phase intervals and the connected base receiver. The ith and $(i+1)$st ranks at two different source bases and phases consequently have opposite actual signs outside the vanishing incomplete products. Dividing by the positive unit-direction factor does not change these signs. These retained adjacent source-pair domains are genuine nonnegative subdomains of (AS).

For the moment take $0 \le \phi \in C^\infty_c(\mathbb{R}^d)$. Denote the normal cutoff just constructed by $\gamma_j(t)$, and choose the containing column cap in proposition 9.6 equal to one on its support. Apply

(AS) to $e_j = (k_j, 1)/\sqrt{1+|k_j|^2}$ and the globally smooth nonnegative physical test

$$\eta_j(X) = \phi((X_x - k_j X_t)/R_j)\gamma_j(X_t).$$

Write $\omega_j = \sqrt{1+|k_j|^2}$. Under the determinant-one pullback $(x,t) \mapsto (x + k_j t, t)$, the unit Jacobi field is $H_j^{\mathrm{sh}}/\omega_j$, where $H_j^{\mathrm{sh}} = (H_j^{\mathrm{o}} + k_j \cdot V_j^{\mathrm{o}}) \circ S_{k_j}$. The exact physical test metric is

$$|D_X \eta_j|^2 \circ S_{k_j} = \frac{\gamma_j^2}{R_j^2}\big(|D\phi|^2 + (k_j \cdot D\phi)^2\big) + \phi^2 |\gamma_j'|^2 - \frac{2\gamma_j \gamma_j'}{R_j}\phi\, k_j \cdot D\phi.$$

Here $\phi, D\phi$ are evaluated at x/R_j. The last two terms are supported on the normal collars, and their absolute values are bounded by CR_j^{-2} on the fixed base support, since $k_j \to 0$. Complete normal-column convergence, core exhaustion, and $\omega_j^{-2} \to 1$ make the plateau term, after division by R_j^{d-2}, tend to $m\sigma \int |D\phi|^2$, once the collar mass is removed as below. This uses the actual unit derivative and its exact pulled-back test metric; no isotropic equation is assigned to the sheared field.

The normal collars of the macroscopic test contribute $o(R_j^{d-2})$. To see this, retain a fixed bounded normalized-height set and fixed physical core radius. On their genuine regular supports, the selected-core phases belong to the fixed interval J. The explicit confinement above therefore places their root centers in $|t| \le \epsilon_j R_j$ and their phase-pinned comparison centers within a further fixed $O(1)$ distance. Their fixed-radius core rectangles are consequently away from the normal collars at distances comparable to R_j, without requiring an additional estimate for the common scalar center. Complete normal-core exhaustion makes the remaining normal-column mass arbitrarily small after the fixed core radius is increased; column uniform integrability removes the bounded-height restriction. Thus the normal energy on the collars is $o(R_j^d)$. Both the squared normal-cutoff derivative and its cross term with the base derivative are bounded by CR_j^{-2} there, giving the asserted $o(R_j^{d-2})$ cost. Core exhaustion also shows that γ_j^2 captures the complete limiting column mass $m\sigma$, so the entire right side of (AS), divided by R_j^{d-2}, tends to $m\sigma \int |D\phi|^2$. This uses a globally smooth macroscopic test and no count, height, or separation screen in its right side.

On the nonnegative left side retain the $m-1$ adjacent opposite-sign pairs, restrict $|z| \le L$, and use normal coarea for the actual directional derivative. More explicitly, when rank i has positive orientation take X on its I_1-phase branch and Y on rank $i+1$'s I_2-phase branch; when its orientation is negative interchange X, Y. These are disjoint subsets of the single ordered positive–negative source integral, including for successive values of i. In shear coordinates $|h_e|dt = dc/\omega_j$. With $x = R_j\xi$, $x' = R_j\xi + D_j z$, the two base Jacobians and the interaction kernel give the physical scale

$$R_j^d D_j^d D_j^{-d-1-p} = R_j^d D_j^{-p-1} = \lambda_j R_j^{d-2}.$$

After division of (AS) by R_j^{d-2}, their integrands are therefore the nonnegative functions

$$\frac{4c_{d+1,p}\lambda_j}{\omega_j^2} \frac{\phi(\xi)\phi(\xi + a_j z)}{\big(|z + k_j r_{j,i}|^2 + r_{j,i}^2\big)^{(d+1+p)/2}}, \qquad r_{j,i} = \frac{T_{j,i+1}(R_j\xi + D_j z, c') - T_{j,i}(R_j\xi, c)}{D_j},$$

integrated in $d\xi\, dz\, dc\, dc'$ on the complete products, with the compact normal cap equal to one there. This also records every scaling and coarea factor before taking limits. Extend each displayed integrand by zero off its complete two-source product. The complements of those products have measure tending to zero. On $\{g_i > \delta\}$, all remaining arguments converge in measure, and the limiting denominator is strictly positive. To justify the resulting lower bound for the full liminf, first choose any subsequence attaining that liminf for the nonnegative

integrated sum, and then extract a further subsequence on which the finitely many ranks and complete-product indicators converge almost everywhere. Fatou gives the desired lower bound on that further subsequence, hence for the original liminf. If the liminf is infinite there is nothing to prove. The horizontal kernel integral is exactly

$$c_{d+1,p}\int_{\mathbb{R}^d}(|z|^2+g^2)^{-(d+1+p)/2}dz = c_{1,p}g^{-p-1},$$

Indeed polar coordinates give the unnormalized integral $\pi^{d/2}\Gamma((1+p)/2)\,g^{-p-1}/\Gamma((d+1+p)/2)$, and the displayed formula for $c_{n,p}$ reduces its coefficient to $c_{1,p}$. The small shear changes the kernel to $K(z+k_jr,r)$; for a fixed positive limiting gap its metric tends to the displayed isotropic one, and the recorded two-source factor ω_j^{-2} tends to one. Taking the record liminf, then sending $L\to\infty$ and exhausting the bounded-height sets, gives the explicit inequality

$$4c_{1,p}\lambda|I_1||I_2|\int\phi^2\sum_{i=1}^{m-1}\mathbf{1}_{\{g_i>\delta\}}g_i^{-p-1}\le m\sigma\int|D\phi|^2.$$

The bounded-height restrictions were used only to establish convergence in measure; the actual stability test and its right side have no such screen. Their exhaustion is legitimate by finiteness of every f_k almost everywhere and positivity of the retained interaction. Now exhaust each phase interval to $(-1,1)$, and then send $\delta\downarrow 0$. Monotone convergence and the already proved positivity of the gaps give

$$16c_{1,p}\lambda\int\phi^2\sum_{i=1}^{m-1}g_i^{-p-1}\le m\sigma\int|D\phi|^2.$$

The factor sixteen is the factor four in (AS) times the product of the two full phase lengths, each equal to two. Dividing by $m\sigma>0$ gives exactly the stated sharp coefficient. In particular the inverse powers are locally integrable after choosing a nonnegative bump equal to one on a compact base set.

All preceding limits are taken for fixed compact phase intervals. Their constants may depend on those intervals. Only after the record limit and bounded-z limit do we increase the intervals to $(-1,1)$, using nonnegativity and monotone convergence on the signed source-pair integral. No estimate uniform as a phase approaches a well is used.

For a real $\phi\in C_c^\infty$, use the nonnegative smooth compact tests $\phi_\varepsilon=(\phi^2+\varepsilon^2)^{1/2}-\varepsilon$, whose supports lie in $\operatorname{spt}\phi$. They converge uniformly to $|\phi|$, and

$$D\phi_\varepsilon=\frac{\phi}{\sqrt{\phi^2+\varepsilon^2}}D\phi\longrightarrow D|\phi|\quad\text{in } L^2.$$

The derivative convergence follows by dominated convergence and Sobolev locality on $\{\phi=0\}$. Thus $\phi_\varepsilon\to|\phi|$ in H^1, with $\int|D|\phi||^2=\int|D\phi|^2$. Dominated convergence against the now locally integrable inverse-gap weight proves the stated inequality for real ϕ. □

10. Nested nonlinear comparison and actual selectors

All base factors remain in the complete nonlinear test. The normal comparison uses scalar-centered macroscopic caps, so their bounded-amplitude cost is $R^{d+1-p}=o(S)$. The source-clearance argument uses globally smooth recovered heights and deweighted actual lifts, and therefore controls source partners on disconnected base components as well.

Lemma 10.1 (Deweighted clipped lifts give nested selectors and global source clearance)**.** *Let $\Omega\Subset\Omega'\subset\mathbb{R}^d$ be bounded Lipschitz domains, and let $m\ge 1$ be an integer. Let $R_j\to\infty$, $D_j\to\infty$, and $a_j=D_j/R_j\to 0$. On a regular open subset $O_j\subset\Omega'$ let $t_{j,1}<\cdots<t_{j,m}$ be*

smooth normalized root heights. Suppose $0 \le \theta_j \le 1$ *is a continuous* $H^1(\Omega')$ *regularity cutoff, tends strongly to one in* $H^1(\Omega')$*, and vanishes in a relative neighborhood of* $\Omega' \setminus O_j$*. Suppose also that for every fixed* L *the zero-extended lifts*

$$F_{j,k,L} = \theta_j T_L(t_{j,k})$$

converge strongly in $H^1(\Omega')$ *to* $T_L(f_k)$*, where* $f_k \in H^1(\Omega')$*. Here* T_L *clips to* $[-L, L]$*. Let* $q_j \ge 0$ *tend strongly in* $H^1(\Omega')$ *to a constant* q_**.*

Fix $M \ge 1$*,* $\delta \in (0,1)$*, a finite set of adjacent gaps to be separated, and a count threshold* $L > q_*$*. There are nested* H^1 *cutoffs* $0 \le \beta_{j,1}, \beta_{j,0} \le 1$*, with compact support in* Ω' *and with* $\beta_{j,0} = 1$ *almost everywhere on* $\{\beta_{j,1} > 0\}$*, having the following properties for fixed smooth-approximation parameter* $\varepsilon > 0$*. The outer nonzero support consists of genuine unclipped roots with* $|t_{j,k}| < 3M$*, each tested gap greater than* δ*, and*

$$|t_{j,k} - f_k^\varepsilon| < 3c\delta, \qquad c = 1/100,$$

for $f_k^\varepsilon \in C_b^\infty(\mathbb{R}^d)$ *converging to* $T_{8M}(f_k)$ *strongly in* $H^1(\Omega')$ *as* $\varepsilon \downarrow 0$*. In particular their gradients are globally bounded for each fixed* ε*. The inner nonzero support has tested gaps greater than* 3δ*. Both cutoffs converge strongly in* H^1 *for fixed parameters. After* $\varepsilon \downarrow 0$*, their inner limit is a bounded height/separation cutoff of* $f_1, \dots, f_m$*, with a fixed compact base factor equal to one on* Ω*; its height factor tends strongly to one as* $M \to \infty$*. The inner cutoff has a factor* $\psi(q_j/L)$*, where* $0 \le \psi \le 1$ *is smooth, equals one on* $[0,1]$*, and equals zero on* $[2, \infty)$*.*

The construction can moreover be chosen with explicit deweighted lifts. For the fixed M*, put* $L_h = 8M$ *and choose a bounded smooth function* $r : [0,1] \to \mathbb{R}$ *equal to* $1/s$ *on* $[2/3, 1]$*. The globally defined bounded functions*

$$h_{j,k} = r(\theta_j) F_{j,k,L_h} \longrightarrow T_{L_h}(f_k) \quad \text{strongly in } H^1(\Omega')$$

may be used for all the outer height, gap and mismatch factors. On every compact tested base, the essential support of the outer factor is contained in $\{\theta_j \ge 3/4,\ |h_{j,k}| \le 3M\ (1 \le k \le m)\}$*. There is an open neighborhood of that support, contained in* O_j*, on which* $h_{j,k} = t_{j,k}$ *for every* k*. Thus their weak gradients there are exactly the ordinary gradients of the actual normalized root heights. If* β_0 *denotes the limit of* $\beta_{j,0}$*, then* $\beta_0 = 0$ *wherever any* $|f_k| \ge 3M$*.*

Write the physical roots as $T_{j,k}(x) = \tau_j + D_j t_{j,k}(x/R_j)$*. For a tested adjacent gap* i*, let*

$$M_j(x) = \frac{T_{j,i}(x) + T_{j,i+1}(x)}{2}, \qquad W_j(x) = \tau_j + \frac{D_j}{2}(f_i^\varepsilon + f_{i+1}^\varepsilon)(x/R_j).$$

Let k *be a smooth step, zero below* $-1/8$ *and one above* $1/8$*. The selectors* $k((t - M_j)/\delta D_j)$ *and* $k((t - W_j)/\delta D_j)$ *have difference supported in* $|t - M_j| < \delta D_j/4$ *on the inner support. If the physical root slopes are at most* $\sqrt{a_j}$ *there, this difference* ζ_j *satisfies*

$$|\partial_t \zeta_j| \le C_\delta / D_j, \qquad |D_x \zeta_j| \le C_\delta \sqrt{a_j}/D_j + C_{\delta,\varepsilon}/R_j.$$

More generally, if the physical root slopes on the tested inner support are bounded by a fixed L_0*, the same construction gives*

$$|\partial_t \zeta_j| \le C_\delta / D_j, \qquad |D_x \zeta_j| \le C_\delta L_0/D_j + C_{\delta,\varepsilon}/R_j.$$

The global source clearance below is unchanged. The corresponding capped bounds replace $\sqrt{a_j}$ *by* L_0 *in the horizontal derivative estimate.*

Every receiver in that corridor with $x/R_j \in \{\beta_{j,1} > 0\}$ *is at Euclidean distance at least* $c'\delta D_j$ *from every source* (z, s) *with* $z/R_j \in \{\beta_{j,0} > 0\}$ *satisfying* $\min_k |s - T_{j,k}(z)| \le C_I$*, for*

all large j, with C_I fixed. The same bound holds for an extension receiver (x,t,y), $y \geq 0$, and a trace source $(z,s,0)$. Let $\rho \in C_c^\infty(\mathbb{R})$ and $\upsilon \in C_c^\infty([0,\infty))$ take values in $[0,1]$, and set

$$\eta_R(t,y) = \rho((t-\tau_j)/R_j)\upsilon(y/R_j).$$

Multiplying both normal selectors by this same scalar-centered cap replaces their difference by $\eta_R\zeta_j$. Its corridor support and Euclidean source clearance persist. Its displayed t and x derivative estimates persist with an additional C/R_j term, and its height derivative is bounded by C/R_j. This operation leaves the base factors $\beta_{j,0},\beta_{j,1}$ unchanged, including their nesting. When nested normal/height cutoffs are required in the nonlinear comparison, choose separate caps η_0,η_1 with $\eta_0 = 1$ on a neighborhood of $\operatorname{supp}\eta_1$.

Proof. Use the deweighting construction in the proof of proposition 9.6, now with clipping level $8M$ and a bounded smooth r equal to $1/s$ on $[2/3,1]$. Put $h_{j,k} = A_{j,k} = r(\theta_j)F_{j,k,8M}$. Since $r(\theta_j) \to 1$ in measure and $\|Dr(\theta_j)\|_2 \leq C\|D\theta_j\|_2 \to 0$, the bounded product argument of lemma 8.3 gives $h_{j,k} \to T_{8M}(f_k)$ strongly in H^1. It uses the given bounded lifts, without presuming their equality to the actual heights. On each compact tested base $K \Subset \Omega'$, the closed set $K \cap \{\theta_j \geq 3/4\}$ is a compact subset of O_j. Consequently the cutoffs constructed below have genuine interior supports and admit fixed-index Sobolev approximation inside the regular region.

Choose all factors smooth and valued in $[0,1]$. Take an outer regularity factor zero on $[0,3/4]$ and one on $[7/8,1]$, and an inner factor zero on $[0,15/16]$ and one near 1. Choose outer height factors equal to one on $[-2M,2M]$ and zero outside $(-3M,3M)$; choose inner height factors equal to one on $[-M,M]$ and supported in $(-2M,2M)$, with derivatives bounded by C/M. Apply these to all $A_{j,k}$. On their outer nonzero support, $A_{j,k} = T_{8M}(t_{j,k})$ and $|A_{j,k}| < 3M < 8M$, so the actual height is unclipped and equals $A_{j,k}$ exactly.

This also proves the exported neighborhood and gradient assertions. The clipped lifts are continuous: on O_j they are products of continuous functions, and they extend continuously by zero because θ_j vanishes in a relative neighborhood of its complement. Hence the deweighted $h_{j,k} = A_{j,k}$ are continuous. The closure of the outer nonzero support is contained in $\theta_j \geq 3/4$ and $|h_{j,k}| \leq 3M$. On the open neighborhood

$$\{\theta_j > 2/3,\ |h_{j,k}| < 4M\ (1 \leq k \leq m)\}$$

one has $r(\theta_j)\theta_j = 1$, so $h_{j,k} = T_{8M}(t_{j,k})$. The strict bound $4M < 8M$ makes clipping inactive there; consequently $h_{j,k} = t_{j,k}$ and $Dh_{j,k} = Dt_{j,k}$. The existing product convergence proves global strong H^1 convergence of these deweighted functions. The limiting outer height factors vanish where $|f_k| \geq 3M$, so the same strict unclipping margin applies to its nonzero limit support. These facts use the selected-phase lifts in the hypotheses, without any selection from a phase-product gradient limit.

Use the gaps $A_{j,i+1} - A_{j,i}$, with outer factors zero below δ and one above 2δ, and inner factors zero below 3δ and one above 4δ. For the Euclidean mismatch $e_j = (\sum_k |A_{j,k} - f_k^\varepsilon|^2)^{1/2}$, use outer factors one for $e \leq 2c\delta$ and zero for $e \geq 3c\delta$, and inner factors one for $e \leq c\delta$ and zero for $e \geq 3c\delta/2$. Add the outer count factor $\psi(q_j/(2L))$ and the inner factor $\psi(q_j/L)$. Choose $0 \leq \chi_0,\chi_1 \leq 1$ in $C_c^\infty(\Omega')$, with $\chi_1 = 1$ near $\overline{\Omega}$ and $\chi_0 = 1$ near $\operatorname{supp}\chi_1$, as the outer and inner base factors. The products are the required $\beta_{j,0},\beta_{j,1}$. Every outer factor is one wherever its corresponding inner factor is nonzero; for the count factors this uses $q_j < 2L$ whenever $\psi(q_j/L) > 0$.

For fixed parameters, the bounded composition and product argument of lemma 8.3 gives strong H^1 convergence of these factors. The Euclidean norm in the mismatch is also strongly continuous in H^1: use its chain rule away from zero and Sobolev locality of the input gradient on its zero set. To obtain the stated global approximants, extend each $T_{8M}(f_k)$ by the

bounded Lipschitz-domain extension operator to $H^1(\mathbb{R}^d)$, clip the extension to $[-8M, 8M]$, and mollify. The restrictions converge strongly in $H^1(\Omega')$, and every mollified function belongs to $C_b^\infty(\mathbb{R}^d)$, with a bounded gradient on all of $\mathbb{R}^d$. This global gradient bound is what is used for two-base source clearance. The limiting mismatch norm tends to zero strongly in H^1 as $\varepsilon \downarrow 0$, so its cutoff tends strongly to one. On a nonzero height factor clipping of f_k is inactive, so the limiting gaps are the actual gaps of f. The count and regularity factors tend to one, since $L > q_*$ and $\theta_j \to 1$. Finally the height-factor exhaustion follows from lemma 8.3, or directly from their almost-everywhere convergence to one and derivative bound $(C/M)\sum_k |Df_k|$. All cutoff bounds here are for fixed $M, \delta, \varepsilon, L$.

On the tested inner support, $|M_j - W_j| \le 3c\delta D_j$. The step transition width is $\delta D_j/4$, so its difference is supported in $|t - M_j| \le (1/8 + 3c)\delta D_j < \delta D_j/4$. Differentiation gives the displayed derivative estimates: $|D_x M_j| \le \sqrt{a_j}$ and $|D_x W_j| \le C_\varepsilon a_j$.

For a fixed physical slope bound L_0, use $|D_x M_j| \le L_0$ in the same differentiation, giving the stated general bound. The subsequent Euclidean clearance uses only the globally smooth recovered midpoint W_j and its Lipschitz constant $C_\varepsilon a_j$, and is independent of the actual midpoint derivative. The cap product rule below applies to either slope bound.

For global source clearance, the ordering and the outer gap imply

$$|s - W_j(z)| \ge (1/2 - 3c)\delta D_j - C_I.$$

For a receiver in the difference corridor, $|t - W_j(x)| \le (1/4 + 3c)\delta D_j$. The globally smooth function W_j has Lipschitz constant $C_\varepsilon a_j \to 0$, so

$$|s - t| + \mathrm{Lip}(W_j)|x - z| \ge (1/4 - 6c)\delta D_j - C_I \ge c''\delta D_j.$$

Cauchy–Schwarz gives

$$\sqrt{|s-t|^2 + |x-z|^2} \ge \frac{c''\delta D_j}{\sqrt{1 + \mathrm{Lip}(W_j)^2}} \ge \frac{c''}{\sqrt{2}}\delta D_j$$

for all sufficiently large j, with ε fixed. Adding the receiver height $y \ge 0$ only increases this distance. This argument is global in the two base points and never infers global clearance from disconnected actual graph charts. Finally η_R is independent of the base variable and satisfies $|\partial_t \eta_R| + |\partial_y \eta_R| \le C/R_j$. The difference of the two capped selectors is exactly $\eta_R \zeta_j$. Its support is contained in the original difference corridor, so the source-clearance estimate just proved is unchanged. The product rule gives

$$|\partial_t(\eta_R\zeta_j)| \le C_\delta/D_j + C/R_j, \qquad |D_x(\eta_R\zeta_j)| \le C_\delta\sqrt{a_j}/D_j + C_{\delta,\varepsilon}/R_j, \qquad |\partial_y(\eta_R\zeta_j)| \le C/R_j.$$

Here the original selectors, and hence ζ_j, are independent of y. Because $D_j/R_j \to 0$, the extra normal derivative term may also be absorbed into C_δ/D_j. The cap multiplies the selectors only; the base factors remain the original nested $\beta_{j,0}, \beta_{j,1}$. For the separate nonlinear-comparison test, choose the scalar-centered outer cap with a larger plateau so it equals one near the support of the inner cap. Together with the unchanged nesting of the base factors, this gives the required outer-test plateau. □

Lemma 10.2 (A finite separated family has a uniform pinned normal secant gap)**.** *Let $1 < p < 2$, and let $q : \mathbb{R} \to (-1, 1)$ be the centered strictly increasing classical solution of $(-\partial_t^2)^{p/2} q = q - q^3$, $q(0) = 0$, $\lim_{t\to\pm\infty} q(t) = \pm 1$, where the operator has Fourier symbol $|\tau|^p$. Put $V_q = 3q^2 - 1$. Fix an integer $m \ge 1$, $A > 0$, $c_* > 0$, and $C_* > 0$. There exist K, $\varepsilon_* > 0$, $\gamma > 0$, and C with this property. Let centers $S_1 < \cdots < S_m$ have gaps at least $8K$, and let anchors satisfy $|T_k - S_k| \le A$. Suppose a real potential c is bounded in absolute value by C_*, obeys $c \ge c_*$ outside the union of $|t - S_k| < K$, and satisfies*

$$|c(t) - V_q(t - S_k)| \le \varepsilon_* \quad (|t - S_k| < K).$$

Then, with the unitary Fourier transform, every $e \in H^{p/2}(\mathbb{R})$ satisfies

$$\int |\tau|^p |\widehat{e}|^2 + \int ce^2 \geq \gamma \|e\|_{H^{p/2}}^2 - C \sum_k |e(T_k)|^2, \qquad \|e\|_{H^{p/2}}^2 := \int (1 + |\tau|^p) |\widehat{e}|^2.$$

Point values use the continuous representative, which exists because $p/2 > 1/2$. Constants are independent of the centers and their large gaps. The core radius K can be required to exceed any prescribed fixed lower bound; the resulting constants may depend on that bound.

For the extension version put

$$\mathscr{E}_p(E) = d_p \int_{\mathbb{R}\times(0,\infty)} y^{1-p} |D_{t,y} E|^2 \, dt \, dy, \qquad d_p = 2^{p-1} \frac{\Gamma(p/2)}{\Gamma(1 - p/2)}.$$

The normal Dirichlet space with trace $e \in H^{p/2}$ consists of $E = P_p e + W$, where $P_p e$ is its normal weighted harmonic Poisson extension and W belongs to the completion of $C_c^\infty(\mathbb{R} \times (0,\infty))$ in the norm $\mathscr{E}_p(W)^{1/2}$. Every such extension satisfies, after possibly decreasing γ,

$$\mathscr{E}_p(E) + \int ce^2 \geq \gamma(\mathscr{E}_p(E) + \|e\|_2^2) - C \sum_k |e(T_k)|^2.$$

This is the finite weighted-energy class with trace e, equivalently its energy completion; no weighted bulk L^2 hypothesis is imposed.

Proof. We first justify the one-layer spectral facts. By lemma 2.4, $h = q'$ is smooth, strictly positive, belongs to $L^1 \cap L^\infty \subset L^2$, and satisfies the differentiated equation

$$((-\partial_t^2)^{p/2} + 2)h = \beta h, \qquad \beta = 3(1 - q^2) \geq 0.$$

Its right side is in L^2. Fourier transformation gives $(2 + |\tau|^p)\widehat{h} = \widehat{\beta h}$, so $h \in H^p \subset H^{p/2}$. Pairing the equation with h in L^2 yields

$$Q_q(h) = 0, \qquad Q_q(e) := \int |\tau|^p |\widehat{e}|^2 + \int V_q e^2.$$

For $e \in C_c^\infty$, test the equation for h against e^2/h. This is a smooth compact test because h has a positive minimum on its support. Symmetric integration of the fractional form and elementary expansion give

$$Q_q(e) = \frac{c_{1,p}}{2} \iint \frac{h(t)h(s)}{|t - s|^{1+p}} \left| \frac{e(t)}{h(t)} - \frac{e(s)}{h(s)} \right|^2 dt \, ds.$$

The diagonal integrals are controlled by smoothness, and the off-diagonal tails by compact support and $h \in L^1 \cap L^\infty$; truncating the kernel first gives the same identity without any singular interchange. For $e \in H^{p/2}$, take compact smooth approximants converging strongly in that space and almost everywhere. Continuity of Q_q and Fatou give

$$0 \leq \frac{c_{1,p}}{2} \iint \frac{h(t)h(s)}{|t - s|^{1+p}} \left| \frac{e(t)}{h(t)} - \frac{e(s)}{h(s)} \right|^2 dt \, ds \leq Q_q(e).$$

If $Q_q(e) = 0$, positivity of the pair weight and Fubini imply that e/h is constant almost everywhere. Together with $Q_q(h) = 0$, this identifies the zero space exactly as $\operatorname{span}\{h\}$.

Since $V_q \to 2$ at both infinities, choose K_0 and $v_0 > 0$ with $V_q \geq v_0$ on $|t| \geq K_0$. Put $\nu = \min\{v_0, c_*\} > 0$ and $C_b = \max\{C_*, c_*, \|V_q\|_\infty\}$. We prove the pinned gap at once for the following auxiliary class: $v = V_q$ on $[-K, K]$, $v \geq \nu$ outside, and $|v| \leq C_b$, with K sufficiently large. This includes $v = V_q$ itself and every auxiliary potential with exterior bound c_* needed below. Write

$$Q_v(e) = \int |\tau|^p |\widehat{e}|^2 + \int ve^2.$$

If no uniform positive gap held for $e(a) = 0$, $|a| \le A$, choose $K_j \ge \max\{j, K_0\}$, potentials v_j in this class, and functions with

$$\|e_j\|_{H^{p/2}} = 1, \qquad e_j(a_j) = 0, \qquad Q_{v_j}(e_j) \le 1/j.$$

Pass to a weak Sobolev subsequence and $a_j \to a \in [-A, A]$. Fourier Cauchy–Schwarz gives bounded point evaluations and a common modulus $|e_j(t)-e_j(s)| \le C|t-s|^{(p-1)/2}$ for $|t-s| \le 1$. Arzela–Ascoli and a diagonal therefore give uniform convergence on compact intervals to the weak limit e; in particular $e(a) = 0$ and convergence is strong in local L^2.

Outside $[-K_0, K_0]$, every $v_j \ge \nu$, so

$$Q_{v_j}(e_j) \ge \min\{1, \nu\}\|e_j\|^2_{H^{p/2}} - (C_b + \nu)\int_{-K_0}^{K_0} e_j^2.$$

Thus $e = 0$ would contradict normalization. For a fixed $L \ge K_0$ and all j with $K_j \ge L$, discard the nonnegative potential integral outside $[-L, L]$. Weak lower semicontinuity of the kinetic energy and strong local convergence give

$$\int |\tau|^p|\widehat{e}|^2 + \int_{-L}^{L} V_q e^2 \le \liminf_j Q_{v_j}(e_j) \le 0.$$

Let $L \to \infty$. Since V_q is bounded and $e \in L^2$, this yields $Q_q(e) \le 0$. The zero-space description forces $e = bh$; its pin and $h(a) > 0$ force $b = 0$, again a contradiction. Consequently there are K_1 and $\gamma_0 > 0$ such that

$$Q_v(e) \ge \gamma_0\|e\|^2_{H^{p/2}} \quad \text{if } e(a) = 0,$$

uniformly for $K \ge K_1$, $|a| \le A$, and this entire auxiliary class.

If v is only ε_* close to V_q on $[-K, K]$, replace it there by V_q, leaving its positive exterior unchanged. The difference of the forms is at most $\varepsilon_*\|e\|_2^2$. Choosing $\varepsilon_* < \gamma_0/2$ gives the same pinned estimate with half the gap. To remove the exact pin, take a fixed compact smooth bump b with $b(0) = 1$ and apply the pinned estimate to $e - e(a)b(\cdot - a)$. The bilinear forms of all these bounded potentials are uniformly bounded on $H^{p/2} \times H^{p/2}$. Expansion and Young's inequality consequently give

$$Q_v(e) \ge \gamma_1\|e\|^2_{H^{p/2}} - C|e(a)|^2$$

with $\gamma_1 > 0$ uniform in the same parameters. These constants remain valid when K is further increased.

We now localize without importing other wells into the auxiliary single-well form. Choose a smooth squared partition of unity $\theta_0^2 + \sum_{k=1}^m \theta_k^2 = 1$, with $\theta_k = 1$ on $|t - S_k| \le K$, supported in $|t - S_k| < 2K$, and all first derivatives bounded by C_m/K. It can be constructed on each disjoint annulus by a sine-cosine pair. The exterior factor θ_0 vanishes on the union of the K-cores. Enlarge K so $K > A + 1$.

The bounded Lipschitz factors preserve $H^{p/2}$, by the increment estimate used below. For each k define the auxiliary potential c_k to equal c on $|t - S_k| < 2K$ and to equal c_* elsewhere. No other K-core meets this $2K$ neighborhood because center gaps are at least $8K$. Thus $c_k \ge c_*$ outside its own K-core, and it is ε_* close to $V_q(t - S_k)$ on that core. Its absolute bound is at most C_b. The preceding single-well estimate therefore applies to $\theta_k e$, centered at S_k, with anchor T_k. Also $Q_{c_k}(\theta_k e) = Q_c(\theta_k e)$ since the localized function vanishes outside the $2K$ neighborhood. At its anchor $\theta_k(T_k) = 1$. The exterior form satisfies

$$Q_c(\theta_0 e) \ge \min\{1, c_*\}\|\theta_0 e\|^2_{H^{p/2}}.$$

The fractional IMS error obeys

$$\left|\sum_{k=0}^{m} Q_c(\theta_k e) - Q_c(e)\right| \le C_m K^{-p}\|e\|_2^2.$$

Indeed expansion leaves the kernel factor $e(t)e(s)\sum_k |\theta_k(t)-\theta_k(s)|^2$. The squared factor sum is bounded by $C_m \min\{|t-s|^2/K^2, 1\}$, whose integral against $|t-s|^{-1-p}$ is $C_m K^{-p}$. Apply $2|e(t)e(s)| \le e(t)^2 + e(s)^2$. The same calculation with zero potential gives

$$\sum_{k=0}^{m} \|\theta_k e\|_{H^{p/2}}^2 \ge \|e\|_{H^{p/2}}^2 - C_m K^{-p}\|e\|_2^2.$$

Put $\delta_0 = \min\{\gamma_1, 1, c_*\} > 0$. Summing the localized gaps and using these two inequalities gives

$$Q_c(e) \ge \delta_0\|e\|_{H^{p/2}}^2 - C_m(1+\delta_0)K^{-p}\|e\|_2^2 - C\sum_k |e(T_k)|^2.$$

Choose K still larger so the middle coefficient is at most $\delta_0/2$. This proves the trace estimate with $\gamma = \delta_0/2$. All constants are independent of center locations and larger separations. Every restriction on K is a fixed lower bound, and the single-well gap is uniform for all sufficiently large K; impose any prescribed additional fixed lower bound at the same time.

Finally, the normalized Fourier mode calculation in (5) applies in one trace dimension as well: its mode equation depends only on $y|\tau|$ and uses precisely the present constant d_p. First for compact smooth traces, and then by strong $H^{p/2}$ approximation, it gives

$$\mathscr{E}_p(P_p e) = \int |\tau|^p |\widehat{e}(\tau)|^2 d\tau.$$

The extension $P_p e$ is weakly weighted harmonic. For a compactly supported smooth interior test W, its cross energy with $P_p e$ vanishes. Cauchy–Schwarz passes this identity to the zero-trace energy completion. Consequently, for every admissible $E = P_p e + W$,

$$\mathscr{E}_p(E) = \mathscr{E}_p(P_p e) + \mathscr{E}_p(W).$$

To identify this completion with the finite-energy class, let W be a locally Sobolev zero-trace remainder with finite weighted gradient energy. For almost every t, its vertical slice is absolutely continuous away from zero and weighted Cauchy–Schwarz gives

$$\varepsilon^{-p}|W(t,\varepsilon)|^2 \le \frac{1}{p}\int_0^{\varepsilon} y^{1-p}|\partial_y W(t,y)|^2 dy \longrightarrow 0.$$

Integrate $y^{-1-p}|W|^2$ by parts on (ε, Y). The upper boundary term is nonpositive; the lower boundary term tends to zero by this bound. Cauchy–Schwarz on the remaining integral $(2/p)\int y^{-p} W\partial_y W$, followed by $\varepsilon \downarrow 0$ and $Y \to \infty$, proves on each slice, and then by Tonelli,

$$\int y^{-1-p}|W|^2 dt\, dy \le \frac{4}{p^2}\int y^{1-p}|\partial_y W|^2 dt\, dy$$

without assuming density in advance. A cutoff changing on $\varepsilon < y < 2\varepsilon$ has additional squared-gradient energy bounded by $C\int_{\varepsilon<y<2\varepsilon} y^{-1-p}|W|^2$, which vanishes. For an outer cutoff changing on $L < \sqrt{t^2+y^2} < 2L$, use $y \le 2L$ to bound its additional energy by the same Hardy integrand on that annulus. It also tends to zero as $L \to \infty$; the truncated gradients converge by the finite gradient energy. On the remaining compact interior region ordinary smooth approximation applies because the weight is bounded above and below. Thus every finite-energy zero-trace remainder belongs to the completion. Conversely the Hardy bound persists under energy completion and identifies a locally Sobolev representative

with zero trace, so the two classes agree. Add $\mathscr{E}_p(W)$ to the trace inequality and decrease its coefficient to at most one. This gives exactly the stated extension inequality. □

Proposition 10.3 (The bounded-height nested actual nonlinear comparison is dimension independent)**.** *Use an actual fine record in definition 9.1, with* $S_j = R_j^{d-2}D_j^2$, $a_j = D_j/R_j \to 0$, *and* $R_j^2/D_j^{p+1} \le \Lambda$. *Assume its actual complete ranks, low-angle strong slopes, column integrability, and full transverse profile are those of proposition 9.6 and proposition 9.7. Fix* $M, \delta > 0$ *and the nested cutoffs of lemma 10.1 on one containing receiver, testing all adjacent gaps, pulled back as functions of* x/R_j; *in particular* $\int |D_x\beta_{j,i}|^2 \le CR_j^{d-2}$. *On their outer support let the actual roots have every adjacent gap at least* δD_j *and a common phase* c_j *with* $\sup_j |c_j| < \sqrt{2/3}$. *The central source here is* $|Du_j|\mathbf{1}_{\{|u_j|<\sqrt{2/3}\}}$. *For Lebesgue-almost every* (x,t) *in the outer macroscopic trace box where the surviving source density is positive and whose base lies in the nonzero support of the outer base factor* $\beta_{j,0}$, *assume fixed normal distance from a selected retained root on its own fiber:* $|t - T_{j,k}(x)| \le C_I$ *for some* k, *with one fixed* C_I *independent of* j, x, t. *Assume fixed-core planar detectors at those roots can be refined to any fixed accuracy, uniformly on this support. Suppose the remaining central variation source has total mass* $o(S_j)$ *and the detector-error trace volume is* $o(S_j)$. *The central gradient is bounded below off that error region. All assertions are for fixed cutoff and smooth-recovery parameters before the record limit.*

Let P_j *be the alternating sum of normal layer extensions, centered at the exact phase-pinned comparison centers* $S_{j,k} = T_{j,k} - \epsilon_{j,k}q^{-1}(c_j)$ *of proposition 9.7. On each regular component choose its first orientation* $\epsilon_1 \in \{-1, 1\}$, *put* $\epsilon_k = \epsilon_1(-1)^{k-1}$, *and set* $c_m = 0$ *for odd* m *and* $c_m = -\epsilon_1$ *for even* m. *Thus* $P_j = c_m + \sum_k \epsilon_k Q(t - S_{j,k}, y)$. *Let* η_0, η_1 *be scalar-centered normal/height caps of size* R_j, *with* $\eta_0 = 1$ *near the support of* η_1, *and let the base cutoffs satisfy* $\beta_{j,0} = 1$ *almost everywhere on* $\{\beta_{j,1} > 0\}$. *Then*

$$\int \beta_{j,1}^2\eta_1^2|D_n(U_j - P_j)|^2 d\mu = o(S_j), \qquad D_n = (\partial_t, \partial_y).$$

There is also a trace correction b_j, $|b_j| \le 2$, *vanishing at the actual selected-phase roots* $T_{j,k}$, *for which*

$$\int |\beta_{j,1}\eta_1(\cdot, 0)(u_j + b_j - P_{j,0})|^2 dX = o(S_j).$$

Let W_{per} *be the period-two* C^2 *continuation of* $W(r) = (1 - r^2)^2/4$ *from* $[-1, 1]$. *On the outer tested trace support there is a state-interpolation exception of volume* $o(S_j)$; *outside it* $b_j \in \{0, -2, 2\}$ *and*

$$W(u_j) = W_{\mathrm{per}}(u_j + b_j), \qquad W'(u_j) = W'_{\mathrm{per}}(u_j + b_j).$$

The profile trace lies in $[-1, 1]$, *so* $W(P_{j,0}) = W_{\mathrm{per}}(P_{j,0})$ *and the same equality holds for the derivatives.*

For the inner normal Poisson comparison, let $U_j^{\mathrm{n}}(x,t,y) = P_y^{(1)} *_t u_j(x,t)$, *at each fixed base in the coordinates being used. For every sufficiently small fixed* $c > 0$ *such that the closed smaller normal window lies in the plateau of the inner caps, one has the explicit full-height estimate*

$$\int_{\mathbb{R}^d\times\{|t-\tau_j|<cR_j\}\times(0,\infty)} \beta_{j,1}^2|D_n(U_j^{\mathrm{n}} - P_j)|^2 d\mu = o(S_j).$$

The same nested comparison and this inner normal Poisson estimate hold after imposing the count cutoff and fixed-M *height factors. The essential surviving-source support has the global Euclidean clearance of lemma 10.1. No bound on* $\int H_j^2|D_x\beta_{j,1}|^2 d\mu$ *is required. An allowed constant shear* $k_j \to 0$ *is retained with its exact coefficient matrix.*

Proof. The actual field satisfies proposition 3.4, including its cubic equation and complete extension bounds. In particular, with $W(r) = (1-r^2)^2/4$ and $d\mu = d_p y^{1-p} dX dy$, its exact weak equation is

$$\int DU \cdot D\phi \, d\mu + \int W'(u)\phi(X,0)dX = 0$$

for compact weighted-energy tests. Indeed its extension is weighted harmonic and $-d_p \lim_{y\downarrow 0} y^{1-p} U_y = u - u^3 = -W'(u)$. The same identity in an allowed determinant-one constant shear has $DU \cdot D\phi$ replaced by $A_j DU \cdot D\phi$, where A_j is the exact matrix in definition 9.1. All integrations below use that equation, with the normal and tangential terms split from this same quadratic form. Uniform smoothness and boundedness allow the compact tests by weighted-energy approximation; normal and height caps are removed only after the stated full-height estimates.

By lemma 2.1 the actual traces have $|u| \le 1$ and uniform bounds for $|Du| + |D^2 u|$. The same range bound is part of proposition 3.4. Positivity of the Poisson kernel gives $|U| \le 1$. These bounds, in particular the upper bound on $|Du|$, will be used quantitatively in the correction-energy estimate.

Write R, D, a, S for one index of the actual record in definition 9.1; its extension and exact constant-shear matrix are those of that definition. Unless an explicitly global normal-line or correction norm is written, the capped actual-field integrals below are on one containing physical receiver $R\Omega' \times (-bR, bR)$ with all extension heights, for fixed b large enough to contain the normal caps. The containing column cap is chosen equal to one on that physical normal interval; only the model Gram integrals are extended to the whole normal line as upper bounds. Use the selected-phase deweighted lifts exported by lemma 10.1. Put $t_{j,k} = (T_{j,k} - \tau_j)/D_j$ in normalized base coordinates, $w_{j,k} = D_\xi t_{j,k}$, and $z_{j,k} = -w_{j,k} = D_x u_j/(a_j u_{j,t})$ evaluated at the selected root. The supplied globally bounded lifts $h_{j,k}$ converge strongly in H^1 to $T_{8M}(f_k)$ and equal $t_{j,k}$ on a neighborhood of the outer cutoff support. Write $\beta_{\infty,0}$ for the limit of $\beta_{j,0}$. Since $0 \le \beta_{j,0} \le 1$, its convergence in measure and the strong lift-gradient convergence give

$$\beta_{j,0} w_{j,k} \longrightarrow \beta_{\infty,0} Df_k \quad \text{strongly in } L^2.$$

Indeed replace $w_{j,k}$ by $Dh_{j,k}$ on that support. The error $\beta_{j,0}(Dh_{j,k} - DT_{8M}(f_k))$ tends to zero in L^2, while $(\beta_{j,0} - \beta_{\infty,0})DT_{8M}(f_k)$ tends to zero by bounded convergence in measure against a fixed L^2 gradient. Finally $\beta_{\infty,0} DT_{8M}(f_k) = \beta_{\infty,0} Df_k$, because the limiting outer height support has $|f_k| < 3M < 8M$. The same argument against Df_k yields $\beta_{j,0}(w_{j,k} - Df_k) \to 0$ strongly in L^2.

For the moment keep the actual orientations $\epsilon_{j,k}$ and put

$$\mathcal{V}_j = -a_j \sum_{k=1}^{m} \epsilon_{j,k} w_{j,k} H_*(t - S_{j,k}, y).$$

The full transverse profile in proposition 9.7 uses the coefficients $-Df_k$. Replacing them by $-w_{j,k}$ has weighted squared error at most

$$C_m a_j^2 R_j^d \sum_k \|\beta_{j,0}(w_{j,k} - Df_k)\|_2^2 = o(S_j),$$

because each translated H_* has normal squared norm σ, and Cauchy–Schwarz bounds the finite Gram matrix by $m\sigma$ independently of the centers. Thus, before any orientation repair,

$$\int \beta_0^2 |V - \mathcal{V}_j|^2 d\mu = o(S), \qquad \int |V|^2 d\mu \le CS.$$

Here and in the subsequent physical-variable integrals $\beta_0 = \beta_{j,0}$ denotes the finite-index outer factor, with its index suppressed; $\beta_{\infty,0}$ is used only for its limit. The nested supports and

global smooth-midpoint clearance are those of lemma 10.1. The normal comparison uses the assumed complete central-source mass bound; its other fixed factors enter through their bounded amplitude, bounded normalized H^1 norms, and these exact nested supports.

We make the detector and source hypotheses quantitative. Write the assumed source split as

$$m_c = |Du|\mathbf{1}_{\{|u|<\sqrt{2/3}\}} = m_* + r_*, \qquad m_*, r_* \geq 0, \qquad \|r_*\|_1 = o(S).$$

Let C_I be the fixed normal proximity bound of the surviving source m_* to the selected roots. Put $A = \sup_j |q^{-1}(c_j)| < \infty$, and fix

$$\sup_j |c_j| < c_0 < c_1 < c_2 < \sqrt{2/3}, \qquad c_1 > 1/\sqrt{3}.$$

Use lemma 10.2 with $c_* = 3c_1^2 - 1 > 0$, a fixed potential bound at least three, and anchor bound A. Require its fixed core radius K also to exceed $C_I + A + 1$ and to be so large that the one-layer tails make every alternating profile satisfy $|P_0 - \varsigma| < 1 - c_2$ outside its K-cores. Here $\varsigma \in \{-1, 1\}$ is the predicted well in the corresponding interval between seams. Since $K > A$, the actual phase seams and comparison centers determine the same well intervals outside these cores. The assertion follows by replacing each $q(t - S_k)$ by its limiting sign: outside the cores the error is at most $C_m K^{-p}$. These are fixed lower bounds for K. Eventually $\delta D > 8K + 2A$.

The fixed-core conclusions of propositions 9.6 and 9.7 already give the needed comparison with the same phase-pinned normalized layer: on $|t - S_k| \leq 2K$, the trace is C^2 close to $\epsilon_{j,k} q(t - S_k)$, and on each fixed normal-height rectangle the actual normal extension derivatives have the corresponding weighted comparison. Quantitatively, if the physical root slope is at most η and the detector error is ε, the fixed-core trace error is at most $C_K(\eta + \varepsilon)$. Its phase shift is fixed by $S_k = T_k - \epsilon_{j,k} q^{-1}(c_j)$. We use this previously proved comparison, rather than infer an extension estimate from trace convergence alone. In a selected shear it concerns the actual pulled-back extension.

For each fixed $\eta > 0$, the bases on the outer support where a selected physical root slope exceeds η have measure $o(S)$. At the actual selected roots, $D_x T_{j,k} = a_j w_{j,k}$ and $z_{j,k} = -w_{j,k}$. The global deweighted lifts $h_{j,k}$ of lemma 10.1 converge strongly in H^1 and have $Dh_{j,k} = w_{j,k}$ near this support. Their squared gradients are uniformly integrable. Consequently

$$|\{x \text{ on the outer support} : |D_x T_{j,k}| > \eta\}| \leq \frac{a_j^2 R_j^d}{\eta^2} \int_{\{|Dh_{j,k}| > \eta/a_j\}} |Dh_{j,k}|^2 d\xi = o(S_j).$$

Call their union B_{ang}. Fix η small and then refine the detector so that $C_K(\eta + \varepsilon)$ is below both the coercivity perturbation threshold and the fixed phase/well margins used below. Detector failures on the retained core supports are already excluded by the assumed uniform fixed-core construction. The separate trace error region, of volume $o(S)$, is denoted by E_{det}.

Predict alternating orientations from the first retained root. Outside B_{ang}, at the first disagreement the two consecutive retained cores have the same actual orientation. The values just after the first core and just before the next core have opposite signs and magnitudes greater than c_2. The segment between those cores therefore crosses every phase of $[-c_2, c_2]$. Both cores and the segment lie in the outer trace box for large j, since their heights are $O_M(D_j) = o(R_j)$. It lies outside every retained K-core. Any central point there is farther than C_I from every selected retained root, since $|S_k - T_k| \leq A$ and $K > C_I + A$. Thus $m_* = 0$ almost everywhere there, and its whole central variation equals r_*. Apply the fundamental theorem to the clipping $T_{c_2}(u)$ on that segment. Its endpoint difference has magnitude $2c_2$, and its total variation is at most $\int_{\{|u|<c_2\}} |u_t| dt \leq \int r_* dt$. This gives a charge at least $2c_2$ to r_*

on almost every such fiber. Integration over bases bounds the pattern-disagreement set B_{pat} by $\|r_*\|_1/(2c_2) = o(S)$. Set $B_{\mathrm{bad}} = B_{\mathrm{ang}} \cup B_{\mathrm{pat}}$. Its finitely many fixed core strips have trace volume at most $2Km|B_{\mathrm{bad}}| = o(S)$. Since $S = o(R^d)$ their normalized base measure tends to zero. The squared gradients of the global $h_{j,k}$ are uniformly integrable and equal $|w_{j,k}|^2$ on the outer support. The normalized base measure of B_{bad} tends to zero. Thus the transverse energy of the actual-orientation model $\mathcal{V}_j$ there is $o(S)$ by the same finite Gram bound. The predicted alternating signs agree with the actual signs off B_{bad}. Changing to those predicted signs therefore changes $\mathcal{V}_j$ by $o(S)$ in squared transverse norm. On each regular component the signs are locally constant, and exact phase pinning gives $D_x S_{j,k} = D_x T_{j,k} = a_j w_{j,k}$. The modified coefficient model is consequently exactly $D_x P$. Together with the already proved approximation to $\mathcal{V}_j$, this establishes

$$\int \beta_0^2 |V - D_x P|^2 d\mu = o(S)$$

for the alternating profile used in the equation test below. Complete-column uniform integrability also makes the actual transverse energy over these bad bases $o(S)$. No unweighted selected-phase gradient convergence has been inferred from convergence on base–phase products.

The center gaps satisfy $S_{k+1} - S_k \geq \delta D - 2A > 8K$ for all sufficiently large indices, so their ordering is unchanged by phase pinning. For fixed t, y, monotonicity gives $Q(t - S_1, y) \geq \cdots \geq Q(t - S_m, y)$, all in $[-1, 1]$. For odd m their alternating sum is between its first and last terms. For even m the sum of paired differences lies between zero and two. The stated baseline c_m and, if needed, an overall sign reversal therefore give $P \in [-1, 1]$. Hence the cubic and its period-two continuation agree exactly at P. The correction constructed next is an integer well shift outside its charged interpolation region, giving both exported periodic identities.

Let $\varsigma(x, t) \in \{-1, 1\}$ be the predicted well between actual root seams, and choose a smooth $0 \leq \Theta \leq 1$, zero below c_0 and one above c_1. Define $b = 2\varsigma\Theta(-\varsigma u)$, so $|b| \leq 2$. At a root seam $u = c_j$, and $\sup |c_j| < c_0$; the uniform gradient bound gives a fixed physical neighborhood on which $b = 0$. Therefore the jumps of ς produce no distributional derivative of b. Off seams,

$$|Db| \leq C|Du| \mathbf{1}_{\{c_0 < -\varsigma u < c_1\}}.$$

On every pattern-consistent, small-angle refined core this derivative is identically zero. The oriented model trace satisfies $-\varsigma \epsilon_k q(t - S_k) \leq |c_j|$ on both sides of its exact phase seam T_k. The actual trace therefore satisfies $-\varsigma u \leq |c_j| + C_K(\eta + \varepsilon) < c_0$, by the chosen fixed accuracy margin. All other core strips are over B_{bad} and have volume $o(S)$.

Let $\mathcal{T}_{\mathrm{out}}$ denote the outer macroscopic tested trace box at this index, containing the trace support of $\beta_0 \eta_0$. All error sets and source charges in the following construction are restricted to this box. Define the outside-core shallow set

$$E_{\mathrm{sh}} = \{(x, t) \in \mathcal{T}_{\mathrm{out}} : x \text{ on the outer support, } |t - S_k(x)| \geq K \ (1 \leq k \leq m), \ |u(x, t)| < c_2\}.$$

Every such point is in the central source band and farther than C_I from every selected root. The proximity hypothesis therefore gives $m_* = 0$ and $|Du| = r_*$ almost everywhere there. The assumed full-gradient floor outside E_{det} gives

$$|E_{\mathrm{sh}}| \leq |E_{\mathrm{det}}| + C \int_{E_{\mathrm{sh}}} r_* = o(S).$$

The entire exception is

$$E = (E_{\mathrm{det}} \cap \mathcal{T}_{\mathrm{out}}) \cup E_{\mathrm{sh}} \cup \{(x, t) \in \mathcal{T}_{\mathrm{out}} : x \in B_{\mathrm{bad}}, \ \min_k |t - S_k(x)| < K\}, \qquad |E| = o(S).$$

Outside cores the support of Db lies in E_{sh}, since $c_1 < c_2$. The uniform upper gradient bound from lemma 2.1 now gives the needed power estimate, rather than inferring it from an L^1 bound alone:

$$\int |Db|^p \le C\|Du\|_\infty^{p-1} \int_{E_{\mathrm{sh}}} r_* + C|E_{\mathrm{det}}| + C_K|B_{\mathrm{bad}}| = o(S).$$

All these integrals are on the fixed outer capped trace support. The same E pays every bounded state-interpolation error; off E the correction is an integer well shift.

The bounded-amplitude interpolation estimate

$$\|v\|^2_{\dot{H}^{p/2}(\mathbb{R}^{d+1})} \le C\|v\|_\infty^{2-p}\|Dv\|_p^p$$

holds for compactly supported bounded $W^{1,p}$ functions. To check it, put $L = \|v\|_\infty$ and $G = \mathcal{M}|Dv|$. At Lebesgue points the usual Sobolev maximal-function estimate gives $|v(X) - v(Y)| \le C|X-Y|(G(X)+G(Y))$; it follows by telescoping averages on concentric balls about the two points. Combine this with $|v(X) - v(Y)| \le 2L$, split the squared minimum into the two symmetric terms involving $G(X)$ and $G(Y)$, and use

$$\int_0^\infty \min\{L^2, r^2G(X)^2\}\, r^{-1-p}dr = C_pL^{2-p}G(X)^p.$$

Integration in X and the L^p boundedness of the maximal operator, valid because $p > 1$, prove the displayed estimate. The case $L = 0$ is immediate. Apply it to $v_b = \beta_0\eta_0(\cdot, 0)b$, zero-extended from the regular root region, and let B be its auxiliary full isotropic Poisson extension in the coordinates of the test. Only its prescribed trace and small Dirichlet energy are used; it is not assigned the actual constant-shear equation. The bounded function b is locally Lipschitz across its seams and has the stated integrable derivative, so the zero extension follows by the defining H_0^1 approximation of the root cutoff and bounded truncation. By (5), we obtain

$$E(B) \le o(S) + CR^{d+1-p} = o(S).$$

Indeed $\int |D_x\beta_0|^p \le CR^{d-p}$ by Holder, and the normal trace cap has length $O(R)$; its normal derivative contributes the same power. Finally

$$\frac{R^{d+1-p}}{S} = \frac{R^{3-p}}{D^2} \le CR^{-(p-1)^2/(p+1)} \longrightarrow 0.$$

The nesting gives $B_0 = b$ exactly wherever the inner test is nonzero.

Set $F = U + B - P$ and $\theta = \beta_1\eta_1$. Define the secant coefficient, including its continuous value when the two arguments agree, by

$$c(x,t) = \int_0^1 W''_{\mathrm{per}}(P_0 + s(u + b - P_0))ds.$$

The continuation is indeed C^2: the endpoint jets are $W(\pm1) = W'(\pm1) = 0$, $W''(\pm1) = 2$. Hence $|W''_{\mathrm{per}}| \le 2$, and the secant coefficient is bounded in absolute value by two. Outside E, b is 0 or 2ς, so periodicity gives $W'(u) = W'_{\mathrm{per}}(u + b)$, while $P_0 \in [-1, 1]$ gives $W'(P_0) = W'_{\mathrm{per}}(P_0)$. On every good core $b = 0$, and the quantitative detector comparison plus the $O(D^{-p})$ remote-layer tails makes c uniformly close to $3q(t - S_k)^2 - 1$.

Outside all cores and outside E one has $|u| \ge c_2$ and $\varsigma P_0 \ge c_2$. If $\varsigma u \ge c_2$ then $b = 0$ and $u + b \in \varsigma[c_2, 1]$. If $\varsigma u \le -c_2$ then $b = 2\varsigma$ and $u + b \in \varsigma[1, 2 - c_2]$. Thus both $u + b$ and P_0 lie in the SAME periodic deep-well interval,

$$[c_1, 2 - c_1] \quad (\varsigma = 1), \qquad [-2 + c_1, -c_1] \quad (\varsigma = -1).$$

On each whole interval $W''_{\mathrm{per}} \ge 3c_1^2 - 1 = c_* > 0$, including across its integer well. This proves the actual outside-core secant bound. Replace c on E by $3q(t - S_k)^2 - 1$ inside a core

and by c_* outside the cores. This modified coefficient is defined on the entire normal line, rather than only on the outer tested trace box: outside that box set it equal to c_*. For fixed M, all the comparison centers satisfy $|S_k - \tau_j| \le C_M D + A$ relative to the common scalar cap center τ_j, and $D/R \to 0$, so every fixed K-core lies strictly inside the plateau of both scalar-centered trace caps for all sufficiently large indices. The extension therefore changes no core and is positive outside the cores. The tested trace $\theta_0 F_0$ is zero outside the capped trace box, so the extended coefficient leaves its quadratic form unchanged. With this explicit global extension it satisfies every hypothesis of lemma 10.2 on each entire normal line. Since $|F_0| \le 4$ and $|E| = o(S)$, this replacement and the possible failure of the periodic identity on E change the tested form by $o(S)$.

Apply lemma 10.2 fiberwise. The layer well and derivative tails used in the anchor and residual estimates below are supplied by lemma 2.4. Choose its fixed core size first, then detector accuracy below its perturbation threshold; the actual separations $\delta D \to \infty$ meet its gap assumption. The anchors lie within a fixed distance of their zero centers. On the orientation-consistent bases the exact phase pinning gives $|F_0(T_k)| \le C D^{-p}$, since $u(T_k) = c_j$ and $b(T_k) = 0$. On the incorrect-pattern bases use the uniform $O(1)$ anchor bound instead. Their physical base measure is $o(S)$ by the orientation charge together with the already controlled high-angle bases. The total anchor penalty is therefore at most $C R^d D^{-2p} + o(S) = o(S)$; the first term divided by S is at most $C\Lambda^2 R^{-2}$. The comparison centers retain their original actual phase-pinned definition on both sets. Thus

$$\int |D_n(\theta F)|^2 d\mu + \int [W'(u) - W'(P_0)]\theta_0^2 F_0 \ge c\left[\int |D_n(\theta F)|^2 d\mu + \int |\theta_0 F_0|^2\right] - o(S).$$

We verify the profile residual and its cancellation, including the two unbounded exterior normal intervals. At a fixed retained base put $q_i(t) = q(t - S_i)$ and

$$r_P = (-\partial_t^2)^{p/2} P_0 + W'(P_0) = W'(P_0) - \sum_i \epsilon_i W'(q_i).$$

Partition the normal line into nearest-center cells V_k, with ties assigned arbitrarily. For $i \ne k$ on V_k let $\ell_i = \operatorname{sgn}(t - S_i)$, $r_i = q_i - \ell_i$, and $\delta_k = \sum_{i \ne k} \epsilon_i r_i$. The ordered alternating signs and the stated baseline give the exact identity $c_m + \sum_{i \ne k} \epsilon_i \ell_i = 0$, so $P_0 = \epsilon_k q_k + \delta_k$ there. Since $W'(v) = v^3 - v$, direct expansion yields

$$r_P = 3(q_k^2 - 1)\delta_k + 3\epsilon_k q_k \delta_k^2 + \delta_k^3 - \sum_{i \ne k} \epsilon_i (3\ell_i r_i^2 + r_i^3). \tag{RC}$$

In particular the remote linear terms cancel: each $W'(\ell_i + r_i)$ has linear part $2r_i$, which combines with $(3q_k^2 - 1)\delta_k$ to give $3(q_k^2 - 1)\delta_k$.

The tails in lemma 2.4 imply $|q(s) - \operatorname{sgn} s| \le C(1 + |s|)^{-p}$ and $\int (1 - q^2) < \infty$, because $p > 1$. Every remote center is at distance at least half the adjacent separation from V_k, hence at least $c_\delta D$. Consequently, uniformly in all the retained center positions,

$$\sup_{V_k} |r_i| \le C_\delta D^{-p}, \qquad \int_{V_k} |r_i| dt \le C_\delta D^{1-p} \quad (i \ne k).$$

These integrals are valid also on V_1 and V_m, whose infinite tails are integrable. Thus the first term of (RC) has integral at most $C_\delta D^{-p} \int (1 - q^2)$; every quadratic term has integral at most $C_\delta D^{-p} D^{1-p}$, and the cubic terms are smaller. Summing the finitely many cells gives

$$\|r_P\|_{L^1(\mathbb{R}_t)} \le C_{\delta,m}(D^{-p} + D^{1-2p}) \le C_{\delta,m} D^{-p}.$$

Integrating over the physical base receiver gives $\|r_P\|_1 \le C R^d D^{-p} = o(S)$, since $R^d D^{-p}/S = \lambda_j / D \to 0$. This is the residual bound used in the compact test pairing below.

For the actual constant-shear equation, compatibility $D_x H = V_t$ gives, for the complete product test Φ,

$$\int Hk \cdot D_x\Phi \, d\mu = \int (k \cdot V)\Phi_t \, d\mu.$$

Subtract the normal profile equation and test with $\Phi = \theta^2 F$. The profile equation has right side $\int r_P \Phi_0$, and compatibility combines the two mixed shear terms. Thus the exact identity is

$$\begin{aligned}
&\int D_n F \cdot D_n(\theta^2 F) \, d\mu + \int [W'(u) - W'(P_0)]\theta_0^2 F_0 \, dX \\
&= -\int (I + kk^T)V \cdot D_x(\theta^2 F) \, d\mu + 2\int (k \cdot V)\partial_t(\theta^2 F) \, d\mu \\
&\quad + \int D_n B \cdot D_n(\theta^2 F) \, d\mu - \int r_P \theta_0^2 F_0 \, dX.
\end{aligned}$$

The normal product rule gives

$$\int D_n F \cdot D_n(\theta^2 F) \, d\mu = \int |D_n(\theta F)|^2 d\mu - \int F^2 |D_n\theta|^2 d\mu.$$

The bounded-amplitude cutoff costs obey

$$\int F^2(|D_x\theta|^2 + |D_n\theta|^2) d\mu \le CR^{d+1-p} = o(S).$$

For the base part this follows from $\int |D_x\beta_1|^2 \le CR^{d-2}$ and the capped normal-height weighted volume $O(R^{3-p})$; the scalar cap derivatives have size C/R on weighted volume $O(R^{d+3-p})$. Also $\int \theta^2 |D_x F|^2 = o(S)$, by transverse approximation, nesting, and $E(B) = o(S)$. Since $\int |V|^2 \le CS$ on the containing receiver, Cauchy–Schwarz makes the entire transverse term $o(S)$. For any fixed $\epsilon > 0$, the mixed shear and correction terms have absolute value at most

$$\epsilon \int |D_n(\theta F)|^2 d\mu + C_\epsilon |k|^2 S + C_\epsilon E(B) + o(S).$$

Here write $D_n(\theta^2 F) = \theta D_n(\theta F) + \theta F D_n\theta$ and use Young's inequality; the same identity applies to its t component. The residual term is $o(S)$, since $|\theta_0^2 F_0| \le 4$ and $\|r_P\|_1 = o(S)$. Choose ϵ smaller than the fixed coercivity constant. This absorbs the normal derivative term and proves the stated comparison. The tests are legitimate in the finite weighted-energy class: their base support is compact inside the regular region, their factors are bounded, and all displayed derivatives have finite weighted energy. Bounded Sobolev approximation within that region, followed by weighted-energy approximation in the normal variables, therefore passes the weak equation to this complete product. Every derivative of every base factor remains in the test throughout the cancellation.

Coercivity gives $\int |D_n(\theta F)|^2 + \int |\theta_0 F_0|^2 = o(S)$. The isotropic correction has $\int |D_n B|^2 d\mu \le E(B) = o(S)$; in a selected shear the same bound has a fixed multiplicative constant. The bounded-amplitude normal/height cutoff term is at most $CR^{d+1-p} = o(S)$. Expanding $D_n(\theta F)$ and removing these two terms therefore proves the claimed weighted normal comparison for $U - P$.

The inner normal Poisson conclusion now follows directly from the previous full-height discrepancy estimate in proposition 9.7; no new trace truncation is needed. Choose $c > 0$ small enough that $\eta_1 = 1$ on $|t - \tau_j| < cR_j$, $0 < y < cR_j$. On this region,

$$\beta_1^2 |D_n(U^{\mathrm{n}} - P)|^2 \le 2\beta_1^2 |D_n(U - P)|^2 + 2|D_n(U - U^{\mathrm{n}})|^2.$$

The first term has integral $o(S)$ by the comparison just proved, and the second by that preceding proposition on a fixed containing physical receiver. Its selected-shear version applies

in the same coordinates as the actual equation. For $y \geq cR$, both U^{n} and P are the one-dimensional normal Poisson extensions of traces in $[-1,1]$. Thus $|D_n(U^{\mathrm{n}} - P)| \leq C/y$, and the remaining full-height integral is at most

$$CR^d R \int_{cR}^{\infty} y^{1-p} y^{-2} dy \leq CR^{d+1-p} = o(S).$$

This proves the displayed inner normal Poisson estimate on every smaller fixed normal window, with all distant trace partners already included in the established discrepancy theorem. The count and fixed-height factors are retained in β_1 throughout; this last passage uses no base derivative. □

Lemma 10.4 (The actual selector replacement has negligible force cost in every base dimension)**.** *Let $d \geq 1$ and $1 < p < 2$. Let $R_j, D_j \to \infty$, $a_j = D_j/R_j \to 0$, $S_j = R_j^{d-2} D_j^2$, and $F_j = S_j/D_j$. Let $H_j = (U_j)_t$, $V_j = D_x U_j$, $Z_j = (V_j, a_j H_j)$, and $\mathcal{A}_j = |DZ_j|^2 - |D|Z_j||^2$, with $d\mu = d_p y^{1-p} dxdtdy$. Suppose*

$$\int \mathcal{A}_j d\mu \leq C a_j^2 R_j^{d-2}.$$

At each fixed index the common normal and height caps have compact support. Assume H_j, V_j and their first physical derivatives are bounded on a containing compact capped product, with bounds allowed to depend on j. Let $0 \leq \beta_j \leq 1$ be bounded H^1 base factors whose zero extensions have compact support inside the tested inner regular base of lemma 10.1, where the selectors are smooth and their stated derivative bounds hold. Let ζ_j be the difference of actual and smooth recovered selectors constructed in lemma 10.1, including common scalar-centered macroscopic caps. Assume the physical root slopes in that construction are bounded by a fixed L_0 on the tested inner regular support. Suppose on its complete support C_j,

$$\int_{C_j} \beta_j^2 H_j^2 d\mu \leq C S_j, \qquad \int_{C_j} \beta_j^2 |V_j|^2 d\mu = o(S_j).$$

Then for any fixed smooth compactly supported base function η,

$$\int \zeta_j H_j V_j \cdot D_x[\eta(x/R_j)\beta_j^2] d\mu = o(F_j).$$

The identity is valid for the complete bounded H^1 product with its original zero extension from the actual regular region.

Proof. Fix all cutoff and smooth-recovery parameters before taking the record limit. The capped selector difference from lemma 10.1 is uniformly bounded and, on the tested inner regular support, satisfies

$$|\zeta_{j,t}| \leq C_\delta/D_j + C/R_j, \qquad |D_x \zeta_j| \leq C_\delta L_0/D_j + C_{\delta,\varepsilon}/R_j.$$

The slope in this estimate is the physical slope $D_x T_{j,k}$. No bound on the normalized slope $D_\xi t_{j,k} = a_j^{-1} D_x T_{j,k}$ is being assumed. Put $q_j = \eta(x/R_j)\beta_j^2$ and suppress the index temporarily. Since $D_x H = \partial_t V$,

$$\mathrm{div}_x(HV) = \partial_t |V|^2 - \mathscr{W}, \qquad \mathscr{W} = \sum_{i=1}^{d} (V_i \partial_i H - H \partial_i V_i).$$

For $Z = (V, aH)$, the $(i, d+1)$ component of $Z \wedge \partial_i Z$ is $a(V_i \partial_i H - H \partial_i V_i)$. Cauchy–Schwarz in i and the angular identity therefore give

$$|\mathscr{W}| \leq \frac{\sqrt{d}}{a} |Z| \sqrt{\mathcal{A}}.$$

This inequality also holds almost everywhere on $\{Z = 0\}$, where the wedges vanish and the angular density has its Sobolev-locality convention. Base integration by parts and then integration in t, using that q is independent of t, y, yield the exact identity

$$\begin{aligned}\int \zeta HV \cdot D_x q \, d\mu &= -\int q \, [\zeta \operatorname{div}_x(HV) + D_x\zeta \cdot HV] d\mu \\ &= \int q \, [\zeta_t |V|^2 + \zeta \mathscr{W} - D_x\zeta \cdot HV] d\mu.\end{aligned}$$

We justify this identity for the actual Sobolev product at each fixed index, before estimating it uniformly. Its base support is a compact subset of the open regular region. The bounded-density construction in the proof of lemma 9.8, or direct convolution of its zero extension inside a fixed compact neighborhood of that support, gives smooth base tests q_ℓ with a common supremum bound, converging strongly in H^1 and almost everywhere to q. Their supports remain inside that regular region. The compact normal and height caps and fixed-index bounds on H, V, D_xH, D_xV, together with the selector derivative bounds, make all base columns on the right bounded on this neighborhood. The weight is integrable at zero because $p < 2$. The left current column is likewise bounded, hence in L^2 on the fixed base support. Strong gradient convergence passes the left pairing; bounded convergence against the fixed integrable right columns passes the right side. Thus the displayed identity is an equality of ordinary integrals for the original zero extension, without any regular-region boundary term. Compact normal support removes the t endpoint term. No integration by parts in height has been made.

Return to the index j, and set

$$e_j = S_j^{-1} \int_{C_j} \beta_j^2 |V_j|^2 d\mu \longrightarrow 0.$$

Constants below may depend on the fixed parameters, on L_0, and on η, but not on j. The first right-hand term is bounded by

$$C \left(D_j^{-1} + R_j^{-1} \right) S_j e_j \le C F_j (1 + D_j/R_j) e_j = o(F_j).$$

For the Wronskian term use $0 \le \beta_j \le 1$, the angular budget, and

$$\int_{C_j} \beta_j^2 |Z_j|^2 d\mu \le S_j (e_j + C a_j^2).$$

Consequently

$$\begin{aligned}\int_{C_j} \beta_j^2 |\mathscr{W}_j| d\mu &\le \frac{C}{a_j} \sqrt{S_j (e_j + C a_j^2)} \sqrt{C a_j^2 R_j^{d-2}} \\ &\le C F_j \sqrt{e_j + C a_j^2} = o(F_j),\end{aligned}$$

since $\sqrt{S_j} R_j^{(d-2)/2} = R_j^{d-2} D_j = F_j$. Finally the horizontal-selector term is at most

$$(C_\delta L_0/D_j + C_{\delta,\varepsilon}/R_j) \sqrt{C S_j} \sqrt{S_j e_j} \le C F_j (L_0 + D_j/R_j) \sqrt{e_j} = o(F_j).$$

Here the physical slope bound L_0 is fixed, while $D_j/R_j = a_j \to 0$. All cap derivatives were included in the first displayed bounds, so none has been discarded separately. Summing these estimates proves the claim for the derivative of the entire base product; no estimate involving $H_j^2 |D_x \beta_j|^2$ is required. □

11. The complete signed force and zero-interaction covariance

Proposition 11.1 (Actual complete ranks yield the full alternating force or zero-interaction covariance system)**.** *Let definition 9.1 hold. The actual limits of proposition 9.6 satisfy the following alternatives. If $\lambda > 0$, all adjacent gaps $g_i = f_{i+1} - f_i$ are positive almost everywhere with $g_i^{-p-1} \in L^1_{\mathrm{loc}}$, and with $\epsilon_k = (-1)^{k-1}$,*

$$\Delta f_k = C_\lambda \sum_{l \neq k} \epsilon_k \epsilon_l \operatorname{sgn}(l-k) |f_l - f_k|^{-p}, \quad C_\lambda = \frac{4c_{1,p}\lambda}{p\sigma},$$

and, for every $\phi \in C_c^\infty(\mathbb{R}^d)$,

$$\int V\phi^2 \leq \int |D\phi|^2, \qquad V = \frac{16c_{1,p}\lambda}{m\sigma} \sum_{i=1}^{m-1} g_i^{-p-1}.$$

If $\lambda = 0$, every f_k is harmonic and all gaps are constant. With $b = -Df_1 = -Df_k$, the common tilt is a constant vector and the complete mixed and transverse covariance limits are exactly

$$B = m\sigma b, \qquad G = m\sigma b \otimes b.$$

The optimizing covariance and the retained normalization give

$$\int_{B_1} B = 0, \qquad 1 \leq \int \varphi \operatorname{tr} G$$

for a fixed nonnegative containing base cutoff $\varphi = 1$ on B_1. These two identities are incompatible.

Proof. The scalar equation, stability, and normalization used here are supplied directly by proposition 3.4 for this actual record. Thus

$$\int DU \cdot D\phi \, d\mu + \int W'(u)\phi_0 = 0, \qquad \int DH \cdot D\phi \, d\mu = \int (1 - 3u^2) h \phi_0,$$

where $H = U_t$, $h = u_t$, $W' = -u + u^3$, and $d\mu = d_p y^{1-p} dX dy$ represents the Fourier multiplier $|\xi|^p$. The second identity follows by differentiating the first in the fixed physical t direction, using uniform smoothness and compact tests. They justify all actual stationarity and Jacobi tests below. In the selected shear both left sides retain the exact constant coefficient matrix of definition 9.1; determinant one and the chain rule give these pulled-back identities without changing the trace terms. The normal caps and their constant component values on every fixed compact phase interval are those of proposition 5.2. They can be chosen in one containing scalar-centered box, and avoid all tested central roots by its whole-central-set confinement.

By lemma 2.1 the first and second derivatives in the physical variables $X = (x, t)$ of the trace, and hence of its Poisson extension, are bounded at every index. The scalar-centered normal/height caps have compact support and the full base products have genuine compact inner regular support. Thus the finite-index boundedness, ordinary-integral, and support hypotheses of lemma 10.4 are satisfied when that lemma is applied below.

Suppress the index and put $R = R_j$, $D = D_j$, $a = a_j = D/R$, $S = a^2 R^d$, and $Z = (V, aH)$. We use the already proved complete-height angular estimate (6): on every fixed enlarged physical receiver,

$$\int |Z|^2 d\mu \leq CS, \qquad \int \mathcal{A}_Z d\mu \leq CS/R^2, \qquad \mathcal{A}_Z = |DZ|^2 - |D|Z||^2.$$

Its proof in proposition 9.6 includes the transfer to the actual selected small shear by its uniformly bounded source and target maps. Thus these estimates hold for the selected-frame

fields used throughout this proof. We use the angular identity $\mathcal{A}_Z = \sum_\alpha |Z \wedge \partial_\alpha Z|^2/|Z|^2$ on $\{Z \neq 0\}$, with value zero almost everywhere on $\{Z = 0\}$. In particular the geometric-defect hypothesis of lemma 10.4 is available before its application.

Choose one fixed enlarged physical trace receiver containing all the normal caps and their derivative supports, and choose the column cap ρ_R to equal one on this receiver's tested normal interval. When the normal comparison is used below, take a common smooth equation-test cap $0 \le \gamma_j(t, y) \le 1$, of scale R_j, whose support, including its derivative supports, lies inside the plateau of the larger cap in proposition 10.3. Use γ_j^2 for both actual and recovered selectors. All these caps equal one near every fixed core on a fixed bounded-height support; the margins are possible because $D_j/R_j \to 0$. This choice leaves the base nesting unchanged and ensures that the comparison norm controls every normal derivative of the equation test.

Use the actual fine record definition 9.1, the complete ranks and normal columns of proposition 9.6, their full transverse profile and selected-frame normal discrepancy in proposition 9.7, and the complete all-phase source decomposition and corridor energy in proposition 9.9. All parameters below are fixed before $j \to \infty$. Heights are truncated at M throughout the finite-record argument. Here the selected-phase hypotheses of lemma 10.1 are checked from the producer statements. Use the choice $\tau_j = 0$ in proposition 9.6. On the regular open set O_j its normalized actual heights are $t_{j,k} = T_{j,k}/D_j$. That proposition supplies one continuous regularity cutoff $0 \le \theta_j \le 1$, with $\theta_j \to 1$ strongly in local H^1 at the same selected phase c_j, vanishing in a relative neighborhood of the complement of O_j. For every fixed L its zero-extended clipped lifts satisfy

$$\theta_j T_L(t_{j,k}) \longrightarrow T_L(f_k) \quad \text{strongly in local } H^1.$$

The same phase sequence and cutoff are used for all clips by its countable diagonal. The all-phase source construction proposition 9.9 supplies the nonnegative count norm $q_j \to q_* = \sqrt{m|I|}$ strongly in local H^1. Take its fixed count threshold larger than q_*. Restrict to a bounded Lipschitz buffered receiver and multiply by the fixed compact base factors. These are precisely the cutoff, lift, regular-support, and count-norm hypotheses of lemma 10.1; finite bounded products and the deweighting chain preserve their strong H^1 convergence. No phase-product convergence is being substituted for a selected-phase statement. It therefore supplies nested factors $\beta_{j,1}, \beta_{j,0}$, strongly convergent in H^1, with every tested actual height at most $C_M D_j$ and every tested outer gap at least $c\delta D_j$. Its globally smooth recovered midpoints prove Euclidean clearance from every surviving source, including partners on other connected base components. The count-adapted screen pays every excluded source-base partner. This is the source decomposition on the entire outer macroscopic trace box, not merely on the same fiber as a receiver.

Write R, D, a, S, F for one record index, so $S = R^{d-2}D^2$ and $F = S/D$. The corridor estimate just cited gives, on each cleared gap of width δD,

$$\int \beta_1^2 \left[W(u) + \int_0^\infty d_p y^{1-p} |DU|^2 dy \right] dX \le C_\delta R^d D^{1-p} + o(S) = C_\delta \lambda_j S + o(S).$$

On the scalar-centered macroscopic caps it gives $CR^{d+1-p} + o(S)$. A normal cap derivative thus costs at most $CR^{d-p} + o(S/R) = o(F)$, because $D/R \to 0$ and the bounded interaction parameter gives

$$\frac{R^{2-p}}{D} = \lambda_j^{1/(p+1)} R^{-p(p-1)/(p+1)} \longrightarrow 0.$$

The complete transverse profile also gives $\int_{\text{cleared gap}} \beta_1^2 |V|^2 d\mu = o(S)$: its finitely many translated core profiles have tails tending to zero at distance $c\delta D \to \infty$, and their fixed L^2 slope coefficients allow dominated convergence. These estimates retain the entire extension

height. Equivalently insert a height cutoff at Y and remove it at each fixed index; $|DU| \le C/y$ makes its final stress contribution tend to zero.

We first establish the current interface used in both alternatives. Fix the height, separation and smooth-recovery parameters. Let ϑ_j be a selector formed from the globally smooth recovered midpoints, multiplied by the common squared normal cap (and by the height cap when present). Its support and derivative supports lie in the plateau of ρ_{R_j}, as chosen above. Its derivatives satisfy

$$|\partial_t \vartheta_j| \le C_\delta/D_j + C/R_j, \qquad |D_x \vartheta_j| \le C_{\delta,\varepsilon}/R_j.$$

On the retained nested supports its core values are the prescribed rank or group constants. Globally they need not be constants; write $\vartheta^\varepsilon(\xi, f_k(\xi), 0)$ for their bounded limiting values on the kth core, with the normal argument scaled by D_j. Then

$$j_j(\xi) := a_j^{-1} \int \rho_{R_j} \vartheta_j H_j V_j d\mu_{t,y} \longrightarrow -\sigma \sum_k \vartheta^\varepsilon(\xi, f_k, 0) Df_k \quad \text{strongly in } L^1_{\mathrm{loc}}.$$

Indeed the full transverse profile first replaces $a_j^{-1} V_j$ in this product by its complete model, by Cauchy–Schwarz and the uniform L^1 bound for n_j. Truncate the fixed coefficients Df_k at a finite magnitude and keep fixed normal core rectangles. Normal core convergence identifies each diagonal product there; its orientation cancels between H_j and V_j. Distinct rectangles are physically separated in measure, so the off-diagonal products vanish. Complete normal-core exhaustion removes the complement as the core radius grows. Finally the discarded coefficient tails cost at most a constant times $(\sum_k \int_{\{|Df_k|>K\}} |Df_k|^2)^{1/2}$ by Cauchy–Schwarz, and tend to zero as $K \to \infty$. The bounded selector core values converge in measure, which suffices against these fixed integrable coefficients. This proves the asserted strong convergence, without an orientation pattern. Its limiting current is in L^2 for the finite selected family.

We check the divergence hypothesis with its physical scaling. Put $\psi_j = \rho_{R_j} \vartheta_j$. The cap margin gives $\rho'_{R_j} \vartheta_j = 0$, and the derivative bounds may be weighted by ρ_{R_j}; thus they are controlled by precisely the containing columns e_j, n_j. At each fixed index its compact normal support and the bounded physical derivatives, together with their C/y large-height bounds, make the current locally bounded and smooth; in particular it is in L^2 on every fixed bounded base receiver. Compatibility gives

$$\operatorname{div}_x(HV) = \partial_t |V|^2 - \mathscr{W}, \qquad \mathscr{W} = V \cdot D_x H - H \operatorname{div}_x V, \qquad |\mathscr{W}| \le \sqrt{d}\, a^{-1} |Z| \sqrt{\mathcal{A}_Z}.$$

Consequently, with $x = R_j \xi$, normal integration by parts gives

$$\operatorname{div}_\xi j_j = \frac{R_j}{a_j} \int [-(\psi_j)_t |V_j|^2 - \psi_j \mathscr{W}_j + D_x \psi_j \cdot H_j V_j] d\mu_{t,y}.$$

The absolute first and third terms are bounded respectively by $C_\delta e_j$ and $C_{\delta,\varepsilon} \sqrt{e_j n_j}$, since $R_j a_j = D_j$. For a measurable scaled base set E, the wedge term satisfies

$$\begin{aligned} \int_E \frac{R_j}{a_j} \int |\psi_j \mathscr{W}_j| d\mu_{t,y} d\xi &\le \frac{C R_j^{1-d}}{a_j^2} \left(\int_{R_j E} \rho_{R_j} |Z_j|^2 d\mu \right)^{1/2} \left(\int \mathcal{A}_{Z_j} d\mu \right)^{1/2} \\ &\le C \left(\int_E (e_j + n_j) d\xi \right)^{1/2}. \end{aligned}$$

All physical integrals here stay in the fixed containing receiver; the second factor can be enlarged to that receiver. The column uniform integrability proves uniform integrability of these divergences, since $\sqrt{e_j n_j} \le (e_j + n_j)/2$. This assertion concerns the smooth recovered selectors at fixed parameters; their actual counterparts have larger horizontal derivatives and are handled by replacement below.

On a bounded buffered base domain, the complete base products belong to $H^1_0 \cap L^\infty$, have common bounds and converge strongly in H^1. Thus every hypothesis of lemma 9.8 has been checked for these currents, and it passes their pairing with the derivative of the entire product. First take $j \to \infty$ at fixed smooth-recovery parameter. On the support of the limiting base product the retained core values are precisely the rank or group constants. Its weak gradient vanishes almost everywhere on its zero set by Sobolev locality, so the limiting pairing there equals the pairing with the corresponding fixed sum of $-\sigma D f_k$. The nested products converge strongly in H^1 as $\varepsilon \downarrow 0$, and their pairing with this fixed L^2 current converges by Cauchy–Schwarz. At each finite index, bounded Sobolev approximation stays inside the genuine interior regular support supplied by lemma 10.1. Hence these pairings are ordinary integrals for the original zero-extended products, without a regular-region boundary term.

The same transport statement holds with HV replaced by HA_jV for every bounded constant symmetric matrix $A_j \to A$. Indeed,

$$\operatorname{div}_x(HA_jV) = \partial_t(V \cdot A_jV) - \mathscr{W}_{A_j}, \qquad \mathscr{W}_{A_j} = \sum_{i,l}(A_j)_{il}(V_l\partial_i H - H\partial_i V_l).$$

Every summand is a two-component wedge of Z and $\partial_i Z$, divided by a. Consequently

$$|\mathscr{W}_{A_j}| \le C_d\|A_j\|a^{-1}|Z|\sqrt{\mathcal{A}_Z}.$$

The preceding column estimates therefore prove uniform integrability of the normalized divergences for these matrix currents too. Their strong L^1 limit is A times the preceding current, so lemma 9.8 applies to the derivative of the same entire bounded H^1 product. In particular it applies directly to $A_j = I + k_j k_j^T \to I$; no multiplication of an uncontrolled pairing by $|k_j|^2$ is used.

The actual rank selectors are finite differences of midpoint steps. Specifically choose a smooth nondecreasing step $k : \mathbb{R} \to [0,1]$, zero below $-1/8$ and one above $1/8$, and let $K_i = k((t-(T_i+T_{i+1})/2)/(\delta D))$, $K_0 = 1$, $K_m = 0$, and take $\gamma_j^2(K_{k-1}-K_k)$ with the common cap chosen above. The ordering gives $0 \le K_{k-1} - K_k \le 1$. Their smooth counterparts replace every T_i by its globally smooth recovered height. The actual-minus-smooth difference is confined to the cleared gap corridors by lemma 10.1. On that difference corridor the normal energy is $O(S)$ and the transverse energy is $o(S)$ by the estimates above. lemma 10.4 consequently makes their pairing with the derivative of the entire base product differ by $o(F)$.

Its proof also applies to these constant matrix currents: the exact identity becomes

$$\int \zeta HA_jV \cdot D_x q \, d\mu = \int q\,[\zeta_t V \cdot A_jV + \zeta\mathscr{W}_{A_j} - D_x\zeta \cdot HA_jV]d\mu.$$

The bounded factor $\|A_j\|$ leaves every $o(F_j)$ estimate unchanged. Thus the replacement and the full-product transport both hold for the displayed $I + k_j k_j^T$ current.

The physical slope bound $|D_x T_{j,k}| \le 1$ on the actual regular supports is supplied by proposition 9.6. The bounded-slope variant of lemma 10.1 gives $|D_x\chi| \le C_\delta/D$ on an actual midpoint transition. Apply lemma 10.4 with $L_0 = 1$. Its pairing with the derivative of the entire base product is $o(F)$, and terms containing the actual selector's own horizontal derivative are bounded by $C_\delta\sqrt{O(S)o(S)}/D = o(F)$. Thus for rank k, after the smooth-recovery limit,

$$F_j^{-1}\int H_j(I + k_j k_j^T)V_j \cdot D_x q_{j,k} d\mu \longrightarrow -\sigma\int Df_k \cdot D(\eta h_M^2 b_\delta^2).$$

Here $q_{j,k} = \eta(x/R_j)\beta_{j,1}^2\gamma_j^2\chi_{j,k}$, where $\chi_{j,k} = K_{k-1} - K_k$; thus it includes the complete base product, rank selector and shared squared normal/height cap; h_M is its limiting bounded

height factor. Finitely many extra fixed count factors tend to one strongly in H^1, as already included in the nested construction.

The pure shear is removed only after assembling this complete product. In this paragraph χ includes the common squared normal/height cap, while the complete bounded factor $b = b(x)$ depends only on the base. Compatibility gives

$$\int H^2 D_x(b\chi)d\mu = -\int b\chi(HV_t - H_tV)d\mu + \int b\chi_t HV\,d\mu.$$

To check it, differentiate $\int \chi H^2$ in x, use $D_xH = V_t$, and integrate $\partial_t(HV)$; the two terms with $D_x\chi$ cancel exactly. The wedge integral is at most $Ca^{-1}\sqrt{S}\sqrt{S/R^2} = CF$. The base factor satisfies $|b| \le C\beta_1^2$ on its support. Thus a gap derivative contributes at most $C_\delta D^{-1}\sqrt{O(S)o(S)} = o(F)$. On a macroscopic normal cap use $|\chi_t| \le C/R$, the complete normal bound $\int H^2 d\mu \le CR^d$, and the transverse bound $\int |V|^2 d\mu \le CS$; its contribution is at most $CR^{-1}\sqrt{R^dS} = CF$. Multiplication by $|k_j| \to 0$ makes the full pure-shear contribution $o(F)$. Thus the canonical current

$$\mathcal{J}_{k_j}(U) = H_j(I + k_jk_j^T)V_j - k_jH_j^2$$

has precisely the same limit. No comparison of $|k_j|$ with a_j is needed.

We record the stress identity with its signs before taking either limit. Put $A = I + kk^T$ and $\mathsf{S}_n(U) = ((H^2 - U_y^2)/2, HU_y)$. For a compact smooth space–height test q, test the pulled-back scalar equation with Hq. Since its coefficient matrix is constant, its terms in which the derivative falls on H equal the t-derivative of half the full quadratic energy. Integration in t, together with $W'(u)u_t = \partial_t W(u)$, therefore gives

$$\int \left[\mathcal{J}_k(U)\cdot D_xq + \mathsf{S}_n(U)\cdot D_nq - \tfrac{1}{2}V\cdot AV\,q_t\right]d\mu - \int W(u)q_{0,t}\,dX = 0.$$

More explicitly, the half quadratic energy is

$$\mathcal{E}_k = \tfrac{1}{2}(V\cdot AV - 2Hk\cdot V + H^2 + U_y^2),$$

and the q_t coefficient after integration by parts is $H(H - k\cdot V) - \mathcal{E}_k = (H^2 - U_y^2)/2 - (V\cdot AV)/2$. This verifies the mixed-term cancellation and the displayed bottom sign. The same calculation is valid for the actual bounded Sobolev base products by the fixed-index approximation already justified. Exhausting the height is also legitimate: the only extra flux is HU_yq_y, and on $Y < y < 2Y$ its absolute integral is at most C_jY^{-p-1}, using $|DU| \le C/y$ and $|q_y| \le C/Y$. Thus it vanishes at every fixed index. This proves the complete-height version without a missing height boundary term.

Suppose first $\lambda > 0$. lemma 9.10 gives $g_i > 0$ almost everywhere, local integrability of g_i^{-p-1}, and the exact stability inequality stated in the proposition. Fix the simultaneous gap factor $b_\delta = \prod_i b(g_i/\delta)$. The outer actual support separates all adjacent gaps, so proposition 10.3 applies. Its central phase, all-phase source mass, detector volume/floor and refined fixed-core hypotheses are supplied respectively by proposition 9.6 and proposition 9.9. It proves, on the full nested product and fixed macroscopic caps,

$$\int \beta_1^2\eta_1^2|D_n(U - P)|^2 d\mu = o(S), \qquad \int |\beta_1\eta_1(\cdot,0)(u + b - P_0)|^2 = o(S),$$

where b is its bounded periodic well-state correction. The actual alternating finite profile is

$$P = c_m + \sum_{k=1}^{m}\epsilon_k Q(t - S_k, y), \qquad \epsilon_k\epsilon_l = (-1)^{k+l},$$

with the exact phase-pinned comparison centers of proposition 9.7. Here $\epsilon_k = \epsilon_1(-1)^{k-1}$, with $\epsilon_1 \in \{-1, 1\}$ constant on each regular component, and $c_m = 0$ for odd m and $c_m = -\epsilon_1$ for even m. Any overall reversal of its orientations has no effect on the products appearing below. The orientation repair in the comparison charges a disagreement to discarded central variation, so no extra untracked transition is omitted.

Use the normal residual

$$r_P = (-\partial_t^2)^{p/2} P_0 + W'(P_0) = W'(P_0) - \sum_i \epsilon_i W'(q(t - S_i)).$$

Its exact cubic decomposition (RC) and full-line estimate $\|r_P\|_{L^1(\mathbb{R}_t)} \le C_{\delta,m} D^{-p}$ were proved in proposition 10.3, including the two unbounded exterior nearest-center cells. We need the following additional tightness for its projection onto an individual layer. In (RC), on the nearest-center cell about S_i, the only remote linear term is $3(q(t - S_i)^2 - 1)\delta_i$, with $|\delta_i| \le C_\delta D^{-p}$. The total integral of every other term is $O_\delta(D^{1-2p})$, by the estimates in that proof. Restricting outside every fixed L-core therefore gives

$$\int_{\min_i |t-S_i|>L} |r_P| dt \le C_\delta D^{-p} \omega(L) + O_\delta(D^{1-2p}), \qquad \omega(L) = \int_{|s|>L} (1 - q(s)^2) ds \longrightarrow 0.$$

Here $p > 1$ makes the layer well deviation integrable and $D^{1-2p} = o(D^{-p})$. All bounds are uniform over the retained center configurations for fixed M, δ. This is the off-core control needed before increasing the projection core radius.

On a fixed core about S_k, the remote correction from the lth layer is

$$\delta_l = \epsilon_l \operatorname{sgn}(l - k) \frac{c_{1,p}}{p} |S_l - S_k|^{-p} + o_\delta(D^{-p}).$$

The error is uniform on that fixed core, because every other center is at distance at least $c_\delta D$; the tail asymptotic and derivative bound in lemma 2.4 apply uniformly at those distances. Its own fractional equation contributes $-2\delta_l + o_\delta(D^{-p})$, and the linear variation of W' at $\epsilon_k q$ contributes $(3q^2 - 1)\delta_l$. Their sum is $-3(1 - q^2)\delta_l$, the remote linear term of (RC). The selected layer selector equals one on this fixed core and zero on the other fixed cores for sufficiently large indices; the exact phase-pinned centers differ from their actual selected roots by only the fixed phase offset. Also $(P_0)_t = \epsilon_k q'(t - S_k) + O_\delta(D^{-1-p})$ there. The integral $\int 3(1 - q^2) q' \, dt = 4$ now yields, after first taking the record limit on fixed cores and then sending $L \to \infty$,

$$\int r_P \chi_k (P_0)_t \, dt = -\frac{4 c_{1,p}}{p} \sum_{l \ne k} \epsilon_k \epsilon_l \operatorname{sgn}(l - k) |S_l - S_k|^{-p} + o_\delta(D^{-p}).$$

Indeed $|(P_0)_t| \le C_m$ and $|\chi_k| \le 1$, so the preceding off-core residual bound makes the normalized projection remainder at most $C_\delta \omega(L) + o_\delta(1)$. This proves the displayed coefficient and its sign with the full unbounded normal line included. The same leading term is obtained after multiplying χ_k by the shared macroscopic trace cap squared: that cap is one on every fixed retained core, and its off-core error has the same bound just proved. The previously established full-line residual bound also gives $\|r_P\|_1 \le C R^d D^{-p}$ after integration over the containing physical base.

We choose the complete product so that its cap derivatives can be estimated directly. Write

$$q = q_{j,k} = \eta(x/R) \beta_1^2 \gamma^2 \chi_k, \qquad \mathcal{L}_q = 2\chi_k D_n \gamma + \gamma D_n \chi_k,$$

where γ is the shared smooth normal/height cap, χ_k is the uncapped actual rank selector, and β_1 contains the entire base product. Choose the comparison caps in proposition 10.3 with their stated nested margins so that $\int \beta_1^2 \gamma^2 |D_n(U - P)|^2 = o(S)$ and the analogous squared

trace estimate hold on this product. The cap is one on every fixed retained core for large indices, and

$$D_n q = \eta(x/R)\beta_1^2\gamma\mathcal{L}_q.$$

This factorization never divides by a cap that may vanish. The normal derivative of χ_k is supported in the two cleared adjacent corridors and is bounded by C_δ/D; its height derivative is zero. The components of $D_n\gamma$ have size at most C/R.

Subtract the normal profile stress identity from the actual vertical stress identity established above, using this same complete product. The profile identity follows by testing its normal equation with $(P_0)_t q_0$ at the trace and $P_t q$ in the bulk; it gives

$$\int \mathsf{S}_n(P)\cdot D_n q\,d\mu - \int W(P_0)q_{0,t}\,dX = \int r_P q_0 (P_0)_t\,dX.$$

Consequently the exact subtraction is

$$\begin{aligned}\int \mathcal{J}_{k_j}(U)\cdot D_x q\,d\mu = &-\int r_P q_0(P_0)_t\,dX + \int [W(u)-W(P_0)]q_{0,t}\,dX\\ &-\int [\mathsf{S}_n(U)-\mathsf{S}_n(P)]\cdot D_n q\,d\mu + \frac{1}{2}\int V\cdot(I+k_jk_j^T)Vq_t\,d\mu,\end{aligned}$$

where $\mathsf{S}_n(U) = ((U_t^2 - U_y^2)/2, U_tU_y)$. Compact normal/height support and fixed-index Sobolev approximation of the whole base product justify these identities before any limit. The mixed shear terms cancel in the normal stress component, as in the preceding actual stress formula.

For the normal-stress error, the quadratic difference is bounded by $C(|D_nU| + |D_nP|)|D_n(U-P)|$. The actual corridor energy is $O_\delta(S)$, by the source estimate established above. The weighted model corridor energy has the same bound: on the support of $D_n\chi_k$,

$$\int \beta_1^2\gamma^2|D_nP|^2d\mu \le 2\int \beta_1^2\gamma^2|D_nU|^2d\mu + 2\int \beta_1^2\gamma^2|D_n(U-P)|^2d\mu = O_\delta(S).$$

The complete actual normal energy is $O(R^d)$. The complete model normal energy is also $O(R^d)$, because each translated layer has the finite full normal energy established in proposition 3.5, and the finite-sum bound is independent of the centers. Thus

$$\int \beta_1^2(|D_nU| + |D_nP|)^2|\mathcal{L}_q|^2d\mu \le C_\delta\left(S/D^2 + R^d/R^2\right) \le C_\delta R^{d-2}.$$

Cauchy–Schwarz against the comparison norm gives

$$\left|\int [\mathsf{S}_n(U)-\mathsf{S}_n(P)]\cdot D_n q\,d\mu\right| = o(\sqrt{S}\,R^{(d-2)/2}) = o(F).$$

For the trace potential put $e = u + b - P_0$. The periodic identity from the comparison gives, with a uniformly bounded error supported on its state-interpolation exception E,

$$W(u) - W(P_0) = W'_{\mathrm{per}}(P_0)e + O(e^2) + O(\mathbf{1}_E), \qquad |E| = o(S).$$

On an inner transition, all centers are at distance at least $c\delta D$, so the layer tails give $|W'_{\mathrm{per}}(P_0)| \le C_\delta D^{-p}$. These finitely many corridors have volume at most $C_\delta R^d D$; therefore

$$\int_{\text{inner gaps}} \beta_1^2\gamma_0^2|W'_{\mathrm{per}}(P_0)|^2dX \le C_\delta R^d D^{1-2p}.$$

The quadratic Taylor error and the exception cost at most $o(S)/D = o(F)$, using the squared trace comparison and $|\chi_{k,t}| \le C_\delta/D$. The linear error is bounded by

$$\frac{C}{D}\,o(\sqrt{S})\,(R^dD^{1-2p})^{1/2}.$$

After division by F, this is $o(1)RD^{-p-1/2} = o(1)\sqrt{\lambda_j}D^{-p/2} \to 0$. On the true macroscopic normal caps, use instead the positive cap energy: the actual potential has integral at most $CR^{d+1-p} + o(S)$. For the model, $|P_0 - \sigma_{\text{well}}| \le CR^{-p}$ there, because its bounded-height centers are $o(R)$; hence its cap potential is at most CR^{d+1-2p}. Multiplication by the cap derivative C/R makes both contributions $o(F)$, since $R^{2-p}/D \to 0$ and $D/R \to 0$, as proved above. This treats the whole potential error without differentiating the state correction.

Finally, the transverse quadratic term is $o(S)/D = o(F)$ on the inner corridors, by the vanishing transverse corridor energy. On the normal caps it is $O(S/R) = o(F)$. Thus every term in the stress subtraction after the residual is $o(F)$.

On the fixed separated supports, $(S_{j,l} - S_{j,k})/D_j \to f_l - f_k$ in measure by the selected-phase rank convergence; the phase-pinning offsets are bounded and $D_j \to \infty$. Each scaled inverse distance is uniformly bounded by C_δ. Thus the residual projection passes against the complete bounded base products by dominated convergence in measure, while their strong H^1 convergence is already incorporated in the current limit. The orientation products are exactly $(-1)^{k+l}$, even if the first orientation varies across regular components. Combining the two limits, with $R_j^d D_j^{-p}/F_j = \lambda_j$, gives

$$\int Df_k \cdot D(\eta h_M^2 b_\delta^2) = -\frac{4c_{1,p}\lambda}{p\sigma} \int \sum_{l\neq k} \epsilon_k \epsilon_l \operatorname{sgn}(l-k)|f_l - f_k|^{-p} \eta h_M^2 b_\delta^2.$$

The force density is locally integrable: for every $k \neq l$, the distance $|f_l - f_k|$ dominates an intervening adjacent gap, and $s^{-p} \le 1 + s^{-p-1}$ for $s > 0$. The inverse adjacent powers are locally integrable by lemma 9.10. Independently of this force calculation, proposition 9.6 already gives $f_k \in W^{2,1}_{\text{loc}}$, so $\Delta f_k \in L^1_{\text{loc}}$. Apply lemma 8.4 to the whole bounded H^1 test $\eta h_M^2 b_\delta^2$: bounded Sobolev approximation first gives its integration-by-parts identity against this existing L^1 Laplacian. Then $M \to \infty$ and $\delta \downarrow 0$ pass by dominated convergence against $|\Delta f_k|$ and the locally integrable force density. The height factors tend to one almost everywhere because every rank is finite, and the separation factors do so because every adjacent gap is positive almost everywhere. This proves the full force law without presupposing that law to justify cutoff removal; no derivative of an individual cutoff is discarded.

Now suppose $\lambda = 0$. Fix a consecutive group $B = [r, s]$ and apply lemma 10.1 with the tested-gap index set

$$J_B = \{r-1, s\} \cap \{1, \dots, m-1\}.$$

Thus only its existing outer gaps are separated; its internal gaps are unrestricted. With the smooth recovered midpoint steps from the common construction, use

$$\chi_B = \gamma_R(t)(K_{r-1} - K_s), \qquad K_0 = 1, \quad K_m = 0,$$

where γ_R is the shared scalar-centered normal test cap, chosen with support inside the plateau of the containing column cap ρ_R. The missing exterior gaps impose no condition. Keep M, δ and the smooth-recovery mismatch parameters fixed before the record limit. Apart from the cap collars, derivatives of this selector occur only in the two tested outer gap corridors. Their complete positive energy is $C_\delta \lambda_j S_j + o(S_j) = o(S_j)$ by proposition 9.9 and the global clearance in lemma 10.1; the cap energy is $CR^{d+1-p} + o(S)$. No comparison profile is used on the internal gaps.

Take $q_j = \eta(x/R_j)\beta_j^2 \chi_B$, independent of extension height, in the complete-height stress identity established above. Its specialization is

$$\int \mathcal{J}_{k_j}(U_j) \cdot D_x q_j \, d\mu = \int q_{j,t} \left[W(u_j) + \frac{d_p}{2} \int_0^\infty y^{1-p} (V_j \cdot (I + k_j k_j^T) V_j + U_{j,y}^2 - H_j^2) dy \right] dX.$$

The outer-gap contribution is $o(S)/D = o(F)$, and the cap contribution is $o(F)$ by the common cap estimate. Remove the pure-shear term for the complete product by the wedge cancellation already proved, before expanding $D_x q_j$. For the remaining matrix current, the part containing $D_x\chi_B$ has coefficient $C_{\delta,\varepsilon}/R$ on the outer corridors and costs $o(S/R) = o(F)$. Apply the smooth-selector current interface directly to χ_B: it equals one near the cores of precisely the ranks in B, and zero near the other cores on the tested support. The internal midpoint steps have telescoped out, so this argument requires no internal separation and does not sum identities that were established only under full separation. lemma 9.8 and then the smooth-recovery limit therefore give

$$\int DF_B \cdot D(\eta h_M^2 b_{B,\delta}^2) = 0, \qquad F_B = \sum_{k=r}^{s} f_k.$$

Here $b_{B,\delta} = \prod_{i\in J_B} b(g_i/\delta)$, with an empty product equal to one and b the fixed separation cutoff, zero near zero and one above its fixed upper threshold. For each fixed δ, lemma 8.4 with zero right side gives

$$\int \Delta F_B\, \eta b_{B,\delta}^2 = 0.$$

As $\delta \downarrow 0$, these bounded factors converge almost everywhere to $\mathbf{1}_{E_B}$, where

$$E_B = \{f_r > f_{r-1}\ (r > 1), \quad f_{s+1} > f_s\ (s < m)\}.$$

Dominated convergence against $|\Delta F_B| \in L^1_{\mathrm{loc}}$ yields $\int(\Delta F_B)\mathbf{1}_{E_B}\eta = 0$ for every compact smooth η. Hence $\Delta F_B = 0$ almost everywhere on E_B, although this measurable set need not be open.

For clarity, apply Sobolev locality twice to $w = f_i - f_k \in W^{2,1}_{\mathrm{loc}}$. First $Dw = 0$ almost everywhere on $\{w = 0\}$. Each component of Dw belongs to $W^{1,1}_{\mathrm{loc}}$, so its gradient vanishes almost everywhere on its own zero set, and therefore on $\{w = 0\}$. Thus $D^2 f_i = D^2 f_k$ almost everywhere on $\{f_i = f_k\}$. There are finitely many groups and pairs, so discard their exceptional null sets once. At every remaining point, the maximal consecutive block of equal ranks has strictly positive existing outer gaps. Its summed Laplacian vanishes by the group identity, and all member Laplacians are equal by locality. Each member Laplacian is therefore zero. Consequently every f_k is distributionally harmonic everywhere. Weyl's lemma gives smooth harmonic representatives. Their almost-everywhere ordering extends everywhere by continuity, so each adjacent difference is a nonnegative entire harmonic function and hence constant.

It remains to retain the complete covariance at a normalized collision. Fix one containing column cap ρ_{R_j} equal to one on $|t| \le 3R_j$, and use this same cap throughout the localization below. Define the normalized columns explicitly by

$$B_j(\xi) = a_j^{-1} \int \rho_{R_j}(t) H_j(R_j\xi, t, y) V_j(R_j\xi, t, y)\, d\mu_{t,y},$$

$$G_j(\xi) = a_j^{-2} \int \rho_{R_j}(t) V_j(R_j\xi, t, y) \otimes V_j(R_j\xi, t, y)\, d\mu_{t,y}.$$

They are finite almost everywhere by the complete normal and transverse column bounds. Put

$$h_{j,k}(\xi, t, y) = \epsilon_{j,k}(\xi) H_*(t - S_{j,k}(R_j\xi), y), \qquad \mathsf{H}_j = \sum_k h_{j,k}, \quad \mathsf{V}_j = -\sum_k Df_k(\xi) h_{j,k}.$$

As in the full profile proposition, extend the centers and signs measurably off the genuine regular bases; the estimates below do not depend on those extensions. The fixed-core normal

comparison and complete normal-core exhaustion in proposition 9.6 give, on every compact base receiver Ω,

$$\int_\Omega \int \rho_{R_j}(t)|H_j(R_j\xi,t,y)-\mathsf{H}_j(\xi,t,y)|^2\,d\mu_{t,y}d\xi \longrightarrow 0.$$

Indeed, first keep disjoint fixed-radius cores and use the uniform normal comparison on them. The other model profiles there contribute only their translated tails. Outside those cores, the actual normal mass is the complete-core remainder, while the model remainder is at most a fixed multiple of the sum of the m tails of H_*. Increase the core radius after the record limit. The omitted regular bases have measure tending to zero; uniform integrability of n_j pays their actual mass, and the bound $\int |\mathsf{H}_j|^2 d\mu_{t,y} \le m^2\sigma$ pays their model mass. The transverse profile of proposition 9.7 similarly gives

$$\int_\Omega \int \rho_{R_j}(t)|a_j^{-1}V_j(R_j\xi,t,y)-\mathsf{V}_j(\xi,t,y)|^2\,d\mu_{t,y}d\xi \longrightarrow 0.$$

The actual and model norms in these two displays are bounded. Cauchy–Schwarz therefore makes B_j differ in local L^1 by $o(1)$ from $\int \rho_{R_j}\mathsf{H}_j\mathsf{V}_j d\mu_{t,y}$, and makes G_j differ similarly from $\int \rho_{R_j}\mathsf{V}_j\otimes\mathsf{V}_j d\mu_{t,y}$.

For the model Gram entries

$$C_{j,ik}(\xi)=\int \rho_{R_j}(t)h_{j,i}h_{j,k}\,d\mu_{t,y}, \qquad |C_{j,ik}|\le\sigma,$$

physical fixed-radius core separation implies $|S_{j,i}-S_{j,k}|\to\infty$ in measure whenever $i\neq k$, including where $f_i=f_k$. The full-line correlation of H_* with a normal translate tends to zero at infinite displacement: truncate H_* in t, so two sufficiently separated truncations have disjoint supports, and bound the discarded terms by its L^2 tail. Hence $C_{j,ik}\to 0$ in measure for $i\neq k$. Diagonal full-line integrals equal σ. Their capped integrals converge to the same value in measure: on fixed bounded-height supports the centers are $o(R_j)$, so the cap equals one on every fixed core, and the remaining L^2 tail can be exhausted. Uniform integrability removes the height restriction for the actual columns; absolute continuity of the fixed model coefficients $1,|Df_k|^2$ does so for the models. The fixed products $|Df_i||Df_k|$ are integrable, and the uniform Gram bound passes their convergence in measure to the required integrals. We obtain the strong local L^1 limits

$$B_j\longrightarrow B=-\sigma\sum_k Df_k, \qquad G_j\longrightarrow G=\sigma\sum_k Df_k\otimes Df_k.$$

All gradients agree because the gaps are constant; writing $b=-Df_1$ gives $B=m\sigma b$ and $G=m\sigma b\otimes b$. For $L\ge 2$, the capped physical receiver over B_L maps under the bounded shear into B_{CLR_j} in the original frame, for a fixed C. Thus the outward same-frame bound in definition 9.1 and the strong covariance limit give

$$m\sigma\int_{B_L}|b|^2=\int_{B_L}\operatorname{tr}G=\lim_j\int_{B_L}\operatorname{tr}G_j\le CL^d(1+\log L)^2.$$

For each fixed z and all sufficiently large L, the interior derivative estimate for the harmonic vector b, applied on $B_L(z)\subset B_{2L}$, yields

$$|Db(z)|\le CL^{-1-d/2}\|b\|_{L^2(B_L(z))}\le C\frac{1+\log L}{L}\longrightarrow 0.$$

Hence b is a constant vector.

Restore the original orthogonal optimizing frame before imposing its exact covariance identity. If $V_j^{\mathrm{o}}, H_j^{\mathrm{o}}$ denote those derivatives, then

$$\int_{B_{R_j}\times(0,\infty)} H_j^{\mathrm{o}} V_j^{\mathrm{o}} d\mu = 0.$$

Under the selected shear $V_j^{\mathrm{sh}} = V_j^{\mathrm{o}} \circ S_{k_j}$ and $H_j^{\mathrm{sh}} = (H_j^{\mathrm{o}} + k_j \cdot V_j^{\mathrm{o}}) \circ S_{k_j}$. Thus after normalization by $a_j R_j^d$, the additional mixed covariance is bounded by

$$\frac{|k_j|}{a_j R_j^d} \int_{B_{CR_j}\times(0,\infty)} |V_j^{\mathrm{o}}|^2 d\mu \le C|k_j| a_j \longrightarrow 0.$$

We give the normal localization and the optimizing-ball passage explicitly. Our selected family has $\tau_j = 0$ by proposition 9.6, so $\tau_j/R_j \to 0$. More generally, on a bounded normalized-height support the shear displacement is bounded by

$$|k_j| \left(|\tau_j|/R_j + C_M a_j + O(R_j^{-1}) \right),$$

which tends to zero with this chosen center. Thus no untracked term $k_j\tau_j/R_j$ is omitted.

For this common column cap, on each fixed base receiver the columns n_j, e_j from proposition 9.6 are uniformly integrable. Their sum dominates the absolute normalized mixed and transverse column densities, since $a_j^{-1}|H_j V_j| \le (H_j^2 + a_j^{-2}|V_j|^2)/2$. For every fixed $\eta > 0$ the complete normal and transverse masses also satisfy

$$R_j^{-d} \int_{\{|x|<CR_j,\ |t|>\eta R_j\}} \rho_{R_j}(t) \left(H_j^2 + a_j^{-2}|V_j|^2 \right) d\mu \longrightarrow 0. \tag{BL}$$

To prove (BL), first keep a fixed bounded-height factor and a fixed physical core radius L. Its m actual core centers lie at $|t| \le C_M D_j + O(1) = o(R_j)$, so these entire fixed-radius rectangles avoid $|t| > \eta R_j$. Complete normal-core exhaustion in proposition 9.6 pays the normal remainder after $L \to \infty$. The full-height transverse profile in proposition 9.7 pays the transverse remainder: its finitely many translated profile tails tend to zero and its coefficient squares $|Df_k|^2$ are integrable. Finally remove the height factors. Their limits tend to one in measure as $M \to \infty$, and uniform integrability of $e_j + n_j$ makes their discarded complete mass tend to zero, first $j \to \infty$ and then $M \to \infty$. This proves (BL) for every fixed η. It also pays the mixed mass on the same normal complement by Cauchy–Schwarz. All these estimates include every extension height.

In selected shear coordinates the original physical ball is

$$\mathcal{B}_j = \{(\xi, t) : |\xi + k_j t/R_j|^2 + (t/R_j)^2 < 1\}.$$

For $|t|/R_j \le \eta$ and large j, its indicator differs from $\mathbf{1}_{B_1}(\xi)$ only on the base annulus $1 - 3\eta < |\xi| < 1 + 3\eta$. The absolute normalized covariance columns there are bounded by $C(e_j + n_j)$. Uniform integrability makes their mass tend to zero uniformly as $\eta \downarrow 0$, because the annulus measure tends to zero. On $|t|/R_j > \eta$, (BL) controls the difference for each fixed η as $j \to \infty$. Thus integration on $\mathcal{B}_j$ and integration of the complete columns over B_1 have the same limit. The already proved strong local L^1 convergence of B_j, G_j permits integration of their limits on B_1.

The previously bounded extra mixed term $k_j \cdot V_j$ tends to zero after its exact normalization. Applying the last localization to the original orthogonal mixed covariance, which is exactly zero, now gives

$$\int_{B_1} B = 0.$$

Because the original frame is optimizing, $E(R_j) = \int_{B_{R_j}\times(0,\infty)} |V_j^o|^2 d\mu$. The transverse derivative is unchanged by the shear pullback, and the column cap is one on this transformed ball. The same localization therefore retains the exact normalization

$$1 = \lim_j \frac{E(R_j)}{S_j} = \int_{B_1} \operatorname{tr} G \le \int \varphi \operatorname{tr} G,$$

for a fixed nonnegative compact smooth base cutoff $\varphi = 1$ on B_1. This transfers the actual optimizing-ball normalization as well as its mixed identity; it does not infer either from an unanchored cylinder. Since $B = m\sigma b$ with $m\sigma > 0$ and b constant, the zero mixed covariance forces $b = 0$, contradicting the retained transverse normalization. This proves every assertion of the zero-interaction alternative and finishes the proof.

□

12. Cluster exclusion and stable endpoint rigidity

The inverse-power test below excludes the actual force–stability coefficients through base dimension six. It needs no local upper bound for the span and retains every coupled interaction before reducing to the scalar inequality.

Lemma 12.1 (Optimized negative powers exclude the scalar cluster in six base dimensions)**.** *Let $1 < p < 2$ and $1 \le d \le 6$. There do not exist a finite almost everywhere positive function $G \in H^1_{\mathrm{loc}}(\mathbb{R}^d) \cap W^{2,1}_{\mathrm{loc}}(\mathbb{R}^d)$, a nonnegative $V \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, and constants $\kappa > 0$, $0 < \rho \le 1/p$, such that*

$$\Delta G \le \rho G V, \qquad V \ge \kappa G^{-p-1} \quad \text{almost everywhere},$$

and

$$\int V\phi^2 \le \int |D\phi|^2 \qquad (\phi \in C_c^\infty(\mathbb{R}^d)).$$

Proof. Suppose otherwise, put $h = G + 1$, and let $t = 2p$. For a compactly supported smooth nonnegative cutoff η, write

$$I = \int V h^{-2t}\eta^2, \quad J = \int h^{-2t-2}|DG|^2\eta^2, \quad K = \int h^{-2t-1}\eta DG \cdot D\eta, \quad T = \int h^{-2t}|D\eta|^2.$$

All four quantities are finite on compact supports: V is locally integrable, $DG \in L^2_{\mathrm{loc}}$, and every inverse power of h is at most one. The nonnegative test $w = h^{-2t-1}\eta^2$ is bounded, belongs to H^1_0 on a containing ball, and has compact support inside that ball. The bounded compact-test approximation in the proof of lemma 8.4 applies to w: convolution of its zero extension with nonnegative mollifiers preserves nonnegativity and the common supremum bound. Its strong H^1 convergence and bounded almost-everywhere convergence give

$$\int \Delta G\, w = -\int DG \cdot Dw,$$

using $DG \in L^2_{\mathrm{loc}}$ and $\Delta G \in L^1_{\mathrm{loc}}$. Only after this identity is established do we multiply the almost-everywhere force inequality by w. No local integrability of GV is needed: the tested right side is integrable because $Gh^{-2t-1} \le h^{-2t} \le 1$. Thus $G/h \le 1$ gives

$$(2t+1)J - 2K \le \rho I.$$

Applying the same bounded approximation to $h^{-t}\eta$, with dominated convergence against V, extends stability to that test and yields

$$I \le t^2 J - 2tK + T.$$

The inverse powers have bounded derivatives for $h \geq 1$, so their Sobolev chain rules are applicable without a local upper bound for G. For $\delta = 1/8$, Cauchy–Schwarz and Young give $|K| \leq \delta J + (4\delta)^{-1}T$. Consequently

$$(2t + 1 - 2\delta)J \leq \rho I + (2\delta)^{-1}T,$$

and

$$I \leq (t^2 + 2t\delta)J + (1 + t/(2\delta))T.$$

The coefficient of I after substitution of the first inequality in the second is at most

$$\frac{\rho(t^2 + 2t\delta)}{2t + 1 - 2\delta} \leq \frac{4p + 1/2}{4p + 3/4} = 1 - \frac{1}{16p + 3} < 1.$$

Absorption gives $I \leq C_p T$. Since $G \leq h$ implies $V \geq \kappa h^{-p-1}$, put $q = 5p + 1$ to obtain

$$\int h^{-q}\eta^2 \leq C_{p,\kappa} \int h^{-4p}|D\eta|^2.$$

Choose $\zeta_R \in C_c^\infty(B_{2R})$, equal to one on B_R, with values in $[0, 1]$ and $|D\zeta_R| \leq C/R$. Let M be an integer at least $q/(p + 1)$ and set $\eta = \zeta_R^M$, $X_R = \int h^{-q}\zeta_R^{2M}$. With $\beta = 4p/q < 1$, we have $1 - \beta = (p + 1)/q$ and

$$h^{-4p}\zeta_R^{2M-2} = (h^{-q}\zeta_R^{2M})^\beta \zeta_R^{2M(1-\beta)-2}.$$

The last exponent is nonnegative by the choice of M. Hölder's inequality with conjugate exponents $1/\beta$ and $1/(1 - \beta)$ therefore gives

$$X_R \leq CR^{-2}X_R^\beta R^{d(1-\beta)}, \qquad X_R \leq CR^{d-2q/(p+1)}.$$

The zero case is immediate; otherwise divide by X_R^β. For $d \leq 6$ the exponent is at most $4(1 - p)/(p + 1) < 0$. Letting $R \to \infty$ forces $\int_{B_1}(G + 1)^{-q} = 0$, although its integrand is positive almost everywhere. This is a contradiction. □

Lemma 12.2 (The exact alternating cluster coefficients are impossible through six base dimensions)**.** *Let $d \in \{3, 4, 5, 6\}$, $1 < p < 2$, let $m \geq 2$ be an integer, and let $C > 0$. There is no finite ordered family $f_1 \leq \cdots \leq f_m$ in $H^1_{\mathrm{loc}} \cap W^{2,1}_{\mathrm{loc}}(\mathbb{R}^d)$ whose adjacent gaps $g_i = f_{i+1} - f_i$ are positive almost everywhere, with $g_i^{-p-1} \in L^1_{\mathrm{loc}}$, satisfying*

$$\Delta f_k = C\sum_{l \neq k}(-1)^{k+l}\operatorname{sgn}(l - k)|f_l - f_k|^{-p}, \qquad \int V\phi^2 \leq \int |D\phi|^2, \quad V = \frac{4pC}{m}\sum_{i=1}^{m-1} g_i^{-p-1},$$

where the equations hold almost everywhere for $k = 1, \ldots, m$, and stability is required for every real $\phi \in C_c^\infty(\mathbb{R}^d)$.

Proof. Set $G = f_m - f_1 = \sum_{i=1}^{m-1} g_i$. It is finite and positive almost everywhere and belongs to $H^1_{\mathrm{loc}} \cap W^{2,1}_{\mathrm{loc}}$. To track both endpoint signs for either parity of m, index the other sheets by their increasing distance from that endpoint. Put

$$A_r = (f_{r+1} - f_1)^{-p}, \qquad B_r = (f_m - f_{m-r})^{-p} \qquad (1 \leq r \leq m - 1).$$

The two endpoint equations are exactly

$$-C^{-1}\Delta f_1 = \sum_{r=1}^{m-1}(-1)^{r-1}A_r, \qquad C^{-1}\Delta f_m = \sum_{r=1}^{m-1}(-1)^{r-1}B_r.$$

Each of the positive sequences A_r, B_r is strictly decreasing at almost every point. For a decreasing positive sequence $a_1, \ldots, a_N$, pairing $(a_1 - a_2), (a_3 - a_4), \ldots$, with the last positive term if present, shows that its alternating sum is nonnegative. Pairing from the second term

shows that this sum is at most a_1. Thus the two endpoint sums are between zero and their first terms, for both odd and even m.

For $m = 2$ the sole gap is G, and the endpoint equations give the exact identities

$$\Delta G = 2CG^{-p}, \qquad V = 2pCG^{-p-1}, \qquad \Delta G = \frac{1}{p}GV.$$

For $m \geq 3$, the first and last gaps are distinct indices, so the alternating bounds give

$$0 \leq \Delta G \leq C(g_1^{-p} + g_{m-1}^{-p}) \leq C \sum_{i=1}^{m-1} g_i^{-p}.$$

The inverse-distance terms in all endpoint equations are locally integrable: each full distance dominates an intervening adjacent gap, and $s^{-p} \leq 1 + s^{-p-1}$ for $s > 0$.

Put $n = m - 1$. The identity

$$G \sum_{i=1}^{n} g_i^{-p-1} - n \sum_{i=1}^{n} g_i^{-p} = \sum_{i<j} (g_i - g_j)(g_j^{-p-1} - g_i^{-p-1}) \geq 0$$

follows by expanding the double sums; every paired term is nonnegative because $s \mapsto s^{-p-1}$ decreases. For $m \geq 3$ it therefore implies

$$\Delta G \leq \frac{C}{m-1} G \sum_i g_i^{-p-1} = \frac{m}{4p(m-1)} GV.$$

Together with the exact $m = 2$ identity, this gives $\Delta G \leq \rho GV$ with

$$\rho = \frac{1}{p} \quad (m = 2), \qquad \rho = \frac{m}{4p(m-1)} \leq \frac{1}{p} \quad (m \geq 3).$$

Also $0 < g_i \leq G$ almost everywhere, so

$$V \geq \kappa G^{-p-1}, \qquad \kappa = \frac{4pC(m-1)}{m} > 0.$$

The assumed inverse-gap integrability gives $V \in L^1_{\mathrm{loc}}$, and the stated stability is exactly the scalar stability hypothesis. Thus G, V, ρ, κ satisfy every hypothesis of lemma 12.1, a contradiction. Its bounded negative-power tests require no local upper bound for any height or for G. □

Theorem 12.3 (Individual stable endpoint rigidity under sharp growth through ambient dimension seven)**.** *Let $3 \leq d \leq 6$ and $1 < p < 2$. Every bounded classical stable solution u of*

$$(-\Delta_{\mathbb{R}^{d+1}})^{p/2} u = u - u^3$$

satisfying $\int_{B_R(z)} |Du| \leq C_u R^d$ for all $z \in \mathbb{R}^{d+1}$ and $R \geq 1$ is a constant well or a translated rotated increasing one-dimensional layer.

Proof. By lemma 2.1, u takes values in $[-1, 1]$, is smooth with uniform physical derivative bounds, and its translates and rotations are compact in every local C^k topology. The equation and stability pass to their limits by that lemma; the stated all-center variation bound passes on each fixed ball by local C^1 convergence with the same constant C_u. Thus the actual-field and translate-compactness hypotheses used by the macroscopic results apply to this individual solution. If u is constant, lemma 6.1 makes it a constant well. Henceforth assume it is nonconstant.

By proposition 3.4, proposition 3.5, and lemma 3.6, its complete energy has one positive integer multiplicity m, independent of the subsequence, and its optimized transverse excess

satisfies $E(r) = o(r^d)$. If $m = 1$, proposition 6.3 gives the layer. Suppose henceforth that $m \geq 2$.

We construct the required records rather than assuming their existence. For the actual Poisson extension U, use the normalized measure $d\mu = d_p y^{1-p} dX\, dy$ of proposition 3.4 and put

$$M(r) = \int_{B_r \times (0,\infty)} D_X U \otimes D_X U \, d\mu, \qquad E(r) = \operatorname{tr} M(r) - \lambda_{\max}(M(r)).$$

The covariance-continuity calculation at the start of the proof of proposition 5.2 uses only the complete local extension-energy bound of proposition 3.4; it shows that $M(r)$ is finite, positive semidefinite and continuous, and that $E : [1,\infty) \to [0,\infty)$ is continuous. The quantization result gives $E(r) = o(r^d)$ for the full radius limit. These preliminary facts require neither an interaction bound nor a fine record.

All hypotheses of lemma 4.1 now hold, with the present $3 \leq d \leq 6$. Apply it to $S(r) = E(r) + r^{d-1}$. It supplies $R_j \to \infty$ and

$$S_j = S(R_j) = a_j^2 R_j^d = R_j^{d-2} D_j^2, \qquad a_j \to 0, \qquad D_j = a_j R_j \geq \sqrt{R_j} \to \infty,$$

together with the exact two record inequalities

$$S(r) \leq S_j (r/R_j)^d \quad (r \geq R_j), \qquad S(r) \leq S_j (r/R_j)^{d-3/2} \quad (1 \leq r \leq R_j).$$

For each j choose a unit top eigenvector of the finite symmetric matrix $M(R_j)$ and complete it to an orthonormal frame. This is an optimizing frame in the exact sense of definition 9.1. The sharp all-center variation bound, actual equation, stability, and positive integer multiplicity required by proposition 5.2 are precisely the hypotheses already established here; its record hypotheses are those just constructed. Apply that proposition, and then proposition 5.3, to these same records. Since $m \geq 2$, the latter gives, after discarding a finite initial segment,

$$\lambda_j := \frac{R_j^2}{D_j^{p+1}} \leq C, \qquad \frac{E(R_j)}{S_j} \to 1, \qquad \int_{B_{LR_j} \times (0,\infty)} |D_x U|^2 d\mu \leq C S_j L^d (1 + \log L)^2 \quad (L \geq 2)$$

in the original optimizing frames, with one constant independent of j, L. Extract a subsequence with $\lambda_j \to \lambda \in [0,\infty)$. The exact identities for S_j, a_j, D_j, their escape and smallness limits, the absorbed true-excess normalization, the optimizing frames, the actual normalized Poisson extension, and this uniform outward same-frame estimate check every hypothesis of definition 9.1. Any later small shear is the determinant-one pullback of this actual extension and equation, with the constant matrix specified in that definition. We have thus obtained an actual fine record to which the following fine-rank results apply.

The normalized one-dimensional kernel coefficient $c_{1,p}$ is positive, and the complete one-layer horizontal energy σ of proposition 9.6 is finite and positive by the layer-energy identity in proposition 3.5. Consequently $C_\lambda = 4c_{1,p}\lambda/(p\sigma) > 0$ whenever $\lambda > 0$.

Apply proposition 9.6 to this actual fine record and then proposition 11.1 to its resulting complete family. The ranks belong to $H^1_{\mathrm{loc}} \cap W^{2,1}_{\mathrm{loc}}(\mathbb{R}^d)$. If $\lambda > 0$, their adjacent gaps are positive almost everywhere with locally integrable g_i^{-p-1}, and their exact alternating force coefficient is $C = C_\lambda > 0$. The accompanying stability potential is exactly

$$\frac{16 c_{1,p} \lambda}{m\sigma} \sum_i g_i^{-p-1} = \frac{4pC_\lambda}{m} \sum_i g_i^{-p-1}.$$

Together with $m \geq 2$ and $3 \leq d \leq 6$, these are all the hypotheses of lemma 12.2, which rules out this alternative. If $\lambda = 0$, the zero-interaction alternative of proposition 11.1 already

proves that the complete covariance and the retained optimizing normalization are incompatible. Thus this alternative is impossible as well. No $m \geq 2$ remains, so the multiplicity-one conclusion proves the theorem. □

13. Endpoint minimality and the directed-cone closure

The final geometric closure is stated with its actual minimizing hypotheses. In dimension eight one cannot invoke a blanket classification of all global minimizers from Savin's theorem through dimension seven. Instead, monotonicity makes the limiting minimizing cone directed, and the next lemma makes that cone flat. Savin's asymptotically flat minimizer theorem then applies.

Definition 13.1 (Localized cubic energy and one-sided minimizing properties)**.** Let $n \geq 1$, $s \in (0,1)$, and let $c_{n,s} > 0$ be the constant for which the principal-value operator

$$(-\Delta)^s v(x) = c_{n,s}\,\mathrm{PV}\int_{\mathbb{R}^n} \frac{v(x)-v(y)}{|x-y|^{n+2s}}\,dy$$

has Fourier multiplier $|\xi|^{2s}$. Set

$$W(t) = \frac{1}{4}(1-t^2)^2, \qquad Q_\Omega = \mathbb{R}^{2n} \setminus (\Omega^c \times \Omega^c),$$

and, for a bounded ball $\Omega \subset \mathbb{R}^n$, define

$$\mathcal{E}(v;\Omega) = \frac{c_{n,s}}{4}\iint_{Q_\Omega} \frac{(v(x)-v(y))^2}{|x-y|^{n+2s}}\,dx\,dy + \int_\Omega W(v)\,dx.$$

An admissible perturbation in Ω is a bounded function $h \in H^s(\mathbb{R}^n)$ that is zero almost everywhere outside Ω. A bounded function v, whose energy is finite in every ball, is *minimizing from above* if $\mathcal{E}(v+h;\Omega) \geq \mathcal{E}(v;\Omega)$ for every ball and every nonnegative admissible perturbation. It is *minimizing from below* when the same assertion holds for nonpositive admissible perturbations. It is a *global minimizer* when the assertion holds for all admissible perturbations.

For a bounded classical solution v of $(-\Delta)^s v = v - v^3$, write

$$Q_v(h) = \frac{c_{n,s}}{2}\iint_{\mathbb{R}^{2n}} \frac{(h(x)-h(y))^2}{|x-y|^{n+2s}}\,dx\,dy + \int_{\mathbb{R}^n} (3v^2-1)h^2\,dx.$$

Call v stable when $Q_v(h) \geq 0$ for every $h \in C_c^\infty(\mathbb{R}^n)$.

The admissible class makes each energy comparison finite. Indeed, if $\mathcal{E}(v;\Omega) < \infty$ and h is admissible, then

$$\begin{aligned}\iint_{Q_\Omega} \frac{|(v+h)(x)-(v+h)(y)|^2}{|x-y|^{n+2s}}\,dx\,dy &\leq 2\iint_{Q_\Omega} \frac{|v(x)-v(y)|^2}{|x-y|^{n+2s}}\,dx\,dy \\ &\quad + 2\iint_{\mathbb{R}^{2n}} \frac{|h(x)-h(y)|^2}{|x-y|^{n+2s}}\,dx\,dy < \infty.\end{aligned}$$

The potential integral is finite because $v+h$ is bounded and Ω has finite measure. Also, for bounded v, the quadratic form Q_v is finite and continuous on $H^s(\mathbb{R}^n)$; thus smooth-test stability extends to that space by strong H^s approximation.

Lemma 13.2 (Exact nonlocal lattice deficit)**.** *In the notation of definition 13.1, let a, b be bounded measurable functions with finite energy in a ball Ω. Put $A = \{a > b\}$, $B = \{a < b\}$, $m = \min(a,b)$ and $M = \max(a,b)$. Then m, M have finite energy and*

$$\mathcal{E}(m;\Omega) + \mathcal{E}(M;\Omega) - \mathcal{E}(a;\Omega) - \mathcal{E}(b;\Omega) = -c_{n,s}\iint_{(A\times B)\cap Q_\Omega} \frac{(a-b)(x)(b-a)(y)}{|x-y|^{n+2s}}\,dx\,dy \leq 0.$$

Proof. Use the energy in definition 13.1. At every point, the unordered pair $\{m, M\}$ is $\{a, b\}$, so the potential contributions cancel. If x, y have the same ordering, sorting changes neither the sum of the two squared increments nor its value. If $x \in A$ and $y \in B$, direct expansion gives

$$(m(x)-m(y))^2+(M(x)-M(y))^2-(a(x)-a(y))^2-(b(x)-b(y))^2=-2(a-b)(x)(b-a)(y).$$

The reversed ordering gives the same expression after interchanging x, y, and an equality $a = b$ at either endpoint contributes zero. The sorted kinetic integrand is nonnegative and no larger than the original sum, so both sorted kinetic energies are finite. Their potential integrals are finite by boundedness. The nonnegative kinetic deficit is also dominated by the original sum and hence integrable. Integrate first with the diagonal removed, use symmetry of Q_Ω and the kernel to combine the two opposite orderings, and multiply by $c_{n,s}/4$. The two orderings each contribute -2 times the displayed product, giving precisely the coefficient $-c_{n,s}$. Monotone convergence of the nonnegative deficit integral, and convergence of each finite kinetic integral, give the identity. □

Lemma 13.3 (Exact energy remainder at a cubic solution)**.** *Let v be a bounded C^2 solution of $(-\Delta)^s v = v - v^3$ on $\mathbb{R}^n$, and let $h \in C_c^\infty(\Omega)$ for a ball Ω. Using definition 13.1, one has*

$$\mathcal{E}(v+h;\Omega)-\mathcal{E}(v;\Omega)=\frac{1}{2}Q_v(h)+\int_\Omega vh^3\,dx+\frac{1}{4}\int_\Omega h^4\,dx.$$

If v is stable and $vh \geq 0$ pointwise, this difference is nonnegative.

Proof. Both energies in the difference are finite. In fact, boundedness of v and its bounded gradient on the closed unit neighborhood of $\overline{\Omega}$ control the near and far kinetic integrals by constants times

$$\int_0^1 r^{1-2s}dr+\int_1^\infty r^{-1-2s}dr<\infty;$$

the potential is bounded on its range. The same finiteness for $v+h$ follows from the admissible-perturbation estimate after definition 13.1.

Since $h = 0$ outside Ω, its increment vanishes on $\Omega^c \times \Omega^c$. Thus the kinetic energy difference is the full-space expression

$$\frac{c_{n,s}}{2}\iint_{\mathbb{R}^{2n}}\frac{(v(x)-v(y))(h(x)-h(y))}{|x-y|^{n+2s}}\,dx\,dy + \frac{c_{n,s}}{4}\iint_{\mathbb{R}^{2n}}\frac{(h(x)-h(y))^2}{|x-y|^{n+2s}}\,dx\,dy.$$

The first term equals $\int h(-\Delta)^s v$ by symmetric integration. All pair integrals are absolutely integrable after pairing: close to the diagonal the product of the increments is $O(|x-y|^2)$ on a compact set, and at infinity v is bounded while h has compact support. To justify the principal-value passage, first perform the symmetric integration with $|x-y| > \varepsilon$. On a fixed neighborhood of $\operatorname{supp} h$, the Hessian of v is bounded. Symmetric second differences therefore bound the omitted part of the operator by $C\int_0^\varepsilon r^{1-2s}\,dr = C'\varepsilon^{2-2s}$, uniformly for $x \in \operatorname{supp} h$. Since $s < 1$, this tends to zero and allows the truncated operator to be integrated against h before taking its limit. On the paired-increment side, absolute integrability justifies the same limit by dominated convergence. The exact polynomial expansion is

$$W(v+h)-W(v)=W'(v)h+\frac{1}{2}(3v^2-1)h^2+vh^3+\frac{1}{4}h^4.$$

Since $W'(v) = v^3 - v = -(-\Delta)^s v$, the two linear terms cancel. Combining the quadratic terms proves the formula. Stability makes its first term nonnegative, and $vh^3 = (vh)h^2 \geq 0$ under the stated sign condition. □

Proposition 13.4 (Outward endpoint minimality gives global minimality)**.** *Let $n \geq 2$ and $1/2 < s < 1$. Suppose that $u \in C^2(\mathbb{R}^n)$ is a solution of*

$$(-\Delta)^s u = u - u^3, \qquad -1 < u < 1, \qquad \partial_n u > 0.$$

Let

$$v^{\pm}(x') = \lim_{t \to \pm\infty} u(x', t), \qquad V^{\pm}(x', x_n) = v^{\pm}(x'),$$

and assume that $V^{\pm}$ are bounded C^2 solutions of the same equation. If V^+ is minimizing from above and V^- is minimizing from below, in the sense of definition 13.1, then u is a global minimizer.

Proof. Fix a ball Ω. Strict monotonicity gives

$$V^-(x) < u(x) < V^+(x) \qquad (x \in \mathbb{R}^n).$$

The energy and admissibility are those of definition 13.1.

The reference energy is finite by the near-diagonal and tail estimates in the proof of lemma 13.3, applied to the bounded C^2 function u. The same estimates apply to $V^{\pm}$.

There is a minimizer z with exterior data u. Here are the direct-method details needed in this setting. Truncation to $[-1, 1]$ decreases both the kinetic term and W, and preserves exterior data because $-1 < u < 1$. It also preserves admissibility: if $\widehat{z}$ is a truncated finite-energy competitor, then $\widehat{z} - u$ vanishes outside Ω, is bounded, and

$$\iint_{\mathbb{R}^{2n}} \frac{|(\widehat{z} - u)(x) - (\widehat{z} - u)(y)|^2}{|x - y|^{n+2s}}\, dx\, dy \leq 2 \iint_{Q_\Omega} \frac{|\widehat{z}(x) - \widehat{z}(y)|^2 + |u(x) - u(y)|^2}{|x - y|^{n+2s}}\, dx\, dy < \infty.$$

Bounded support supplies the L^2 norm. We may therefore choose a minimizing sequence z_j valued in that interval. Its kinetic energies are bounded by the finite energy of u plus a fixed constant. The functions $z_j - u$ vanish outside the bounded ball; the inequality

$$|(z_j - u)(x) - (z_j - u)(y)|^2 \leq 2|z_j(x) - z_j(y)|^2 + 2|u(x) - u(y)|^2$$

on Q_Ω, together with their uniform L^2 bound, bounds them in $H^s(\mathbb{R}^n)$. Weak compactness and compact embedding into $L^2(\Omega)$ give a weakly convergent subsequence whose limit also converges almost everywhere, equals u outside, and belongs to $[-1, 1]$. Fatou's lemma for the kinetic term and bounded convergence for the potential give a minimizer. This is also an unconstrained minimizer, because truncation lowers the energy of every bounded competitor.

Apply lemma 13.2 to z, V^+. The function $\max(z, V^+) - V^+ = (z - V^+)_+$ is a nonnegative admissible perturbation in Ω: it vanishes outside; the lattice inequality gives finite localized energy for $\max(z, V^+)$, and the squared-increment estimate just used, now subtracting V^+, gives its finite global fractional seminorm. Bounded support gives its L^2 norm. The same argument shows that $\min(z, V^+) - u$ is admissible, since $\min(z, V^+) = u$ outside the ball. The hypothesis at the upper endpoint and the lattice inequality therefore give

$$\mathcal{E}(\min(z, V^+); \Omega) \leq \mathcal{E}(z; \Omega).$$

Replace z by this clipped minimizer. Next apply the same lemma to z, V^-. Now $\min(z, V^-) - V^-$ is nonpositive and admissible, so clipping z from below also does not raise its energy; the clipped function again differs from u by an admissible perturbation. The resulting minimizer satisfies

$$V^- \leq z \leq V^+ \qquad \text{on } \mathbb{R}^n.$$

Bounded smooth compact variations are allowed for this unconstrained minimizer, and differentiating their finite kinetic and potential energies gives

$$\frac{c_{n,s}}{2} \iint_{Q_\Omega} \frac{(z(x) - z(y))(\phi(x) - \phi(y))}{|x - y|^{n+2s}}\, dx\, dy = \int_\Omega (z - z^3)\phi\, dx \qquad (\phi \in C_c^\infty(\Omega)).$$

The classical equation for u has the same weak form, by the symmetric-increment argument in lemma 2.1, with $p = 2s$. Subtract the two identities. Their difference has a global H^s function on the left and a bounded right side, so both sides are continuous for the H^s norm of tests vanishing outside Ω. Such tests can be approximated in H^s by smooth compact tests in the ball: first contract their zero extensions towards its center, then mollify within the resulting positive boundary margin. Consequently $w = z - u \in H^s(\mathbb{R}^n)$ is a weak solution, with $w = 0$ outside the ball, of

$$(-\Delta)^s w = (u + w) - (u + w)^3 - (u - u^3) \quad \text{in } \Omega.$$

The right side is bounded. Moreover, considered as a function of (x, w) on bounded w intervals, it is locally Lipschitz because u is C^2. Ros-Oton and Serra, *The Dirichlet problem for the fractional Laplacian: regularity up to the boundary*[9], arXiv:1207.5985v1, Proposition 1.1 (PDF p. 2) and Corollary 1.6(a) (PDF p. 6), apply: a ball is bounded and $C^{1,1}$, and w is a bounded weak solution with zero exterior data. They give continuity on all of $\mathbb{R}^n$ and interior C^β regularity for $\beta < 1+2s$. Since $s > 1/2$, choose a noninteger $\beta \in (2, 1+2s)$. Thus z has a continuous representative on all of $\mathbb{R}^n$, equals u on Ω^c, including the boundary, and is classical in Ω. The almost-everywhere endpoint inequalities above therefore hold pointwise.

The endpoint inequalities are strict on $\overline{\Omega}$. At the boundary this follows from $z = u$. At an interior contact, for example $z(x_0) = V^+(x_0)$, the nonnegative function $d = V^+ - z$ would have

$$(-\Delta)^s d(x_0) = -c_{n,s} \int_{\mathbb{R}^n} \frac{d(y)}{|x_0 - y|^{n+2s}}\, dy < 0,$$

because $d = V^+ - u > 0$ outside the ball. This contradicts the two equations, whose right sides agree at the contact point. The integral is finite near x_0 since d is C^2 there and has a minimum; the tail is finite since d is bounded. The lower contact is excluded in the same way. Compactness now gives a positive minimum for each endpoint gap on $\overline{\Omega}$.

Write $u_t(x) = u(x + te_n)$. As $t \to +\infty$, these functions increase to V^+ uniformly on $\overline{\Omega}$: apply Dini's theorem to integer translations and use monotonicity to pass to all real t. Consequently $u_t > z$ on $\overline{\Omega}$ for large positive t; outside Ω, $u_t \geq u = z$ for every $t \geq 0$. Let τ be the infimum of nonnegative t such that $u_t \geq z$ everywhere. This set is a nonempty upper interval, and continuity implies $u_\tau \geq z$. If $\tau > 0$, compactness forces contact in $\overline{\Omega}$, since otherwise uniform positivity on the closure permits a slightly smaller translation. Boundary contact is impossible since $z = u$ and $u_\tau > u$ there. Interior contact is impossible by the same nonlocal integral argument, using $u_\tau - z > 0$ outside Ω. Thus $\tau = 0$ and $u \geq z$. Apply the identical argument to $-u(x', -x_n)$, $-z(x', -x_n)$ on the reflected ball; its positive translates correspond to negative translates of u. It gives $u \leq z$. Hence $u = z$, so u minimizes in every ball. □

Lemma 13.5 (A directed minimal cone with only an isolated possible singularity is flat)**.** *Let $n \geq 3$. Let $E \subset \mathbb{R}^n$ be a set of locally finite perimeter invariant, up to null sets, under every positive dilation about the origin. Let $C = \operatorname{supp} |D\mathbf{1}_E|$ be nonempty. Suppose $C \setminus \{0\}$ is a smooth embedded hypersurface without boundary, with zero mean curvature, and that near each of its points E agrees up to a null set with one side of that hypersurface. Suppose there is a unit vector e such that its inward unit normal ν satisfies $\nu \cdot e \geq 0$ everywhere on $C \setminus \{0\}$. Then C is a hyperplane through the origin.*

Proof. Dilation invariance and the change of variables for the distributional derivative give

$$|D\mathbf{1}_E|(rA) = r^{n-1} |D\mathbf{1}_E|(A) \qquad (r > 0)$$

for every relatively compact Borel set A. Thus C is a closed cone. It contains a nonzero point: the reduced-boundary representation of the perimeter measure gives $|D\mathbf{1}_E| = \mathcal{H}^{n-1} \llcorner \partial^* E$,

which has no atoms for $n \geq 2$. A nonzero measure cannot have support contained in the singleton $\{0\}$.

The link $\Sigma = C \cap S^{n-1}$ is therefore a nonempty compact smooth embedded hypersurface of the unit sphere, of dimension $n-2$ and without boundary. The intersection with the sphere is transverse since $\nabla_C |x| \neq 0$: the unit radial vector is tangent to C. The link has finitely many connected components: the components of a manifold are open, and their cover of the compact link has a finite subcover. Each component is compact. The locally occupied side of E fixes the inward normal smoothly and consistently on overlapping charts. Positive dilations preserve that occupied side, so this normal is homogeneous of degree zero: $\nu(r\omega) = \nu(\omega)$. The cone parametrization $(r, \omega) \mapsto r\omega$ gives the metric $dr^2 + r^2 g_\Sigma$ and $|A_C|^2 = r^{-2}|A_\Sigma|^2$, where A_Σ is the second fundamental form in the sphere. Zero mean curvature of the cone implies zero mean curvature of the link in the sphere.

For a minimal Euclidean hypersurface its normal satisfies $\Delta_C \nu + |A_C|^2 \nu = 0$: differentiate the Weingarten equation in a tangent orthonormal frame geodesic at the point; the tangential part is the gradient of the mean curvature, which vanishes, and the normal part is $-|A_C|^2 \nu$. The conical metric gives

$$\Delta_C w = \partial_r^2 w + \frac{n-2}{r} \partial_r w + r^{-2} \Delta_\Sigma w.$$

Taking the normal component along e, its degree-zero homogeneity therefore gives, for $\phi = \nu \cdot e \geq 0$,

$$\Delta_\Sigma \phi + |A_\Sigma|^2 \phi = 0.$$

On each compact connected component, integration yields $\int |A_\Sigma|^2 \phi = 0$. The integrand is continuous and nonnegative, so it vanishes identically. Thus $\Delta_\Sigma \phi = 0$ there, and integration against ϕ proves that ϕ is constant on that component.

If this constant is positive on some component, then $A_\Sigma = 0$ there. Its normal is consequently constant, so the component lies in the great sphere perpendicular to that normal. Equality of dimensions makes it open in that great sphere, and compactness makes it closed. Since $n \geq 3$, the great sphere is connected, so the component is the entire great sphere. Any other connected component would be disjoint from it and therefore contained in one of its two open hemispheres. If a is the pole of that hemisphere, the coordinate function $h(\omega) = a \cdot \omega$ would be strictly positive on that component. For a minimal $(n-2)$-dimensional submanifold of the unit sphere, $\Delta_\Sigma h + (n-2)h = 0$: the Euclidean position vector has Laplacian $-(n-2)\omega$, since the spherical mean curvature is zero. Integration contradicts $h > 0$ and $n - 2 > 0$. There is therefore no other component, and the cone is a hyperplane.

It remains to treat the case $\phi = 0$ on all components. On the smooth part of C, the distributional derivative $e \cdot D\mathbf{1}_E$ is the inward normal component times perimeter measure, and hence zero. Off C the perimeter measure is zero by definition of support; the origin has zero perimeter measure as above. Thus $\partial_e \mathbf{1}_E = 0$ on all of $\mathbb{R}^n$ in distributions. Convolve with a compactly supported smooth approximate identity. Each convolution has zero directional derivative and is therefore invariant under every translation by te. For each fixed $t \in \mathbb{R}$, passage to the L^1_{loc} limit gives $\mathbf{1}_E(\,\cdot + te) = \mathbf{1}_E$ almost everywhere. Translation commutes with distributional differentiation and total variation, so the perimeter measure, and hence its closed support C, are exactly invariant under every such translation.

Closedness and conicality imply $0 \in C$, so $e \in C$. The hypersurface is smooth near e by hypothesis. Translating by $-e$ now shows that it is smooth near the origin as well. Let T be its tangent hyperplane at the origin. For each $x \in C \setminus \{0\}$, all points rx for $r > 0$ lie in C. Differentiability at the origin implies $\operatorname{dist}(rx, T) = o(r|x|)$ as $r \downarrow 0$. Dividing by r shows that $x \in T$. Hence $C \subset T$. Near the origin the smooth hypersurface C has the same dimension as

T and is contained in it, so it contains a relative neighborhood of the origin in T. Dilation then gives $T \subset C$. Thus $C = T$, completing the proof. □

Lemma 13.6 (Range of a bounded monotone cubic solution)**.** *Let $0 < s < 1$. If $u \in C^2(\mathbb{R}^8)$ is bounded and satisfies $(-\Delta)^s u = u - u^3$ with Fourier symbol $|\xi|^{2s}$ and $\partial_8 u > 0$, then $-1 < u < 1$ everywhere.*

Proof. We do not assume attainment of the supremum. Put $\psi(x) = (1+|x|^2)^{s/2}$. This barrier is admissible in the singular-integral formula although it is unbounded: it grows like $|x|^s$, and $s < 2s$ gives an integrable fractional tail. Its fractional Laplacian is bounded on $\mathbb{R}^8$. Indeed, write the operator with symmetric second differences and split its integral at $r_x = (1+|x|)/2$. On $|z| < r_x$ the Hessian estimate is $|D^2\psi(x+\tau z)| \le C(1+|x|)^{s-2}$ for $|\tau| \le 1$; integration gives a bound $C(1+|x|)^{-s}$. The map $x \mapsto (1+|x|^2)^{1/2}$ is one-Lipschitz, and $|a^s - b^s| \le |a-b|^s$ for $a, b \ge 0$. Thus $|\psi(x+z) - \psi(x)| \le |z|^s$, and integration on $|z| \ge r_x$ gives the same tail bound. Thus $|(-\Delta)^s \psi| \le C$. Because u is bounded and $\psi(x) \to \infty$, the function $u - \delta\psi$ attains a global maximum at some x_δ. Its symmetric-difference integral is finite by local C^2 regularity and the preceding tail bound. At that maximum its fractional Laplacian is nonnegative, and hence

$$u(x_\delta) - u(x_\delta)^3 \ge -C\delta.$$

For each fixed x, $u(x_\delta) \ge u(x) - \delta\psi(x) + \delta\psi(x_\delta) \ge u(x) - \delta\psi(x)$, so $u(x_\delta) \to \sup u$ as $\delta \downarrow 0$. A supremum greater than 1 contradicts the displayed inequality. Apply this argument to $-u$ for the lower bound. Finally strict monotonicity gives $-1 \le u(x - e_8) < u(x) < u(x + e_8) \le 1$ for every x, proving both strict inequalities. This argument uses no derivative bootstrap and is valid on the entire stated range $0 < s < 1$. □

Theorem 13.7 (Monotone global minimizers in dimension eight are layers)**.** *Let $1/2 < s < 1$. Suppose that $u \in C^2(\mathbb{R}^8)$ is bounded,*

$$(-\Delta)^s u = u - u^3 \quad \text{in } \mathbb{R}^8, \qquad \partial_8 u > 0,$$

where the operator has Fourier symbol $|\xi|^{2s}$. Suppose also that u is a global minimizer in the sense of definition 13.1. Then there are $e \in S^7$, $e_8 > 0$, $a \in \mathbb{R}$, and a strictly increasing classical solution $q : \mathbb{R} \to (-1, 1)$ such that

$$(-\partial_t^2)^s q = q - q^3, \qquad q(0) = 0, \qquad \lim_{t \to \pm\infty} q(t) = \pm 1, \qquad u(x) = q(e \cdot x - a).$$

The one-dimensional operator has the same Fourier normalization.

Proof. All energies and admissible perturbations are those of definition 13.1.

By lemma 13.6, $-1 < u < 1$. We use the following external phase-transition results. Write $c = c_{8,s}$ and

$$\widetilde{W} = 2W/c, \qquad K(v;\Omega) = \frac{1}{2} \iint_{Q_\Omega} \frac{|v(x) - v(y)|^2}{|x-y|^{8+2s}}\, dx\, dy.$$

Then $\mathcal{E} = (c/2)(K + \int \widetilde{W})$. The potential $\widetilde{W}$ is C^2, positive between its two zero minima ± 1, has zero derivative there, and has positive second derivative there. These are the hypotheses of Savin–Valdinoci, Density estimates for a variational model driven by the Gagliardo norm[13], arXiv:1007.2114v3, Theorems 1.3–1.4 and Corollary 1.7 (printed pp. 4–5). Our minimizing property supplies their competitor class: if $v = u$ outside a bounded domain and $|v| \le 1$ has finite localized energy, then $v - u$ is bounded, supported in a containing ball, and belongs to H^s by the squared-increment inequality and bounded support. Energy differences in that domain and the containing ball agree. Competitors with infinite energy require no comparison. Consequently, for $R \ge 1$,

$$\mathcal{E}(u; B_R) \le CR^7.$$

Furthermore, whenever $u(x_0) > \theta_1$ with $\theta_1 \in (-1,1)$, each $\theta_2 \in (-1,1)$ satisfies $|\{u > \theta_2\} \cap B_R(x_0)| \geq c_0 R^8$ for sufficiently large R. Apply this also to $-u$, whose energy is the same. Since $u(x_0) \in (-1,1)$, these two estimates show

$$\sup_{\mathbb{R}^8} u = 1, \qquad \inf_{\mathbb{R}^8} u = -1.$$

Continuity therefore gives a zero. Translate it to the origin for the rest of the proof.

The first blowdown is a perimeter minimizer, without any assertion yet that it is a cone. Specifically, put $u_R(x) = u(Rx)$. For $\varepsilon = R^{-1}$, their rescaled functional satisfies

$$\mathcal{F}_\varepsilon(v;\Omega) = \varepsilon^{2s-1} K(v;\Omega) + \varepsilon^{-1} \int_\Omega \widetilde{W}(v), \qquad \mathcal{F}_{1/R}(u_R; B_r) = \frac{2}{c} R^{-7} \mathcal{E}(u; B_{Rr}).$$

The compactness and minimizing-limit theorem of Savin–Valdinoci, Gamma-convergence for nonlocal phase transitions[12], arXiv:1007.1725v3, Theorem 1.3 (its compactness assertion and part (ii)), printed p. 3, applies on each fixed ball: the normalized energies are bounded by the preceding R^7 estimate, and $s > 1/2$. Their limit is a positive constant times classical perimeter. A diagonal subsequence $R_j \to \infty$ thus gives

$$u_{R_j} \longrightarrow 2\mathbf{1}_E - 1 \quad \text{in } L^1_{\mathrm{loc}}(\mathbb{R}^8),$$

where E minimizes perimeter against every compactly supported change. The lower-bound inequality in their Theorem 1.2(i) and the energy estimate give

$$P(E; B_r) \leq C_1 r^7 \qquad (r > 0).$$

For each fixed r, the displayed rescaling identity applies, and $R_j r \geq 1$ for sufficiently large j.

The two phase-density estimates with $u(0) = 0$, applied to u and $-u$ with thresholds $-1/2, 1/2$, imply

$$|E \cap B_r| \geq c_0 r^8, \qquad |B_r \setminus E| \geq c_0 r^8 \qquad (r > 0).$$

Indeed, the indicators of $\{u_{R_j} > 1/2\}$ converge in local L^1 to $\mathbf{1}_E$, and those of $\{u_{R_j} < -1/2\}$ to its complement; this follows directly from convergence to values ± 1. These inequalities in particular put the origin in the support of $|D\mathbf{1}_E|$: if that measure vanished in a ball, the characteristic function would be constant almost everywhere there.

For each $t \geq 0$, strict monotonicity before the limit gives $u_{R_j}(x + te_8) \geq u_{R_j}(x)$. Local L^1 convergence at the two translated arguments yields

$$\mathbf{1}_E(x + te_8) \geq \mathbf{1}_E(x) \quad \text{for almost every } x.$$

This is an assertion for every fixed real $t \geq 0$. Taking the density-one representative of E upgrades the corresponding inclusion $E + te_8 \subset E$ to an exact set inclusion for each t: the almost-everywhere inequality compares the measures of every pair of equally translated balls, so density one passes to the translated point. Equivalently, $\partial_8 \mathbf{1}_E$ is a nonnegative distribution, and therefore a nonnegative Radon measure. No global finite-valued graph representation of E is required.

We now take a second blowdown. We use classical minimizing-boundary compactness and regularity in Leon Simon, Introduction to Geometric Measure Theory[16], Chapter 7, Theorem 5.3 (printed p. 198) and Theorem 5.8 (printed pp. 200–201), and the stationary monotonicity identity in Chapter 4, section 3, (3.6) (printed p. 90). Its nonnegative remainder gives the equality case used below. In Euclidean space, a perimeter-minimizing set defines a minimizing boundary current. Indeed, a compactly supported integral filling is represented in full dimension by an integer-valued function of bounded variation. Adding that function to $\mathbf{1}_E$ represents the competing boundary. Truncating the sum to $[0,1]$ does not increase its total variation and, because the sum is integer-valued, produces the characteristic function of a set agreeing with E outside the filling's support. Perimeter minimality of E therefore

gives the current comparison. These results apply to all the dilations of E. Choose $\rho_k \to \infty$ and pass to a subsequence such that, for $E_k = \rho_k^{-1} E$,

$$\mathbf{1}_{E_k} \to \mathbf{1}_F \text{ in } L^1_{\mathrm{loc}}, \qquad |D\mathbf{1}_{E_k}| \rightharpoonup |D\mathbf{1}_F|.$$

The limit F again minimizes perimeter. The phase-density bounds pass to F, so $0 \in \operatorname{supp}|D\mathbf{1}_F|$ and that support is nonempty. Translation monotonicity also passes to F.

Here is why this second limit is conical. The ratio $r^{-7}P(E; B_r)$ is nondecreasing by stationarity and is bounded above by C_1, hence has a finite limit L at infinity. At every $r > 0$ for which $|D\mathbf{1}_F|(\partial B_r) = 0$, measure convergence gives

$$r^{-7}P(F; B_r) = \lim_k (r\rho_k)^{-7} P(E; B_{r\rho_k}) = L.$$

There are at most countably many exceptional radii; monotonicity then makes the ratio constant at every radius. Equality in the stationary monotonicity formula says that the radial normal component vanishes almost everywhere on the reduced boundary of F. In distributional form, $x \cdot D\mathbf{1}_F = 0$ away from the origin. This implies dilation invariance, including the Jacobian in the change of variables: with $x = e^t\omega$, test this distribution against $\psi(e^t\omega) = e^{-8t}\eta(t, \omega)$, where η is compactly supported and smooth. Since $\operatorname{div}(\psi x) = e^{-8t}\partial_t\eta$ and $dx = e^{8t}\, dt\, d\omega$, one obtains

$$\int_{S^7} \int_{\mathbb{R}} \mathbf{1}_F(e^t\omega)\, \partial_t\eta(t, \omega)\, dt\, d\omega = 0.$$

Consequently $\mathbf{1}_F(e^t\omega)$ is independent of t almost everywhere, and F is invariant under every positive dilation, up to null sets. Its density-one representative is dilation invariant as an actual set.

Let $C = \operatorname{supp}|D\mathbf{1}_F|$. The regularity theorem just cited, with hypersurface dimension seven and ambient dimension eight, makes its singular set locally finite. A nonzero singular point of a cone would generate an entire ray of singular points, contradicting local finiteness. Hence $C \setminus \{0\}$ is a smooth embedded minimal hypersurface without boundary, and F locally occupies one side. On this smooth part the inward normal ν satisfies

$$\partial_8 \mathbf{1}_F = (\nu \cdot e_8)|D\mathbf{1}_F|.$$

The left side is nonnegative by the inherited translation monotonicity. Continuity of the normal upgrades the resulting almost-everywhere inequality to $\nu \cdot e_8 \geq 0$ everywhere on the smooth part. All the hypotheses of lemma 13.5 now hold. It follows that $C = \{x \cdot \xi = 0\}$ for a unit vector ξ.

We next obtain slabs for the original zero set, retaining the two scales of the argument. The boundaries of E_k converge locally to C in the direction needed here:

$$\eta_k := \frac{1}{k} + \sup\{|z \cdot \xi| : z \in \operatorname{supp}|D\mathbf{1}_{E_k}| \cap \overline{B}_2\} \longrightarrow 0.$$

To justify this from the stated measure convergence, a point of a minimizing boundary has the perimeter lower bound $P(E_k; B_r(z)) \geq \omega_7 r^7$ by stationary monotonicity and unit density at reduced-boundary points, which are dense in the support. If a sequence of such points in $\overline{B}_2$ stayed a fixed distance from C, a subsequence would converge to a point off C. A small compact ball around that limit point has zero limiting perimeter measure but uniformly positive approximating measure, contradicting weak convergence. The supremum is over a nonempty compact set because the origin belongs to every support.

Corollary 1.7(iii) of the density-estimate paper gives the local uniform level-set inclusion. We justify the needed form with perimeter support directly. Otherwise some rescaled zero points in a fixed compact ball would converge to a point a positive distance from $\operatorname{supp}|D\mathbf{1}_E|$.

In a small ball about that point, $D\mathbf{1}_E = 0$, so $\mathbf{1}_E$ is constant almost everywhere. The two phase-density estimates, centered at the moving zero points, put a fixed positive fraction of both phases in a smaller fixed-radius ball for all large j. This contradicts local L^1 convergence to that constant. Thus $\{u_{R_j} = 0\}$ is contained in every prescribed neighborhood of the perimeter support on each fixed ball for all sufficiently large j. For each fixed $k \geq 2$, choose $j(k)$, increasing in k, such that $R_{j(k)} \geq k$ and

$$\{u_{R_{j(k)}} = 0\} \cap B_{\rho_k} \subset \{y : \operatorname{dist}(y, \operatorname{supp} |D\mathbf{1}_E|) < \rho_k/k\}.$$

If $u(R_{j(k)}\rho_k z) = 0$ and $|z| < 1$, the inclusion supplies $p \in \operatorname{supp} |D\mathbf{1}_E|$ with $|\rho_k z - p| < \rho_k/k$. In particular $p/\rho_k \in B_2$, and hence

$$|z \cdot \xi| < \eta_k + 1/k.$$

With

$$\ell_k = R_{j(k)}\rho_k, \qquad \theta_k = \ell_k(\eta_k + 2/k),$$

we have $\ell_k \to \infty$, $\theta_k/\ell_k \to 0$, and

$$\{u = 0\} \cap B_{\ell_k} \subset \{|x \cdot \xi| < \theta_k\}.$$

Apply Savin, Rigidity of minimizers in nonlocal phase transitions[11], arXiv:1610.09295v1, Theorem 1.1 and its precise sequential formulation in Theorem 6.2 (printed pp. 1 and 16). Its variational hypothesis is our global minimality, its exponent range is $1/2 < s < 1$, and its geometric hypothesis is the preceding slab inclusion. In Savin's pair-energy convention the kinetic coefficient is $1/4$, so his potential here is W/c and his energy is exactly $\mathcal{E}/c$; this differs from the $\widetilde{W} = 2W/c$ used with K above. For the extension formulation of Theorem 6.2, the pair-energy/minimizing-extension equivalence on printed p. 3 applies. After normalizing its Dirichlet coefficient, the boundary potential is W/d_p, with $p = 2s$. This is the same cubic extension normalization as in the proof of proposition 6.3. The fixed-dilation adjustment verified there therefore also meets Savin's simplifying unit-scale convexity convention on printed p. 4. A fixed dilation preserves global minimality and the vanishing slab ratios, and can be undone after the conclusion. The theorem gives $u(x) = g(e \cdot x)$ for some $e \in S^7$ and a classical function g.

Since $\partial_8 u = e_8 g'(e \cdot x) > 0$, $e_8 \neq 0$. Reverse e and the parameter of g if necessary, obtaining $e_8 > 0$ and $g' > 0$. The one-dimensional fractional equation for g has exactly the stated Fourier normalization. For example, integrate the symmetric second-difference kernel in the seven perpendicular variables; the resulting kernel constant is $c_{1,s}$, because the original and reduced operators both have multiplier $|\xi|^{2s}$ on functions of the single unit coordinate. The integrals are justified near zero by $g \in C^2$ and at infinity by its boundedness. Monotonicity of g and the already established global supremum and infimum show that its limits are 1 and -1. It has a unique zero b. Put $q(t) = g(t + b)$. Undoing the earlier spatial translation changes only the constant a, and gives $u(x) = q(e \cdot x - a)$ with all the required properties. □

Corollary 13.8 (Outward minimizing endpoints suffice for the dimension-eight conclusion)**.** *Let $1/2 < s < 1$, and let $u \in C^2(\mathbb{R}^8)$ be a bounded solution of $(-\Delta)^s u = u - u^3$ with $\partial_8 u > 0$, in the Fourier normalization $|\xi|^{2s}$. Let*

$$v^{\pm}(x') = \lim_{t \to \pm\infty} u(x', t), \qquad V^{\pm}(x', x_8) = v^{\pm}(x').$$

Suppose $V^{\pm}$ are bounded C^2 solutions of the same equation, and, in the sense of definition 13.1, V^+ is minimizing from above and V^- is minimizing from below. Then there are

$e \in S^7$ with $e_8 > 0$, $a \in \mathbb{R}$, and a strictly increasing classical solution $q : \mathbb{R} \to (-1,1)$ such that

$$(-\partial_t^2)^s q = q - q^3, \qquad q(0) = 0, \qquad \lim_{t\to\pm\infty} q(t) = \pm 1, \qquad u(x) = q(e \cdot x - a).$$

The one-dimensional operator also has Fourier symbol $|\xi|^{2s}$.

Proof. By lemma 13.6, $-1 < u < 1$. The endpoint hypotheses and proposition 13.4 therefore imply that u is a global minimizer in the sense of definition 13.1. All hypotheses of theorem 13.7 now hold, and its conclusion is the asserted one. □

Lemma 13.9 (Directional endpoints are stable solutions with the reduced Fourier normalization)**.** *Let $1/2 < s < 1$, and let v be a bounded classical solution in $\mathbb{R}^8$ of $(-\Delta)^s v = v - v^3$ with $\partial_8 v > 0$. Then*

$$u_\pm(x') = \lim_{t\to\pm\infty} v(x', t)$$

are bounded smooth stable solutions of $(-\Delta_{\mathbb{R}^7})^s u = u - u^3$, with Fourier symbol $|\xi'|^{2s}$. Their lifts $U_\pm(x', x_8) = u_\pm(x')$ solve the eight-dimensional equation, and $U_- < v < U_+$ pointwise.

Proof. Put $p = 2s \in (1,2)$. The range bound lemma 13.6 gives $-1 < v < 1$, so monotonicity supplies the finite limits $u_\pm$. The all-dimensional uniform regularity and translate-closure conclusions of lemma 2.1 apply at this p. Thus every sequence of vertical translates has a locally smoothly convergent subsequence. For translates tending in either direction, the pointwise monotone limit is necessarily $U_\pm(x', t) = u_\pm(x')$. Uniqueness of that subsequential limit proves local C^k convergence along the full translating family for every k. The same closure result shows that each $U_\pm$ is a bounded smooth solution of the eight-dimensional equation.

We record the normalized dimensional reduction in a form that will also apply to later invariant lifts. For any integers $m > n \geq 1$ and $0 < s < 1$, the kernel constants for the Fourier symbol $|\xi|^{2s}$ are

$$c_{m,s} = \frac{4^s \Gamma(m/2 + s)}{\pi^{m/2} |\Gamma(-s)|}.$$

Polar coordinates and the beta integral, with $k = m - n$, give

$$\int_{\mathbb{R}^k} (|h|^2 + |z|^2)^{-(m+2s)/2} dz = \pi^{k/2} \frac{\Gamma((n+2s)/2)}{\Gamma((m+2s)/2)} |h|^{-n-2s} \qquad (h \neq 0).$$

Consequently the exact marginal identity is

$$c_{m,s} \int_{\mathbb{R}^{m-n}} (|h|^2 + |z|^2)^{-(m+2s)/2} dz = c_{n,s} |h|^{-n-2s}. \tag{7}$$

For a bounded C^2 function w on $\mathbb{R}^n$, put $\widetilde{w}(x, z) = w(x)$ on $\mathbb{R}^m$. At a fixed x, its symmetric second difference

$$\delta_h^2 w(x) = 2w(x) - w(x + h) - w(x - h)$$

is bounded by $C_x |h|^2$ for $|h| \leq 1$ and by $4\|w\|_\infty$ otherwise. Therefore its absolute integral against $|h|^{-n-2s}$ is finite: the two radial bounds are $\int_0^1 r^{1-2s} dr$ and $\int_1^\infty r^{-1-2s} dr$. Tonelli for its absolute value and (7) justify Fubini in the higher-dimensional symmetric formula, proving pointwise

$$\begin{aligned} (-\Delta_{\mathbb{R}^m})^s \widetilde{w}(x, z) &= \frac{c_{m,s}}{2} \int_{\mathbb{R}^n} \int_{\mathbb{R}^{m-n}} \frac{\delta_h^2 w(x)}{(|h|^2 + |\zeta|^2)^{(m+2s)/2}} d\zeta dh \\ &= (-\Delta_{\mathbb{R}^n})^s w(x). \end{aligned} \tag{8}$$

This calculation requires no integrable Fourier transform of w. Apply it with $m = 8, n = 7$ to $U_\pm$. It gives the claimed seven-dimensional equation with exactly the reduced Fourier normalization. Moreover

$$U_-(x',t) \le v(x',t-1) < v(x',t) < v(x',t+1) \le U_+(x',t),$$

so the endpoint ordering is strict at every point.

For stability, $w = \partial_8 v > 0$ solves the differentiated equation $(-\Delta)^s w = (1-3v^2)w$; its smoothness and bounded derivatives are already supplied by lemma 2.1. Fix a real compact smooth ϕ, and test this Jacobi equation with $\zeta = \phi^2/w \in C_c^\infty$. Such a test is legitimate because w has a positive minimum on a compact neighborhood of the support; no global positive lower bound is used. The pointwise algebra

$$(\phi_X - \phi_Y)^2 - (w_X - w_Y)\left(\frac{\phi_X^2}{w_X} - \frac{\phi_Y^2}{w_Y}\right) = w_X w_Y \left(\frac{\phi_X}{w_X} - \frac{\phi_Y}{w_Y}\right)^2$$

and symmetric integration give

$$Q_v(\phi) = \frac{c_{8,s}}{2} \iint w(X)w(Y)\left(\frac{\phi(X)}{w(X)} - \frac{\phi(Y)}{w(Y)}\right)^2 |X-Y|^{-8-2s} dX dY \ge 0,$$

with the quadratic form of definition 13.1. Near the diagonal, on that compact neighborhood, all differences are controlled by their first derivatives, giving an integrable $|X-Y|^{2-8-2s}$ bound. When one point is outside the neighborhood and the other is in $\operatorname{supp}\phi$, the nonzero ground-state term is $\phi(X)^2 w(Y)/w(X)$, up to exchanging the points. Boundedness of w, its positive minimum on the support, and the integrable kernel tail justify exhaustion and the displayed identity. Where both test values vanish the integrand is zero. Thus v, and every translate of it, is stable. Local smooth convergence passes the potential term in the fixed-test quadratic form to each $U_\pm$, while its kinetic term is unchanged. Hence both lifts are stable in $\mathbb{R}^8$.

We verify descent rather than presuming it. For $\phi \in C_c^\infty(\mathbb{R}^7)$ and $\chi \in C_c^\infty(\mathbb{R})$ nonzero, test stability of $U_\pm$ with $\phi(x')\chi(t/L)$ and divide by $L\|\chi\|_2^2$. Plancherel expresses the kinetic part as

$$\frac{1}{\|\chi\|_2^2} \int_{\mathbb{R}^7} \int_{\mathbb{R}} (|\xi'|^2 + |\tau|^2/L^2)^s |\widehat{\phi}(\xi')|^2 |\widehat{\chi}(\tau)|^2 d\tau d\xi'.$$

It converges to the seven-dimensional kinetic form by dominated convergence; for $L \ge 1$, the multiplier is bounded by $|\xi'|^{2s} + |\tau|^{2s}$, whose product with the two Schwartz squares is integrable. The potential term is exactly the seven-dimensional one after division. The resulting limiting inequality is the required stability. □

Lemma 13.10 (Every well and planar layer minimizes the full ambient nonlocal energy)**.** *For $1/2 < s < 1$ and any $n \ge 1$, the constants ± 1 and every translated rotated increasing one-dimensional layer, considered in $\mathbb{R}^n$, are global minimizers in the sense of definition 13.1.*

Proof. Use the potential, normalized kernel, localized pair set, and admissible perturbations of definition 13.1. Here an increasing one-dimensional layer means a bounded classical solution $q : \mathbb{R} \to (-1,1)$ of

$$(-\partial_t^2)^s q = q - q^3, \qquad q'(t) > 0, \qquad \lim_{t\to-\infty} q(t) = -1, \qquad \lim_{t\to+\infty} q(t) = 1,$$

with Fourier symbol $|\tau|^{2s}$. We may center it by $q(0) = 0$, absorbing its unique zero into the translation parameter. Write $u(X) = q(e \cdot X - a)$, where $|e| = 1$. A well has zero energy in every ball, whereas every energy integrand is nonnegative; this proves its assertion.

We check the equation and the hypotheses for the layer. Set $p = 2s \in (1, 2)$. The constant $c_{n,s}$ of definition 13.1 is exactly the earlier $c_{n,p}$, since both operators have Fourier symbol $|\xi|^p$. Apply lemma 2.1 in dimension one to q: it is smooth with uniformly bounded derivatives of every fixed order. In particular $\|q'\|_\infty + \|q''\|_\infty < \infty$. Rotation reduces the equation to $u(X) = q(X_n - a)$. For $n > 1$, apply the normalized marginal identity (7) and its operator consequence (8), proved in lemma 13.9, with higher dimension n and lower dimension one. Their symmetric-second-difference argument is justified by the bounded second derivative near zero and boundedness at infinity. It gives $(-\Delta_{\mathbb{R}^n})^s u = u - u^3$ with exactly the normalization of definition 13.1. For $n = 1$ this is the defining equation for q.

The translates $u^t(X) = q(X_n - a + t)$ form a C^2 field of classical solutions with $\partial_t u^t = q'(X_n - a + t) > 0$ and

$$\sup_{X,t}(|u^t(X)| + |\partial_t u^t(X)|) \le 1 + \|q'\|_\infty < \infty.$$

Take $F = -W \in C^\infty(\mathbb{R})$. Apply Corollary 1.4 of X. Cabré, I. U. Erneta and J.-C. Felipe-Navarro, *A Weierstrass extremal field theory for the fractional Laplacian*[5], arXiv:2211.16536v2, PDF page 9, (1.13). Its energy (1.8), with coefficient $c_{n,s}/4$ and potential $-F = W$, is exactly ours. The corollary compares u in every bounded ball B with bounded continuous competitors w satisfying $-1 \le w \le 1$ and $w = u$ outside B.

To check the endpoint provision explicitly, use $w_\theta = (1 - \theta)u + \theta w$, $0 < \theta < 1$, as in that corollary's proof, PDF page 29, (4.20). Then $-1 < w_\theta < 1$, so its graph lies in the translation field's region. For finite-energy w, the quadratic kinetic term and bounded potential give $\mathcal{E}(w_\theta; B) \to \mathcal{E}(w; B)$ as $\theta \uparrow 1$; infinite energy is immediate. Thus competitors attaining either endpoint are included.

We extend that inequality to the exact admissible class. Fix a ball Ω and a bounded $h \in H^s(\mathbb{R}^n)$ with $h = 0$ almost everywhere on Ω^c. Put $v = u + h$. The layer has finite energy in every ball: bounded local gradients make the near-diagonal kinetic integral finite, and boundedness makes its tail finite. The same holds for v, by the H^s bound on h and the inequality $(A + B)^2 \le 2A^2 + 2B^2$; its potential is integrable on a ball because v is bounded.

Let $T(r) = \max\{-1, \min\{1, r\}\}$ and put $v^* = T(v)$. Pointwise,

$$|T(r) - T(t)| \le |r - t|, \qquad W(T(r)) \le W(r).$$

Also $T(u) = u$. Thus $v^* = u$ outside Ω and $\mathcal{E}(v^*; \Omega) \le \mathcal{E}(v; \Omega)$. To prove that $h^* = v^* - u$ really belongs to H^s, choose $\eta \in C_c^\infty(\mathbb{R}^n)$ with $0 \le \eta \le 1$, equal to one on a neighborhood of $\overline{\Omega}$. The smooth compactly supported function ηu belongs to H^s, and

$$h^* = T(\eta u + h) - \eta u.$$

The identity holds inside Ω because $\eta = 1$ there, and outside it because $h = 0$ and $|\eta u| \le 1$. A Lipschitz composition fixing zero preserves H^s for $0 < s < 1$, directly by its L^2 and difference-quotient bounds. Hence $h^* \in H^s$, with support in $\overline{\Omega}$.

Choose a larger ball $B \supset \overline{\Omega}$ and a nonnegative standard mollifier. Set $h_j = \rho_{\varepsilon_j} * h^*$ with $\varepsilon_j \downarrow 0$ small enough that every h_j has support in a single compact subset of B. Then $h_j \to h^*$ strongly in H^s, and $\|h_j\|_\infty \le \|h^*\|_\infty$. The functions

$$w_j = T(u + h_j)$$

are bounded continuous competitors in $[-1, 1]$ and equal u outside B. Their perturbations converge strongly in H^s to h^*. Here is a justification that does not require the layer itself to lie in global H^s. Choose $\widetilde{\eta} \in C_c^\infty$ with $0 \le \widetilde{\eta} \le 1$, equal to one on a neighborhood of the common supports of all h_j and h^*. Then

$$w_j - u = T(\widetilde{\eta} u + h_j) - \widetilde{\eta} u, \qquad h^* = T(\widetilde{\eta} u + h^*) - \widetilde{\eta} u.$$

The Lipschitz map T is strongly continuous on H^s. For completeness, if $z_j \to z$ in H^s, the L^2 conclusion is its contraction property. Along an almost-everywhere convergent subsequence, the fractional differences of $T(z_j) - T(z)$ tend to zero almost everywhere on pairs. Their squares are bounded by

$$8|z(X) - z(Y)|^2 + 8|(z_j - z)(X) - (z_j - z)(Y)|^2.$$

After division by $|X-Y|^{n+2s}$ the first term is integrable and the second has integral tending to zero. Dominated truncation therefore proves convergence of the seminorm. The subsequence criterion gives the assertion for the full sequence. Apply it to $z_j = \widetilde{\eta}u + h_j$ and $z = \widetilde{\eta}u + h^*$.

It follows that the kinetic energies of w_j on Q_B converge to that of v^*: use the strong global H^s convergence of their differences and Cauchy–Schwarz against the finite-energy difference quotient of v^*. The potentials converge as well, since W is Lipschitz on $[-1, 1]$ and $w_j - v^* \to 0$ in $L^2(B)$. Apply the continuous minimizing inequality in B and pass to the limit to obtain

$$\mathcal{E}(u; B) \le \mathcal{E}(v^*; B).$$

Finally, $v^* = u$ outside Ω, so the energy differences in B and in Ω are identical. Indeed every pair added to Q_Ω when forming Q_B has both endpoints outside Ω, where the two functions agree, and the added potential difference is zero. Hence

$$\mathcal{E}(u; \Omega) \le \mathcal{E}(v^*; \Omega) \le \mathcal{E}(v; \Omega).$$

This holds for every admissible perturbation from definition 13.1, and proves the global minimizing property. Rotation invariance restores arbitrary e. □

Proposition 13.11 (Seven-dimensional stable classification closes the monotone problem through dimension eight)**.** *Fix $s \in (1/2, 1)$. Assume every bounded stable classical cubic solution u in $\mathbb{R}^7$ satisfying an all-center estimate $\int_{B_R(z)} |Du| \le C_u R^6$ is a well or a planar layer. Assume also that every bounded stable classical cubic solution in $\mathbb{R}^7$ does satisfy this estimate. Then every bounded classical cubic solution v in $\mathbb{R}^N$, $1 \le N \le 8$, with $\partial_N v > 0$ admits $v(x) = q(e \cdot x - a)$ for $e \in S^{N-1}$ with $e_N > 0$, $a \in \mathbb{R}$, and a strictly increasing centered profile q satisfying $(-\partial_t^2)^s q = q - q^3$, $q(0) = 0$, and $\lim_{t \to \pm\infty} q(t) = \pm 1$, with Fourier symbol $|\xi|^{2s}$.*

Proof. We use the classification terminology in the precise sense covered by lemma 13.10: a well is one of the constants $+1, -1$, and a planar layer is $q(e \cdot x - a)$ with $|e| = 1$, where q solves the normalized one-dimensional cubic equation, is strictly increasing, has limits $-1, +1$, and is centered by $q(0) = 0$. A reversal is included by reversing e.

Put $p = 2s \in (1, 2)$. By lemma 2.1, every bounded classical cubic solution in each dimension used below is smooth, with derivative bounds uniform under translation. In particular the C^2 regularity needed for the endpoint, comparison and lifting arguments is already available.

First suppose $N = 8$. By lemma 13.9 the directional endpoints are bounded smooth stable solutions in $\mathbb{R}^7$, their lifts solve the same normalized eight-dimensional equation, and those lifts lie strictly below and above v. The assumed all-center growth and stable classification apply to each endpoint, so each is one of the wells or layers just specified. An endpoint layer $q(e' \cdot x' - a)$ lifts to the same layer with ambient unit direction $(e', 0) \in \mathbb{R}^8$, while a well remains a well. By lemma 13.10 both lifted endpoints are global eight-dimensional minimizers for the energy of definition 13.1. In particular the upper endpoint is minimizing from above and the lower endpoint is minimizing from below. The range bound lemma 13.6 gives $-1 < v < 1$. These C^2 bounds, strict monotonicity, and the endpoint equations and smoothness in lemma 13.9 verify all hypotheses of proposition 13.4. Hence v is a global minimizer. Apply theorem 13.7 to this bounded C^2 solution in dimension eight. Its conclusion

is exactly the required representation, including $e_8 > 0$, strict increase, centering, limits ± 1, and Fourier symbol $|\xi|^{2s}$.

For $1 \le N < 8$, write $k = 8 - N$ and first arrange coordinates as $(x, z) \in \mathbb{R}^N \times \mathbb{R}^k$. The normalized marginal identity (7) and its operator consequence (8), proved with symmetric second differences in lemma 13.9, apply with $m = 8$ and $n = N$. Thus for every bounded C^2 function w and its constant lift $\widetilde{w}(x, z) = w(x)$,

$$(-\Delta_{\mathbb{R}^8})^s \widetilde{w}(x, z) = (-\Delta_{\mathbb{R}^N})^s w(x).$$

The regularity just cited applies to v, so the identity holds classically for its lift with the same Fourier normalization.

Permute the eight coordinates so that the original monotonicity coordinate x_N becomes X_8. The preceding identity and rotation invariance show that

$$\widetilde{v}(X_1, \dots, X_8) = v(X_1, \dots, X_{N-1}, X_8)$$

solves the same normalized eight-dimensional cubic equation, is bounded and C^2, and satisfies $\partial_8 \widetilde{v} > 0$. Apply the eight-dimensional conclusion already proved. Thus $\widetilde{v} = q(e \cdot X - a)$ with q strictly increasing and $e_8 > 0$. Differentiating in the actual monotonicity coordinate gives $0 < \partial_8 \widetilde{v}(X) = e_8 q'(e \cdot X - a)$. The affine argument ranges over all of $\mathbb{R}$ because $|e| = 1$, so $q'(t) > 0$ for every t. For each added coordinate X_i, differentiation gives $0 = e_i q'(e \cdot X - a)$, so $e_i = 0$. The remaining components form a unit vector in the original N variables, with positive final component. Restriction yields the stated representation. The reduction includes $N = 1$, since (8) also covers $n = 1$. □

Proof of theorem 1.1. Fix the exponent s of the main theorem and put $p = 2s \in (1, 2)$. In dimension seven the assumed estimate is precisely the all-center bound with sheet dimension $d = 6$. Thus theorem 12.3 supplies the stable seven-dimensional classification required by proposition 13.11. The second hypothesis of that proposition is exactly the assumed sharp growth bound in dimension seven. Apply it to every $1 \le N \le 8$. Its conclusion gives the asserted unit direction, positive final component, scalar translation, centered strictly increasing profile, reduced Fourier normalization, and both well limits. No extra directional-limit hypothesis is introduced. □

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